

Integrated Science

Basic 8

Complete BECE Preparation
Examination Papers & Marking Schemes

Beacon Educational Consult

2026 Edition

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Preface

This comprehensive Integrated Science textbook for Basic 8 has been carefully prepared to support Junior High School students in their preparation for the Basic Education Certificate Examination (BECE).

The book contains:

- Multiple choice questions with complete answer keys
- Theory questions with detailed marking schemes
- Curriculum-aligned content following the 2019 Standards-Based Curriculum
- Practice papers from BECE Mock examinations

Each question has been verified for accuracy and aligned with the appropriate curriculum objectives. The marking schemes provide detailed explanations to help students understand the reasoning behind each answer.

We hope this book will be a valuable resource in your academic journey.

— Beacon Educational Consult

Accra, Ghana

2026

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BECE Mock Examinations (22 Papers)

Papers 1-22: BECE Mock Examinations

Note: Use the running headers at the top of each page to navigate the book. Each examination paper begins with a title page showing the paper number and topic.

BEACON EDUCATIONAL SERIES

Integrated Science for Basic 8

A Comprehensive Question Bank
with Marking Schemes

Curriculum-Aligned Practice Papers
for Basic 8 (JHS 2) Students and Teachers

Beacon Educational Consult
2026 Edition

Integrated Science for Basic 8
First Edition, 2026

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This book is designed to support the Ghana Education Service (GES) curriculum for Junior High School. It is intended as a supplementary resource for teachers and students.

The questions in this book are original practice materials created for educational purposes. They are not official BECE or GES examination questions.

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PREFACE

This book has been developed to support Junior High School Basic 8 (JHS 2) students and teachers in their study of Integrated Science, as prescribed by the Ghana Education Service (GES) and the National Council for Curriculum and Assessment (NaCCA).

The book contains a comprehensive collection of practice questions organised according to the curriculum's strand and sub-strand structure. Each paper includes both multiple-choice questions (Section A) and structured theory questions (Section B), reflecting the format of BECE examinations.

Key Features

- Curriculum-Aligned Content — All questions mapped to specific content standards.
- Comprehensive Coverage — Topic-based papers covering all major strands.
- Detailed Marking Schemes — Complete answer keys with method marks and marking guidance.
- Professional Diagrams — Clear, scientifically accurate illustrations.
- Cognitive Demand — Questions spanning all levels of Bloom's taxonomy.

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ANSWER KEY AND MARKING SCHEMES

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Paper 1: Agriculture

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. An agricultural tool is best described as
 - A. a machine that works without any person operating it
 - B. a chemical sprayed on crops to kill pests
 - C. a simple implement used by hand to carry out work on a farm or garden
 - D. a crop grown for sale in the market
2. Which of the following is an agricultural tool?
 - A. Cutlass
 - B. Fertiliser bag
 - C. Maize cob
 - D. Poultry feed
3. The tool with a long wooden handle and a broad metal blade fixed at an angle, used mainly for tilling the soil, is the
 - A. watering can
 - B. hoe
 - C. cutlass
 - D. rake
4. A tool with a head carrying several short teeth, used for levelling a seedbed and gathering dry leaves, is the
 - A. shovel
 - B. trowel
 - C. rake
 - D. garden fork
5. The tool used to carry water and sprinkle it gently over seedlings is the
 - A. head pan
 - B. knapsack sprayer
 - C. watering can
 - D. wheelbarrow
6. A garden fork is a tool with
 - A. several strong prongs used to loosen and turn the soil
 - B. a flat blade used to cut grass
 - C. a wheel used to carry loads
 - D. a closed container used to store water
7. The small hand tool with a short handle and a scoop-shaped blade, used for planting seedlings in a nursery, is the
 - A. sickle
 - B. trowel
 - C. axe
 - D. mattock
8. A wheelbarrow is used on a farm mainly for
 - A. spraying chemicals on crops
 - B. carrying manure, soil and farm produce from one place to another
 - C. measuring the size of the plot
 - D. cutting down tall grass
9. Which of the following tools is used for pruning the branches of a hedge or a shrub?
 - A. Watering can

- B. Head pan
- C. Secateurs (pruning shears)
- D. Garden line

11. A knapsack sprayer is used on a farm to

- A. carry seedlings to the field
- B. apply liquid chemicals such as weedicides and insecticides to crops
- C. loosen the soil before planting
- D. harvest ripe fruits from a tall tree

10. The tool used for breaking hard ground and digging out tree stumps is the

- A. garden line
- B. trowel
- C. pickaxe (mattock)
- D. rake

- 13.** Making ridges and mounds on a cleared plot is done with a
- hoe
 - head pan
 - secateurs
 - rake
- 15.** A garden line (planting line) is used to
- carry manure to the plot
 - sharpen the blade of the cutlass
 - water the young seedlings
 - mark out straight rows for planting
- 17.** Transplanting young seedlings from the nursery to the field is best done with a
- knapsack sprayer
 - pickaxe
 - wheelbarrow
 - hand trowel
- 19.** A dibber is used on a farm to
- make holes in the soil for planting seeds or seedlings
 - carry water to the nursery
 - weigh the harvested produce
 - cut the branches of a hedge
- 21.** When a hoe is used to till the soil, the pupil should
- drag the blade behind while walking backwards
 - hold the blade and push the handle into the soil
 - hold the handle firmly with both hands and swing it to bring the blade into the soil
 - throw the hoe onto the ground from a distance
- 23.** Before using a cutlass to clear weeds, the farmer should first
- heat the blade over a fire
 - soak it in water overnight
 - sharpen it and check that the handle is firm
 - paint the blade with oil paint
- 12.** Clearing tall weeds and grass from a plot before planting is best done with a
- watering can
 - cutlass
 - garden line
 - trowel
- 14.** Which activity is a rake used for?
- Measuring the spacing between rows
 - Spraying insecticide on vegetables
 - Cutting the stems of ripe plantain
 - Levelling the seedbed and gathering dry leaves and stones
- 16.** The tool used to cut down a tree on a farm is the
- sprinkler
 - axe
 - rake
 - trowel
- 18.** Harvesting cereals such as rice by hand on a small farm is done with a
- dibber
 - sickle
 - head pan
 - watering can
- 20.** Which pair of tools would a pupil need in order to clear a weedy plot and then till it?
- Watering can and rake
 - Cutlass and hoe
 - Secateurs and trowel
 - Garden line and head pan
- 22.** A watering can is fitted with a rose (sprinkler head) so that the water
- falls gently in many fine drops and does not wash away the soil and seedlings
 - becomes warm before it reaches the plants
 - becomes cleaner as it leaves the can
 - comes out in one strong jet to reach far rows

- 25.** A pupil who is passing a tool with a sharp edge to another pupil should
- throw it across to the other pupil
 - hold the handle and point the blade forward
 - hold the blade and offer the handle first
 - place it on the ground and walk away
- 27.** Watering a vegetable bed is best done
- immediately after a heavy rain
 - at noon when the sun is hottest
 - only when the leaves have turned yellow
 - in the early morning or late evening
- 29.** A plot is covered with tall weeds. The most suitable tool for the first stage of the work is the
- cutlass
 - trowel
 - watering can
 - rake
- 31.** Manure has to be moved from the compost heap to a bed thirty metres away. The most suitable tool is the
- garden line
 - wheelbarrow
 - dibber
 - trowel
- 33.** Choosing the right tool for a farm activity is important because it
- makes the work easier, faster and safer, and protects the crops and the tool
 - makes the work last much longer
 - increases the price of the produce at the market
 - removes the need to water the crops
- 24.** The correct way to carry a cutlass while walking to the farm is
- by throwing it ahead and picking it up again
 - with the blade held in front of the chest
 - with the blade pointing upwards over the shoulder
 - with the blade pointing downwards and away from the body
- 26.** Which protective wear is most suitable for pupils working in a school garden?
- A woollen cardigan and a cap
 - Sandals and a wristwatch
 - Sunglasses and a necktie
 - Boots and gloves
- 28.** A pupil uses a garden fork rather than a hoe to turn a bed of heavy, wet soil because the fork
- has prongs that enter the heavy soil easily and lift it without packing it together
 - cannot be used on dry soil at all
 - does not need to be cleaned after use
 - is lighter than every other garden tool
- 30.** The soil of a nursery bed has been dug and must now be made level and fine before sowing. The most suitable tool is the
- pickaxe
 - rake
 - knapsack sprayer
 - secateurs
- 32.** A hedge around the school garden has grown tall and uneven. The most suitable tool for trimming it is the
- rake
 - pickaxe
 - watering can
 - pair of secateurs (shears)
- 34.** After using a hoe on a muddy plot, a pupil should
- wash off the soil, dry the blade and store the tool in the shed
 - leave it in the field until the next

lesson

- C. soak it in a bucket of water till the next day
- D. bury it in the moist soil to keep it cool

35. Metal parts of agricultural tools are rubbed with oil after cleaning in order to

- A. change their colour for identification
- B. make them blunt for safety
- C. prevent them from rusting

D. make them heavier to use

37. A cutting tool such as a cutlass is sharpened regularly because a blunt tool

- A. makes the soil too soft to plant
- B. rusts faster than a sharp one
- C. cuts the crops too quickly
- D. needs more effort to use and can slip and cause an injury

39. Which of the following is a safety rule in a school garden?

- A. Never play with or run while carrying a sharp tool
- B. Use a cutlass to dig holes for seedlings
- C. Leave the tools lying with their blades facing upwards
- D. Share one pair of gloves among all the pupils

36. Agricultural tools are kept in a dry store or shed mainly to

- A. hide them from the pupils who need them
- B. make them sharper without filing
- C. allow the handles to grow longer
- D. protect them from rain and dew, which cause rust, and to keep them safe

38. A wooden handle that has become loose on a hoe should be

- A. used carefully as it is until it breaks
- B. soaked in water so that it swells permanently
- C. fixed firmly in the head again before the tool is used
- D. removed completely so that the head is used alone

40. A farmer who cares for his tools well benefits because the tools

- A. can be used without any skill
- B. no longer need to be sharpened at all
- C. become heavier each season
- D. last longer, work better and cost less to replace

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Write the names of tools in full. Do not use sketches in place of the names asked for.

Question 7

- (a) Use a pencil and a ruler where a plan or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 2: Animal Production

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Animal feed is
 - A. the water given to farm animals only
 - B. the food given to farm animals to keep them alive, healthy and productive
 - C. the medicine given to sick farm animals
 - D. the house in which farm animals are kept
2. Animals that feed only on plant material are described as
 - A. carnivores
 - B. omnivores
 - C. herbivores
 - D. decomposers
3. Which of the following farm animals is a herbivore?
 - A. Hawk
 - B. Cat
 - C. Cattle
 - D. Dog
4. Pigs and fowls eat both plant and animal material. They are therefore
 - A. parasites
 - B. herbivores
 - C. carnivores
 - D. omnivores
5. The main feed of cattle, sheep and goats on a Ghanaian farm is
 - A. grass and other forage
 - B. maize grains only
 - C. milk only
 - D. fish meal only
6. Which of the following is the most suitable feed for layers on a poultry farm?
 - A. Pig grower pellets
 - B. Broiler starter mash
 - C. Cattle salt lick
 - D. Layer mash
7. The feed given to young chicks within their first few weeks is called
 - A. silage
 - B. chick mash (starter mash)
 - C. finisher pellets
 - D. layer mash
8. Grass that has been cut and dried for storage as animal feed is called
 - A. silage
 - B. concentrate
 - C. mash
 - D. hay
9. Green fodder that is stored in a pit or silo where it ferments is known as
 - A. hay
 - B. bran
 - C. silage
 - D. pellets
10. Feeds such as fish meal, groundnut cake and soya bean meal are added to animal feed mainly to supply
 - A. protein
 - B. fibre
 - C. vitamin C
 - D. water
11. Maize, cassava and rice bran are included in animal feed mainly to supply
 - A. carbohydrates for energy

- B. proteins for growth
- C. roughage for chewing
- D. minerals for strong bones

- 12.** Bone meal and oyster shell are added to poultry feed mainly to supply
- A. proteins
 - B. minerals such as calcium and phosphorus

- C. carbohydrates
D. fats and oils
- 13.** A hen that is laying eggs needs plenty of calcium in its feed mainly for the formation of
A. bright feathers
B. a loud voice
C. sharp claws
D. strong egg shells
- 15.** Vitamins are needed in animal feed mainly to
A. provide the greatest amount of energy
B. build the bones and the teeth
C. protect animals from diseases and keep the body working properly
D. add weight to the animal quickly
- 17.** A concentrate is a feed that
A. contains a high proportion of nutrients in a small quantity
B. contains mostly water and little food value
C. is given to animals only when they are sick
D. is made only of dried grass
- 19.** The label on a bag of feed shows the
A. price of meat in the market
B. names of all the farm workers
C. number of animals on the farm
D. proportions of the different nutrients in the feed
- 21.** A 50 kg bag of feed contains 22% protein. The mass of protein in the bag is
A. 11 kg
B. 50 kg
C. 2.2 kg
D. 22 kg
- 23.** Which of the following is NOT a nutrient found in animal feed?
A. Vitamin
B. Protein
C. Soil
D. Mineral salt
- 14.** Which nutrient is needed in the largest proportion for the growth of body tissues and muscle in young animals?
A. Roughage
B. Protein
C. Mineral salt
D. Vitamin
- 16.** Roughage (fibre) in the feed of cattle is important because it
A. helps in digestion and the proper working of the gut
B. supplies all the protein the animal needs
C. is stored in the body as fat
D. kills the germs inside the stomach
- 18.** Which of the following pairs of feed and animal is correctly matched?
A. Silage - day-old chick
B. Layer mash - laying hen
C. Milk only - adult goat
D. Fish meal only - cattle
- 20.** A farmer should read the label on a bag of feed before buying it mainly in order to
A. find out the colour of the bag
B. know the name of the lorry driver
C. count the number of bags in the store
D. check that the nutrients suit the type and age of his animals
- 22.** Feed that is stored in a damp room may become mouldy. Mouldy feed should be
A. thrown away because it can poison the animals
B. mixed with fresh feed and given to the animals
C. given only to the youngest animals
D. dried in the sun and sold to other farmers

- 25.** Which of the following is a function of water in the body of a farm animal?
- A. It is stored in the bones to make them hard
 - B. It helps in digestion and the transport of nutrients in the blood
 - C. It builds the feathers and the hair
 - D. It gives the animal the greatest amount of energy
- 27.** A dairy cow needs a large amount of water each day mainly because
- A. water helps it to run faster
 - B. water changes the colour of its skin
 - C. milk is made up largely of water
 - D. water makes its horns grow longer
- 29.** Which of the following is a sign that a farm animal is not getting enough water?
- A. Dry nose and muzzle with sunken eyes
 - B. Playing actively with other animals
 - C. Steady increase in body mass
 - D. Smooth, shiny coat of hair
- 31.** Drinking troughs on a poultry farm should be washed regularly mainly to
- A. stop the birds from drinking too much
 - B. reduce the amount of feed the birds eat
 - C. prevent the spread of germs and diseases among the birds
 - D. make the farm look attractive to visitors
- 33.** Overfeeding farm animals with energy-giving feed may cause the animals to
- A. produce more eggs than usual
 - B. become too fat and reproduce poorly
 - C. need much less water
 - D. grow longer and stronger bones
- 24.** Clean water should be provided for farm animals
- A. only once a week
 - B. only during the rainy season
 - C. only when they are sick
 - D. at all times
- 26.** Water helps a farm animal to control its body temperature mainly through
- A. eating more grass
 - B. sweating and panting
 - C. growing more hair
 - D. laying more eggs
- 28.** A farmer who does not give his animals enough water should expect the animals to
- A. lay bigger eggs than before
 - B. grow faster and produce more meat
 - C. grow slowly and produce less milk, meat and eggs
 - D. become fatter within a few days
- 30.** If farm animals are deprived of water for several days, they will finally
- A. sleep and wake up healthy
 - B. grow new teeth
 - C. produce more milk
 - D. die
- 32.** Animals kept on a farm should be fed
- A. only when the farmer has time
 - B. only when they are seen to be thin
 - C. regularly, with the right quantity at the right time
 - D. once a month in large quantities
- 34.** Two groups of chicks are given the same feed, but one group is given water freely while the other is given very little water. The best conclusion from this investigation is that
- A. feed alone is enough for animals to grow well
 - B. water is necessary for the proper growth of animals
 - C. the type of feed does not matter at all
 - D. chicks grow better when they drink less water

- 35.** In the investigation above, the factor that is changed is the
- A. type of feed given to the chicks
 - B. size of the cages used
 - C. amount of water given to the chicks
 - D. breed of the chicks used
- 37.** The best way for a farmer to find out whether a new feed improves growth is to
- A. ask his neighbours what they think of the feed
 - B. count the number of bags bought in a month
 - C. look at the colour of the bag of feed
 - D. weigh the animals regularly and compare the results with those of another group
- 39.** A poultry farmer in Dormaa finds that his birds are growing slowly although they are given plenty of maize. The most likely reason is that the feed is low in
- A. protein
 - B. carbohydrate
 - C. water
 - D. fibre
- 36.** Akosua keeps goats which browse on grass all day but produce very little milk. The best advice to give her is to
- A. add a protein-rich concentrate and clean water to the feeding routine
 - B. stop the goats from eating grass altogether
 - C. keep the goats in a dark room all day
 - D. give the goats only water and no feed
- 38.** Which of the following statements about animal feeding is TRUE?
- A. Young animals need less protein than adult animals
 - B. Different animals, and animals of different ages, need different types of feed
 - C. Feed alone, without water, is enough for good growth
 - D. All farm animals should be given exactly the same feed
- 40.** Apart from good feed and clean water, which of the following would most improve the growth of animals on a farm?
- A. More workers standing around the pens
 - B. Playing music loudly near the animals all day
 - C. A larger signboard at the entrance of the farm
 - D. A clean, well-ventilated house and regular treatment against diseases

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show your working clearly in all questions that require calculations.

Question 7

- (a) Do not use correction fluid. Cancel any mistake neatly with a single line.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 3: Atomic Structure

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The smallest particle of an element that can take part in a chemical reaction is called
 - A. a molecule.
 - B. a compound.
 - C. an atom.
 - D. an electron.
2. The three main sub-atomic particles present in an atom are
 - A. protons, neutrons and electrons.
 - B. protons, neutrons and ions.
 - C. atoms, molecules and ions.
 - D. electrons, shells and ions.
3. Which of the following particles is found in the nucleus of an atom?
 - A. Electron
 - B. Proton
 - C. Shell
 - D. Molecule
4. The particle that carries a negative charge is the
 - A. proton.
 - B. neutron.
 - C. shell.
 - D. electron.
5. Which of the following sub-atomic particles is electrically neutral?
 - A. Proton
 - B. Electron
 - C. Neutron
 - D. Ion
6. The atomic number of an atom is the number of
 - A. protons in the atom.
 - B. neutrons in the atom.
 - C. protons plus neutrons in the atom.
 - D. shells in the atom.
7. Neutrons are found
 - A. in the shells, together with the electrons.
 - B. in the nucleus, together with the protons.
 - C. outside the atom, together with the ions.
 - D. in the nucleus, together with the electrons.
8. An atom has 13 protons, 14 neutrons and 13 electrons. Its mass number is
 - A. 40.
 - B. 14.
 - C. 13.
 - D. 27.
9. An atom of element Z has atomic number 8 and mass number 16. The number of neutrons in the atom is
 - A. 16.
 - B. 24.
 - C. 8.
 - D. 4.
10. An atom of sodium has 11 protons and 12 neutrons. The atomic number of sodium is
 - A. 11.
 - B. 23.
 - C. 12.
 - D. 1.
11. An atom of aluminium has atomic number 13 and mass number 27.

How many electrons are in the atom?

- A. 14
- B. 13
- C. 27
- D. 40

12. Atoms are electrically neutral because the number of

- A. protons equals the number of neutrons.
- B. neutrons equals the number of

- electrons.
- C. protons is greater than the number of electrons.
- D. protons equals the number of electrons.
- 13.** What charge will an atom of sodium (2, 8, 1) carry when it loses its outermost electron?
- A. No charge
- B. -1
- C. +1
- D. +11
- 15.** Ionic bonding takes place when
- A. electrons are transferred from one atom to another.
- B. electrons are shared between two atoms.
- C. protons are transferred from one atom to another.
- D. neutrons move from one atom to another.
- 17.** Which of the following is ****not**** a noble gas?
- A. Argon
- B. Helium
- C. Neon
- D. Nitrogen
- 19.** The chemical symbol for calcium is
- A. Cl.
- B. C.
- C. Ca.
- D. Cu.
- 21.** The electron configuration of an atom is the arrangement of its electrons
- A. in the nucleus.
- B. in the shells around the nucleus.
- C. on its protons.
- D. on its neutrons.
- 23.** An atom of chlorine has 17 electrons. Its electron configuration is
- A. 2, 8, 7.
- B. 2, 7, 8.
- C. 8, 8, 1.
- D. 2, 8, 8.
- 14.** The charge on an ion formed when a chlorine atom (2, 8, 7) gains one electron is
- A. +1.
- B. -1.
- C. -7.
- D. 0.
- 16.** Covalent bonding involves
- A. the transfer of a pair of electrons between two atoms.
- B. the sharing of protons between two atoms.
- C. the transfer of neutrons between two atoms.
- D. the sharing of a pair of electrons between two atoms.
- 18.** The chemical symbol for potassium is
- A. P.
- B. K.
- C. Po.
- D. Pt.
- 20.** An atom of element L has atomic number 12 and mass number 24. The number of neutrons in the atom is
- A. 12.
- B. 24.
- C. 36.
- D. 2.
- 22.** The maximum number of electrons that can occupy the first shell of an atom is
- A. 8.
- B. 1.
- C. 2.
- D. 18.
- 24.** The number of valence electrons in an atom of potassium (2, 8, 8, 1) is
- A. 8.
- B. 19.
- C. 2.
- D. 1.

- 25.** An atom of element W has 12 protons and 12 neutrons. The number of electrons in the atom is
- 12.
 - 24.
 - 0.
 - 6.
- 27.** The atoms of elements in the same period of the Periodic Table have the same number of
- neutrons.
 - valence electrons.
 - shells (occupied energy levels).
 - protons.
- 29.** A sodium atom (2, 8, 1) loses its outermost electron. The electron configuration of the sodium ion formed is
- 2, 8, 1.
 - 2, 8, 2.
 - 2, 8.
 - 2, 9.
- 31.** A molecule of element E contains two atoms. Its formula is written as
- 2E.
 - E₂.
 - E₂O.
 - 2E₂.
- 33.** Two atoms of element X combine with three atoms of element Y. The chemical formula of the compound formed is
- X₂Y₃.
 - X₃Y₂.
 - XY.
 - X₂Y.
- 35.** A sodium ion has 11 protons and 10 electrons. The charge on the ion is
- 1.
 - 0.
 - +1.
 - +21.
- 37.** Elements in the same group of the Periodic Table have similar
- masses.
 - chemical properties.
 - numbers of neutrons.
- 26.** Which of the following determines the group of an element in the Periodic Table?
- The number of shells in the atom
 - The number of valence electrons in the atom
 - The number of neutrons in the atom
 - The mass number of the atom
- 28.** The number of electrons in an atom of element Q (2, 8, 4) that are in the second shell is
- 14.
 - 4.
 - 2.
 - 8.
- 30.** Which of the following pairs of atoms will form an **ionic** bond?
- A sodium atom and a chlorine atom
 - Two chlorine atoms
 - Two hydrogen atoms
 - Two oxygen atoms
- 32.** How many atoms of hydrogen are present in one molecule of water, H₂O?
- 3
 - 1
 - 0
 - 2
- 34.** Which of the following pairs forms a molecule?
- Sodium and chlorine ions
 - Two oxygen atoms
 - Sodium ion and chloride ion
 - An oxygen ion and a chloride ion
- 36.** Which of the following statements about the Periodic Table is **true**?
- It contains only man-made elements.
 - All the elements in it are metals.
 - The elements in it are arranged according to their colours.
 - It is a chart in which elements are arranged according to their atomic numbers.

D. colours.

- 39.** The kind of bond formed when electrons are transferred from a metal atom to a non-metal atom is
- A. a covalent bond.
 - B. a metallic bond.
 - C. an ionic bond.
 - D. a hydrogen bond.
- 38.** An element has atomic number 19. Its electron configuration is
- A. 2, 8, 8, 1.
 - B. 2, 8, 9.
 - C. 2, 8, 7.
 - D. 8, 8, 3.
- 40.** An aluminium ion is represented as Al^{3+} . This shows that the aluminium atom has
- A. gained three electrons.
 - B. shared three electrons.
 - C. lost three protons.
 - D. lost three electrons.

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Show all working where a calculation is required.

Question 6

- (a) Use the copy of the Periodic Table provided in the classroom where you need it.

Question 7

- (a) Read every question carefully before you answer it.

Question 8

- (a) Do not use correction fluid. Cancel any mistake neatly with a single line.

Question 9

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 4: Bacterial D

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Bacteria are best described as
 - A. tiny grains of sand found in dirty water
 - B. very small living things, each made of a single cell
 - C. poisonous chemicals released by rotting food
 - D. the smallest animals with a backbone
2. Bacteria can be seen only with the aid of a
 - A. pair of spectacles
 - B. hand lens held close to the eye
 - C. telescope
 - D. microscope
3. Which of the following diseases is caused by bacteria?
 - A. Measles
 - B. Hepatitis B
 - C. Ringworm
 - D. Meningitis
4. Which of the following diseases is NOT caused by bacteria?
 - A. Gonorrhoea
 - B. Cholera
 - C. HIV and AIDS
 - D. Tuberculosis
5. A disease caused by bacteria is usually treated with
 - A. antibiotics prescribed by a health worker
 - B. a drug used against malaria parasites
 - C. vitamin tablets alone
 - D. any painkiller bought from a hawker
6. Not all bacteria are harmful. Some bacteria are useful because they
 - A. make the soil hard so that it holds water
 - B. change a virus into a vitamin
 - C. destroy all the nutrients in food before it is eaten
 - D. help to decay dead matter and are used in making foods such as yoghurt
7. Bacteria multiply fastest when they are in a place that is
 - A. boiled and then covered
 - B. exposed to strong sunlight all day
 - C. cold, dry and salty
 - D. warm, moist and has food
8. The best reason why a wound should be washed and covered is that
 - A. the skin will grow a new layer of hair
 - B. bacteria from dirt can enter the body through the broken skin
 - C. the blood will change from red to white
 - D. the bone under the wound will become weak
9. Food poisoning is an illness that a person gets after eating food that
 - A. contains harmful bacteria or the poison they produce
 - B. contains a large amount of vitamin C
 - C. has been cooked with too much

salt

- D. has been eaten while it is still hot

11. A common symptom of food poisoning is

- A. the loss of hearing in one ear
- B. a yellow colour in the whites of the eyes
- C. vomiting, stomach pains and diarrhoea a few hours after eating
- D. a stiff neck with a dislike of bright light

10. Which of the following bacteria is a common cause of food poisoning?

- A. Amoeba
- B. Plasmodium
- C. Salmonella
- D. Anopheles

- 13.** The first help that should be given at home to a person suffering from food poisoning is to
- give oral rehydration salts solution and send the person to a health facility
 - give the person more of the same food to eat
 - ask the person to sleep until the pain goes away
 - give the person a strong antibiotic kept in the house
- 15.** Which practice by a food vendor is MOST likely to cause food poisoning?
- Handling money and then serving food without washing the hands
 - Covering the food with a clean net
 - Keeping the food hot over a fire
 - Serving the food with a clean ladle
- 17.** Before a food vendor is allowed to sell food in a market, the Environmental Health Unit of the assembly requires the vendor to
- undergo a medical examination and obtain a health certificate
 - pay a fine to the chief of the town
 - buy all the food items from one shop
 - prepare the food inside the market office
- 19.** Gonorrhoea is caused by
- a roundworm
 - a virus
 - a bacterium
 - a fungus
- 21.** A common symptom of gonorrhoea is
- a yellow colour of the whites of the eyes
 - white patches on the tongue only
 - a sudden swelling of the knee joints
 - pain or a burning feeling when passing urine, with a discharge
- 12.** The greatest danger to a child who is vomiting and passing watery stools after food poisoning is
- dehydration, the loss of too much water and salts from the body
 - the loss of the sense of smell
 - the sudden growth of extra teeth
 - a permanent change in the colour of the skin
- 14.** Cooked rice that is left uncovered on a table from morning till evening is unsafe to eat because
- the rice loses all its colour
 - the rice becomes too hard for the teeth
 - bacteria have had warmth, moisture and time to multiply in it
 - the starch in the rice changes into a vitamin
- 16.** Which of the following is the BEST way to prevent food poisoning in a school canteen?
- Wash the plates in a bucket of water used all week
 - Keep the left-over stew in the open until the next day
 - Cook food thoroughly, keep it covered and serve it while it is still hot
 - Prepare all the food a day before it is needed
- 18.** Gonorrhoea is a disease that is spread mainly through
- eating food prepared by an infected person
 - unprotected sexual contact with an infected person
 - the bite of an infected mosquito
 - sharing a desk with an infected person
- 20.** A disease such as gonorrhoea, which is passed on during sexual contact, is called a
- hereditary disease
 - deficiency disease
 - vector-borne disease
 - sexually transmitted infection

- 23.** If gonorrhoea is not treated, a serious effect on a young person later in life may be
- A. a permanent change in the colour of the hair
 - B. difficulty in having children
 - C. the loss of the sense of balance
 - D. the inability to digest fat
- 25.** Apart from abstaining from sex, which of the following helps to prevent the spread of gonorrhoea?
- A. Washing the body with hot water after sexual contact
 - B. Early reporting to a health facility so that the patient and the partner are both treated
 - C. Keeping the infection a secret from the health worker
 - D. Buying antibiotics from a hawker and taking half of them
- 27.** Meningitis is spread from one person to another mainly through
- A. droplets from the nose and the throat during coughing, sneezing and close talking
 - B. sharing a reading book
 - C. walking barefoot on a dusty road
 - D. drinking water from a borehole
- 29.** A patient with meningitis often turns away from a window because the patient
- A. has lost the use of both legs
 - B. cannot bear bright light
 - C. is afraid of the people outside
 - D. needs more fresh air than other patients
- 31.** Outbreaks of meningitis are common in the northern parts of Ghana during the Harmattan because
- A. mosquitoes breed faster in the dry season
 - B. the food sold in the markets is cheaper at that time
 - C. the air is hot, dry and dusty, which injures the lining of the nose and throat and helps the bacteria to enter
 - D. the people drink more water in the dry season
- 22.** Gonorrhoea is often not noticed early in many women because
- A. the bacteria cannot live in the body of a woman
 - B. the disease changes into malaria in women
 - C. a woman may have mild symptoms or none at all while still carrying the bacteria
 - D. women are treated with antibiotics every month
- 24.** A baby born to a mother who has untreated gonorrhoea may suffer from
- A. an extra finger on each hand
 - B. a shortage of red blood cells only
 - C. a serious infection of the eyes, which can lead to blindness
 - D. the loss of all its milk teeth
- 26.** Meningitis is an inflammation of the
- A. covering of the brain and the spinal cord
 - B. wall of the heart
 - C. air sacs of the lungs
 - D. lining of the stomach
- 28.** Which of the following is a well-known symptom of meningitis?
- A. A painless swelling of the big toe
 - B. A slow loss of weight over two years
 - C. A rash on the palms only
 - D. A sudden high fever with a severe headache and a stiff neck
- 30.** A person who recovers from meningitis that was treated late may be left with
- A. a permanent yellow colour of the skin
 - B. deafness or damage to the brain
 - C. an inability to grow adult teeth
 - D. a longer arm on one side
- 32.** Sleeping in a crowded, poorly aired room increases the risk of meningitis because
- A. crowded rooms have more food for the bacteria
 - B. the bacteria are carried by the mosquitoes in the room
 - C. the floor of such a room is

always wet

- D. droplets from an infected person are easily breathed in by those nearby

- 33.** The most effective way of protecting a whole community against meningitis during an outbreak is
- A. asking everybody to drink plenty of hot tea
 - B. a vaccination campaign organised by the health authorities
 - C. spraying the rooms against mosquitoes
 - D. closing all the boreholes in the town
- 35.** Which of the following is mainly the duty of the government in the control of bacterial diseases?
- A. Sweeping the compound every morning
 - B. Washing one's own hands before eating
 - C. Providing vaccines, health facilities and trained health workers
 - D. Covering food in the kitchen at home
- 37.** A pupil who wants to educate the community about meningitis would find a poster useful mainly because a poster
- A. removes the need for vaccination
 - B. can be understood only by health workers
 - C. is cheaper than speaking to one person
 - D. passes a short, clear message with pictures to many people, including those who read little
- 39.** A patient stops taking the antibiotics for a bacterial disease as soon as he feels better. The likely result is that
- A. the bacteria that are left may survive, become resistant and make the illness return
 - B. the patient will never fall ill again
 - C. the remaining drugs will change into vitamins
 - D. the disease will be passed on only to animals
- 34.** Which of the following is the duty of an individual in the control of bacterial diseases?
- A. Washing the hands with soap and reporting early to a health facility when ill
 - B. Passing laws on food safety
 - C. Supplying vaccines to all the regions
 - D. Building hospitals in every district
- 36.** Communal labour to clear choked gutters and clean a market helps to control disease because it
- A. makes the town larger
 - B. removes the dirt and stagnant water in which germs and pests thrive
 - C. keeps the traders away from the market
 - D. increases the price of food in the market
- 38.** Which of the following is the BEST message for a poster on the prevention of food poisoning?
- A. Buy food only from your own relatives
 - B. Boil all fruits before eating them
 - C. Eat only once a day
 - D. Cook it well, cover it, eat it hot
- 40.** A town has many cases of food poisoning after a funeral. The MOST useful first action by the health team is to
- A. close the town's only school for a month
 - B. find out what the people ate and how the food was prepared and stored
 - C. arrest the family that organised the funeral
 - D. ask everybody in the town to take antibiotics

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show your working clearly in all the questions that require a calculation.

Question 7

- (a) Use a pencil and a ruler where a plan, a poster or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 5: Carbon Cycle

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Which of the following gases contains carbon?
 - A. oxygen
 - B. nitrogen
 - C. hydrogen
 - D. carbon (IV) oxide
2. Carbon is an element that is found in all living things. A simplified name for the gas carbon (IV) oxide is
 - A. carbon monoxide
 - B. carbonate
 - C. carbonic acid
 - D. carbon dioxide
3. The process by which green plants make their own food is called
 - A. respiration
 - B. digestion
 - C. photosynthesis
 - D. transpiration
4. During photosynthesis, green plants take in
 - A. oxygen
 - B. carbon (IV) oxide
 - C. nitrogen
 - D. hydrogen
5. During photosynthesis, green plants give out
 - A. carbon (IV) oxide
 - B. oxygen
 - C. nitrogen
 - D. carbon
6. Which of the following is ****not**** a raw material needed for photosynthesis?
 - A. nitrogen
 - B. carbon (IV) oxide
 - C. water
 - D. sunlight
7. The gas carbon (IV) oxide is used by green plants during photosynthesis to make
 - A. water
 - B. minerals
 - C. chlorophyll
 - D. food (carbohydrates)
8. In which part of the green plant does photosynthesis mainly take place?
 - A. the roots
 - B. the stem
 - C. the flowers
 - D. the green leaves
9. The process by which living things obtain energy from food, with the release of carbon (IV) oxide, is called
 - A. respiration
 - B. photosynthesis
 - C. decay
 - D. burning
10. During respiration, living things take in
 - A. carbon (IV) oxide
 - B. nitrogen
 - C. water vapour
 - D. oxygen
11. During respiration, living things give out
 - A. carbon (IV) oxide
 - B. oxygen
 - C. nitrogen

D. hydrogen

- 12.** Which of the following processes adds carbon (IV) oxide to the air?
- A. photosynthesis in green plants
 - B. the formation of clouds
 - C. burning of firewood

13. When a farmer burns a pile of rubbish, the carbon in the rubbish is released into the air mainly as
- oxygen
 - nitrogen
 - carbon (IV) oxide
 - water vapour
15. The main greenhouse gas that is released when fossil fuels are burnt is
- oxygen
 - carbon (IV) oxide
 - nitrogen
 - helium
17. The main gases given out when dead plants and animals decay are
- carbon (IV) oxide and methane
 - oxygen and nitrogen
 - hydrogen and helium
 - oxygen and water vapour
19. Which of the following processes removes carbon (IV) oxide from the air?
- the burning of bushes
 - respiration in animals
 - photosynthesis in green plants
 - the decay of dead leaves
21. The carbon cycle is described as a repeated pattern of change because
- carbon is destroyed after use
 - carbon is found only in plants
 - carbon is produced only by people
 - carbon is circulated continuously in the environment
23. Fossil fuels such as coal, crude oil and natural gas were formed from
- rocks that melted deep inside the earth
 - rain water stored under the ground
 - the bones of living animals in the bush
 - the remains of dead plants and animals buried for a very long time
- D. the cooling of the air at night
14. The burning of fuels such as petrol and diesel to release energy is called
- respiration
 - combustion
 - photosynthesis
 - condensation
16. Decay (decomposition) is the breakdown of dead plants and animals by
- bacteria and fungi
 - wind and rain
 - sunlight and heat
 - insects and birds only
18. Which of the following organisms help to break down dead plant and animal material in the soil?
- cattle and goats
 - bacteria and fungi
 - maize and cassava
 - fishes and frogs
20. A ****repeated pattern of change in nature**** means a process that
- goes round and round and repeats itself
 - happens once and stops
 - happens only at night
 - happens only during the rainy season
22. Which of the following is ****not**** part of the carbon cycle?
- photosynthesis
 - the formation of nitrogen gas in the soil
 - respiration
 - the burning of fossil fuels
24. Burning fossil fuels is a matter of concern to the environment because it
- adds oxygen to the air
 - reduces the amount of carbon in the air
 - adds a lot of carbon (IV) oxide to the air
 - cleans the air of dust

- 25.** When the amount of carbon (IV) oxide in the atmosphere increases too much, it leads to
- A. an increase in rainfall only
 - B. global warming
 - C. the cooling of the whole earth
 - D. an increase in the amount of oxygen
- 27.** Which of the following human activities adds the ****least**** carbon (IV) oxide to the air?
- A. riding a bicycle
 - B. driving a car
 - C. burning a bush
 - D. cooking with firewood
- 29.** Which of the following practices helps to keep the balance of gases in the air?
- A. setting fire to the bush every dry season
 - B. throwing rubbish into the river
 - C. planting trees to replace those that are cut down
 - D. cutting down trees without replacing them
- 31.** In a food chain, animals that feed on the green plants are called
- A. producers
 - B. primary consumers
 - C. decomposers
 - D. secondary consumers
- 33.** In the food chain ****maize plant - rat - snake****, the snake is the
- A. secondary consumer
 - B. producer
 - C. primary consumer
 - D. decomposer
- 35.** The carbon in the bodies of animals was first taken from the air by
- A. the animals themselves during respiration
 - B. decomposers in the soil
 - C. green plants during photosynthesis
 - D. the burning of fossil fuels
- 37.** Which of the statements below is correct?
- 26.** Global warming is a problem because it leads to
- A. more oxygen in the air
 - B. stronger plants
 - C. fewer bacteria in the soil
 - D. changes in the climate
- 28.** Which of the following is the best way to reduce the amount of carbon (IV) oxide released into the air?
- A. burning more rubbish in the compound
 - B. planting more trees in the community
 - C. cutting down trees for firewood
 - D. using more petrol in vehicles
- 30.** In a food chain, green plants are called
- A. consumers
 - B. decomposers
 - C. producers
 - D. predators
- 32.** Animals that feed on other animals in a food chain are called
- A. producers
 - B. primary consumers
 - C. decomposers
 - D. secondary consumers
- 34.** In the food chain ****grass - goat - lion****, which organism is the producer?
- A. goat
 - B. lion
 - C. grass
 - D. none of them
- 36.** Which of the following processes release carbon (IV) oxide into the air?
- A. I, II, III and IV
 - B. II and IV only
 - C. I and IV only
 - D. I, II and III only

- A. Photosynthesis removes carbon (IV) oxide from the air while respiration, burning and decay add it.
- B. Photosynthesis and respiration both add carbon (IV) oxide to the air.
- C. Photosynthesis adds carbon (IV) oxide to the air while respiration removes it.
- D. Burning removes carbon (IV) oxide from the air.

38. Why does the carbon cycle continue in nature without the carbon being used up?

- A. Because the same carbon atoms are used again and again by living things and in the air
- B. Because plants make new

- carbon each year.
- C. Because carbon cannot enter the air.
 - D. Because animals absorb carbon from the soil.
- 40.** Which of the following is the best reason for planting trees around your school?
- A. The trees add carbon (IV) oxide to the air.
 - B. The trees remove carbon (IV) oxide from the air and give out oxygen.
 - C. The trees prevent the water cycle from happening.
 - D. The trees kill the bacteria in the soil.
- 39.** A forest is cleared and burnt, and the land is left bare. The most likely result is that
- A. the amount of carbon (IV) oxide in the air decreases.
 - B. green plants in the area will make more food.
 - C. the amount of carbon (IV) oxide in the air increases
 - D. the total amount of carbon on earth will reduce to half.

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 7 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Draw and label diagrams where you are asked to.

Question 6

- (a) Read every question carefully before you answer it.

Question 7

- (a) Do not use correction fluid. Cancel any mistake neatly with a single line.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 6: Climate C

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Climate is best described as
 - A. the condition of the atmosphere of a place at one moment
 - B. the temperature recorded at noon on one day
 - C. the average pattern of weather of a place taken over many years
 - D. the amount of rain that falls in a single storm
2. Climate change means
 - A. the change from morning coolness to afternoon heat
 - B. the difference in temperature between two towns
 - C. the change of wind direction during a storm
 - D. a long-term change in the average pattern of temperature and rainfall of a place
3. Which of the following is a sign of climate change?
 - A. A cold morning during the Harmattan
 - B. Seasons that no longer begin at the times farmers expect
 - C. A heavy downpour in the rainy season
 - D. A cloudy sky in the afternoon
4. Which of the following is NOT a sign of climate change?
 - A. The daily rise in temperature from morning to midday
 - B. The rise in the average temperature of the earth over many years
 - C. The retreat of ice at the poles
 - D. The rise in the level of the sea
5. The melting of ice at the poles and on high mountains is evidence that
 - A. the earth is moving closer to the moon
 - B. the sun has stopped shining on the poles
 - C. the oceans are becoming shallower
 - D. the average temperature of the earth is rising
6. A rise in the level of the sea along the coast of Ghana is caused mainly by
 - A. the melting of ice and the expansion of sea water as it warms
 - B. the pumping of water out of the lagoons
 - C. the building of harbours at Tema and Takoradi
 - D. the digging of boreholes in the coastal towns
7. Which record would be MOST useful in showing that the climate of a town has changed?
 - A. The temperature taken every hour for one day
 - B. The amount of rain that fell last week
 - C. The wind speed recorded during one storm
 - D. Rainfall and temperature figures collected over the past fifty

years

- 9.** The gases that trap heat in the atmosphere of the earth are known as
- A. inert gases
 - B. noble gases
 - C. greenhouse gases
 - D. respiratory gases

- 8.** A farmer says, 'The rains used to come in March, but for the past ten years they begin in May.' This statement describes
- A. a change in climate
 - B. a change in crop variety
 - C. a single weather event
 - D. a change in soil type

- 11.** Carbon dioxide is added to the atmosphere mainly by
- the evaporation of water from lakes
 - the burning of wood, charcoal, petrol, diesel and other fuels
 - the breathing of green plants at night only
 - the weathering of rocks on a hillside
- 13.** Cutting down large areas of forest adds to climate change chiefly because
- the soil becomes too soft for farming
 - the remaining trees grow much taller
 - fewer trees are left to take in carbon dioxide from the air
 - the rivers begin to flow more slowly
- 15.** Bush burning before the planting season adds to climate change because it
- releases carbon dioxide and destroys the plants that would absorb it
 - increases the number of earthworms in the soil
 - cools the surface of the soil
 - brings the rains earlier in the year
- 17.** A likely effect of climate change on farming in Ghana is
- an increase in the fertility of every soil
 - poor harvests because the rains come late or stop early
 - a fall in the number of pests on the farm
 - the ripening of all crops at the same time
- 19.** Which of the following is an effect of a rise in sea level on a coastal town such as Keta?
- Flooding of houses and the washing away of land by the sea
 - An increase in the area of land for building
- 10.** Which of the following is the greenhouse gas released in the greatest quantity by human activity?
- Hydrogen
 - Carbon dioxide
 - Oxygen
 - Nitrogen
- 12.** Methane, another greenhouse gas, comes mainly from
- the burning of solar panels
 - the grinding of maize in a mill
 - rotting refuse in landfill sites, cattle and flooded rice fields
 - the freezing of water in a refrigerator
- 14.** Which of the following human activities adds LEAST to climate change?
- Burning refuse in the open
 - Drying clothes in the sun
 - Running a diesel generator all night
 - Clearing a forest by fire for farming
- 16.** Which of the following best explains why the earth is warming?
- The earth has moved nearer to the sun
 - The atmosphere has lost all its oxygen
 - Human activity has increased the greenhouse gases, so more heat is held in the atmosphere
 - The earth now spins faster on its axis
- 18.** Longer and more severe dry spells in northern Ghana are likely to lead to
- a rise in the number of fish in the rivers
 - a fall in the price of water in the towns
 - an increase in the depth of the White Volta
 - the drying up of streams and a shortage of water for people and animals

- C. A fall in the salt content of the lagoon
- D. A rise in the number of trees on the shore

- 21.** Heavy, unusual rainfall in a Ghanaian city often leads to
- A. a rise in the quality of the roads
 - B. a permanent improvement in the drainage system
 - C. floods that destroy property and lives, especially where gutters are choked
 - D. a fall in the number of people in the city
- 23.** Climate change threatens the production of cocoa in Ghana mainly because cocoa
- A. is harvested only during the Harmattan
 - B. grows only on land covered by sea water
 - C. cannot grow where there are any trees
 - D. needs steady rainfall and shade, and suffers when the dry season is long and hot
- 25.** Which statement about the effects of climate change is TRUE?
- A. Only the countries that produce the gases are affected
 - B. The countries that produce the least greenhouse gas often suffer its effects most
 - C. The effects are felt only at the poles
 - D. The effects can be seen within a single week
- 27.** Adaptation to climate change means
- A. changing the way we live and work so that we can cope with the changes already happening
 - B. measuring the temperature of the earth every day
 - C. moving every community to another country
 - D. stopping the release of all gases from factories
- 20.** Climate change can increase disease in a community because
- A. germs can no longer live in warm places
 - B. warmer, wetter conditions allow mosquitoes and other disease carriers to spread
 - C. people stop drinking water when it is hot
 - D. hospitals close during the dry season
- 22.** Which of the following groups of people in Ghana is MOST at risk from climate change?
- A. Office workers in air-conditioned buildings
 - B. Small-scale farmers and fishers who depend directly on the rains and the sea
 - C. Traders who sell goods in a covered market
 - D. Pupils who attend school in the morning
- 24.** The drying up of a dam that supplies a hydroelectric station leads to
- A. an increase in the amount of power produced
 - B. an improvement in the quality of the water
 - C. a shortage of electricity for homes and industries
 - D. a fall in the price of petrol in the country
- 26.** A direct effect of climate change on biodiversity is that
- A. all animals move to the same habitat
 - B. every species produces more young than before
 - C. some plants and animals die out because their habitats change faster than they can adapt
 - D. plants no longer need sunlight to grow
- 28.** Which of the following is an adaptation measure for a farming community?
- A. Using more firewood for cooking
 - B. Cutting down the trees along the stream

- C. Burning the bush before every planting season
- D. Planting early-maturing and drought-resistant varieties of maize

- 29.** The building of small dams and the harvesting of rain water in northern Ghana is an example of
- A. adaptation, because it stores water for the longer dry season
 - B. pollution, because it changes the river

- C. mitigation, because it reduces greenhouse gases
- D. deforestation, because land is cleared
- 31.** A farmer who now keeps part of his harvest in an improved store, instead of selling it all at once, is adapting because he
- is preparing for a season in which the harvest may fail
 - wants to reduce the carbon dioxide in the air
 - is obeying a law about bush burning
 - expects the price of maize to fall
- 33.** Which of the following shows a community adapting to a longer dry season?
- Burning refuse in the open every evening
 - Clearing more forest for new farms
 - Buying more firewood for the kitchen
 - Changing the planting calendar and using irrigation from a small dam
- 35.** Which of the following is a mitigation measure?
- Replacing a diesel generator with solar panels
 - Storing rain water in a tank
 - Building a higher wall round a flooded house
 - Planting a crop that ripens earlier
- 37.** Which of the following is a green job?
- Felling timber without replanting
 - Installing and repairing solar panels
 - Burning old tyres to recover steel wire
 - Mining gold in the bed of a river
- 30.** Planting trees along the shore and rebuilding sea defences at Keta is mainly intended to
- provide timber for export
 - protect the town from the advancing sea
 - raise the temperature of the lagoon
 - increase the number of fish caught
- 32.** Legislation against bush burning helps a country to deal with climate change because it
- increases the number of farms in the forest
 - makes the rains fall at the same time every year
 - removes the need for weeding
 - protects the trees and grass that take in carbon dioxide and hold the soil
- 34.** Mitigation of climate change means
- reducing the amount of greenhouse gas released, or taking it out of the air
 - storing grain for a season of poor harvest
 - learning to live with the effects of a warmer world
 - moving a village away from a flooded river
- 36.** A green economy is one in which
- only green vegetables are grown for sale
 - all factories are closed down permanently
 - people earn a living in ways that do not destroy the environment
 - every building in the town is painted green
- 38.** Planting trees helps to slow climate change chiefly because growing trees
- lower the level of the sea directly
 - prevent the sun from reaching

the earth

- C. take in carbon dioxide and store the carbon in their wood
- D. release large amounts of oxygen at night

39. Which of the following household changes would do MOST to reduce a family's greenhouse gas release?

- A. Sweeping the compound twice a day

- B. Washing plates in cold water instead of warm
- C. Painting the kitchen a lighter colour
- D. Cooking with an improved, fuel-saving stove or gas instead of an open wood fire

- 40.** A Basic 8 class wants to help its community act on climate change. The MOST effective action is to
- A. write the words 'stop climate change' on the school wall
 - B. ask the pupils to stay at home on hot days
 - C. wait until the government solves the problem
 - D. plant and care for trees on the school compound and teach others why it matters

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show your working clearly in all the questions that require a calculation.

Question 7

- (a) Use a pencil and a ruler where a plan or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 7: Conservation

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The law of conservation of energy states that energy
 - A. can be stored only in fuels and in food
 - B. can be created whenever a machine is switched on
 - C. can neither be created nor destroyed, but only changed from one form to another
 - D. is destroyed each time it is used by a machine
2. Energy conservation, as used in daily life, means
 - A. using energy carefully so that none of it is wasted
 - B. creating new energy for the home
 - C. using as much energy as possible every day
 - D. changing energy from one form into another
3. Which of the following is an example of energy conversion?
 - A. Switching off a lamp that is not in use
 - B. Buying a cheaper electricity meter
 - C. Closing the windows of a room
 - D. A gas stove burning to cook rice
4. Which of the following is an example of energy conservation?
 - A. Plugging a kettle into the socket to boil water
 - B. Switching off the fan when leaving the classroom
 - C. Burning charcoal in a coal pot
 - D. Using a dynamo to light a bicycle lamp
5. In every energy conversion, the total energy after the change is
 - A. equal to the total energy before the change
 - B. less than the total energy before the change
 - C. greater than the total energy before the change
 - D. equal to zero
6. The useful energy given out by an electric fan is
 - A. sound energy from the blades
 - B. chemical energy in the wires
 - C. heat energy in the motor
 - D. kinetic energy of the moving air
7. A machine is said to be efficient when it
 - A. converts a large part of the energy supplied into the useful form
 - B. works without being supplied with any energy
 - C. produces a great deal of heat and sound
 - D. uses a very large amount of energy each time
8. A lamp is supplied with 100 J of energy and gives out 20 J as light. The energy wasted is
 - A. 120 J
 - B. 20 J
 - C. 100 J
 - D. 80 J

9. Which of the following household appliances usually uses the MOST electricity in a month?

- A. A radio used for one hour a day
- B. An air conditioner used every night
- C. A torch with two dry cells
- D. A mobile-phone charger

10. Replacing filament bulbs with LED bulbs in a school helps mainly because LED bulbs

- A. give out far more heat than filament bulbs

11. Drying clothes in the sun instead of using an electric dryer is an example of
- A. converting heat energy into chemical energy
 - B. wasting energy that could be stored
 - C. conserving energy by using a free natural source
 - D. creating new energy for the home
13. Covering a cooking pot with a lid saves energy because it
- A. keeps the heat in the pot, so the food cooks faster
 - B. adds extra heat energy to the food
 - C. prevents the pot from becoming hot at all
 - D. changes the heat energy into light energy
15. Which of the following is the BEST reason for a school to conserve energy?
- A. It reduces the school's bills and protects the environment
 - B. It makes the classrooms darker and quieter
 - C. It stops the school from needing any electricity
 - D. It increases the amount of fuel burnt at the power station
17. The gas produced in the greatest amount when petrol or charcoal is burnt is
- A. oxygen
 - B. hydrogen
 - C. carbon dioxide
 - D. nitrogen
- B. make the classroom cooler by producing cold
- C. give the same light while using much less electricity
- D. need no electricity at all to work
12. Which of the following habits wastes electricity in a home?
- A. Covering a pot while cooking
 - B. Switching off the lights before travelling
 - C. Using daylight for reading in the afternoon
 - D. Leaving the television on when nobody is watching
14. A refrigerator should not be opened too often mainly because
- A. opening it adds heat energy to the electricity supply
 - B. warm air enters and the refrigerator uses more electricity to cool again
 - C. the food inside changes into energy
 - D. the cold inside escapes and is destroyed
16. Burning fossil fuels to generate electricity harms the environment mainly because it
- A. produces large amounts of clean water
 - B. releases carbon dioxide and other harmful gases into the air
 - C. makes the rivers flow faster than normal
 - D. removes all the oxygen from the soil
18. A build-up of carbon dioxide in the atmosphere leads mainly to
- A. the complete loss of the atmosphere
 - B. an increase in the oxygen of the air
 - C. global warming and changes in the climate
 - D. a fall in the temperature of the Earth

- 19.** Cutting many trees to make charcoal for cooking leads to
- deforestation, soil erosion and loss of habitats
 - more oxygen being produced by the land
 - cooler rivers and more rainfall every year
 - an increase in the number of forests
- 21.** Smoke from a wood fire inside a kitchen is harmful mainly because it
- cools the food that is being prepared
 - causes coughing and chest diseases in the people who breathe it
 - increases the amount of oxygen in the room
 - prevents the fire from burning properly
- 23.** A family changes from a petrol generating set to a solar home system. The main benefit to the environment is that
- the family can use more electricity than before
 - the solar panel produces energy without any conversion
 - the home becomes cooler during the day
 - no fuel is burnt, so no smoke or harmful gases are produced
- 25.** A pupil says, "Energy saved today is energy available tomorrow." This statement supports the idea that
- energy can be created by saving it
 - energy conservation makes resources last longer for future use
 - energy stops being used when it is saved
 - saving energy destroys part of the energy
- 27.** The main reason why a country encourages the use of energy-saving bulbs is to
- increase the heat produced in homes
- 20.** Which of the following energy sources causes the LEAST air pollution?
- Petrol
 - Coal
 - Solar energy
 - Charcoal
- 22.** Which of the following shows energy being wasted?
- A fan switched on in a room full of pupils
 - A lamp lit in a dark classroom at night
 - A solar panel charging a battery during the day
 - An electric iron left switched on with nobody using it
- 24.** Which of the following is the best example of using energy efficiently in a school kitchen?
- Cooking for the whole school at one time in covered pots
 - Cooking small quantities many times a day
 - Leaving the gas burner on between meals
 - Boiling a full pot of water for one cup of tea
- 26.** Which of the following is NOT a way of conserving energy?
- Servicing appliances so that they work efficiently
 - Leaving the security lights on throughout the day
 - Using natural daylight in the classroom
 - Switching off the computer when it is not in use

- B. make people use more electricity at night
 - C. make electricity more expensive for everybody
 - D. reduce the total demand for electricity and the fuel burnt to produce it
- 29.** A household bill shows that 300 kWh of electricity was used in a month. If one kWh costs GH¢1.50, the bill for the month is
- A. GH¢450.00
 - B. GH¢200.00
 - C. GH¢301.50
 - D. GH¢45.00
- 28.** The energy used to run a refrigerator is finally given out to the surroundings mainly as
- A. heat
 - B. sound
 - C. chemical energy
 - D. light
- 30.** In a hydroelectric station, the energy that does not become electricity is lost mainly as
- A. chemical energy stored in the

- water
- B. nuclear energy in the dam wall
 - C. heat and sound in the turbines and generators
 - D. light from the surface of the lake
- 32.** A school fits solar panels and also teaches pupils to switch off fans. This programme shows
- A. the conversion of energy only
 - B. both the conversion of energy and the conservation of energy
 - C. the conservation of energy only
 - D. neither conversion nor conservation of energy
- 33.** Which of the following best explains why no machine can be 100% efficient?
- A. Machines create new energy as they work
 - B. Some of the energy supplied is always changed into forms that are not useful, such as heat
 - C. The energy supplied disappears into the air
 - D. Some of the energy supplied is destroyed inside the machine
- 34.** A town decides to replace its old street lamps with LED street lamps. The BEST reason for the decision is that it will
- A. make the streets brighter than daylight
 - B. remove the need for a power station
 - C. give the same lighting at a much lower cost and with less fuel burnt
 - D. allow the lamps to work without electricity
- 35.** Two families use the same number of appliances, but one pays a much smaller electricity bill. The MOST likely reason is that the family
- A. lives in a house without a meter
 - B. uses appliances that create their own energy
 - C. switches off appliances when they are not in use and uses energy-saving bulbs
 - D. uses appliances that destroy less energy
- 36.** A poster on energy conservation in a school should contain mainly
- A. advertisements for expensive appliances
 - B. long paragraphs copied from a textbook
 - C. a clear message, simple pictures and practical tips that pupils can follow
 - D. the names of all the pupils in the school
- 37.** The energy from the Sun that falls on a school roof is best described as
- A. a free and renewable supply that can be converted into electricity
 - B. a non-renewable supply that will soon be used up
 - C. a waste product of the atmosphere
 - D. a form of chemical energy stored in the roof

37. Before using information from the internet for a project on energy, a pupil should FIRST

- A. print every page that mentions energy
- B. change the figures so that they look better
- C. copy the first page that appears in the search
- D. check that the source is reliable and the information is up to date

39. A factory manager claims that his machine produces more energy than it uses. A pupil should conclude that the claim is

- A. true, because the machine is very modern
- B. false, because energy cannot be created
- C. true, because machines can create energy
- D. false, because machines destroy energy

38. A community uses firewood for cooking. Which measure would BOTH save energy and protect the environment?

- A. Cooking in the open air without any stove
- B. Cutting more trees so that the fire burns longer
- C. Buying more pots for each household
- D. Using improved cook-stoves that need less firewood, and planting trees to replace those cut

- 40.** Which of the following actions would reduce a school's energy use the MOST over a whole term?
- A. Buying one extra refrigerator for the staff room
 - B. Asking pupils to speak quietly in class
 - C. Painting the classroom doors a lighter colour
 - D. Fitting energy-saving bulbs and switching off all fans and lights at the close of school

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show your working clearly in all the questions that require a calculation.

Question 7

- (a) Use a pencil and a ruler where a plan or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 8: Crop Production

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A seed bed is
 - A. a basket used for carrying seeds to the farm
 - B. a store room where seeds are kept after harvest
 - C. a bed on which a farmer sleeps on the farm
 - D. a piece of land that has been prepared for planting crops
2. The main reason why a farmer prepares a seed bed before planting is to
 - A. make the farm look beautiful to visitors
 - B. make the soil loose and suitable for seeds to germinate and grow
 - C. make the land easy to measure in metres
 - D. increase the number of weeds on the land
3. A seed bed in which the soil is heaped above the level of the surrounding ground is called a
 - A. flat bed
 - B. sunken bed
 - C. raised bed
 - D. ridge furrow
4. A seed bed in which the soil is dug below the level of the surrounding ground is a
 - A. sunken bed
 - B. flat bed
 - C. raised bed
 - D. nursery bed
5. A seed bed whose surface is at the same level as the surrounding ground is a
 - A. flat bed
 - B. sunken bed
 - C. mound
 - D. raised bed
6. A small bed on which seeds are sown closely and cared for before they are moved to the main farm is a
 - A. ridge
 - B. sunken bed
 - C. nursery bed
 - D. flat bed
7. Raised beds are most useful in an area where the land is
 - A. very dry and sandy all year
 - B. waterlogged and poorly drained
 - C. sloping steeply down a hill
 - D. rocky and cannot be dug at all
8. Sunken beds are most suitable in areas that are
 - A. dry, because the beds hold water around the crops
 - B. swampy, because the beds drain water quickly
 - C. rocky, because the beds hold the stones together
 - D. forested, because the beds keep out sunlight
9. Kofi's garden gets flooded whenever it rains heavily. The best seed bed for him to use is a
 - A. sunken bed
 - B. nursery bed
 - C. flat bed
 - D. raised bed

- 11.** Cassava is normally planted using
- seeds
 - roots
 - leaves
 - stem cuttings
- 13.** Yam is planted using
- setts cut from the tuber
 - leaves
 - flowers
 - stem cuttings
- 15.** The young plant that is moved from the nursery bed to the main field is called a
- cutting
 - seedling
 - sucker
 - sett
- 17.** Seedlings are best transplanted
- immediately the seeds are sown
 - at midday when the sun is hottest
 - in the evening or on a cool, cloudy day
 - only after the plants have begun to flower
- 19.** Which of the following plant parts can be used to plant a new crop?
- A fallen flower of a tomato plant
 - A stem cutting of cassava
 - A piece of bark from an orange tree
 - A dried maize cob without grains
- 21.** The practice of removing extra seedlings so that the remaining ones have enough space is called
- transplanting
 - pruning
 - weeding
 - thinning
- 10.** Which of the following crops is usually raised on a nursery bed before transplanting?
- Maize
 - Tomato
 - Cowpea
 - Groundnut
- 12.** Which of the following crops is planted using seeds?
- Maize
 - Cassava
 - Sugar cane
 - Pineapple
- 14.** Plantain and banana are usually planted using
- cuttings of the leaf stalk
 - leaves
 - suckers
 - seeds
- 16.** Moving seedlings from a nursery bed to the main field is known as
- transplanting
 - mulching
 - harvesting
 - thinning
- 18.** Why are seedlings watered immediately after transplanting?
- To make the soil heavy so that the wind does not blow it
 - To kill any insects living in the soil
 - To help the roots settle in the soil and prevent wilting
 - To wash the leaves and make them green
- 20.** Sweet potato is commonly planted using
- vine cuttings
 - seeds only
 - tubers only
 - flowers
- 22.** Crops are planted at a correct spacing mainly so that each plant gets enough
- shade from the other plants
 - sunlight, water and nutrients
 - weeds around its roots

D. insects to visit its flowers

24. Before planting, a farmer clears, ploughs and harrows the land. These activities are known as

- A. harvesting
- B. marketing
- C. land preparation
- D. storage

23. Which of the following is NOT a type of seed bed?

- A. Raised bed
- B. Compost heap
- C. Nursery bed
- D. Flat bed

25. Which tool is most suitable for making a raised bed on a small school

garden?

- A. Cutlass only
- B. Watering can
- C. Rake only
- D. Hoe

27. Two beds are prepared side by side. One is given manure and the other is not. The plants on the bed with manure are expected to

- A. die within one week of planting
- B. grow shorter and produce no flowers
- C. grow taller and produce more fruits
- D. grow at exactly the same rate as the other bed

29. The results of the heights of crops on three seed beds can best be compared at a glance by drawing a

- A. map of the school
- B. labelled diagram of a seed
- C. bar chart
- D. plan of the classroom

31. Which of the following is measured in order to compare the growth of crops on different seed beds?

- A. The number of flowers on each plant
- B. The name given to each bed
- C. The day of the week of planting
- D. The colour of the farmer's shirt

33. Plants on one bed flower earlier than plants of the same crop on another bed. The most likely reason is that the first bed had

- A. better soil nutrients, water and drainage
- B. a longer name written on its label
- C. more stones in the soil
- D. fewer hours of daylight

26. Covering the soil surface with dry grass after planting is called mulching. Mulching helps mainly to

- A. attract more insects to the garden
- B. make the soil hotter during the day
- C. stop the seeds from germinating
- D. keep the soil moist and control weeds

28. A class measured the heights of crops on different seed beds every week. The best way to record their results is in a

- A. poem
- B. letter
- C. story
- D. table

30. The instrument most suitable for measuring the height of a young crop is a

- A. measuring cylinder
- B. thermometer
- C. beam balance
- D. metre rule or tape measure

32. Five tomato plants have heights 20 cm, 22 cm, 24 cm, 26 cm and 28 cm. The average height of the plants is

- A. 22 cm
- B. 24 cm
- C. 26 cm
- D. 120 cm

34. In a fair test comparing two seed beds, which of the following must be kept the same?

- A. The depth to which each bed is dug
- B. The amount of manure put on each bed
- C. The type of seed bed being

compared

- D. The type of crop and the number of times it is watered

- 35.** Plants growing on a waterlogged bed are often yellow and short. This is mainly because
- A. the roots cannot get enough air in the waterlogged soil
 - B. the leaves receive too much sunlight

- C. the stem grows faster than the roots
- D. the plants have been watered too little
- 37.** A school garden records 18 cm, 26 cm and 21 cm as the average heights of maize on beds A, B and C. The bed on which the crop grew best is
- A. C
 - B. A
 - C. B
 - D. none of the beds
- 39.** A farmer in a swampy area in the Volta Region keeps losing his pepper plants to rotting roots. The best advice to give him is to
- A. stop watering his crops completely for a month
 - B. plant on raised beds so that excess water drains away
 - C. plant on sunken beds so that more water is held
 - D. plant the pepper deeper in the same wet soil
- 36.** Adjoa planted okro on a raised bed and on a sunken bed in a dry area. The plants on the sunken bed grew better. The most likely reason is that the sunken bed
- A. was closer to the farmer's house
 - B. contained no soil organisms at all
 - C. held more water for the plants during the dry weather
 - D. received more sunlight than the raised bed
- 38.** Which of these statements about seed beds is TRUE?
- A. Seed beds are only used for growing flowers
 - B. Different crops and different soils may need different types of seed beds
 - C. A sunken bed is always better than a flat bed
 - D. Every crop in Ghana must be planted on a raised bed
- 40.** Why should pupils write down and present the results of their seed bed experiment to the class?
- A. So that nobody else plants the same crop again
 - B. So that the teacher can give everybody the same mark
 - C. So that the seeds will germinate more quickly
 - D. So that others can learn from the findings and repeat the investigation

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Read every question carefully before you answer it.

Question 7

- (a) Do not use correction fluid. Cancel any mistake neatly with a single line.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 9: Ecosystems

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. An ecosystem is
 - A. the soil and rocks found in a particular place
 - B. a community of living things together with the non-living things in their environment
 - C. a group of animals of the same kind living together
 - D. a farm on which only one type of crop is grown
2. Which of the following is a non-living part of an ecosystem?
 - A. Mushroom
 - B. Water
 - C. Maize plant
 - D. Grasshopper
3. The original source of energy for almost all ecosystems on Earth is the
 - A. Sun
 - B. soil
 - C. wind
 - D. rain
4. Green plants are able to make their own food by the process of
 - A. transpiration
 - B. respiration
 - C. photosynthesis
 - D. digestion
5. An organism that makes its own food is called a
 - A. producer
 - B. predator
 - C. decomposer
 - D. consumer
6. Which of the following is a producer?
 - A. Earthworm
 - B. Cassava plant
 - C. Hawk
 - D. Goat
7. An organism that cannot make its own food and must feed on other organisms is a
 - A. mineral
 - B. producer
 - C. green plant
 - D. consumer
8. An animal that feeds directly on plants is called a
 - A. tertiary consumer
 - B. producer
 - C. secondary consumer
 - D. primary consumer
9. In the food chain: grass to goat to lion, the goat is the
 - A. primary consumer
 - B. producer
 - C. secondary consumer
 - D. decomposer
10. In the food chain: maize to rat to snake, the snake is the
 - A. secondary consumer
 - B. producer
 - C. primary consumer
 - D. decomposer
11. A food chain always begins with a
 - A. decomposer

- B. secondary consumer
- C. producer
- D. primary consumer

12. The arrows in a food chain show

- A. the direction in which energy flows from one organism to the next
- B. the order in which the organisms were discovered
- C. the way the wind blows across the farm

- D. the direction in which the animals walk
- 13.** In the food chain: grass to grasshopper to lizard to hawk, how many consumers are there?
- A. Three
 - B. One
 - C. Two
 - D. Four
- 15.** Which of the following best describes a food web?
- A. A set of food chains that are linked together in the same ecosystem
 - B. A net used by fishermen to catch fish
 - C. A single straight line of five organisms
 - D. A list of all the foods eaten in a home
- 17.** Animals that feed only on other animals are described as
- A. omnivores
 - B. carnivores
 - C. producers
 - D. herbivores
- 19.** The breaking down of dead organisms by decomposers is important in an ecosystem because it
- A. removes all the water from the soil
 - B. produces light energy for green plants
 - C. returns nutrients to the soil for plants to use again
 - D. increases the number of predators
- 21.** In the relationship between a hawk and a rat, the rat is the
- A. prey
 - B. decomposer
 - C. producer
 - D. predator
- 23.** Why are food chains usually short, with only four or five organisms?
- A. There are not enough names for the organisms
 - B. The Sun shines for only a few
- 14.** Many food chains joined together in an ecosystem form a
- A. carbon cycle
 - B. food store
 - C. food web
 - D. life cycle
- 16.** Animals that feed only on plants are described as
- A. producers
 - B. omnivores
 - C. carnivores
 - D. herbivores
- 18.** Organisms such as bacteria and fungi that break down dead plants and animals are called
- A. producers
 - B. decomposers
 - C. predators
 - D. primary consumers
- 20.** An animal that hunts and kills other animals for food is a
- A. producer
 - B. decomposer
 - C. predator
 - D. prey
- 22.** Energy is lost at each stage of a food chain mainly as
- A. sound
 - B. water
 - C. heat
 - D. light

hours each day

- C. Animals get tired of eating the same food
- D. Much of the energy is lost at each stage, so little is left for the organisms at the top

25. Which of the following statements about energy in an ecosystem is correct?

- A. Energy is made by the animals as they move about
- B. Energy flows from the consumers to the producers
- C. Energy moves round and round among the animals only
- D. Energy flows from the Sun to the producers and then to the consumers

24. In a food chain, the organism with the greatest amount of available energy is the

- A. primary consumer
- B. secondary consumer
- C. producer
- D. top carnivore

26. If all the green plants in an ecosystem were destroyed, the first organisms to suffer would be the

- A. carnivores
- B. top predators
- C. decomposers

- D. herbivores
- 28.** Which of the following would NOT happen if the Sun stopped shining?
- A. Food chains would break down
 - B. Green plants would grow faster than before
 - C. The Earth would become extremely cold
 - D. Photosynthesis would stop
- 30.** The removal of all the grasshoppers from an ecosystem would most directly affect the
- A. fishes in a nearby river
 - B. clouds in the sky
 - C. rocks in the soil
 - D. lizards that feed on them
- 27.** If there were no Sun, life on Earth would be impossible mainly because
- A. green plants could not make food, so there would be no food for other organisms
 - B. animals would not be able to sleep at night
 - C. there would be too much oxygen in the air
 - D. the Earth would spin more quickly on its axis
- 29.** In a maize farm, farmers kill all the snakes. The most likely immediate result is that the number of
- A. hawks will increase
 - B. rats will decrease
 - C. maize plants will increase
 - D. rats will increase
- 31.** Organisms in an ecosystem are said to be interdependent because they
- A. never meet one another at all
 - B. depend on one another for food and other needs
 - C. live in different countries
 - D. all look exactly the same
- 33.** In the food web of a pond, algae are eaten by small fishes, which are eaten by bigger fishes. The algae are the
- A. secondary consumers
 - B. decomposers
 - C. producers
- 32.** Which of the following is the correct order of a food chain?
- A. Bird, green plant, hawk, caterpillar
 - B. Hawk, bird, caterpillar, green plant
 - C. Caterpillar, green plant, bird, hawk
 - D. Green plant, caterpillar, bird, hawk

D. primary consumers

35. Which of the following animals is an omnivore?

- A. A cow
- B. A hawk
- C. A goat
- D. A fowl

37. A farmer sprays a chemical that kills all the insects on his farm. Which of the following is the most likely long-term effect?

- A. The Sun will stop shining on the farm
- B. Birds that feed on the insects will reduce in number
- C. The soil will turn into rock
- D. The maize plants will stop making food

34. An omnivore is an animal that feeds on

- A. plants only
- B. animals only
- C. both plants and animals
- D. dead organisms only

36. In the food chain: maize to rat to snake to hawk, which organism is both a predator and a prey?

- A. The rat only
- B. The hawk
- C. The snake
- D. The maize

38. Which of the following best explains why a food web gives a better picture of feeding relationships than a single food chain?

- A. A food web is easier to draw than a food chain
- B. Most organisms feed on more than one kind of food, and a web

- shows all those links
- C. A food web contains only producers
 - D. A food chain shows the names of all the organisms in the world
- 40.** The most important reason for protecting all the organisms in an ecosystem is that
- A. organisms must be counted every year
 - B. the removal of one organism can upset the whole balance of the ecosystem
 - C. every organism is beautiful to look at
 - D. all organisms can be sold for money
- 39.** Pupils want to study the feeding relationships in the grass around their school. The best method would be to
- A. observe the organisms carefully over some days and record what each one feeds on
 - B. copy a food chain from another country's textbook
 - C. ask their friends to guess what the animals eat
 - D. collect all the organisms and keep them in one jar

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use arrows correctly when you are asked to write or draw a food chain.

Question 7

- (a) Do not use correction fluid. Cancel any mistake neatly with a single line.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 10: Electricity and

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Electricity transmission means the
 - A. storing of electricity inside a dry cell
 - B. changing of electricity into heat in a wire
 - C. carrying of electricity from the power station to the places where it is used
 - D. measuring of the electricity used in a home
2. The place where electricity is produced in large quantities is the
 - A. distribution pole
 - B. substation
 - C. consumer unit
 - D. power station
3. Which of the following is the correct order of the stages of electricity transmission?
 - A. Generation, distribution, step-up, consumer, transmission lines, step-down
 - B. Consumer, distribution, transmission lines, generation, step-up, step-down
 - C. Generation, step-down, distribution, step-up, transmission lines, consumer
 - D. Generation, step-up, transmission lines, step-down, distribution, consumer
4. The device used to raise or lower the voltage of electricity is the
 - A. transformer
 - B. capacitor
 - C. ammeter
 - D. generator
5. A step-up transformer is used at the power station in order to
 - A. change alternating current into direct current
 - B. store the electricity until it is needed
 - C. raise the voltage before the electricity is sent along the transmission lines
 - D. lower the voltage before it reaches the homes
6. Electricity is transmitted over long distances at a very high voltage mainly to
 - A. make the cables heavier and stronger
 - B. reduce the energy lost as heat in the cables
 - C. increase the amount of heat produced in the cables
 - D. allow birds to perch safely on the wires
7. The tall steel towers that carry high-voltage transmission cables are called

- A. insulators
 - B. pylons
 - C. penstocks
 - D. turbines
- 9.** The wires that carry electricity from a substation to houses and shops in a town are the
- A. telephone lines
 - B. distribution lines
 - C. transmission lines
 - D. earth wires only
- 11.** The electricity meter in a house is used to
- A. store electricity for use at night
 - B. measure the amount of electrical energy used by the household
 - C. raise the voltage supplied to the house
 - D. change direct current into alternating current
- 8.** A substation in a town contains mainly
- A. water pumps for the town
 - B. capacitors for charging phones
 - C. turbines and boilers
 - D. step-down transformers and switchgear
- 10.** The glass or porcelain fittings that hold the cables away from the metal pylon are the
- A. insulators
 - B. fuses
 - C. transformers
 - D. conductors

- 13.** Which of the following is a reason why cables are supported high above the ground on pylons?
- A. To keep the high-voltage cables out of reach of people and vehicles
 - B. To make the cables cool faster in the rain
 - C. To allow the cables to carry water as well
 - D. To reduce the length of cable needed
- 15.** The earth wire in a house wiring system is provided mainly to
- A. store charge in the appliance
 - B. increase the brightness of the lamps
 - C. reduce the cost of electricity used
 - D. carry current safely to the ground if a fault occurs
- 17.** The national grid is best described as
- A. the generator used in a hospital
 - B. a single cable joining two towns
 - C. the meter box fixed in each house
 - D. the network of power stations, transmission lines and substations that supplies a country
- 19.** Which of the following is the MOST important safety rule near a fallen high-voltage cable?
- A. Pour water on it to cool it down
 - B. Lift it carefully with a wooden stick
 - C. Keep far away and report it at once to the electricity company
 - D. Cover it with sand and continue working
- 12.** The kind of current supplied by the national grid to homes in Ghana is
- A. direct current
 - B. static charge
 - C. alternating current
 - D. chemical current
- 14.** A fuse or circuit breaker is fitted in a house mainly to
- A. break the circuit when too large a current flows
 - B. increase the current flowing in the circuit
 - C. store energy for use during a power cut
 - D. measure the voltage at the socket
- 16.** Which of the following best explains why some energy is lost during transmission?
- A. Electricity leaks out of the cables into the air as light
 - B. Birds perching on the cables take away some energy
 - C. The cables have resistance, so some electrical energy becomes heat in them
 - D. The transformers destroy part of the energy
- 18.** A town at the end of a very long distribution line receives a lower voltage than one near the substation because
- A. electricity becomes weaker the longer it is stored
 - B. the far town uses a different kind of current
 - C. the cables near the substation are made of plastic
 - D. some voltage is lost along the resistance of the line
- 20.** A community wants fewer power cuts. Which of the following would help MOST?
- A. Regular maintenance of the lines and transformers that serve the community
 - B. Connecting more houses to one small transformer
 - C. Painting all the pylons a bright

colour

D. Asking everybody to switch on more appliances

22. The unit in which the resistance of a resistor is measured is the

- A. ampere
- B. ohm
- C. volt
- D. farad

21. A resistor is a component used in a circuit to

- A. store electrical charge
- B. limit or control the size of the current flowing
- C. allow current in one direction only
- D. change the voltage of the supply

23. A diode is a component that

- A. allows current to flow equally in both directions
 - B. measures the current in a circuit
 - C. allows current to flow through it in one direction only
 - D. stores charge on two metal plates
- 25.** An LED is used in a circuit mainly to
- A. change alternating current into heat
 - B. store charge for later use
 - C. increase the resistance of the circuit
 - D. give out light when current flows through it in the correct direction
- 27.** A capacitor is a component used to
- A. measure the resistance of a circuit
 - B. produce a steady current of its own
 - C. store electrical charge and release it later
 - D. allow current to pass in one direction only
- 29.** The unit of capacitance is the
- A. watt
 - B. farad
 - C. joule
 - D. ohm
- 31.** While a capacitor is charging, the current in the circuit
- A. flows backwards through the cell
 - B. is zero at first and then rises steadily
 - C. is large at first and then falls to zero
 - D. stays exactly the same all the time
- 24.** LED stands for
- A. light energy detector
 - B. light emitting diode
 - C. linked electrical diode
 - D. low energy device
- 26.** A small resistor is usually connected in series with an LED in order to
- A. limit the current so that the LED is not damaged
 - B. make the LED brighter than normal
 - C. change the colour of the light given out
 - D. store charge in the LED
- 28.** A simple capacitor is made of
- A. two liquids in a glass jar
 - B. a single plate of carbon
 - C. a long coil of copper wire only
 - D. two metal plates separated by an insulator
- 30.** When a capacitor is connected to a DC supply through a switch, it becomes charged when
- A. the supply is disconnected completely
 - B. the capacitor is removed from the circuit
 - C. the switch is closed and charge flows on to its plates
 - D. the switch is left open all the time
- 32.** A fully charged capacitor in a steady DC circuit
- A. allows no further current to flow through that branch
 - B. produces heat in the same way as a resistor
 - C. changes the direct current into alternating current
 - D. allows a large steady current to

flow for ever

- 33.** When a charged capacitor is disconnected from the cell and joined to an LED, the LED
- A. does not glow at all under any condition
 - B. glows steadily for many hours
 - C. becomes hotter and hotter until it bursts
 - D. glows brightly for a moment and

then fades out

35. Increasing the resistance in series with a capacitor makes the capacitor

- A. charge up more slowly
- B. change into a diode
- C. store no charge at all
- D. charge up more quickly

37. An LED connected the wrong way round in a DC circuit will

- A. store charge like a capacitor
- B. not light, because it conducts in one direction only
- C. light but give out a different colour
- D. light more brightly than before

39. A pupil says, "A capacitor is the same as a cell because both store energy." The statement is

- A. wrong, because a capacitor cannot store any energy
- B. completely correct, since both are sources of current
- C. correct, because both change chemical energy into electrical energy
- D. partly wrong, because a cell produces energy from chemicals while a capacitor only stores charge given to it

34. The LED in the discharge circuit fades because

- A. the charge stored in the capacitor is used up as it flows through the LED
- B. the resistance of the LED falls to zero
- C. the capacitor gains more charge from the LED
- D. the LED stores the charge and keeps it

36. Which of the following uses of a capacitor is correct?

- A. Changing chemical energy into electrical energy
- B. Preventing a lamp from lighting at all
- C. Storing charge to provide a flash in a camera
- D. Measuring the current flowing in a circuit

38. In a circuit containing a cell, a switch, a resistor, a capacitor and an LED, closing the switch makes the LED flash briefly and then go out. The best explanation is that

- A. the LED has burnt out completely
- B. current flows only while the capacitor is charging, and stops when it is fully charged
- C. the cell becomes flat within a second
- D. the resistor stops working after a short time

40. In designing a small torch that must flash for a moment each time a button is pressed, the components needed are

- A. a cell, a switch, a resistor, a capacitor and an LED
- B. a generator, a turbine and a penstock
- C. a cell, a transformer and a pylon
- D. two cells and a thermometer only

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler for the flow chart and for any circuit diagram.

Question 7

- (a) Do not use correction fluid. Cancel any mistake neatly with a single line.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 11: Energy Conversion

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Energy is best defined as
 - A. the ability to do work
 - B. the speed of a moving body
 - C. the weight of an object at rest
 - D. the force that holds matter together
2. The SI unit in which energy is measured is the
 - A. joule
 - B. newton
 - C. ampere
 - D. watt
3. Energy conversion means
 - A. the complete destruction of energy after use
 - B. the measuring of the energy used by a machine
 - C. the storing of energy in a container
 - D. the changing of energy from one form into another
4. The energy stored in a stretched catapult is
 - A. chemical energy
 - B. sound energy
 - C. kinetic energy
 - D. elastic potential energy
5. The energy stored in food, fuel and a dry cell is
 - A. chemical energy
 - B. nuclear energy
 - C. sound energy
 - D. elastic energy
6. The energy possessed by a body because of its motion is called
 - A. chemical energy
 - B. electrical energy
 - C. kinetic energy
 - D. potential energy
7. Water held in a reservoir high above a valley possesses mainly
 - A. chemical energy
 - B. light energy
 - C. kinetic energy
 - D. gravitational potential energy
8. An electric kettle changes electrical energy mainly into
 - A. chemical energy
 - B. heat energy
 - C. light energy
 - D. sound energy
9. A dry cell in a torch changes
 - A. light energy into heat energy
 - B. sound energy into electrical energy
 - C. chemical energy into electrical energy
 - D. electrical energy into chemical energy
10. An electric fan converts electrical energy mainly into
 - A. chemical energy
 - B. nuclear energy
 - C. kinetic (mechanical) energy
 - D. light energy
11. A solar panel on a roof changes
 - A. heat energy into chemical

- energy
- B. solar (light) energy into electrical energy
- C. electrical energy into light energy
- D. sound energy into electrical energy

13. Green plants carry out photosynthesis. In this process, energy is converted from

- A. chemical energy into light energy
- B. light energy into chemical energy
- C. electrical energy into chemical energy
- D. heat energy into kinetic energy

12. Which device changes electrical energy into sound energy?

- A. A loudspeaker
- B. A torch bulb
- C. A solar panel
- D. An electric iron

- 15.** A boy rubs his palms together quickly and they become warm. The energy change is
- potential energy into chemical energy
 - heat energy into kinetic energy
 - chemical energy into sound energy
 - kinetic energy into heat energy
- 17.** In a bicycle dynamo that lights a lamp, the complete energy chain is
- light energy to electrical energy to kinetic energy
 - chemical energy to kinetic energy to sound energy
 - kinetic energy to electrical energy to light energy
 - electrical energy to kinetic energy to light energy
- 19.** In a hydroelectric power station, the part that is turned by the falling water is the
- turbine
 - transformer
 - generator
 - penstock
- 21.** The part of a hydroelectric station that changes the movement of the turbine into electricity is the
- transmission line
 - generator
 - turbine
 - reservoir
- 23.** Which of the following gives the correct order of energy changes in a hydroelectric station?
- Potential energy to kinetic energy to mechanical energy to electrical energy
 - Chemical energy to heat energy to kinetic energy to electrical energy
 - Electrical energy to kinetic energy to potential energy to heat energy
 - Kinetic energy to potential energy to chemical energy to electrical energy
- 14.** When a mobile phone is being charged, the energy change taking place is
- electrical energy into chemical energy
 - heat energy into electrical energy
 - chemical energy into electrical energy
 - light energy into electrical energy
- 16.** A mango falls from a tree. As it falls, its energy changes from
- heat energy into potential energy
 - kinetic energy into potential energy
 - potential energy into kinetic energy
 - chemical energy into kinetic energy
- 18.** In a hydroelectric power station, the energy stored in the water behind the dam is
- chemical energy
 - nuclear energy
 - sound energy
 - potential energy
- 20.** The large pipe that carries water from the reservoir down to the turbine is the
- shaft
 - pylon
 - penstock
 - spillway
- 22.** The main purpose of building the dam wall across a river is to
- clean the water for drinking purposes
 - store a large volume of water at a high level
 - cool the water before it is used
 - prevent fish from swimming upstream

- 25.** In a thermal power station, the fuel is burnt to boil water. The steam produced then
- A. is used to wash the coal before burning
 - B. cools the generator so that it works faster
 - C. turns the turbine, which drives the generator
 - D. is changed directly into electricity in the boiler
- 27.** Biogas is produced when
- A. bacteria break down animal dung and other waste in a digester
 - B. water is passed over hot coal
 - C. crude oil is heated in a refinery
 - D. wind is trapped in a large tank
- 29.** Which of the following is the best example of harnessing a natural form of energy?
- A. Using petrol in a generating set at home
 - B. Using a wind turbine at Ada to pump water from a well
 - C. Using a dry cell to light a torch at night
 - D. Using a car battery to start an engine
- 31.** In every energy conversion, some of the energy is
- A. created out of nothing
 - B. lost to the surroundings, usually as heat
 - C. changed into matter in the machine
 - D. completely destroyed
- 33.** An electric iron is left switched on with nobody using it. The main result is that
- A. the electrical energy is stored for later use
 - B. electrical energy is converted to heat and wasted
 - C. the iron converts heat back into electricity
 - D. no energy conversion takes place at all
- 24.** The Akosombo Dam on the Volta River is important to Ghana mainly because it
- A. produces coal for export
 - B. provides crude oil for the refinery
 - C. supplies natural gas to the north
 - D. generates electricity for homes and industries
- 26.** A windmill (wind turbine) harnesses energy by changing
- A. heat energy of the air into chemical energy
 - B. electrical energy into the motion of the air
 - C. sound energy of the wind into light energy
 - D. the kinetic energy of moving air into electrical energy
- 28.** A solar water heater on a roof changes
- A. chemical energy into light energy
 - B. electrical energy into chemical energy
 - C. heat energy into solar energy
 - D. solar energy into heat energy in the water
- 30.** Energy conversion is important mainly because
- A. it reduces the total amount of energy in the world
 - B. energy can be created whenever it is needed
 - C. most of the energy around us is not in the form in which we can use it directly
 - D. it makes machines work without any energy at all
- 32.** An electric bulb is described as inefficient because
- A. it gives out no light at all
 - B. it destroys some of the energy supplied to it
 - C. it uses chemical energy instead of electrical energy
 - D. much of the electrical energy supplied to it is given out as heat rather than light

- 35.** A pupil says, "The generator in a power station creates electricity." The statement is wrong because the generator
- A. only converts the energy supplied to it into electrical energy
 - B. changes electricity into movement
 - C. destroys the energy of the moving turbine
 - D. stores the electricity that is already in the water
- 37.** A school wishes to light its classrooms during the day without using electricity from the national grid. The most suitable arrangement is
- A. solar panels on the roof connected to lamps and a battery
 - B. a petrol generating set running all day
 - C. a windmill placed inside the classroom
 - D. a dry-cell torch in each classroom
- 39.** Two torches are the same, but one has a filament bulb and the other an LED bulb. The LED torch gives light for much longer on the same cells because it
- A. changes light energy back into chemical energy
 - B. creates extra energy inside the bulb
 - C. uses no energy from the cells at all
 - D. converts a greater part of the electrical energy into light
- 34.** Which of the following energy chains correctly describes a petrol generating set lighting a bulb?
- A. Kinetic energy to chemical energy to heat energy to light energy
 - B. Chemical energy to heat energy to kinetic energy to electrical energy to light energy
 - C. Electrical energy to chemical energy to light energy to heat energy
 - D. Light energy to chemical energy to kinetic energy to electrical energy
- 36.** During a long drought the level of water in a dam falls sharply. The station produces less electricity because
- A. the turbines change into pumps
 - B. electricity cannot travel in dry weather
 - C. there is less stored water to fall and turn the turbines
 - D. the generator becomes too cold to work
- 38.** Which of the following best explains why a community near a fast-flowing stream may build a small water-wheel?
- A. The sound of the water can be changed into electricity
 - B. The kinetic energy of the moving water can be used to turn machines or a small generator
 - C. Water wheels produce energy without any energy input
 - D. Moving water contains chemical energy that can be burnt
- 40.** A town is choosing between a diesel generating plant and a solar plant of the same output. Which reason BEST supports choosing the solar plant?
- A. It produces energy without any conversion taking place
 - B. It can work at exactly the same rate all through the night
 - C. Its energy source is free, renewable and does not pollute the air

D. It needs no space on the ground at all

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) When an energy change is asked for, write it in the form 'A energy to B energy'.

Question 7

- (a) Use a pencil and a ruler where a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 12: Force and Magnetism

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A magnet is a material that
 - A. dissolves in water to form a solution
 - B. conducts heat better than every other material
 - C. attracts iron and other magnetic materials
 - D. burns easily when it is heated
2. Which of the following materials is attracted by a magnet?
 - A. Plastic bucket
 - B. Rubber eraser
 - C. Wooden ruler
 - D. Iron nail
3. Which of the following is NOT a magnetic material?
 - A. Iron
 - B. Steel
 - C. Copper
 - D. Nickel
4. The ends of a bar magnet where the magnetic force is strongest are called the
 - A. poles
 - B. edges
 - C. sides
 - D. centres
5. When the north pole of one magnet is brought near the north pole of another magnet, they
 - A. become hot and bend
 - B. join together and lose their magnetism
 - C. repel each other
 - D. attract each other
6. A freely suspended bar magnet finally comes to rest pointing
 - A. north and south
 - B. east and west
 - C. in any direction at all
 - D. straight upwards
7. Which of the following is a simple way of making a magnet?
 - A. Rubbing a steel bar with palm oil
 - B. Placing a steel bar in the sunshine for one hour
 - C. Soaking a steel bar in clean water overnight
 - D. Stroking a steel bar in one direction with one pole of a bar magnet
8. A magnet made by winding insulated wire round a soft iron core and passing an electric current through the wire is called
 - A. an electromagnet
 - B. a compass needle
 - C. a permanent magnet
 - D. a dry cell
9. In the single-touch (stroking) method of making a magnet, the magnet must be stroked along the bar
 - A. to and fro as quickly as possible
 - B. while the bar is being heated strongly

- C. once and then kept in water
- D. in one direction only, many times

11. A magnet made from steel is described as a permanent magnet because it

- A. can only attract other steel magnets
- B. loses its magnetism as soon as it is made
- C. keeps its magnetism for a long time
- D. works only when a current is flowing

10. Soft iron is used for the core of an electromagnet because it

- A. cannot be magnetised at all by an electric current
- B. keeps its magnetism for ever once the current has flowed
- C. is lighter than every other metal
- D. becomes a strong magnet when the current flows and loses its magnetism when the current is switched off

- 13.** A magnet can lose its magnetism when it is
- painted with a thin layer of oil
 - heated strongly or hammered repeatedly
 - wrapped in a piece of clean cloth
 - kept in a dry box with a keeper
- 15.** The instrument that uses a small magnet to show direction to a traveller is the
- compass
 - voltmeter
 - barometer
 - thermometer
- 17.** A loudspeaker produces sound because the magnet in it
- changes the sound into electricity for the radio
 - heats the air inside the box
 - makes a coil and a cone vibrate when a current passes through the coil
 - stores the sound until it is needed
- 19.** The magnetic door catch of a cupboard is an example of the use of magnetic force in
- printing activities
 - domestic activities
 - fishing activities
 - mining activities
- 21.** A dynamo on a bicycle uses a magnet to
- cool the wheel while the bicycle moves
 - store electricity for the next journey
 - change the movement of the wheel into electrical energy for the lamp
 - change the light of the lamp into movement
- 12.** Which of the following will make an electromagnet stronger?
- Opening the switch in the circuit
 - Using fewer dry cells in the circuit
 - Using a shorter piece of plastic as the core
 - Increasing the number of turns of wire round the core
- 14.** The space around a magnet in which its force can be felt is called the
- magnetic circuit
 - magnetic field
 - magnetic keeper
 - magnetic pole
- 16.** Magnetic force is used in a scrap yard to
- cut the metal sheets into small pieces
 - paint the old vehicles before they are sold
 - weigh the lorries at the gate
 - lift and drop pieces of iron and steel without touching them by hand
- 18.** In an electric bell, the electromagnet
- changes the sound of the bell into light
 - stores current for the next ringing
 - pulls the armature so that the hammer strikes the gong
 - cools the wires so that they do not burn
- 20.** Magnetic force is used in a security alarm at a door mainly to
- detect when the door has been opened and set off the warning sound
 - lock the door with a key
 - light the room when the door is opened
 - weigh the person entering the room
- 22.** Newton's Second Law of motion states that the acceleration of a body is
- the same for all bodies whatever the force applied
 - equal to the mass of the body

- divided by the force applied
- C. directly proportional to the force applied and inversely proportional to the mass of the body
 - D. directly proportional to the mass and inversely proportional to the force

- 24.** The unit of force is the
- A. joule
 - B. newton
 - C. metre per second
 - D. kilogramme

- 23.** Newton's Second Law of motion is written in symbols as
- A. $F = m \div a$
 - B. $F = ma$
 - C. $m = F \times a$
 - D. $a = m \times F$

- 25.** A force of 12 N acts on a body of

mass 3 kg. The acceleration produced is

- A. 0.25 m/s^2
- B. 15 m/s^2
- C. 36 m/s^2
- D. 4 m/s^2

27. When the same force is applied to a light trolley and to a heavy trolley, the light trolley

- A. does not move at all
- B. gains a smaller acceleration
- C. gains a greater acceleration
- D. gains exactly the same acceleration

29. Two loaded trucks are pushed with the same force. Truck X moves off faster than Truck Y. This shows that Truck X

- A. has a smaller mass than Truck Y
- B. has no mass at all
- C. is made of a magnetic material
- D. has a larger mass than Truck Y

31. In an experiment, a magnetic force of 2 N gave a trolley an acceleration of 1 m/s^2 . When the force was raised to 6 N, the acceleration became 3 m/s^2 . The results show that, for a fixed mass,

- A. the mass of the trolley rises with the force
- B. the acceleration is directly proportional to the force applied
- C. the acceleration does not depend on the force applied
- D. the acceleration falls as the force rises

26. A trolley of mass 5 kg is given an acceleration of 3 m/s^2 . The force acting on it is

- A. 8 N
- B. 15 N
- C. 45 N
- D. 1.7 N

28. A magnet is moved nearer to a trolley carrying a steel bar, so that the magnetic force on the trolley becomes larger. The acceleration of the trolley will

- A. fall to zero
- B. remain exactly the same
- C. decrease
- D. increase

30. A pupil pushes an empty wheelbarrow and then the same wheelbarrow loaded with sand, using the same force each time. The loaded wheelbarrow moves off more slowly because

- A. the force becomes smaller when sand is added
- B. its mass is larger, so the acceleration produced by the same force is smaller
- C. the wheel stops turning when the barrow is loaded
- D. sand is a magnetic material

32. A simple machine is a device that

- A. works only when electricity is supplied
- B. produces energy for the home
- C. makes work easier by changing

the size or the direction of a force

D. destroys the force applied to it

34. A pair of scissors works mainly as

A. two screws turning together

B. two wheels on one axle

C. two pulleys joined by a rope

D. two levers joined at a pivot

33. Which of the following is a simple machine?

A. An inclined plane

B. A motor vehicle

C. A corn mill

D. A sewing machine

35. A complex machine is best described as a machine

A. made up of two or more simple machines working together

B. that works without any force being applied

C. that is too heavy to be carried by

- one person
D. made of one simple machine used twice
- 37.** The simple machine found in the pedal and crank of a bicycle is the
- A. wedge
 - B. wheel and axle
 - C. screw
 - D. inclined plane
- 39.** A complex machine such as a corn mill improves the quality of life mainly because it
- A. does heavy work quickly, so that people save time and energy
 - B. removes the need for any energy supply
 - C. makes the food more expensive to buy
 - D. replaces the need for farmers to grow maize
- 36.** Which of the following is a complex machine?
- A. A single fixed pulley
 - B. A bicycle
 - C. A ramp
 - D. A screw
- 38.** A grinding mill in a Ghanaian market is a complex machine because it
- A. is kept inside a large building
 - B. costs more money than a wheelbarrow
 - C. is operated by more than one person
 - D. is built from several simple machines, such as wheels and axles, screws and belts, working together
- 40.** A disadvantage of depending on complex machines is that
- A. they need fuel or electricity and must be repaired and maintained
 - B. they never break down at all
 - C. they cannot be used in the home
 - D. they always do work more slowly than people

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures and the table carefully before you answer the questions on them.

Question 6

- (a) Show your working clearly in all the questions that require a calculation.

Question 7

- (a) Use a pencil and a ruler where a plan or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 13: Human Body

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The three main regions of a mammalian tooth are the
 - A. enamel, dentine and pulp
 - B. gum, jaw bone and cement
 - C. incisor, canine and molar
 - D. crown, neck and root
2. The part of a tooth that is seen above the gum is the
 - A. root
 - B. crown
 - C. pulp cavity
 - D. neck
3. The part of a tooth that is fixed in the jaw bone is the
 - A. root
 - B. crown
 - C. neck
 - D. enamel
4. The narrow part of the tooth found at the level of the gum is the
 - A. neck
 - B. root
 - C. crown
 - D. cement
5. The hardest substance in the human body is
 - A. bone
 - B. enamel
 - C. dentine
 - D. cement
6. Enamel covers which part of a tooth?
 - A. The jaw bone
 - B. The crown
 - C. The pulp cavity
 - D. The root
7. The layer of a tooth found just beneath the enamel is the
 - A. pulp
 - B. cement
 - C. gum
 - D. dentine
8. The soft central part of a tooth which contains nerves and blood vessels is the
 - A. cement
 - B. pulp cavity
 - C. enamel
 - D. dentine
9. The layer that covers and protects the root of a tooth is the
 - A. plaque
 - B. cement
 - C. enamel
 - D. dentine
10. Blood vessels in the pulp cavity of a tooth are important because they
 - A. hold the tooth firmly in the jaw
 - B. grind the food into smaller pieces
 - C. supply food and oxygen to the living cells of the tooth
 - D. make the tooth white in colour
11. A person feels pain when a decayed tooth reaches the pulp cavity mainly because the pulp contains
 - A. nerves
 - B. enamel
 - C. cement
 - D. plaque
12. The teeth are held firmly in position in the jaw bone by the
 - A. gum and the cement, with fibres in the socket
 - B. enamel and the dentine
 - C. saliva and the tongue

D. pulp cavity and the nerves

- 13.** Which of the following is NOT a part of a mammalian tooth?
- A. Enamel
 - B. Dentine
 - C. Pulp cavity
 - D. Cartilage
- 15.** The chisel-shaped teeth found at the front of the mouth are the
- A. incisors
 - B. canines
 - C. premolars
 - D. molars
- 17.** Pointed teeth used for tearing food are the
- A. premolars
 - B. molars
 - C. incisors
 - D. canines
- 19.** The broad teeth at the back of the mouth with large surfaces for grinding food are the
- A. molars
 - B. milk teeth
 - C. incisors
 - D. canines
- 21.** An adult human being normally has
- A. 28 teeth
 - B. 20 teeth
 - C. 32 teeth
 - D. 36 teeth
- 23.** Which of the following best explains why molars have a broad surface?
- A. It helps the tongue to taste food
 - B. It allows the teeth to grow faster
 - C. It provides a large area for grinding food
 - D. It makes the teeth look attractive
- 25.** The breaking down of a tooth by acids produced in the mouth is called
- A. tooth replacement
 - B. tooth decay
 - C. tooth growth
 - D. tooth eruption
- 14.** The four types of teeth found in a human being are
- A. incisors, canines, molars and milk teeth
 - B. crown, neck, root and cement
 - C. incisors, canines, premolars and molars
 - D. enamel, dentine, pulp and cement
- 16.** The main function of the incisors is to
- A. hold food in the mouth only
 - B. tear flesh from bones
 - C. grind and chew food
 - D. cut and bite food
- 18.** The canines are very well developed in
- A. a sheep
 - B. a goat
 - C. a dog
 - D. a cow
- 20.** The main function of the premolars and molars is to
- A. push the food into the throat
 - B. cut the food into two halves
 - C. tear flesh away from bones
 - D. grind and chew food into smaller pieces
- 22.** The first set of teeth that a child develops is called the
- A. milk teeth
 - B. wisdom teeth
 - C. false teeth
 - D. permanent teeth
- 24.** Chewing food properly with the premolars and molars is important because it
- A. removes all the germs from the food
 - B. increases the surface area of the food for digestion
 - C. turns the food into blood immediately
 - D. makes the food taste sweeter in the stomach

- 27.** Tooth decay is caused mainly by
- brushing the teeth twice every day
 - eating fruits and vegetables daily
 - acids produced by bacteria acting on sugary food left on the teeth
 - drinking too much clean water
- 29.** Which part of a tooth is attacked first when tooth decay begins?
- The nerve
 - The pulp cavity
 - The root
 - The enamel
- 31.** Which of the following is the best way to prevent tooth decay?
- Eating sweets slowly throughout the day
 - Rinsing the mouth with water once a week
 - Chewing only on one side of the mouth
 - Brushing the teeth properly at least twice a day, especially before going to bed
- 33.** Visiting the dentist regularly is important mainly because
- problems can be found and treated early before they become serious
 - it prevents a person from eating sugar
 - the dentist will remove all the teeth
 - the teeth will grow longer and stronger
- 35.** Which of the following habits is most likely to damage the teeth?
- Rinsing the mouth after meals
 - Using the teeth to open bottle tops
 - Changing the toothbrush regularly
 - Eating raw carrots and fruits
- 26.** The soft, sticky layer of food particles and bacteria that forms on the surface of teeth is called
- enamel
 - saliva
 - plaque
 - cement
- 28.** Which of the following foods is most likely to cause tooth decay if eaten often and the teeth are not cleaned?
- Toffees and sweet biscuits
 - Roasted groundnuts
 - Boiled kontomire
 - Fresh garden eggs
- 30.** Bleeding and swollen gums are signs of
- a healthy mouth
 - normal tooth growth
 - gum disease
 - strong enamel
- 32.** A chewing stick or a toothbrush should be used to brush the teeth
- on the front teeth alone, since they are the ones seen
 - upwards and downwards, reaching all surfaces of the teeth
 - from side to side across the front teeth only
 - only when there is pain in the mouth
- 34.** Eating foods rich in calcium, such as milk and small fishes eaten with their bones, helps mainly to
- clean the plaque off the teeth
 - make the gums bleed less often
 - reduce the amount of saliva produced
 - build strong teeth and bones
- 36.** Kwame eats toffees every evening and goes to bed without brushing his teeth. The most likely result after some time is that he will
- grow an extra set of teeth
 - produce more saliva than normal
 - develop tooth decay and toothache
 - have whiter and stronger enamel

- 37.** Why is it especially important to brush the teeth before going to bed?
- A. The teeth grow only during the night
 - B. Bacteria die on their own during the day
 - C. Brushing at night makes a person sleep deeply
 - D. Less saliva is produced during sleep, so food left on the teeth is attacked by bacteria for many hours
- 39.** Which of the following statements about teeth is TRUE?
- A. All the teeth in the human mouth have the same shape
 - B. Different types of teeth perform different functions in feeding
 - C. The enamel of a decayed tooth grows back by itself
 - D. Permanent teeth that fall out are always replaced
- 38.** A pupil has a hole in a molar but feels no pain. The best advice to give the pupil is to
- A. report to a dentist early so that the tooth can be filled before the decay reaches the pulp
 - B. chew only on that side so that the hole closes
 - C. wait until the tooth begins to pain before doing anything
 - D. pull out the tooth at home immediately
- 40.** The most important reason for taking good care of the teeth is that
- A. healthy teeth prevent a person from feeling hungry
 - B. healthy teeth make a person run faster
 - C. healthy teeth allow proper chewing and digestion of food, and prevent pain and disease
 - D. healthy teeth stop a person from talking too much

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Read every question carefully before you answer it.

Question 7

- (a) Do not use correction fluid. Cancel any mistake neatly with a single line.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 14: Human Health

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A communicable disease is a disease that
 - A. can be passed from one person to another
 - B. cannot be treated by any drug
 - C. is caused by a shortage of food nutrients
 - D. is inherited from one's parents only
2. Which of the following is a communicable disease?
 - A. Measles
 - B. Night blindness
 - C. Diabetes
 - D. Sickle-cell disease
3. Which of the following is NOT a communicable disease?
 - A. Hepatitis B
 - B. Tuberculosis
 - C. Hypertension (high blood pressure)
 - D. Cholera
4. Communicable diseases are caused by
 - A. the hardening of the walls of the arteries
 - B. a shortage of vitamins in the diet
 - C. the inheritance of faulty genes
 - D. germs such as viruses, bacteria, fungi and protozoa
5. Cholera and typhoid fever are spread mainly through
 - A. food and water contaminated with human faeces
 - B. the bite of a tsetse fly
 - C. the sharing of school desks
 - D. breathing the air in a cold room
6. Measles and tuberculosis are spread mainly through
 - A. drinking water from a covered pot
 - B. walking barefoot on wet grass
 - C. eating food that has been well cooked
 - D. droplets released when an infected person coughs or sneezes
7. Malaria is spread from one person to another by
 - A. sharing a cup with a patient
 - B. breathing the same air as a patient
 - C. the bite of an infected female Anopheles mosquito
 - D. shaking hands with a patient
8. Hepatitis B is caused by
 - A. a worm
 - B. a virus
 - C. a bacterium
 - D. a fungus
9. The organ of the body chiefly attacked by the hepatitis B virus is the
 - A. heart
 - B. liver
 - C. kidneys
 - D. lungs
10. Which of the following is a common symptom of hepatitis B?
 - A. Bowing of the legs
 - B. Loss of hearing
 - C. Yellowing of the eyes and the skin
 - D. Swelling of the gums only

- 11.** The yellowing of the eyes and the skin seen in a hepatitis patient is known as
- dehydration
 - jaundice
 - inflammation of the throat
 - anaemia
- 13.** Hepatitis B is commonly spread through
- eating fruits that have not been washed
 - sitting beside an infected person in class
 - contact with the blood or body fluids of an infected person
 - breathing the air in a crowded market
- 15.** A mother infected with the hepatitis B virus can pass it on to her baby
- during pregnancy or at birth
 - by talking to the baby every day
 - by carrying the baby on her back
 - only when the child is ten years old
- 17.** The most effective way of protecting a child against hepatitis B is
- drinking plenty of palm wine
 - eating only roasted foods
 - bathing twice every day
 - immunisation with the hepatitis B vaccine
- 19.** Blood collected for transfusion is screened before it is used in order to
- increase the amount of blood collected
 - make the blood thicker
 - change the blood group of the donor
 - make sure that it does not carry hepatitis B, HIV and other germs
- 21.** AIDS is best described as
- a disease that is inherited from the parents
 - the condition that develops when HIV has weakened the body's defence system
 - a disease of the bones caused by poor diet
 - an allergy to certain kinds of
- 12.** Apart from the yellowing of the eyes, another symptom of hepatitis B is
- dark urine with tiredness and loss of appetite
 - a sudden increase in body weight
 - an itching rash on the palms only
 - the growth of extra teeth
- 14.** Which of the following practices can spread hepatitis B?
- Playing football with an infected friend
 - Sharing razor blades, needles and toothbrushes
 - Eating from a clean bowl with a friend
 - Sharing a textbook in class
- 16.** A serious long-term effect of hepatitis B on a patient is
- damage to the liver, which may lead to liver cancer
 - the breaking of the bones of the legs
 - the complete loss of all the teeth
 - permanent loss of the sense of smell
- 18.** A barber can prevent the spread of hepatitis B in his shop by
- washing the floor of the shop once a week
 - sterilising the clippers and using a new blade for each customer
 - charging a higher price for a haircut
 - giving each customer a cold drink
- 20.** The letters HIV stand for
- harmful internal virus
 - human immunodeficiency virus
 - human infectious viral anaemia
 - high internal viral fever

food

- 22.** HIV weakens the body because it attacks the
- A. enamel of the teeth
 - B. bones, which support the body
 - C. white blood cells, which defend the body against disease
 - D. red blood cells, which carry

23. Which of the following can spread HIV?

- A. Being bitten by a housefly
- B. Sharing a seat with an infected person
- C. Transfusion with unscreened infected blood
- D. Eating food prepared by an infected person

25. A person living with HIV should be

- A. blamed for the illness in public
- B. sent away from the community
- C. kept away from all school activities
- D. treated with care and respect, and supported to take the prescribed medicines

27. A common sign of measles in a child is

- A. a deep cut on the skin
- B. a sudden loss of all the hair
- C. the swelling of one leg only
- D. fever with a rash over the body and red watery eyes

29. Measles can best be prevented by

- A. immunising children with the measles vaccine
- B. giving children a daily dose of salt
- C. keeping children indoors all day
- D. feeding children on only one kind of food

31. A patient with cholera loses a lot of water from the body through severe diarrhoea and vomiting. The first aid given is to

- A. give oral rehydration salt solution and send the patient to hospital at once
- B. ask the patient to sleep until the next day
- C. give the patient a bottle of soft drink and wait
- D. give the patient a cold bath only

33. Which of the following best prevents the spread of communicable diseases in a school?

- A. Sharing drinking cups in the

oxygen

24. HIV is NOT spread by

- A. an infected mother breastfeeding her baby
- B. unprotected sexual intercourse with an infected person
- C. sharing needles and other skin-piercing instruments
- D. hugging or shaking hands with an infected person

26. Measles is caused by

- A. a virus
- B. a bacterium
- C. a protozoan
- D. a mineral deficiency

28. A serious effect of measles in young children is that it can lead to

- A. the growth of stronger bones
- B. an increase in height
- C. blindness, pneumonia and even death
- D. a permanent change of eye colour

30. A child suffering from measles should be kept away from other children in order to

- A. make the child recover more slowly
- B. punish the child for falling sick
- C. allow the rash to grow larger
- D. prevent the disease from spreading to them

32. Tuberculosis is caused by a bacterium and mainly attacks the

- A. liver
- B. skin
- C. lungs
- D. kidneys

classroom

- B. Washing the hands with soap under running water at the right times
- C. Sweeping the classroom once a term
- D. Keeping all the windows closed during lessons

34. Immunisation protects a person from a disease because it

- A. replaces the blood of the person with new blood
- B. makes the person too strong to fall sick at all
- C. causes the body to produce antibodies against the germ before the person meets it
- D. kills every germ that enters the

body for life

- 35.** In a community where many people fall sick with typhoid fever, the most likely source of the infection is
- A. the sharing of school books
 - B. the noise from a nearby factory
 - C. a contaminated well or water supply
 - D. the hot weather in the afternoon
- 37.** When pupils report the data they have collected on patients, they must
- A. change the figures so that the report looks better
 - B. publish the names of all the patients on the notice board
 - C. tell their friends which families are affected
 - D. keep the names of the patients confidential
- 39.** A record showing that more males than females suffer from hepatitis B in a community is most useful because it
- A. means that the males should be sent out of the community
 - B. shows that the disease is not serious at all
 - C. proves that females can never catch the disease
 - D. helps the health workers to direct their education and screening to the group at greater risk
- 36.** Pupils compiling data on communicable diseases from a health centre should obtain the figures
- A. from the records of the centre, with the permission of the staff in charge
 - B. by copying the figures from another district
 - C. by guessing from the number of people they see outside
 - D. by asking patients to show their prescriptions to the class
- 38.** In one year a health centre recorded 40 hepatitis B cases out of 100 cases of communicable diseases. The percentage of the cases that were hepatitis B is
- A. 4%
 - B. 40%
 - C. 140%
 - D. 60%
- 40.** The best plan for a school that wants to reduce hepatitis B among its pupils is to
- A. stop all sporting activities in the school
 - B. organise health education, encourage vaccination and stop the sharing of blades and needles
 - C. serve all meals in the classroom instead of the canteen
 - D. ask every sick pupil to stay at home for the whole year

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show your working clearly in all the questions that require a calculation.

Question 7

- (a) Use a pencil and a ruler where a plan or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 15: Life Cycle A

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A vector is best described as
 - A. an organism that cannot cause any disease.
 - B. a medicine used to treat malaria.
 - C. a chemical used to kill insects.
 - D. an organism that carries a disease-causing organism from one host to another
2. Which of the following diseases is spread by the female Anopheles mosquito?
 - A. cholera
 - B. malaria
 - C. typhoid fever
 - D. sleeping sickness
3. Malaria is caused by
 - A. a bacterium.
 - B. a virus.
 - C. a fungus.
 - D. a parasite called plasmodium
4. The main reason why the female Anopheles mosquito bites human beings is to
 - A. obtain blood that she needs for the development of her eggs
 - B. obtain nectar from flowers.
 - C. make a home inside the body.
 - D. get water for her eggs.
5. How many stages are there in the life cycle of the mosquito?
 - A. two
 - B. four
 - C. three
 - D. five
6. The stages of the life cycle of the mosquito follow the order
 - A. egg, pupa, larva, adult
 - B. larva, egg, pupa, adult
 - C. egg, larva, pupa, adult
 - D. egg, larva, adult, pupa
7. The Anopheles mosquito lays its eggs
 - A. in a raft (a group) on the surface of water.
 - B. singly on the surface of water
 - C. inside the human body.
 - D. on dry ground and in sand.
8. The stage of the mosquito that lives in water and is called a wriggler is the
 - A. egg
 - B. larva
 - C. pupa
 - D. adult
9. Which stage of the mosquito does ****not**** feed at all?
 - A. the egg
 - B. the larva
 - C. the pupa
 - D. the adult
10. The stage of the mosquito that has wings and lives on land is the
 - A. egg
 - B. adult
 - C. larva
 - D. pupa
11. Which of the following is ****not**** a stage in the life cycle of the mosquito?
 - A. maggot
 - B. egg
 - C. larva

D. pupa

- 13.** The larva of the mosquito takes in air through a tube at the
- A. bottom of the water
 - B. leaves of plants
 - C. skin of other animals
 - D. surface of the water
- 15.** Which of the following will ****not**** help to reduce mosquito breeding in a compound?
- A. keeping old tyres and broken bottles filled with rain water
 - B. pouring away water collected in discarded containers.
 - C. covering all water storage containers.
 - D. clearing the weeds and bushes around the house.
- 17.** The best way to get rid of stagnant water in a compound is to
- A. store the water in open barrels.
 - B. add soap to the water and leave it standing.
 - C. cover the ground with dry leaves.
 - D. drain the water away and fill up the hollows with sand
- 19.** The female Anopheles mosquito bites human beings mainly
- A. at noon
 - B. only at sunrise
 - C. at night
 - D. only in the dry season
- 21.** Malaria is described as an economic burden on Ghana mainly because it
- A. makes the weather hotter.
 - B. destroys the crops on farms.
 - C. keeps workers and pupils away from work and school and costs money for treatment
 - D. kills the fish in the rivers.
- 23.** The part of the mosquito that pierces the skin to suck blood is the
- A. antenna
 - B. wing
 - C. leg
 - D. proboscis
- 12.** The adult mosquito comes out of the
- A. egg
 - B. larva
 - C. pupa
 - D. nymph
- 14.** Mosquitoes breed (lay their eggs and develop) mainly in
- A. fast-flowing rivers
 - B. stagnant water
 - C. dry sandy soil
 - D. tall grasses
- 16.** The best way to protect yourself from mosquito bites when you are sleeping is to
- A. sleep with all the windows wide open.
 - B. sleep on the bare floor.
 - C. sleep under a treated mosquito net
 - D. drink plenty of water before sleeping.
- 18.** The chemical that is used to kill mosquitoes is called an
- A. herbicide
 - B. fungicide
 - C. fertiliser
 - D. insecticide
- 20.** One way in which mosquitoes are important in nature is that
- A. they produce honey.
 - B. their larvae and adults are food for fishes and other animals
 - C. they add nutrients to the soil.
 - D. they pollinate maize flowers.
- 22.** Which of the following is ****not**** a symptom of malaria?
- A. high fever
 - B. headache
 - C. a rash all over the body
 - D. chills and shivering
- 24.** The pair of antennae on the head of the mosquito is used for
- A. sensing (feeling) the surroundings
 - B. sucking blood
 - C. flying to far places
 - D. laying eggs in water

- 25.** An *Anopheles* mosquito can be recognised when it is resting because it rests
- A. with its body at an angle to the surface
 - B. with its body flat (parallel) to the surface.
 - C. hanging upside down from the ceiling.
 - D. with its wings folded out on the ground.
- 27.** Only the female *Anopheles* mosquito spreads malaria because
- A. only the female bites people and takes blood, which she needs for her eggs
 - B. the female mosquito is bigger than the male.
 - C. the male mosquito has no proboscis.
 - D. the female mosquito lives longer than the male.
- 29.** The stage that hatches out of the egg of the mosquito is the
- A. pupa
 - B. adult
 - C. nymph
 - D. larva
- 31.** Fishes are sometimes put into ponds to control mosquitoes. This method works because the fishes
- A. kill the adult mosquitoes in the air.
 - B. lay eggs that destroy mosquito eggs.
 - C. feed on the mosquito larvae and pupae in the water
 - D. make the water too hot for the mosquitoes.
- 33.** The larva of the mosquito stays near the surface of the water mainly to
- A. hide from the fishes
 - B. lay its eggs
 - C. catch fishes to eat
 - D. breathe in air
- 35.** A pupil kept some pond water in a jar and counted 20 wrigglers and 5 pupae. Which of the following statements is correct?
- 26.** The male *Anopheles* mosquito feeds mainly on
- A. human blood
 - B. animal blood
 - C. nectar (plant juice)
 - D. water from the soil
- 28.** The mosquito is said to undergo complete metamorphosis because
- A. the young looks exactly like the adult.
 - B. the young (larva) is very different from the adult and there is a pupa stage
 - C. it lays many eggs at a time.
 - D. all the stages live in water.
- 30.** Which of the following insects also undergoes complete metamorphosis, like the mosquito?
- A. the grasshopper
 - B. the cockroach
 - C. the termite
 - D. the housefly
- 32.** A pupil lives in a room without a mosquito net. Which of the following is the **best** way to keep her from being bitten at night?
- A. Spray the room with insecticide and close the windows with wire mesh
 - B. Leave the door and windows open all night.
 - C. Sleep in the open compound at night.
 - D. Put on a thick sweater before sleeping.
- 34.** Clearing the weeds and bushes around a house helps to control mosquitoes because
- A. the bushes hold the mosquito eggs.
 - B. weeds attract fishes to the compound.
 - C. weeds make the soil hold more water.
 - D. the bushes give shelter to the adult mosquitoes

- A. The pupae will grow into more wigglers.
- B. The wigglers and pupae will develop into adult mosquitoes
- C. The wigglers are already adult mosquitoes.
- D. The wigglers will lay eggs in the jar.

- 37.** Malaria is very common in many parts of Ghana mainly because
- there are many places with stagnant water where mosquitoes breed
 - the sun is too hot in Ghana.
 - Ghanaians eat too much oily food.
 - the rivers in Ghana flow too fast.
- 39.** Which of the following is the best action for a school club to take every week to control mosquitoes on the school compound?
- Pour insecticide into the school well.
 - Cut down all the trees in the school.
 - Empty and turn over every container that can collect water
 - Store water in open drums near the classrooms.
- 36.** Which of the following statements about mosquitoes is correct?
- Every mosquito spreads malaria.
 - Not all mosquitoes spread malaria; it is mainly the female Anopheles that does
 - Only the male mosquito bites human beings.
 - Mosquitoes are useful only as pests.
- 38.** The most important reason why mosquitoes must be controlled in Ghana is that malaria
- causes much suffering, and the death of children and pregnant women, and slows down work and school
 - makes the mosquitoes too many in number.
 - spoils the taste of drinking water.
 - causes the rain to fall too often.
- 40.** Which of the following groups of people suffers most from malaria in Ghana?
- children under five years and pregnant women
 - old men only
 - teachers only
 - traders at the market only

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Draw and label diagrams where you are asked to.

Question 6

- (a) Read every question carefully before you answer it.

Question 7

- (a) Do not use correction fluid. Cancel any mistake neatly with a single line.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 16: Living Cells

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The basic unit of all living things is the
A. atom.
B. cell.
C. molecule.
D. organ.
2. Which of the following is a living thing?
A. a stone
B. a mango tree
C. a bucket of water
D. a piece of iron
3. The part of a cell that controls all the activities of the cell is the
A. nucleus.
B. cell membrane.
C. cytoplasm.
D. vacuole.
4. The living material of a cell that surrounds the nucleus is called the
A. cytoplasm.
B. cell wall.
C. cell membrane.
D. nuclear membrane.
5. The cell membrane is described as **partially permeable** because it
A. lets everything pass through it.
B. does not allow anything to pass through it.
C. allows substances to pass through it only when the cell is dead.
D. lets only certain substances pass through it.
6. The part of a plant cell that is **not** found in an animal cell is the
A. cell membrane.
B. cell wall.
C. nucleus.
D. cytoplasm.
7. Which of the following structures is found in an animal cell but **not** in a plant cell?
A. Centriole
B. Chloroplast
C. Large central vacuole
D. Cell wall
8. The green pigment in a chloroplast that traps light energy for photosynthesis is
A. cytoplasm.
B. chlorophyll.
C. cellulose.
D. cell sap.
9. The cell wall of a plant cell is made mainly of
A. protein.
B. cellulose.
C. fat.
D. starch.
10. The membrane that surrounds the nucleus is called the
A. cell wall.
B. cell membrane.
C. nuclear membrane.
D. vacuole.
11. Which of the following is a **prokaryotic** cell?

- A. a cell of a mushroom
- B. a cell of a bacterium
- C. a cell of a grasscutter
- D. a cell of a mango plant

12. Organisms made up of cells that have ****no true nucleus**** and ****no membrane-bound organelles**** are called

- A. eukaryotic organisms.
- B. multicellular organisms.

13. Which of the following organisms is made up of eukaryotic cells?
- A. a bacterium
 - B. a blue-green alga
 - C. Amoeba
 - D. a virus
15. Which of the following pairs is a pair of disease-causing bacteria in humans?
- A. Plasmodium and Trypanosoma
 - B. Escherichia coli and yeast
 - C. mushroom and Mucor
 - D. Mycobacterium tuberculosis and Vibrio cholerae
17. Which of the following is a disease caused by a bacterium?
- A. malaria
 - B. measles
 - C. sleeping sickness
 - D. cholera
19. The science tool that is used to observe very small organisms and cells is the
- A. telescope.
 - B. periscope.
 - C. stethoscope.
 - D. microscope.
21. When a plant cell is placed in pure water, the water enters the cell and the cell becomes
- A. dry.
 - B. dead at once.
 - C. firm and swollen (turgid).
 - D. smaller.
23. Which of the following is **not** an organelle of the cell?
- A. mitochondrion
 - B. vacuole
 - C. water
 - D. nucleus
- C. vertebrate organisms.
D. prokaryotic organisms.
14. In which part of a prokaryotic cell is the genetic material found?
- A. In the nucleus
 - B. In the mitochondria
 - C. In the cytoplasm (as a single loop of DNA)
 - D. In the cell wall
16. Some bacteria are useful to humans because they
- A. cause tuberculosis.
 - B. cause cholera.
 - C. help to ferment milk into yoghurt.
 - D. spoil food.
18. Which of the following is the best way to protect yourself against diseases caused by bacteria?
- A. Share drinking cups with your friends.
 - B. Leave cooked food uncovered on the table.
 - C. Wash your hands with soap under running water after visiting the toilet.
 - D. Drink water from any stream without boiling it.
20. Which organelle in the cell is described as the **powerhouse** of the cell because it releases energy from food?
- A. Vacuole
 - B. Mitochondrion
 - C. Ribosome
 - D. Chloroplast
22. The main difference between a plant cell and an animal cell is that a plant cell has
- A. a cell wall while an animal cell has no cell wall.
 - B. a nucleus while an animal cell does not.
 - C. cytoplasm while an animal cell does not.
 - D. a cell membrane while an animal cell does not.

- 25.** Single-celled organisms such as Amoeba and Paramecium are
- prokaryotic organisms.
 - non-living things.
 - cells without cytoplasm.
 - eukaryotic organisms.
- 27.** Blue-green algae are grouped as prokaryotic organisms because they
- have no enclosed nucleus.
 - are blue in colour.
 - live only in water.
 - are very small.
- 29.** Which of the following is a raw material that green plants use during photosynthesis?
- oxygen
 - carbon dioxide
 - protein
 - food from the soil
- 31.** Which of the following is ****not**** a part of a prokaryotic cell?
- mitochondria
 - cell membrane
 - cytoplasm
 - cell wall
- 33.** A cell that has a true nucleus enclosed by a nuclear membrane, and membrane-bound organelles, is described as
- eukaryotic.
 - prokaryotic.
 - bacterial.
 - a virus.
- 35.** Which of the following diseases is spread mainly through drinking water that is contaminated?
- malaria
 - measles
 - guinea worm disease
 - cholera
- 24.** Which of the following is a similarity between plant cells and animal cells?
- Both have a nucleus.
 - Both have chloroplasts.
 - Both have a cell wall.
 - Both have a large central vacuole.
- 26.** Which of the following statements about bacteria is correct?
- Bacteria have no cytoplasm.
 - Bacteria are always harmful to humans.
 - Bacteria are prokaryotic organisms.
 - Bacteria are multicellular organisms.
- 28.** The process by which green plants make their own food, using light energy trapped by chlorophyll, is called
- photosynthesis.
 - respiration.
 - digestion.
 - excretion.
- 30.** A pupil examined a cell and found that it had a cell wall, a nucleus and chloroplasts. The cell is most likely from
- a human cheek.
 - a bacterium.
 - a goat's liver.
 - a mango leaf.
- 32.** Bacteria are described as unicellular organisms because they
- have no cell membrane.
 - are made up of only one cell.
 - always cause disease.
 - can live only inside other living things.
- 34.** Which of the following is an example of a prokaryotic organism?
- Plasmodium
 - yeast
 - Escherichia coli
 - Mucor

- 37.** Which statement best explains why a bacterium is classified as a prokaryotic organism?
- A. It is too small to be seen with the naked eye.
 - B. It is made up of only one cell.
 - C. It has a cell membrane.
 - D. Its genetic material is not enclosed in a nucleus.
- 39.** A pupil wants to observe cells from an onion. Which of the following should the pupil do first?
- A. Put the whole onion under the microscope.
 - B. Peel a thin layer of the onion skin and mount it on a glass slide.
 - C. Boil the onion for a long time.
 - D. Put the onion in water and filter it.
- 36.** Which of the following is the best way to make drinking water safe in a community at risk of cholera?
- A. Boiling the water before drinking it.
 - B. Adding sugar to the water.
 - C. Pouring the water from a height.
 - D. Keeping the water in an open bucket.
- 38.** Which of the following pairs of organisms is made up of cells that have a **cell wall**, a **nucleus** and **cytoplasm**?
- A. a goat and a fowl
 - B. Amoeba and a bacterium
 - C. a bacterium and a maize plant
 - D. a maize plant and a mushroom
- 40.** Which of the following statements is **true**?
- A. All bacteria are harmful to humans.
 - B. All cells in a living thing are exactly the same.
 - C. The cell is the basic unit of all living things.
 - D. Only animals are made up of cells.

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 7 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Draw and label diagrams where you are asked to.

Question 6

- (a) Read every question carefully before you answer it.

Question 7

- (a) Do not use correction fluid. Cancel any mistake neatly with a single line.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 17: Mixtures and S

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A mixture is best described as
 - A. a single pure substance made of one kind of particle.
 - B. two or more substances that are combined without joining chemically.
 - C. a substance that can only be separated by chemical means.
 - D. a substance that always looks the same throughout.
2. Which statement about the particles of a mixture is correct?
 - A. The particles of a mixture keep their own properties.
 - B. The particles of a mixture are chemically joined together.
 - C. The particles of a mixture lose all their properties.
 - D. The particles of a mixture can only be seen under a microscope.
3. Which of the following is a **homogeneous** mixture?
 - A. Sand and water
 - B. Milk
 - C. Sugar solution
 - D. Seawater with floating seaweed
4. Which of the following is a **heterogeneous** mixture?
 - A. Salt solution
 - B. Palm wine
 - C. Kerosene and water
 - D. Fresh orange juice with pulp
5. A mixture in which the solute particles are too large to pass through filter paper is called
 - A. a solution.
 - B. a colloid.
 - C. a suspension.
 - D. an alloy.
6. Which of the following mixtures is a colloid?
 - A. Milk
 - B. Kerosene
 - C. Sugar solution
 - D. Muddy water
7. In a salt solution, salt is the
 - A. solvent and water is the solute.
 - B. solute and water is the solvent.
 - C. solute and water is also a solute.
 - D. solvent and water is also a solvent.
8. A pupil dissolves two spoons of sugar in a glass of water. The mixture formed is
 - A. a suspension.
 - B. a colloid.
 - C. an emulsion.
 - D. a solution.
9. The path of light from a torch is clearly visible when it is shone through milk because milk
 - A. is a clear solution.
 - B. is a suspension of large stones.
 - C. scatters the light.
 - D. absorbs all the light.
10. Which of the following is **NOT** a property of a suspension?
 - A. It is always clear and transparent.
 - B. Its particles settle on standing.
 - C. It scatters a beam of light.
 - D. Its particles can be separated by filtration.

- 11.** Which of the following mixtures is ****homogeneous****?
- A. Fufu and soup
 - B. Salt solution
 - C. A mixture of rice and stones
 - D. Sand and water
- 12.** Which of these substances will ****not**** dissolve in water?
- A. Sugar
 - B. Common salt
 - C. Glucose
 - D. Chalk powder
- 13.** A technique used to separate a solid that does not dissolve in a liquid from the liquid is
- A. distillation.
 - B. decantation.
 - C. filtration.
 - D. sublimation.
- 14.** In filtration, the solid collected on the filter paper is called the
- A. filtrate.
 - B. residue.
 - C. solvent.
 - D. solution.
- 15.** In filtration, the liquid that passes through the filter paper into the conical flask is called the
- A. filtrate.
 - B. residue.
 - C. solute.
 - D. distillate.
- 16.** The method that quickly separates a mixture of fine sand and water into two separate layers is
- A. evaporation.
 - B. distillation.
 - C. sublimation.
 - D. decantation.
- 17.** Which of the following mixtures can be separated by ****magnetic separation****?
- A. A mixture of iron filings and sulfur powder
 - B. A mixture of salt and sugar
 - C. A mixture of chalk powder and water
 - D. A mixture of kerosene and water
- 18.** The sieving of gari to remove fine cassava flour and larger particles works mainly because the two substances
- A. have different colours.
 - B. have different particle sizes.
 - C. have different melting points.
 - D. are both solids.
- 19.** Which of the following ****is not**** used as a method of separating mixtures?
- A. Evaporation
 - B. Filtration
 - C. Decantation
 - D. Fermentation
- 20.** Substances that can undergo sublimation change from
- A. liquid to solid directly.
 - B. vapour to liquid directly.
 - C. solid to vapour directly.
 - D. liquid to vapour directly.
- 21.** One difference between evaporation and distillation is that during evaporation
- A. water is recovered as a liquid.
 - B. the salt inside the water is dissolved.
 - C. the water escapes into the air as a vapour.
 - D. the mixture is heated until it boils.
- 22.** In distillation, the liquid obtained after condensation is called the
- A. distillate.
 - B. solvent.
 - C. residue.
 - D. solute.
- 23.** The method most suitable for obtaining pure drinking water from

seawater is

- A. filtration.
- B. sieving.
- C. sedimentation.
- D. distillation.

25. Separating cream from milk at the factory uses

- A. a centrifuge.
- B. a sieve.
- C. filter paper.
- D. sublimation.

24. A mixture of two liquids that do not mix is called

- A. a solution.
- B. an immiscible mixture.
- C. a homogeneous mixture.
- D. a colloidal mixture.

- 27.** Which of the following statements about a mixture is **true**?
- A. A mixture has a fixed boiling point.
 - B. A mixture can be separated by physical means.
 - C. The parts of a mixture are joined in fixed ratios.
 - D. A mixture is always a solid.
- 29.** A beam of light is shone through four mixtures: muddy water, milk, kerosene and palm oil. In how many of them will the path of the light be clearly seen?
- A. One
 - B. Two
 - C. Three
 - D. Four
- 31.** A pupil adds sugar to a glass of water. After stirring for a long time, some sugar remains undissolved at the bottom of the glass. The best way to dissolve the remaining sugar is to
- A. add more sugar.
 - B. pour the mixture into another glass.
 - C. warm the mixture and stir it.
 - D. filter the mixture.
- 33.** Osman dissolves a spoon of salt in a cup of water and a spoon of sugar in another cup of water. Both solids dissolve, but the sugar solution tastes sweet while the salt solution tastes salty. This shows that
- A. the solids have dissolved completely and disappeared.
 - B. the solids are still present and have kept their own properties.
 - C. the water has changed the nature of the solids.
 - D. the two mixtures are now pure substances.
- 26.** The conversion of a liquid into vapour by heating, followed by cooling the vapour back into a liquid, is called
- A. melting.
 - B. evaporation.
 - C. condensation.
 - D. distillation.
- 28.** A student is given a beaker containing salt solution and asked to obtain dry, pure salt. Which combination of methods should the student use?
- A. Filtration followed by decantation
 - B. Evaporation followed by filtration
 - C. Evaporation, then drying the residue
 - D. Distillation followed by sedimentation
- 30.** Equal volumes of sand, salt, iron filings and water are mixed together. Which of the following correctly shows the first two steps needed to obtain dry sand, dry salt and the iron filings separately?
- A. Magnetic separation, then filtration
 - B. Evaporation, then filtration
 - C. Filtration, then sieving
 - D. Sieving, then distillation
- 32.** Amoako wants to obtain clean water from muddy water at home. He performs the following steps: (i) he allows the water to stand, (ii) he pours the clear water on top into another container, (iii) he pours the water through a clean white cloth. Which method has Amoako used in step (ii)?
- A. Filtration
 - B. Distillation
 - C. Evaporation
 - D. Decantation

- 35.** Which of the following pairs of apparatus are used together during filtration?
- Filter paper and funnel
 - Evaporating dish and tripod
 - Liebig condenser and round-bottom flask
 - Magnet and sieve
- 37.** Which of the following mixtures is a suspension?
- A mixture of petrol and water shaken together
 - A mixture of salt and water
 - A mixture of vinegar and water
 - A mixture of milk and water
- 39.** The difference between a solution and a colloid is that
- the particles of a solution are larger than those of a colloid.
 - a colloid settles on standing, while a solution does not.
 - the particles of a colloid are large enough to scatter light, while those of a solution are not.
 - a solution can be filtered, while a colloid cannot.
- 34.** Akosua and Yaw were given a clear liquid. Akosua said it was pure water; Yaw said it was a salt solution. Which of the following is the best test to settle the argument?
- Filtering the liquid
 - Shining a torch through the liquid
 - Smelling the liquid
 - Boiling the liquid to dryness
- 36.** Filtration can be used to separate
- salt from a salt solution.
 - ethanol from water.
 - chalk powder from a chalk-water mixture.
 - iron filings from sulfur powder.
- 38.** A cook adds oil to boiling water in a pot. After the pot is left to stand, the oil floats on top of the water. This is because the oil
- dissolves in the water.
 - is less dense than water.
 - is denser than water.
 - changes into a solid.
- 40.** Which sequence of processes correctly describes the changes of state during distillation?
- Liquid → solid → liquid
 - Liquid → vapour → solid
 - Solid → liquid → vapour
 - Liquid → vapour → liquid

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 5 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Draw and label diagrams or tables where you are asked to.

Question 6

- (a) Read every question carefully before you answer it.

Question 7

- (a) Do not use correction fluid. Cancel any mistake neatly with a single line.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 18: Renewable Energy

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A renewable source of energy is one that
 - A. is found only deep under the ground
 - B. can be used only once in a year
 - C. is replaced by nature and does not run out when it is used
 - D. is used up completely and cannot be replaced
2. A non-renewable source of energy is best described as a source that
 - A. is formed again within a few days of being used
 - B. can never be used to produce electricity
 - C. is found only in the sunlight
 - D. takes millions of years to form and is used up faster than it is replaced
3. Which of the following is a renewable source of energy?
 - A. Crude oil
 - B. Natural gas
 - C. Wind
 - D. Coal
4. Which of the following is a non-renewable source of energy?
 - A. Solar
 - B. Coal
 - C. Biogas
 - D. Hydro
5. The energy obtained from falling or flowing water is called
 - A. geothermal energy
 - B. wind energy
 - C. solar energy
 - D. hydro energy
6. Energy obtained directly from the Sun is called
 - A. solar energy
 - B. biogas energy
 - C. nuclear energy
 - D. tidal energy
7. Crude oil, coal and natural gas are together called
 - A. fossil fuels
 - B. renewable fuels
 - C. biofuels
 - D. nuclear fuels
8. Which of the following sets contains ONLY renewable sources of energy?
 - A. Crude oil, hydro, biogas
 - B. Solar, wind, hydro
 - C. Natural gas, wind, coal
 - D. Coal, solar, wind
9. Which of the following sets contains ONLY non-renewable sources of energy?
 - A. Crude oil, biogas, hydro
 - B. Coal, wind, crude oil
 - C. Coal, crude oil, natural gas
 - D. Natural gas, solar, coal
10. Biogas is obtained mainly from
 - A. the burning of coal in a closed tank
 - B. the breakdown of animal dung and other organic waste by bacteria
 - C. the boiling of crude oil in a refinery
 - D. the drying of firewood in the sun

- 11.** The main gas present in biogas is
- A. nitrogen
 - B. oxygen
 - C. methane
 - D. chlorine
- 13.** Firewood and charcoal are described as renewable sources of energy only if
- A. they are sold at a low price
 - B. the trees that are cut are replaced by planting new ones
 - C. they are burnt in a closed room
 - D. they are cut from the forest reserve
- 15.** Which of the following is the best example of a renewable energy project in Ghana?
- A. The hydroelectric station at Akosombo
 - B. The crude oil refinery at Tema
 - C. A charcoal market in Accra
 - D. A diesel generating set in a shop
- 17.** A disadvantage of solar energy as a source of electricity is that
- A. it cannot be changed into electrical energy
 - B. it will be used up within a few years
 - C. it produces a great deal of smoke
 - D. it is not available at night and is reduced on cloudy days
- 19.** Which of the following is a challenge in the management of hydroelectric power in Ghana?
- A. Water cannot be stored behind a dam
 - B. The level of water in the reservoir falls during a long drought
 - C. Moving water contains no energy
 - D. The river water is used up completely by the turbines
- 21.** Protecting the forests along a river that feeds a dam is important because it
- A. prevents the turbines from
- 12.** In a biogas digester, the dung must be kept
- A. frozen until the gas is needed
 - B. open to the air and to sunlight
 - C. mixed with petrol before use
 - D. in a closed tank away from air
- 14.** Which device is used to change solar energy directly into electrical energy?
- A. A steam turbine
 - B. A solar panel
 - C. A wind vane
 - D. A solar cooker
- 16.** One advantage of using renewable sources of energy is that they
- A. do not pollute the air with smoke and harmful gases
 - B. work equally well at all times, day and night
 - C. are completely free of any cost at all
 - D. can be stored easily in a bottle
- 18.** A major challenge in managing wind energy as a source of electricity is that
- A. wind is a non-renewable source
 - B. the wind cannot turn any machine
 - C. the wind does not blow steadily at all times
 - D. wind turbines produce harmful smoke
- 20.** Which of the following practices helps to manage a renewable energy source sustainably?
- A. Allowing the reservoir to silt up with eroded soil
 - B. Cutting all the trees around the village for charcoal
 - C. Leaving solar panels dirty and unrepaired
 - D. Planting new trees to replace those cut for firewood

- turning too fast
- B. makes the water in the reservoir hotter
- C. increases the amount of coal under the ground
- D. reduces erosion and keeps the river flowing, so the reservoir stays full

- 22.** A community installs solar lamps. To keep them working for many years, the people should
- A. use the lamps only once a month
 - B. clean the panels regularly and repair or replace worn-out

- batteries
- C. leave the panels under a shed away from the Sun
- D. connect the panels to a petrol engine
- 23.** The main reason why a country should develop renewable sources of energy is that
- A. renewable sources can be exported in barrels
 - B. fossil fuels will one day run out and they pollute the environment
 - C. renewable sources are the cheapest machines to build
 - D. fossil fuels cannot be used to generate electricity
- 25.** Heat is best described as
- A. a substance contained inside hot objects
 - B. the degree of hotness or coldness of a body
 - C. the mass of a hot body
 - D. a form of energy that flows from a hotter body to a cooler one
- 27.** The SI unit of heat is the
- A. joule
 - B. kelvin
 - C. newton
 - D. degree Celsius
- 29.** The instrument used to measure temperature is the
- A. thermometer
 - B. ammeter
 - C. hydrometer
 - D. barometer
- 31.** When a body is heated, its temperature usually
- A. falls
 - B. becomes zero
 - C. rises
 - D. remains exactly the same at all times
- 24.** Which of the following best explains why biogas is useful on a mixed farm?
- A. It removes the need to keep any animals on the farm
 - B. It allows crops to be grown without soil
 - C. Animal dung that would be wasted is turned into fuel, and the slurry left is used as manure
 - D. It changes the dung into crude oil for sale
- 26.** Temperature is best described as
- A. the rate at which a body is burning
 - B. the degree of hotness or coldness of a body
 - C. the amount of matter in a body
 - D. the total energy stored in a body
- 28.** The SI unit of temperature is the
- A. calorie
 - B. joule
 - C. watt
 - D. kelvin
- 30.** Heat always flows
- A. from a region of lower temperature to a region of higher temperature
 - B. in all directions at the same rate regardless of temperature
 - C. from a smaller body to a larger body only
 - D. from a region of higher temperature to a region of lower temperature
- 32.** Which of the following statements about heat and temperature is correct?
- A. Temperature is a form of energy, while heat measures hotness
 - B. Both heat and temperature are measured in kelvin
 - C. Heat is a form of energy, while temperature is a measure of hotness

D. Heat and temperature mean exactly the same thing

- 33.** A cup of tea at 80°C is poured into a cold cup. After some time the tea becomes cooler because

- A. temperature flows out of the tea into the cup
 - B. heat flows from the tea to the cup and the surrounding air
 - C. cold flows from the cup into the tea
 - D. the tea loses some of its mass to the cup
- 35.** A spoon of boiling water at $100\text{ }^{\circ}\text{C}$ is added to a bucket of water at $30\text{ }^{\circ}\text{C}$. The temperature of the bucket of water hardly changes because
- A. the heat from the small quantity of boiling water is shared among a very large mass of water
 - B. the boiling water loses its temperature in the air first
 - C. water at $100\text{ }^{\circ}\text{C}$ contains no heat energy
 - D. heat cannot move from hot water to cold water
- 37.** The temperature of a classroom is $30\text{ }^{\circ}\text{C}$. Expressed in kelvin, this is
- A. 303 K
 - B. 330 K
 - C. 243 K
 - D. 30 K
- 39.** A nurse takes the temperature of two patients. Patient X reads $37\text{ }^{\circ}\text{C}$ and patient Y reads $39.5\text{ }^{\circ}\text{C}$. It is correct to say that
- A. patient Y has a higher temperature and is likely to have a fever
 - B. patient Y has lost heat to the surroundings
 - C. patient X contains more heat energy than patient Y
 - D. both patients have exactly the same degree of hotness
- 34.** Two beakers contain water at the same temperature, but one holds twice as much water as the other. The beaker with more water
- A. has a higher temperature
 - B. has a lower temperature
 - C. contains less heat energy
 - D. contains more heat energy
- 36.** Clinical thermometers used in a clinic are marked mainly in
- A. watts
 - B. joules
 - C. degrees Celsius
 - D. newtons
- 38.** Two equal masses of water are heated for the same time, one over a strong flame and the other over a weak flame. The water over the strong flame has a higher temperature because
- A. it contains more water than the other
 - B. its temperature was higher before heating began
 - C. it has received more heat energy in the same time
 - D. a strong flame removes heat from the water
- 40.** A teacher says, "A burning match has a higher temperature than a bucket of warm water, but the bucket contains more heat energy." This statement is
- A. correct, because heat depends on the mass of the body as well as on its temperature
 - B. correct, because the match contains no heat energy at all
 - C. wrong, because a hotter body must always contain more heat
 - D. wrong, because heat and temperature are the same thing

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Where a table is asked for, rule it neatly with a pencil and a ruler and write the headings in full.

Question 7

- (a) Do not use correction fluid. Cancel any mistake neatly with a single line.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 19: Science and

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Science is best described as
 - A. the making of machines for factories only
 - B. the selling of goods produced in a country
 - C. the orderly study of nature to find out how and why things happen
 - D. the repair of broken tools in a workshop
2. Technology is best described as
 - A. the writing down of the results of an experiment
 - B. the study of living things in their habitat
 - C. the use of scientific knowledge to make tools, machines and methods that meet human needs
 - D. the collection of taxes from factories
3. Innovation means
 - A. repeating an old experiment exactly as before
 - B. copying a machine without changing anything
 - C. writing a report on a scientific discovery
 - D. introducing a new or improved product, method or idea that people actually use
4. Which of the following shows the usual order in which new things reach the public?
 - A. Society, then innovation, then technology
 - B. Innovation, then scientific knowledge, then technology
 - C. Technology, then society, then scientific knowledge
 - D. Scientific knowledge, then technology, then innovation used by society
5. The discovery that an electric current in a wire produces a magnetic field is an example of
 - A. an innovation in a market
 - B. scientific knowledge
 - C. a business plan
 - D. a government policy
6. Using the knowledge above to build an electric fan is an example of
 - A. a natural resource
 - B. pure science
 - C. technology
 - D. an environmental hazard
7. Which of the following best explains why science and technology depend on each other?
 - A. Science supplies the knowledge technology uses, and the tools technology makes allow science to investigate further
 - B. Science is practised only in schools and technology only in factories
 - C. Science and technology are two names for the same activity

- D. Technology was invented before any science was known

- 8.** A farmer who fixes a plastic bottle to a rope to make a simple drip waterer for his seedlings is showing
- A. a scientific law, because the bottle holds water
 - B. innovation, because he has created a new way of solving a problem with what he has
 - C. pure science, because he has carried out an experiment
 - D. a natural process, because water

- always flows downwards
- 9.** Which of the following is the MAIN benefit of science and technology to a society?
- A. They help to solve problems and improve the quality of people's lives
 - B. They make all work unnecessary
 - C. They remove the need for education
 - D. They make every citizen wealthy at once
- 11.** Mobile money services have improved life in Ghana chiefly because they
- A. remove the need for people to work
 - B. stop thieves from entering any house
 - C. make all goods in the market free
 - D. allow people to send and receive money quickly without travelling to a bank
- 13.** A borehole fitted with a hand pump improves the health of a village mainly because it
- A. keeps mosquitoes away from the houses
 - B. provides safe drinking water, so fewer people catch water-borne diseases
 - C. makes the village land more fertile
 - D. supplies electricity for the houses
- 15.** A cassava processing machine helps a farming community mainly by
- A. making the cassava grow faster on the farm
 - B. removing the need to plant cassava again
 - C. turning the cassava into gari quickly, so less of the crop is wasted
 - D. changing cassava into maize
- 17.** A refrigerator preserves food mainly because the low temperature
- A. adds extra nutrients to the food
- 10.** The use of a tractor instead of a hoe on a large farm mainly
- A. prevents rain from falling on the farm
 - B. increases the area that can be ploughed in a day
 - C. makes the soil lose all its nutrients
 - D. reduces the amount of food the farm can produce
- 12.** Which technology has done most to improve the storage of vaccines at a rural clinic?
- A. The solar-powered refrigerator
 - B. The mobile telephone
 - C. The bicycle
 - D. The loudspeaker
- 14.** Which of the following is an example of technology used in health care?
- A. The periodic table
 - B. The weather forecast
 - C. The multiplication table
 - D. The X-ray machine
- 16.** Which of the following best shows how technology has changed communication in Ghana?
- A. A message that once took days by letter is now sent in seconds by phone
 - B. People now travel more slowly than before
 - C. Letters are now carried by donkeys
 - D. Only the government may send messages

- B. removes all the water from the food
- C. kills every germ in the food at once
- D. slows down the growth of the bacteria that spoil food

19. Which of the following is a harmful effect of technological advancement on the environment?

- A. A rise in the amount of clean water in rivers
- B. An increase in the number of trees in the forest
- C. Pollution of air, water and land by wastes from industry
- D. A fall in the amount of waste produced

18. The MAIN reason why a factory in Tema uses machines rather than hand labour alone is that machines

- A. never need any maintenance
- B. work without any source of energy
- C. work faster and produce goods of a more even quality
- D. remove the need for safety rules

- 21.** Old televisions, computers and mobile phones that have been thrown away are known as
- organic manure
 - industrial fertiliser
 - scrap timber
 - electronic waste (e-waste)
- 23.** Which of the following is the BEST way to deal with old computers and phones?
- Collect them for proper recycling so that the metals are recovered safely
 - Throw them into a stream
 - Bury them whole in a farm
 - Burn them in an open yard
- 25.** The MAIN effect of clearing a large area of forest for a new factory is
- an immediate improvement in the fertility of the soil
 - the loss of habitats for plants and animals and an increase in soil erosion
 - a fall in the temperature of the whole country
 - an increase in the yearly rainfall of the area
- 27.** Which of the following gases, released when fuels are burnt, is mainly responsible for the warming of the earth?
- Water vapour from a kettle
 - Nitrogen
 - Oxygen
 - Carbon dioxide
- 29.** Technology can also help to repair the damage it causes. Which of the following is an example?
- Using machines to cut down more trees
 - Pumping untreated waste water into a lagoon
 - Using machines to recycle plastic waste into roofing sheets and pavement blocks
 - Burning refuse in the open to save space
- 20.** Smoke from factories and vehicles pollutes the air chiefly because it contains
- oxygen only
 - harmless dust that settles at once
 - water vapour only
 - harmful gases and tiny particles that people breathe in
- 22.** Burning the plastic covering of old cables to get the copper inside is dangerous because it
- releases poisonous fumes that damage the lungs and pollute the air
 - cools the air around the fire
 - turns the copper into gold
 - makes the copper useless
- 24.** Illegal small-scale mining with heavy machines, known as *galamsey*, has harmed Ghana's rivers chiefly by
- making the water safe to drink without treatment
 - making them flow faster to the sea
 - increasing the number of fish in them
 - filling them with mud and chemicals such as mercury
- 26.** Plastic sachets and bottles thrown into gutters cause flooding in Ghanaian towns because they
- make the water in the gutters evaporate faster
 - block the gutters and drains so that rain water cannot flow away
 - attract rain to the town
 - dissolve and thicken the water
- 28.** Noise from a corn mill sited close to a classroom block is a form of
- conservation, because the mill saves time
 - pollution, because it disturbs learning and can damage hearing
 - recycling, because the maize is reused
 - innovation, because the mill is modern

- 31.** The chief advantage of using a solar dryer instead of drying fish in the open is that the dryer
- A. adds salt to the fish automatically
 - B. keeps out flies, dust and rain while drying the fish faster
 - C. makes the fish grow bigger
 - D. removes the need to catch fish
- 33.** Before a new machine is brought into a community, the MOST important question to ask is
- A. what effect it will have on the people, their work and their environment
 - B. what colour the machine has been painted
 - C. how heavy the machine is
 - D. who owns the shop that sells it
- 35.** Which of the following is the best evidence that a piece of information about a new technology is reliable?
- A. It promises results that sound surprising
 - B. It is written in very large letters
 - C. It comes from a recognised institution and agrees with other trusted sources
 - D. It has been shared by many people online
- 37.** Which of the following best describes sustainable use of technology?
- A. Using technology to meet today's needs without destroying the resources of future generations
 - B. Using only the machines made in other countries
 - C. Replacing all workers with machines as quickly as possible
 - D. Using every machine until the fuel of the country is finished
- 30.** Which of the following sources of electricity does LEAST harm to the environment?
- A. A petrol engine
 - B. A diesel generator
 - C. A solar panel
 - D. A coal-fired station
- 32.** A school that collects rain water from its roof into a tank is using technology to
- A. store a free supply of water for use in the dry season
 - B. prevent lightning from striking the roof
 - C. cool the classrooms during lessons
 - D. increase the rainfall over the school
- 34.** A pupil reads on social media that a certain herb cures every disease. The BEST action is to
- A. accept it because many people have shared it
 - B. buy the herb and try it immediately
 - C. forward the message to all the pupils at once
 - D. check the claim with a health worker or a trusted scientific source before believing it
- 36.** Science, technology and society are said to be connected because
- A. society has no influence at all on science
 - B. the needs of society direct what is studied and invented, and inventions in turn change society
 - C. scientific knowledge never changes the way people live
 - D. technology is practised only by governments
- 38.** A factory sited beside a river discharges its waste water into the river untreated. The MOST useful first step for the community is to
- A. ask the factory to work only at night
 - B. fetch the water and boil it for drinking

- C. report the matter to the Environmental Protection Authority and the assembly for action
- D. move the whole village to another place

- 39.** A group of pupils wants to design a device that will warn a farmer when his water tank is full. The FIRST step in their design is to
- A. buy the most expensive materials they can find
 - B. state clearly the problem to be solved and what the device must

- do
- C. paint the device in bright colours
- D. give the device a name

40. Which of the following statements about technological advancement in Ghana is MOST balanced?

- A. It brings real benefits, but it must be managed so that it does not damage health and the environment
- B. It is wholly bad and should be stopped at once
- C. It affects only the people who live in the cities
- D. It is wholly good and should never be questioned

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) In Question 5 you are asked to design something. Marks are given for your thinking, not for the beauty of your drawing.

Question 7

- (a) Use a pencil and a ruler where a plan or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 20: Soils and F

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Soil is best described as
 - A. the loose material on the surface of the earth in which plants grow
 - B. the water held in a stream bed
 - C. the solid rock that lies deep under the ground
 - D. the dead leaves lying on a forest floor only
2. Which of the following is a physical property of a soil?
 - A. The name of the farmer who owns it
 - B. The number of farms in the village
 - C. The price of the land on which it lies
 - D. Its texture, that is how coarse or fine it feels
3. The four main constituents of a fertile soil are
 - A. clay, plastic, metal and water
 - B. mineral particles, humus, water and air
 - C. rock, petrol, air and humus
 - D. sand, salt, sugar and water
4. Humus in a soil is formed from
 - A. the broken pieces of plastic in the soil
 - B. the decayed remains of plants and animals
 - C. the water held between the soil particles
 - D. the smallest grains of sand
5. The MAIN importance of humus in a soil is that it
 - A. gives the soil its red colour only
 - B. makes the soil heavier to dig
 - C. supplies plant nutrients and helps the soil to hold water
 - D. prevents earthworms from entering the soil
6. The air in a soil is important to a plant because it
 - A. supplies oxygen for the respiration of the roots and of soil organisms
 - B. pushes the roots deeper into the soil
 - C. dissolves the rock particles into nutrients
 - D. keeps the soil cool during the night
7. A soil is described as having a coarse texture when its particles are
 - A. equal in size to the particles of clay
 - B. large and can be felt separately between the fingers
 - C. so fine that they stick together when wet
 - D. made entirely of humus
8. A pupil who wants to compare the texture of three soils should
 - A. rub a moist sample of each soil between the fingers and feel the particles
 - B. count the stones in each sample
 - C. measure the temperature of each sample
 - D. weigh each sample on a balance
9. Which of the following soils has the largest particles?
 - A. Clay soil
 - B. Sandy soil
 - C. Silty soil
 - D. Loamy soil

- 11.** Water drains fastest through
- clay soil
 - loamy soil
 - sandy soil
 - waterlogged soil
- 13.** Loamy soil is the best soil for most crops because it
- contains no air at all
 - holds enough water and nutrients while still allowing water and air to pass
 - is made only of very fine particles
 - never needs any manure
- 15.** Clay soil is difficult to work on a farm chiefly because it
- has no mineral particles in it
 - is always too light to hold a plant
 - becomes sticky when wet and very hard when dry
 - contains too many large stones
- 17.** Adding sand and well-rotted manure to a heavy clay soil helps because it
- turns the clay into rock
 - removes all the nutrients from the soil
 - makes the soil hold even more water
 - opens up the soil so that water can drain and air can enter
- 19.** The movement of water from the soil into a root hair takes place by
- evaporation
 - combustion
 - osmosis
 - respiration
- 21.** A plant wilts on a hot afternoon mainly because
- its roots have stopped growing
 - the leaves have taken in too much oxygen
 - the soil has become too rich in humus
 - it is losing water from the leaves faster than the roots can take it in
- 10.** Clay soil holds the most water because its particles are
- made entirely of humus
 - coated with oil that traps water
 - very fine and tightly packed, leaving tiny spaces
 - very large with wide spaces between them
- 12.** Which soil is described as the best for growing most crops?
- Loamy soil
 - Gravelly soil
 - Sandy soil
 - Clay soil
- 14.** A soil that is waterlogged harms a crop mainly because
- the soil becomes too light for the roots
 - the water fills the air spaces, so the roots cannot get oxygen
 - the roots take in too much humus
 - the water washes the leaves off the plant
- 16.** A farmer can improve a sandy soil by
- adding more sand to it
 - watering it only once in a season
 - burning the grass on it every year
 - adding organic manure or compost to help it hold water and nutrients
- 18.** Plants take in water from the soil mainly through the
- lower surface of the leaves
 - root hairs
 - stem bark
 - flowers
- 20.** Water passes into the root hair because the solution inside the root hair is
- more concentrated than the soil water outside it
 - at a higher temperature than the soil
 - held in a vacuum
 - the same strength as the soil water

- 23.** A farmer who waters a sandy soil more often than a loamy soil does so because sandy soil
- loses water quickly by drainage
 - holds water for a very long time
 - contains more humus than loam
 - has no air spaces at all
- 25.** A rock is best described as
- a piece of concrete broken from a building
 - any hard object buried in the ground
 - a lump of dried clay from a river bed
 - a naturally occurring solid material made of one or more minerals
- 27.** Igneous rocks are formed when
- layers of sediment are pressed together
 - molten rock material cools and becomes solid
 - plant remains decay in a swamp
 - a rock is changed by heat and pressure
- 29.** Sedimentary rocks are formed when
- soil is ploughed and left in the sun
 - an existing rock is changed by great heat and pressure
 - hot liquid rock cools inside the earth
 - layers of sand, mud and shell are deposited and pressed together over a long time
- 31.** A rock that shows clear layers, or bands of different materials pressed together, is most likely
- volcanic
 - igneous
 - sedimentary
 - molten
- 33.** When a pupil makes a rock identification guide, the MOST useful features to record for each sample are
- the day on which it was collected only
 - the name of the pupil who found it
- 22.** Besides water, a plant also obtains from the soil
- the energy it needs to grow
 - carbon dioxide for photosynthesis
 - mineral salts (nutrients) dissolved in the soil water
 - the chlorophyll in its leaves
- 24.** In an experiment, maize seedlings were grown in sandy, loamy and clay soils and watered equally. The seedlings in the loamy soil grew tallest. The BEST conclusion is that
- maize can grow only in loamy soil
 - sandy and clay soils contain no water at all
 - loamy soil supplied the balance of water, air and nutrients that the seedlings needed
 - the seedlings in loam were given more water
- 26.** The three main types of rock are
- igneous, sedimentary and metamorphic
 - wet, dry and damp
 - hard, soft and medium
 - surface, middle and deep
- 28.** Granite is an example of
- a sedimentary rock
 - an igneous rock
 - a metamorphic rock
 - a mineral fuel
- 30.** Which of the following is a sedimentary rock?
- Marble
 - Basalt
 - Granite
 - Limestone
- 32.** Metamorphic rocks are formed when
- humus mixes with sand in a garden
 - lava cools quickly on the surface
 - sediments settle at the bottom of a lake
 - an existing rock is changed by great heat and pressure inside the earth

- C. its price in the market and its owner
- D. its colour, hardness, texture and whether it shows grains or layers

- 35.** Which of the following is the parent material of a soil?
- A. The air trapped between its particles
 - B. The humus added by farmers
 - C. The rock beneath it, broken down over a long time
 - D. The water held in its spaces
- 37.** The roots of a tree growing in the crack of a rock help in weathering because they
- A. carry the rock away to the river
 - B. widen the crack as they grow and force pieces of the rock apart
 - C. dissolve the whole rock into water
 - D. make the rock harder as they grow
- 39.** Which of the following gives the correct order in the formation of a soil?
- A. Solid rock, cracked rock, broken pieces, fine particles mixed with humus
 - B. Broken pieces, humus, solid rock, fine particles
 - C. Humus, fine particles, broken pieces, solid rock
 - D. Fine particles, solid rock, humus, broken pieces
- 34.** Weathering is best described as
- A. the breaking down of rocks into smaller pieces where they lie
 - B. the burning of grass on a hillside
 - C. the ploughing of a field before planting
 - D. the carrying away of soil by running water
- 36.** A rock splits when it is heated strongly by day and cooled at night because the rock
- A. expands and contracts repeatedly until it cracks
 - B. absorbs oxygen from the atmosphere
 - C. dissolves in the cold night air
 - D. is eaten away by soil organisms
- 38.** Rain water helps to break down some rocks chiefly because it
- A. is always boiling hot when it falls
 - B. freezes solid in tropical countries
 - C. contains sand that scratches the rock
 - D. is slightly acidic and slowly dissolves certain minerals in the rock
- 40.** Soil formation is described as a very slow process because
- A. rocks do not break down at all in the tropics
 - B. only farmers can make soil from rock
 - C. it may take hundreds of years for a thin layer of soil to form from rock
 - D. soil forms only during the rainy season

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show your working clearly in all the questions that require a calculation.

Question 7

- (a) Use a pencil and a ruler where a plan, a table or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 21: Solar Sys

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The solar system is made up of
 - A. the Sun and the stars nearest to it
 - B. the Sun and all the bodies that move round it
 - C. the Earth and the Moon only
 - D. all the stars that are seen in the night sky
2. The body at the centre of the solar system is the
 - A. Sun
 - B. Earth
 - C. Jupiter
 - D. Moon
3. The Sun is best described as a
 - A. star
 - B. satellite
 - C. comet
 - D. planet
4. The path along which a planet moves round the Sun is called its
 - A. rotation
 - B. axis
 - C. orbit
 - D. equator
5. How many planets are there in the solar system?
 - A. Twelve
 - B. Nine
 - C. Six
 - D. Eight
6. The planets of the solar system are divided into inner and outer planets by the
 - A. asteroid belt
 - B. Milky Way
 - C. equator
 - D. Moon
7. The outer planets of the solar system are
 - A. Earth, Mars, Uranus and Neptune
 - B. Jupiter, Saturn, Uranus and Neptune
 - C. Mercury, Venus, Earth and Mars
 - D. Mercury, Mars, Jupiter and Saturn
8. Which of the following is an inner planet?
 - A. Neptune
 - B. Jupiter
 - C. Mars
 - D. Saturn
9. The largest planet in the solar system is
 - A. Neptune
 - B. Earth
 - C. Jupiter
 - D. Saturn
10. The planet best known for its bright and beautiful rings is
 - A. Saturn
 - B. Jupiter
 - C. Mercury
 - D. Uranus
11. The farthest planet from the Sun is
 - A. Uranus
 - B. Jupiter
 - C. Saturn

D. Neptune

- 13.** The great red spot, which is a huge storm, is found on
- A. Neptune
 - B. Uranus
 - C. Jupiter
 - D. Saturn

- 12.** The outer planet that comes immediately after Saturn is
- A. Jupiter
 - B. Mars
 - C. Uranus
 - D. Neptune

- 15.** Which of the following is a property common to all the four outer planets?
- They are all much larger than the Earth
 - They are all nearer to the Sun than the Earth
 - They all have exactly one moon each
 - They all have solid rocky surfaces
- 17.** The outer planets are very cold mainly because they are
- covered with thick ice from the sea
 - surrounded by many small moons
 - very far away from the Sun
 - always facing away from the Sun
- 19.** Uranus and Neptune both appear blue or blue-green mainly because of the gas
- water vapour from their oceans
 - methane in their atmospheres
 - carbon dioxide in their atmospheres
 - oxygen in their atmospheres
- 21.** An outer planet takes a long time to complete one orbit mainly because
- it is made up of gases only
 - its orbit round the Sun is very long
 - it has many moons holding it back
 - it rotates very slowly on its axis
- 23.** Compared with the Earth, the outer planets have
- exactly one moon each
 - no moons at all
 - many more moons
 - only two moons each
- 25.** There is no life on the outer planets mainly because they
- have too much oxygen in their atmospheres
 - are covered with thick forests and swamps
 - are too close to the Sun and too hot
 - are extremely cold and have no liquid water or breathable air
- 14.** The outer planets are often called gas giants because they are
- very small and made up of solid rock
 - very hot and made up of burning coal
 - very near to the Sun and full of water
 - very large and made up mainly of gases
- 16.** Compared with the inner planets, the outer planets are
- larger and hotter
 - smaller and hotter
 - smaller and colder
 - larger and colder
- 18.** The planet that rotates on its side, so that it appears to roll along its orbit, is
- Jupiter
 - Saturn
 - Uranus
 - Neptune
- 20.** Which of the following planets takes the longest time to go round the Sun once?
- Jupiter
 - Uranus
 - Saturn
 - Neptune
- 22.** A natural body that moves round a planet is called a
- moon (satellite)
 - star
 - comet
 - galaxy
- 24.** Which of the following statements about the rings of Saturn is correct?
- They are roads built round the planet
 - They are clouds of smoke from volcanoes on the planet
 - They are made up of pieces of ice, dust and rock moving round the planet
 - They are made of solid metal joined to the planet

- 27.** One reason why an astronaut cannot stand on the surface of Saturn is that the planet
- A. is too small to carry a person
 - B. is covered with deep sea water
 - C. moves too quickly round the Sun
 - D. has no solid surface to stand on
- 29.** Which of the following is NOT a member of the solar system?
- A. The Moon
 - B. The Milky Way galaxy
 - C. The asteroid belt
 - D. The planet Uranus
- 31.** Planets can be seen in the night sky because they
- A. reflect light from the Sun
 - B. produce their own light like the Sun
 - C. are lit by electricity from the Earth
 - D. are burning gases like a candle flame
- 33.** A pupil wants to make a model of the outer solar system. The most suitable materials would be
- A. balls of different sizes, wire and a cardboard base
 - B. water, salt and a measuring cylinder
 - C. only one ball and a sheet of plain paper
 - D. seeds, soil and a watering can
- 35.** The two outer planets whose diameters are closest to each other in size are
- A. Saturn and Neptune
 - B. Jupiter and Saturn
 - C. Jupiter and Uranus
 - D. Uranus and Neptune
- 26.** Which of the following conditions found on Earth makes life possible here but is absent on Jupiter?
- A. A very strong force of gravity
 - B. Liquid water and oxygen in the air
 - C. A thick atmosphere of hydrogen
 - D. A long time to go round the Sun
- 28.** Which of the following is NOT a reason why there is no life on Neptune?
- A. Neptune has no liquid water on a surface
 - B. Neptune has no orbit round the Sun
 - C. Neptune is extremely cold
 - D. The air on Neptune cannot be breathed
- 30.** The small rocky bodies that move round the Sun between Mars and Jupiter are called
- A. asteroids
 - B. stars
 - C. comets
 - D. moons
- 32.** Which instrument would best help pupils to observe the planets in the night sky?
- A. A hand lens
 - B. A telescope
 - C. A microscope
 - D. A pair of spectacles
- 34.** In a model of the outer solar system, the balls representing Jupiter and Saturn should be
- A. left out because they are gas giants
 - B. much smaller than those representing Uranus and Neptune
 - C. much bigger than those representing Uranus and Neptune
 - D. the same size as all the others

- 37.** Ama says all the planets are hot because they move round the Sun. The best response to Ama is that
- the planets are hot only at night
 - planets do not receive any heat from the Sun at all
 - she is correct, since all planets receive the same heat
 - the planets far from the Sun are very cold, because they receive little heat from it
- 39.** Which of the following best explains why the study of the solar system is important?
- It makes the Sun shine more brightly on Ghana
 - It helps farmers to plant crops on Jupiter
 - It allows people to travel to Neptune for holidays
 - It helps us to understand the Earth and its place among the other bodies in space
- 36.** The diameter of Jupiter is about 143 000 km and that of Neptune about 49 000 km. Jupiter is therefore about
- ten times as wide as Neptune
 - three times as wide as Neptune
 - the same width as Neptune
 - twice as wide as Neptune
- 38.** Why do scientists send space probes rather than people to study the outer planets?
- The journey is extremely long and the conditions there would kill a human being
 - The outer planets are too small for people to see
 - Space probes are more intelligent than people
 - People are not interested in the outer planets
- 40.** Which of these statements about the outer planets is TRUE?
- All the outer planets look exactly the same
 - Each of the outer planets has its own peculiar features, although they are all large and cold
 - The outer planets are all smaller than the Earth
 - The outer planets are all nearer the Sun than Mars

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show your working clearly in all questions that require calculations.

Question 7

- (a) Do not use correction fluid. Cancel any mistake neatly with a single line.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 22: Waste Management

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Waste is best described as
 - A. any material that is bought from a market
 - B. any material that can never be used again by anybody
 - C. any material that is no longer wanted or useful to the person who has it
 - D. any material that is buried in the soil
2. Waste management is
 - A. the burning of all waste in the open air
 - B. the buying and selling of waste in the market
 - C. the collection, treatment and safe disposal of waste
 - D. the storing of waste in the home for many months
3. Which of the following is an example of solid waste?
 - A. Water from the bathroom
 - B. Fumes from a factory
 - C. Smoke from a vehicle
 - D. Empty sachet-water bags
4. Water from the kitchen sink and the bathroom is an example of
 - A. electronic waste
 - B. liquid waste
 - C. gaseous waste
 - D. solid waste
5. Smoke released from a factory chimney is an example of
 - A. hazardous solid waste
 - B. liquid waste
 - C. solid waste
 - D. gaseous waste
6. Biodegradable waste is waste that
 - A. burns without producing any smoke
 - B. can be broken down by micro-organisms into simple substances
 - C. can never rot however long it is kept
 - D. is produced only in hospitals
7. Which of the following is a non-biodegradable waste?
 - A. Plastic bottle
 - B. Cassava peel
 - C. Banana skin
 - D. Vegetable leaves
8. Old mobile phones, broken television sets and used batteries are classified as
 - A. medical waste
 - B. agricultural waste
 - C. electronic waste
 - D. gaseous waste
9. Used syringes, bandages and expired drugs from a clinic are classified as
 - A. agricultural waste
 - B. electronic waste
 - C. domestic waste
 - D. medical waste
10. Which of the following is an example of agricultural waste?
 - A. Old computers from an office
 - B. Broken bottles from a chemical shop
 - C. Used exercise books from a classroom
 - D. Maize husks and animal

droppings from a farm

- 11.** The waste produced in the home from cooking and cleaning is called
- A. domestic waste
 - B. electronic waste

- C. industrial waste
D. medical waste
- 13.** Washing and using a glass bottle again to store drinking water is an example of
- A. incineration
 - B. reuse
 - C. recycling
 - D. landfilling
- 15.** Turning kitchen peelings, leaves and animal droppings into manure is called
- A. recycling of metals
 - B. landfilling
 - C. composting
 - D. incineration
- 17.** Separating waste into different bins for plastics, paper, metals and food waste is known as
- A. waste collection
 - B. waste dumping
 - C. waste incineration
 - D. waste segregation (separation at source)
- 19.** A sanitary landfill is a place where waste is
- A. stored in the rooms of unused buildings
 - B. left in open heaps for the wind to scatter
 - C. poured into a river that flows to the sea
 - D. tipped into a prepared pit, compacted and covered with soil in layers
- 21.** One disadvantage of burning rubbish in the open in a community is that it
- A. turns all the waste into useful manure
 - B. makes the waste too heavy to carry
 - C. releases smoke and harmful gases that pollute the air and cause coughing
 - D. removes the need for any waste bins
- 12.** The best place to begin in managing waste is to
- A. bury all the waste produced each day
 - B. reduce the amount of waste that is produced
 - C. burn all the waste produced each day
 - D. pour the waste into the nearest gutter
- 14.** Collecting used plastic bottles and sending them to a factory to be made into new plastic items is an example of
- A. burying
 - B. reusing
 - C. composting
 - D. recycling
- 16.** Compost made from biodegradable waste is useful because it
- A. kills all the crops growing on the land
 - B. adds plant nutrients and humus to the soil
 - C. prevents rain from reaching the soil
 - D. makes the soil hard and stony
- 18.** Why is waste separated into different containers at home?
- A. So that the waste becomes heavier before collection
 - B. So that the waste can be stored for several years
 - C. So that the materials that can be recycled or composted are not mixed with the rest
 - D. So that the waste can be poured into the gutter easily
- 20.** Incineration as a method of waste disposal means
- A. spreading the waste over a farm as manure
 - B. burning the waste at a high temperature in a special furnace
 - C. packing the waste into sacks for sale
 - D. soaking the waste in water for several days

- 23.** Choked gutters filled with plastic waste in a town commonly lead to
- A. an increase in the number of trees
 - B. cooler weather throughout the year
 - C. flooding when it rains
 - D. a better supply of pipe-borne water
- 25.** Which disease is most likely to spread in a community where human waste is not properly disposed of?
- A. Cholera
 - B. Rickets
 - C. Sickle-cell disease
 - D. Short-sightedness
- 27.** Plastic waste left on farmland is harmful because it
- A. attracts rain to the farmland
 - B. rots quickly and adds nutrients to the soil
 - C. does not rot, and it blocks the soil and can kill animals that swallow it
 - D. makes the soil easier to plough
- 29.** Which of the following practices is the most sustainable?
- A. Sorting waste at home and taking the plastics and metals for recycling
 - B. Dumping household waste in an open field at night
 - C. Burning all household waste behind the kitchen
 - D. Throwing household waste into the nearest gutter
- 22.** A refuse dump close to a house is dangerous to health mainly because it
- A. breeds flies, cockroaches and rats that carry disease germs to food
 - B. makes the house look taller
 - C. increases the supply of clean water
 - D. lowers the temperature of the rooms
- 24.** Stagnant water that collects in discarded tins and containers is a danger because it
- A. keeps the compound free from dust
 - B. provides a breeding place for mosquitoes, which spread malaria
 - C. supplies drinking water to the community
 - D. prevents the growth of weeds nearby
- 26.** Liquid waste from a factory poured into a stream affects the environment because it
- A. increases the oxygen dissolved in the water
 - B. poisons the water and kills the fish and other organisms in it
 - C. makes the stream flow much faster
 - D. turns the stream water into drinking water
- 28.** Waste management is described as sustainable when it
- A. meets the needs of the community today without damaging the environment for future generations
 - B. costs the district assembly no money at all
 - C. depends only on burning and burying waste
 - D. removes all waste from the community in one day
- 30.** A community wishes to reduce the amount of plastic waste it produces. The best first step is to
- A. build a larger open dump

- outside the town
- B. encourage people to use baskets and reusable bottles instead of single-use plastics
 - C. burn the plastic waste every evening
 - D. bury the plastic waste in the school field
- 32.** Which of these is the most suitable way of collecting information for a survey of waste practices in a community?
- A. Copying the results from another school's project
 - B. Guessing what the households usually do
 - C. Interviewing household members with a questionnaire and observing what they do
 - D. Asking only the pupils in your own class
- 31.** Before a survey of waste management practices is carried out in a community, the pupils should first
- A. buy new bins for every household
 - B. write the report of the survey
 - C. decide which households to blame
 - D. decide on the aim of the survey and prepare the questions they will ask

- 33.** In a survey of 200 households, 20 households dump their waste in gutters. The percentage of households that dump waste in gutters is
- A. 10%
 - B. 40%
 - C. 2%
 - D. 20%
- 35.** When pupils write a report on a survey, the information they collected from people should be used
- A. honestly, and the sources should be acknowledged
 - B. only if it agrees with the pupils' own opinion
 - C. after changing the figures to look better
 - D. without telling anybody where it came from
- 37.** A clean-up exercise organised in a community is most effective when
- A. the people are educated on waste separation and the waste collected is properly disposed of
 - B. only the pupils of one class take part
 - C. it is done once and never repeated
 - D. the waste collected is pushed into the gutters
- 34.** A survey report on waste management should end with
- A. the names of all the pupils in the school
 - B. a list of the prices of waste bins only
 - C. a promise never to repeat the survey
 - D. conclusions and recommendations for improving the practices found
- 36.** A school wishes to manage its waste better. The most useful first activity is to
- A. burn the waste in the school compound daily
 - B. place labelled bins for different kinds of waste at key points in the school
 - C. dig a pit in the middle of the playing field
 - D. ask pupils to take all their waste home
- 38.** Which body is mainly responsible for organising waste collection in a district in Ghana?
- A. The district or municipal assembly, working with waste management companies
 - B. The national football association

- C. The owners of the market stalls alone
- D. The parent-teacher association alone

40. A pupil argues that waste is useless and should always be destroyed. The best reply is that

- A. all waste should be thrown into the sea instead
- B. only the district assembly may touch waste
- C. much waste can be reused, recycled or composted, and so has value
- D. waste should be kept in the house for many years

39. The best way of judging whether a community's new waste management practice has worked is to

- A. ask the people whether they like the colour of the bins
- B. count the number of people living in the community
- C. measure the distance from the community to the landfill
- D. compare the state of the gutters, dumps and streets before and after the practice began

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show your working clearly in all the questions that require a calculation.

Question 7

- (a) Use a pencil and a ruler where a plan or a diagram is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

ANSWER KEY

and Marking Schemes

This section contains correct answers and detailed marking schemes.

Agricultural Tools

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	18.1 Types and Names of Agricultural Tools	Understand
2	A	18.1 Types and Names of Agricultural Tools	Remember
3	B	18.1 Types and Names of Agricultural Tools	Remember
4	C	18.1 Types and Names of Agricultural Tools	Remember
5	C	18.1 Types and Names of Agricultural Tools	Remember
6	A	18.1 Types and Names of Agricultural Tools	Understand
7	B	18.1 Types and Names of Agricultural Tools	Remember
8	B	18.1 Types and Names of Agricultural Tools	Understand
9	C	18.1 Types and Names of Agricultural Tools	Remember
10	C	18.1 Types and Names of Agricultural Tools	Remember
11	B	18.2 Matching Tools with On-Farm Activities	Understand
12	B	18.2 Matching Tools with On-Farm Activities	Understand
13	A	18.2 Matching Tools with On-Farm Activities	Understand
14	D	18.2 Matching Tools with On-Farm Activities	Understand
15	D	18.2 Matching Tools with On-Farm Activities	Understand
16	B	18.2 Matching Tools with On-Farm Activities	Remember
17	D	18.2 Matching Tools with On-Farm Activities	Apply
18	B	18.2 Matching Tools with On-Farm Activities	Remember
19	A	18.2 Matching Tools with On-Farm Activities	Understand
20	B	18.2 Matching Tools with On-Farm Activities	Apply
21	C	18.3 Correct Use of Tools on the Farm or i...	Apply
22	A	18.3 Correct Use of Tools on the Farm or i...	Understand
23	C	18.3 Correct Use of Tools on the Farm or i...	Apply
24	D	18.3 Correct Use of Tools on the Farm or i...	Understand
25	C	18.3 Correct Use of Tools on the Farm or i...	Apply
26	D	18.3 Correct Use of Tools on the Farm or i...	Apply
27	D	18.3 Correct Use of Tools on the Farm or i...	Understand
28	A	18.4 Selecting the Appropriate Tool for a ...	Analyse
29	A	18.4 Selecting the Appropriate Tool for a ...	Apply
30	B	18.4 Selecting the Appropriate Tool for a ...	Apply
31	B	18.4 Selecting the Appropriate Tool for a ...	Apply
32	D	18.4 Selecting the Appropriate Tool for a ...	Apply
33	A	18.4 Selecting the Appropriate Tool for a ...	Evaluate
34	A	18.5 Care, Maintenance and Safe Use of Agr...	Apply
35	C	18.5 Care, Maintenance and Safe Use of Agr...	Understand
36	D	18.5 Care, Maintenance and Safe Use of Agr...	Understand

Q	Ans	Curriculum Area	Cognitive Level
37	D	18.5 Care, Maintenance and Safe Use of Agr...	Analyse
38	C	18.5 Care, Maintenance and Safe Use of Agr...	Apply
39	A	18.5 Care, Maintenance and Safe Use of Agr...	Understand
40	D	18.5 Care, Maintenance and Safe Use of Agr...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 2 (5%) | Apply: 12 (30%) | Evaluate: 2 (5%) | Remember: 9 (22%) | Understand: 15 (37%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 9 marks]

(a) [2 marks]

- An agricultural tool is a **simple implement**, usually worked by hand, used to carry out activities on a farm or in a garden (1 mark).
- Farmers use tools because they **make the work easier, faster, neater and safer** than using the bare hands (1 mark).
- **Marking note:** 2 marks, 1 mark each.*

(b) [4 marks]

- **Cutlass** - for **clearing weeds and grass** from the plot.
- **Hoe** - for **tilling the soil and making ridges or mounds**.
- **Watering can** - for **watering seedlings and vegetable beds**.
- **Wheelbarrow** - for **carrying manure, soil and farm produce** (accept rake, garden fork, trowel, secateurs, knapsack sprayer, sickle, axe, pickaxe, head pan or dibber with a correct use).
- **Marking note:** 1 mark each for any FOUR tools, each with a correct use (4 marks). A tool named with no use, or with a wrong use, scores nothing for that item. This is the curriculum exemplar on collecting

(c) [3 marks]

- **The activity to be done** - clearing, tilling, planting, watering, pruning or carrying.
- **The condition of the land or the crop** - how hard the soil is, how tall the weeds are, how delicate the seedlings are.
- **The size, weight and strength of the pupil** who will use it, and whether the tool is **sharp, sound and safe to use** (accept: whether it is available in the store).
- **Marking note:** 1 mark each for any THREE sound points (3 marks).*

Question 2 [Total: 12 marks]

(a) [6 marks]

- **A - cutlass** (machete) (1 mark).
- **B - hoe** (1 mark).
- **C - rake** (1 mark).
- **D - watering can** (1 mark).
- **E - garden fork** (1 mark).
- **F - wheelbarrow** (1 mark).
- **Marking note:** 1 mark each (6 marks). Accept 'machete' for A and 'digging

fork' for E. Do not accept 'knife' for A or 'spade' for E.*

(b) [3 marks]

- **A (cutlass)** - **clearing weeds, grass and shrubs** from the plot (1 mark).
- **D (watering can)** - **watering seedlings and vegetable beds** (1 mark).
- **F (wheelbarrow)** - **carrying manure, soil, refuse and harvested produce** about the garden (1 mark).
- *Marking note: 1 mark each for a correct activity (3 marks).*

(c) [2 marks]

- **B - the hoe** , which **breaks up and tills the soil** into small lumps (accept **E, the garden fork** , which loosens and turns the soil) (1 mark).
- **C - the rake** , which **levels the bed, makes the soil fine and gathers off the stones and dry leaves** (1 mark).
- *Marking note: 2 marks: 1 mark for each tool correctly named WITH what it does. The rake must be one of the two tools given.*

(d) [1 marks]

- **F - the wheelbarrow**.
- *Marking note: 1 mark. Accept the letter alone or the name alone.*

Question 3 [Total: 10 marks]

(a) [3 marks]

- **Hold the handle firmly with both hands, stand with the feet apart** and **swing the hoe so that the blade enters the soil** (1 mark).
- **Pull the loosened soil towards you, turn it over and break the lumps** , removing roots and stones (1 mark).
- **Work backwards in strips across the plot** until the whole bed is dug, then **draw the soil together into a level bed or into ridges** (1 mark).
- *Marking note: 3 marks as shown. Accept any sound, well-ordered description of the digging, the turning of the soil and the shaping of the bed. This is the curriculum exemplar on explaining how the

(b) [3 marks]

- **Fill the can with clean water, fit the rose and hold the can by the handle** (1 mark).
- **Walk slowly along the bed, tilting the can so that the water falls over the seedlings** , moving it to and fro so that every plant is reached and the soil is wetted evenly (1 mark).
- **The rose breaks the water into many fine drops** , so it **does not wash away the soil or beat down and damage the young seedlings** (1 mark).
- *Marking note: 3 marks: 2 marks for the method and 1 mark for the reason. The reason mark needs the gentle, fine drops and the protection of the seedlings or the soil.*

(c) [2 marks]

- The work is **done more quickly and with less effort**, and the job is **done neatly and properly**.
- The **crops, the soil and the tool itself are not damaged**, and there is **less risk of injury** to the user.
- **Marking note: 1 mark each for any TWO sound reasons (2 marks).**

(d) [2 marks]

- **The injury:** a **cut on the hand or the foot from a cutlass or a hoe** (accept a blister, a wound from a tool left blade upwards, or poisoning from a sprayer) (1 mark).
- **The prevention:** **keep a safe working distance from other pupils, carry the cutlass with the blade pointing down and away from the body, wear boots and gloves, pass a tool handle first and never play with tools** (1 mark).
- **Marking note: 2 marks: 1 mark for a genuine injury and 1 mark for a prevention that matches it.**

Question 4 [Total: 10 marks]**(a) [4 marks]**

- **W - cutlass** (machete) (1 mark).
- **X - hoe** (1 mark).
- **Y - watering can** (1 mark).
- **Z - secateurs** (pruning shears) (1 mark).
- **Marking note: 1 mark each (4 marks).** Accept 'sickle' for W and 'pruning knife' for Z. Do not accept the hoe for W, since the weeds must first be cut down.

(b) [4 marks]

- **W:** the weeds are **tall and woody**, and a **sharp cutlass cuts them down quickly at ground level with a swinging stroke** (1 mark); a small hand tool such as a trowel could not do this work, and the plot must be cleared before it can be tilled (1 mark).
- **Z:** the hedge must be **cut evenly on the top and the sides** (1 mark); **secateurs or shears cut the small branches cleanly without tearing them** or damaging the rest of the hedge (1 mark).
- **Marking note: 2 marks for each reason (4 marks).** Award the second mark of each pair only where the candidate goes beyond repeating the na

(c) [2 marks]

- **The wheelbarrow** (accept the rake or the watering can) (1 mark).
- It would be needed in **more than one area**: for example, to **cart away the cut weeds from W, bring manure to the ridges in X and carry off the trimmings from Z** (1 mark).

- *Marking note: 2 marks: 1 mark for the tool and 1 mark for a reason that covers at least two of the four areas. A tool named with only one use scores 1 mark.*

Question 5 [Total: 9 marks]

(a) [4 marks]

- **Wash or scrape off the soil after use and dry the tool well.**
- **Rub the metal parts with oil or grease** so that they do not rust.
- **Sharpen the cutting edges regularly with a file**, and **tighten or replace loose or broken handles**.
- **Store the tools in a dry store or shed**, hung up or arranged neatly, and never leave them lying in the field.
- *Marking note: 1 mark each for any FOUR correct ways (4 marks).*

(b) [3 marks]

- **They last longer**, so the school **saves the money it would spend replacing them**.
- **They work well**, so the work is **easier, faster and neater**.
- **They are safe to use** - blunt blades and loose handles slip and cause injuries
- and they are **always ready for the next lesson**.
- *Marking note: 1 mark each for any THREE sound reasons (3 marks).*

(c) [2 marks]

- **Carry a cutlass or any sharp tool with the blade pointing downwards and away from the body**, and **pass a sharp tool handle first**.
- **Keep a safe distance from other pupils when swinging a tool**, never play with or run while carrying one, and **never leave a tool lying with its blade or prongs facing upwards** (accept: wear boots and gloves).
- *Marking note: 1 mark each for any TWO genuine safety rules (2 marks).*

Question 6 [Total: 10 marks]

(a) [4 marks]

- **Visit the garden store, the school farm and nearby farms and list all the tools found** (1 mark).
- **Collect a picture of each tool** - draw it, photograph it or cut one from a catalogue or a magazine (1 mark).
- **Paste each picture on its own page and write the name, the use and the care of the tool beside it** (1 mark).
- **Arrange the pages in groups** - cutting tools, digging tools, carrying tools, watering tools - and add a **cover, a title page and a list of contents** (1 mark).
- *Marking note: 1 mark each for any FOUR sensible steps in a reasonable order (4 marks). This is the

(b) [3 marks]

- **Name of the tool** (with its local name).
- **The farm activity it is used for**, and **how it is used**.
- **How it is cared for and stored** (accept: the parts of the tool, or the safety rules for using it).
- **Marking note:** 1 mark each for any THREE suitable headings (3 marks).*

(c) [2 marks]

- Two sources: **the agricultural science textbook and the school library**, and **the internet** (accept farm catalogues, newspapers, the garden store itself, a farmer, a hardware seller or the agricultural extension officer) (1 mark for any two).
- Checking: **compare the information with a second source or the textbook**, or **show it to the teacher, an experienced farmer or the agricultural extension officer** for confirmation (1 mark).
- **Marking note:** 2 marks: 1 mark for naming two sources and 1 mark for a sound way of checking. This part also carries the digital-literacy competency.*

(d) [1 marks]

- **Display it in the garden store, the library or at the school open day, and present it to the class or at assembly**, so that it serves as a guide for the pupils who use the tools.
- **Marking note:** 1 mark for any one sensible way of presenting the album.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 4: Forces and Energy | **Sub-strand:** 5: Agricultural Tools |
Class: Basic 8 (JHS 2)

Content Standard: **B8/JHS2.4.5.1** - Demonstrate knowledge and skills in the use of basic and simple agricultural tools for basic on-farm activities.

> Source: NaCCA Science Common Core Pr

Animal Production

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	8.1 Types of Animal Feed and the Animals t...	Remember
2	C	8.1 Types of Animal Feed and the Animals t...	Remember
3	C	8.1 Types of Animal Feed and the Animals t...	Remember
4	D	8.1 Types of Animal Feed and the Animals t...	Understand
5	A	8.1 Types of Animal Feed and the Animals t...	Remember
6	D	8.1 Types of Animal Feed and the Animals t...	Understand
7	B	8.1 Types of Animal Feed and the Animals t...	Remember
8	D	8.1 Types of Animal Feed and the Animals t...	Remember
9	C	8.1 Types of Animal Feed and the Animals t...	Remember
10	A	8.2 Nutrients in Animal Feed and their Sou...	Understand
11	A	8.2 Nutrients in Animal Feed and their Sou...	Understand
12	B	8.2 Nutrients in Animal Feed and their Sou...	Understand
13	D	8.2 Nutrients in Animal Feed and their Sou...	Understand
14	B	8.2 Nutrients in Animal Feed and their Sou...	Remember
15	C	8.2 Nutrients in Animal Feed and their Sou...	Understand
16	A	8.2 Nutrients in Animal Feed and their Sou...	Understand
17	A	8.2 Nutrients in Animal Feed and their Sou...	Understand
18	B	8.1 Types of Animal Feed and the Animals t...	Analyse
19	D	8.3 Reading and Using Feed Labels	Remember
20	D	8.3 Reading and Using Feed Labels	Understand
21	A	8.3 Reading and Using Feed Labels	Apply
22	A	8.5 Feeding, Watering and Management of Fa...	Evaluate
23	C	8.2 Nutrients in Animal Feed and their Sou...	Understand
24	D	8.4 Importance of Water to the Growth of A...	Remember
25	B	8.4 Importance of Water to the Growth of A...	Understand
26	B	8.4 Importance of Water to the Growth of A...	Understand
27	C	8.4 Importance of Water to the Growth of A...	Analyse
28	C	8.4 Importance of Water to the Growth of A...	Understand
29	A	8.4 Importance of Water to the Growth of A...	Apply
30	D	8.4 Importance of Water to the Growth of A...	Remember
31	C	8.5 Feeding, Watering and Management of Fa...	Understand
32	C	8.5 Feeding, Watering and Management of Fa...	Understand
33	B	8.5 Feeding, Watering and Management of Fa...	Analyse
34	B	8.4 Importance of Water to the Growth of A...	Analyse
35	C	8.4 Importance of Water to the Growth of A...	Analyse
36	A	8.5 Feeding, Watering and Management of Fa...	Evaluate
37	D	8.5 Feeding, Watering and Management of Fa...	Evaluate

Q	Ans	Curriculum Area	Cognitive Level
38	B	8.1 Types of Animal Feed and the Animals t...	Evaluate
39	A	8.2 Nutrients in Animal Feed and their Sou...	Analyse
40	D	8.5 Feeding, Watering and Management of Fa...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 6 (15%) | Apply: 2 (5%) | Evaluate: 5 (12%) | Remember: 11 (27%) | Understand: 16 (40%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 9 marks]

(a) [2 marks]

- Animal feed is the **food given to farm animals** (1 mark)
- to keep them **alive, healthy and productive** - for growth, reproduction and the production of meat, milk and eggs (1 mark).
- **Marking note:** 2 marks, 1 mark each. Award 1 mark for 'food given to farm animals' alone. Do NOT accept 'grass' by itself.*

(b) [4 marks]

- (i) Cattle - **grass, forage, hay or silage**
- (ii) Laying hen - **layer mash**
- (iii) Pig - **pig grower mash or pellets** (accept kitchen and farm waste)
- (iv) Day-old chick - **chick mash (starter mash)**
- **Marking note:** 1 mark each (4 marks). The feed must suit the animal AND its age. Do NOT accept 'layer mash' for a day-old chick or 'silage' for a chick.*

(c) [3 marks]

- **Proteins**
- **Carbohydrates**
- **Fats and oils**
- **Mineral salts**
- **Vitamins** (also accept **water** and **roughage/fibre**)
- **Marking note:** 1 mark each for any THREE correct classes (3 marks). Do NOT accept named feedstuffs such as 'maize' or 'fish meal' here - those are sources, not classes of nutrients.*

Question 2 [Total: 11 marks]

(a) [1 marks]

- **Carbohydrate** (55 %)
- **Marking note:** 1 mark. Accept 'carbohydrate' with or without the percentage.*

(b) [2 marks]

- Protein = 22 % of 50 kg = $(22 \div 100) \times 50$ (1 mark)
- = **11 kg** (1 mark)
- **Marking note:** 2 marks: 1 method mark for the correct working and 1 mark for the answer with the unit. Award the method mark even if the arithmetic is wrong. A bare correct answer of 11 kg scores both marks.*

(c) [2 marks]

- **Chicks are growing rapidly**, and protein is the nutrient that **builds body tissues, muscles and feathers** (1 mark).
- **Adult birds have finished growing**, so they need mainly energy and maintenance feed and need a smaller proportion of protein (1 mark).
- **Marking note:** 2 marks, 1 mark each. The first mark is for linking protein to growth or body building; the second is for the comparison with adult birds.*

(d) [2 marks]

- To check that the **proportions of nutrients suit the type and age** of his animals (for example starter mash for chicks).
- To **compare the quality and price** of different brands, and to check the **net mass** and the **expiry (or manufacturing) date** so that stale feed is not bought.
- **Marking note:** 1 mark each for any TWO sound reasons (2 marks). This is the curriculum exemplar on selecting feed from the proportions shown on the label.*

(e) [2 marks]

- **Mineral salts** - calcium and phosphorus (1 mark)
- used by a laying hen to form **strong egg shells** (also accept **strong bones**) (1 mark).
- **Marking note:** 2 marks, 1 mark each. Accept 'calcium' for the nutrient mark. The egg-shell answer is the expected use for a laying hen.*

(f) [2 marks]

- To stop the feed from **taking up moisture and going mouldy** - mouldy feed can **poison the animals**.
- To **keep out rats, weevils and other pests**, and to **preserve the nutrients** so that the feed stays fresh until it is used.
- **Marking note:** 1 mark each for any TWO correct reasons (2 marks).*

Question 3 [Total: 10 marks]**(a)** [4 marks]

- (i) Protein - **builds and repairs body tissues**, for growth, muscle and the production of milk and eggs
- (ii) Carbohydrate - **supplies energy** for movement, work and warmth
- (iii) Mineral salts - **form strong bones, teeth and egg shells** and keep the blood healthy
- (iv) Vitamins - **protect the animal from diseases** and keep the body working properly
- **Marking note:** 1 mark each (4 marks). Accept any one correct use for each nutrient. Do NOT accept 'it makes the animal strong' for more than one of them.*

(b) [2 marks]

- (i) Protein - **fish meal**, groundnut cake, soya bean meal or blood meal
- (ii) Carbohydrate - **maize**, cassava, rice bran, wheat bran or millet
- **Marking note:** 1 mark each (2 marks). Any one correct feedstuff per nutrient earns the mark.*

(c) [2 marks]

- **Hay is dried** grass or fodder, while **silage is kept moist (green)**.
- Hay is **stored dry in a barn or stack**, while silage is **stored in a pit or silo where it ferments**.
- **Marking note:** 1 mark each for any TWO correct differences (2 marks). The answer must compare the two; a statement about only one of them earns no mark.*

(d) [2 marks]

- A concentrate supplies a **high proportion of nutrients (especially protein)** in a small quantity, making up for what grass and forage lack.
- It makes the animals **grow faster and produce more meat, milk and eggs**, and keeps them healthier.
- **Marking note:** 1 mark each for any TWO correct reasons (2 marks).*

Question 4 [Total: 12 marks]**(a)** [1 marks]

- **1000 g** (1 kg)
- **Marking note:** 1 mark. The unit must be given. Accept 1 kg.*

(b) [1 marks]

- **750 g**
- **Marking note:** 1 mark. The unit must be given. Accept 0.75 kg.*

(c) [2 marks]

- Group P = 1500 g and Group Q = 750 g at the end of week 5 (1 mark)
- Difference = 1500 g - 750 g = **750 g** (1 mark)
- **Marking note:** 2 marks: 1 method mark for the subtraction and 1 mark for the answer with the unit. A bare correct answer of 750 g scores both marks.*

(d) [3 marks]

- They **grew much more slowly** and gained far less mass than Group P (1 mark).
- Their growth **almost stopped (levelled off)** after about week 3, while Group P kept growing steadily (1 mark).
- Without enough water they **could not digest and absorb the feed properly**, so they would be weak, dull and unhealthy (1 mark).
- **Marking note:** 3 marks, 1 mark each. At least one mark must come from the

graph itself (slower growth, or the levelling off of the Group Q curve).*

(e) [2 marks]

- They would **stop growing and begin to lose mass**, stop feeding and become very weak, with sunken eyes and dry skin (1 mark).
- Within a few days the chicks would **die** (1 mark).
 - *Marking note: 2 marks, 1 mark each. This is the curriculum exemplar on predicting what happens to animals not provided with adequate water; death must appear for the second mark.*

(f) [2 marks]

- **Water is necessary for the proper growth of animals** (1 mark).
- Animals given **enough clean water grow much faster** than those given too little, **even when they are given the same feed** (1 mark).
 - *Marking note: 2 marks, 1 mark each. The conclusion must refer to water AND to growth. Do NOT accept 'Group P is better than Group Q', which merely restates the result.*

(g) [1 marks]

- The **type and quantity of feed** given to both groups (also accept the **breed, age and number of chicks** or the **housing and temperature**).
- *Marking note: 1 mark. Do NOT award a mark for 'the amount of water' - that is the factor being changed.*

Question 5 [Total: 9 marks]

(a) [4 marks]

- It **helps in the digestion** of food and softens the feed in the gut.
- It **dissolves nutrients and carries them round the body** in the blood.
- It helps to **control the body temperature** of the animal through sweating and panting.
- It forms a large part of the **blood, milk and body fluids**, and helps in **removing waste** from the body as urine.
 - *Marking note: 1 mark each for any FOUR correct uses (4 marks). Do NOT award two marks for the same idea stated twice in different words.*

(b) [3 marks]

- **Dry nose, muzzle and mouth**, with **sunken eyes** and a dull, rough coat.
- **Loss of appetite and loss of body mass**, with weakness and slow movement.
- **Little urine passed, and it is dark and strong-smelling**; the animals crowd round the trough whenever water is brought.
- A **fall in the production of milk or eggs**.
 - *Marking note: 1 mark each for any THREE correct signs (3 marks). Accept any observable sign of dehydration.*

(c) [2 marks]

- **Wash the drinking troughs daily** and **change the water regularly** (at least twice a day).
- **Place the troughs where droppings, dust and litter cannot fall in** (raise them slightly), and **cover the water storage tank**.
- **Marking note: 1 mark each for any TWO practical ways (2 marks).**

Question 6 [Total: 9 marks]**(a) [4 marks]**

- **Give the right feed for the age of the birds** (starter mash first, then grower and finisher) at **fixed times each day**.
 - **Measure out the correct quantity** of feed for the fifty birds, so that there is neither shortage nor waste.
 - **Wash the drinkers and feeders** and **provide clean, fresh water at all times**, changing it at least twice a day.
 - **Remove stale, wet or mouldy feed** and soiled litter, and **observe the birds**, reporting any that look sick or are not eating.
 - **Marking note: 1 mark each for any FOUR sensible daily practices (4 marks).**
- This is a practical design t

(b) [2 marks]

- The **quantity of feed used** each day or week and the **money spent** on it.
- The **body mass of the birds each week** (to check their growth), and the **number of birds that are alive, sick or dead**.
- **Marking note: 1 mark each for any TWO useful records (2 marks). Accept a record of sales or of vaccination dates.**

(c) [3 marks]

- Keep the bags in a **cool, dry, well-ventilated room** so that the feed does not absorb moisture and go mouldy.
 - **Stack the bags on pallets or planks off the floor** and away from the wall, and **keep rats and insects out** of the store.
 - **Use the oldest bags first** and check the expiry date; **never store feed together with insecticides, fertilisers or other chemicals**.
 - **Marking note: 1 mark each for any THREE correct precautions (3 marks).**
- Accept any sound storage practice that protects the feed from damp, pests or contamination.*

CURRICULUM ASSESSMENT BLUEPRINT

**Stran

Atomic Structure

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	2.1 The Atom and the Sub-Atomic Particles	Remember
2	A	2.1 The Atom and the Sub-Atomic Particles	Remember
3	B	2.1 The Atom and the Sub-Atomic Particles	Understand
4	D	2.1 The Atom and the Sub-Atomic Particles	Remember
5	C	2.1 The Atom and the Sub-Atomic Particles	Remember
6	A	2.2 Atomic Number and Mass Number	Remember
7	B	2.1 The Atom and the Sub-Atomic Particles	Understand
8	D	2.2 Atomic Number and Mass Number	Apply
9	C	2.2 Atomic Number and Mass Number	Apply
10	A	2.2 Atomic Number and Mass Number	Apply
11	B	2.2 Atomic Number and Mass Number	Apply
12	D	2.2 Atomic Number and Mass Number	Understand
13	C	2.4 Ions and How They Are Formed	Apply
14	B	2.4 Ions and How They Are Formed	Apply
15	A	2.5 Molecules, Chemical Formulae and Bonding	Understand
16	D	2.5 Molecules, Chemical Formulae and Bonding	Understand
17	D	2.6 The Periodic Table: Groups, Periods an...	Understand
18	B	2.6 The Periodic Table: Groups, Periods an...	Remember
19	C	2.6 The Periodic Table: Groups, Periods an...	Remember
20	A	2.2 Atomic Number and Mass Number	Apply
21	B	2.3 Electron Configuration, Shells and Val...	Remember
22	C	2.3 Electron Configuration, Shells and Val...	Remember
23	A	2.3 Electron Configuration, Shells and Val...	Apply
24	D	2.3 Electron Configuration, Shells and Val...	Apply
25	A	2.2 Atomic Number and Mass Number	Apply
26	B	2.6 The Periodic Table: Groups, Periods an...	Understand
27	C	2.6 The Periodic Table: Groups, Periods an...	Understand
28	D	2.3 Electron Configuration, Shells and Val...	Apply
29	C	2.4 Ions and How They Are Formed	Apply
30	A	2.5 Molecules, Chemical Formulae and Bonding	Apply
31	B	2.5 Molecules, Chemical Formulae and Bonding	Understand
32	D	2.5 Molecules, Chemical Formulae and Bonding	Understand
33	A	2.5 Molecules, Chemical Formulae and Bonding	Apply
34	B	2.5 Molecules, Chemical Formulae and Bonding	Understand
35	C	2.4 Ions and How They Are Formed	Apply
36	D	2.6 The Periodic Table: Groups, Periods an...	Understand

Q	Ans	Curriculum Area	Cognitive Level
37	B	2.6 The Periodic Table: Groups, Periods an...	Understand
38	A	2.3 Electron Configuration, Shells and Val...	Apply
39	C	2.5 Molecules, Chemical Formulae and Bonding	Remember
40	D	2.4 Ions and How They Are Formed	Understand

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Apply: 16 (40%) | Remember: 10 (25%) | Understand: 14 (35%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 7 marks]****(a)** [3 marks]

- The **proton**
- The **neutron**
- The **electron**
- *Marking note: 1 mark each (3 marks). All three must be named. Do NOT accept 'nucleus' or 'shell' - these are parts of the atom, not sub-atomic particles.**

(b) [2 marks]

- (i) **Positive** (+1)
- (ii) **Nucleus**
- (iii) **No charge** (neutral / 0)
- *Marking note: 1 mark for (i) with (ii) and 1 mark for (iii), giving 2 marks. A candidate who writes 'positive' for the neutron or 'outside the nucleus' for the proton earns no mark.**

(c) [1 marks]

- An **atom is electrically neutral** (equal numbers of protons and electrons); an **ion carries a charge** because the numbers are unequal.
- An atom may **gain or lose electrons to become an ion**.
- *Marking note: 1 mark for any one correct difference. Do NOT accept 'an ion is smaller' or 'an ion is charged' alone without saying why.**

(d) [1 marks]

- The number of **protons equals the number of electrons**, so the **positive and negative charges cancel out**.
- *Marking note: 1 mark. The answer must mention both protons and electrons and the idea of the charges cancelling.**

Question 2 [Total: 6 marks]**(a)** [2 marks]

- (i) The atomic number is the number of **protons in one atom** of the element (accept 'number of protons', or 'number of electrons in a neutral atom') - 1 mark.
- (ii) **16** protons - 1 mark.
- *Marking note: 1 mark each (2 marks). In (ii) the figure 16 must appear; 'sixteen' in words is accepted.**

(b) [1 marks]

- Neutrons = mass number – atomic number = $32 - 16 = 16$ neutrons
- *Marking note: 1 mark for 16. Award the method mark for correct working even if the final figure is wrong, but do not award a mark for a wrong figure with no working.*

(c) [1 marks]

- 16 electrons (the atom is neutral, so electrons = protons)
- *Marking note: 1 mark for 16. Do NOT accept 32, which confuses the mass number with the number of electrons.*

(d) [1 marks]

- 2, 8, 6
- *Marking note: 1 mark. The shells must be filled in order: 2, then 8, then the remainder. Do NOT accept 8, 8 or 2, 6, 8.*

(e) [1 marks]

- 6 valence electrons
- *Marking note: 1 mark. Do NOT accept 16, which is the total number of electrons.*

Question 3 [Total: 10 marks]**(a)** [2 marks]

- Atom X: atomic number 11
- Atom Y: atomic number 17
- *Marking note: 1 mark each (2 marks). The figures must be matched to the correct atom; do NOT accept 23 or 35 here - those are mass numbers.*

(b) [2 marks]

- Atom X: mass number = $11 + 12 = 23$
- Atom Y: mass number = $17 + 18 = 35$
- *Marking note: 1 mark each (2 marks). Award the mark for the correct total; accept the correct addition shown without a final figure only if the figure is then given.*

(c) [2 marks]

- Atom X: 2, 8, 1
- Atom Y: 2, 8, 7
- *Marking note: 1 mark each (2 marks). Each configuration must be matched to the correct atom.*

(d) [2 marks]

- Atom X: **1** valence electron
- Atom Y: **7** valence electrons
- *Marking note: 1 mark each (2 marks). Do NOT accept the total number of electrons (11 and 17).*

(e) [2 marks]

- Atom X **loses its single (valence) electron** from the outermost shell (1 mark).
- It then has **11 protons but only 10 electrons**, so the charge on the ion is **+1** (1 mark).
- *Marking note: 2 marks, 1 mark each. The reason must refer to the numbers of protons and electrons, or state that a positive ion forms because a negative electron has been lost. Do NOT accept 'it becomes positive' with no reason.*

Question 4 [Total: 12 marks]**(a)** [1 marks]

- A molecule is the **smallest particle of a substance that can exist on its own**, made up of **two or more atoms chemically joined together**.
- *Marking note: 1 mark. Award the mark for the idea that two or more atoms are chemically combined. Do NOT accept 'a small particle' alone.*

(b) [1 marks]

- An **atom** is a single particle; a **molecule** is made up of **two or more atoms joined together**.
- A molecule of an element contains atoms of the same kind; a molecule of a compound contains atoms of different kinds.
- *Marking note: 1 mark for any one correct difference, stated as a comparison.*

(c) [3 marks]

- (i) **H₂**
- (ii) **O₂**
- (iii) **CO₂**
- *Marking note: 1 mark each (3 marks). Accept the formulae with correct subscripts only; do NOT accept 2H, 2O or Co₂.*

(d) [3 marks]

- (i) Carbon (C) = **6 atoms**; Hydrogen (H) = **12 atoms**; Oxygen (O) = **6 atoms** (2 marks for all three correct).
- (ii) **6 + 12 + 6 = 24 atoms** altogether (1 mark).
- *Marking note: In (i) award 2 marks only when all three figures are correct; award 1 mark if only carbon and oxygen are right. (ii) earns 1 mark for 24.*

(e) [2 marks]

- (i) Hydrogen molecule, H_2 - **covalent bond**
- (ii) Sodium chloride, $NaCl$ - **ionic bond**
- **Marking note:** 1 mark each (2 marks). A candidate who reverses the two earns no mark.*

(f) [2 marks]

- Each chlorine atom has **7 valence electrons** and needs **one more** to complete its outermost shell (1 mark).
- The two atoms each contribute one electron, forming a **shared pair** which is **attracted by both nuclei** - this is a single covalent bond (1 mark).
- **Marking note:** 2 marks, 1 mark each. Look for the idea of sharing (not transfer) and for both atoms reaching a complete outer shell. Do NOT accept 'chlorine gives an electron to the other chlorine'.*

Question 5 [Total: 12 marks]**(a)** [5 marks]

- (i) Sodium - **2, 8, 1**
- (ii) Magnesium - mass number **24**
- (iii) Chlorine - atomic number **17**
- (iv) Chlorine - **17** electrons
- (v) Chlorine - **2, 8, 7**
- **Marking note:** 1 mark each for all FIVE blanks (5 marks). The blank number must be matched to the correct answer. In (ii) accept 24 from '12 protons + 12 neutrons'.*

(b) [2 marks]

- (i) Magnesium is in **Period 3** (its electrons occupy 3 shells) - 1 mark.
- (ii) Sodium is in **Group 1** (it has 1 electron in its outermost shell) - 1 mark.
- **Marking note:** 1 mark each (2 marks). Accept the reason drawn from the electron configuration: 2, 8, 2 has three shells; 2, 8, 1 has one outer electron. Do NOT accept Group 3 for magnesium.*

(c) [5 marks]

- (i) Neutrons = $39 - 19 = 20$ (1 mark)
- (ii) Electron configuration = **2, 8, 8, 1** (1 mark)
- (iii) Valence electrons = **1** (1 mark)
- (iv) The charge on the ion is **+1** (1 mark)
- (iv) Reason: the atom has only **one electron in its outermost shell**, and it loses this electron easily to form a stable ion, K^+ (1 mark)
- **Marking note:** 5 marks: 1 mark each for (i), (ii) and (iii); in (iv) 1 mark for the charge and 1 mark for the reason. Do NOT accept -1 or $+19$.*

Question

(a) [10 marks]

- (i) Sodium atom: **1** valence electron; chlorine atom: **7** valence electrons (2 marks, 1 each).
- (ii) The sodium atom **loses (transfers) its single outermost electron** to the chlorine atom (1 mark); the chlorine atom **gains** this electron and now has a complete outer shell of 8 (1 mark); the **positive and negative ions formed are held together by their opposite charges** (1 mark).
- (iii) Sodium ion: **+1** (Na^+); chloride ion: **-1** (Cl^-) (2 marks, 1 each).
- (iv) Type of bond: **ionic (electrovalent) bond** (1 mark); formula: **NaCl** (1 mark).
- (v) Any one of: flavouring food;

(b) [3 marks]

- The compound is **magnesium chloride** (1 mark).
- A magnesium atom (2, 8, 2) **loses 2 outermost electrons** to form **Mg^{2+}** , while **each** chlorine atom gains 1 electron to form **Cl^-** (1 mark).
- **Two** chlorine atoms are therefore needed for every magnesium atom, giving the formula **MgCl_2** (1 mark).
 - **Marking note:** 3 marks, 1 mark each. The last mark requires the formula MgCl_2 WITH the 1 : 2 ratio explained. Do NOT accept MgCl or Mg_2Cl .

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 1: Diversity of Matter | **Sub-strand:** 1: Materials | **Class:** Basic 8 (JHS 2)

Con

Bacterial Disease

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	21.1 The Nature of Bacteria and Bacterial ...	Remember
2	D	21.1 The Nature of Bacteria and Bacterial ...	Remember
3	D	21.1 The Nature of Bacteria and Bacterial ...	Remember
4	C	21.1 The Nature of Bacteria and Bacterial ...	Understand
5	A	21.1 The Nature of Bacteria and Bacterial ...	Understand
6	D	21.1 The Nature of Bacteria and Bacterial ...	Understand
7	D	21.1 The Nature of Bacteria and Bacterial ...	Understand
8	B	21.1 The Nature of Bacteria and Bacterial ...	Apply
9	A	21.2 Food Poisoning	Remember
10	C	21.2 Food Poisoning	Remember
11	C	21.2 Food Poisoning	Remember
12	A	21.2 Food Poisoning	Understand
13	A	21.2 Food Poisoning	Apply
14	C	21.2 Food Poisoning	Understand
15	A	21.2 Food Poisoning	Analyse
16	C	21.2 Food Poisoning	Evaluate
17	A	21.2 Food Poisoning	Remember
18	B	21.3 Gonorrhoea	Remember
19	C	21.3 Gonorrhoea	Remember
20	D	21.3 Gonorrhoea	Remember
21	D	21.3 Gonorrhoea	Understand
22	C	21.3 Gonorrhoea	Analyse
23	B	21.3 Gonorrhoea	Understand
24	C	21.3 Gonorrhoea	Understand
25	B	21.3 Gonorrhoea	Evaluate
26	A	21.4 Meningitis	Remember
27	A	21.4 Meningitis	Remember
28	D	21.4 Meningitis	Understand
29	B	21.4 Meningitis	Understand
30	B	21.4 Meningitis	Understand
31	C	21.4 Meningitis	Analyse
32	D	21.4 Meningitis	Understand
33	B	21.4 Meningitis	Evaluate
34	A	21.5 Managing Bacterial Diseases: the Indi...	Understand
35	C	21.5 Managing Bacterial Diseases: the Indi...	Understand
36	B	21.5 Managing Bacterial Diseases: the Indi...	Understand
37	D	21.5 Managing Bacterial Diseases: the Indi...	Evaluate

Q	Ans	Curriculum Area	Cognitive Level
38	D	21.5 Managing Bacterial Diseases: the Indi...	Create
39	A	21.5 Managing Bacterial Diseases: the Indi...	Analyse
40	B	21.5 Managing Bacterial Diseases: the Indi...	Apply

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 4 (10%) | Apply: 3 (7%) | Create: 1 (2%) | Evaluate: 4 (10%) | Remember: 12 (30%) | Understand: 1

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 9 marks]

(a) [3 marks]

- **P** - rod-shaped, like short sticks (accept *bacillus*, rod-like, cylindrical) (1 mark).
 - **Q** - round or spherical, joined together in pairs (accept *coccus*, *cocci*, ball-shaped, in pairs) (1 mark).
 - **R** - spiral, coiled or wavy (accept *spirillum*, corkscrew-shaped, curved) (1 mark).
- **Marking note:** 3 marks, 1 mark each. Accept the everyday description or the scientific name; the candidate does not need both. Do not award a mark for naming a disease here - the question asks for shape.*

(b) [3 marks]

- **Food poisoning** (from *Salmonella*, *Staphylococcus* or *Escherichia coli*).
 - **Gonorrhoea**.
 - **Meningitis**.
 - Also accept: **cholera**, typhoid fever, tuberculosis, tetanus, whooping cough, syphilis, leprosy.
- **Marking note:** 1 mark each for any THREE correct bacterial diseases (3 marks). Refuse the mark for diseases that are not bacterial - malaria, measles, hepatitis B, HIV and AIDS, ringworm and athlete's foot are caused by protozoa, viruses or fungi.*

(c) [3 marks]

- **Warmth** - the food is kept at room temperature, which is warm enough for the bacteria to grow.
 - **Moisture** - cooked food such as rice, stew or kenkey holds water, which the bacteria need.
 - **Food (nutrients)** - the bacteria feed on the starch and the protein in the meal.
 - Also accept: **enough time** for the numbers to double again and again, and the **food left uncovered** so that bacteria from the air, from dust and from flies settle on it.
- **Marking note:** 1 mark each for any THREE conditions (3 marks). 'Dirtiness' alone is too vague; it must be linked to germs reaching the food.*

Question**(a)** [3 marks]

- **Food poisoning is an illness that follows the eating of food or the drinking of water that contains harmful bacteria or the poisons (toxins) that the bacteria have produced** (1 mark).
- **Salmonella** (1 mark).
- **Staphylococcus** (accept *Escherichia coli*, *Clostridium*, *Shigella*, *Campylobacter*) (1 mark).
- **Marking note:** 3 marks: 1 for the explanation, 1 for each of TWO named bacteria. The explanation must mention the germ or its poison in the food; 'food that is bad' alone earns nothing.*

(b) [4 marks]

- **Vomiting.**
- **Diarrhoea - frequent, watery stools.**
- **Stomach (abdominal) pain or cramps.**
- **Fever**, with chills.
- Also accept: **nausea, headache, weakness and tiredness, dry mouth and great thirst.**
- **Marking note:** 1 mark each for any FOUR symptoms (4 marks). Mark only the first four given. 'Sickness' on its own is too vague.*

(c) [2 marks]

- **The food was cooked in the morning and served in the afternoon**, so it stood for several hours in a warm, moist state in which any surviving bacteria multiplied many times over.
- **The pan was left open**, so dust, flies and unwashed hands could carry bacteria onto the food.
- Also accept: **the food was not reheated thoroughly before it was served**, and **there was no cold store or refrigerator** to keep it safe.
- **Marking note:** 1 mark each for any TWO sound reasons (2 marks). The reason must explain how the bacteria got in or how they multiplied.*

(d) [3 marks]

- **Wash the hands with soap under running water** before handling food, after using the toilet and after taking money, and keep the fingernails short.
- **Cook the food thoroughly and serve it hot**, and reheat any left-over food until it is steaming before it is sold again.
- **Keep the food covered** with a clean net or lid so that dust and flies cannot reach it.
- Also accept: **use clean water and clean, well-washed utensils**; **do not sell food that was prepared the day before**; **go for the medical examination and hold a valid health certificate from the Environmental Health Unit**; *

Question 3

(a) [2 marks]

- **March** (1 mark).
- **32 cases** (1 mark).
- *Marking note: 2 marks, 1 each. The number must match the month named; a correct number read against the wrong month earns only the number mark if the month is wrong.*

(b) [2 marks]

- **$8 + 20 + 32 + 24 + 10 + 6$** (1 mark for the correct working).
- **$= 100$ cases** (1 mark).
- *Marking note: 2 marks. Award the method mark for a correct addition statement even where the arithmetic slips.*

(c) [2 marks]

- **$32 / 100 \times 100$** (1 mark for the working).
- **$= 32\%$** (1 mark).
- *Marking note: 2 marks. Insist on the per cent sign. A candidate who used a wrong total in (b) but divides correctly here keeps the method mark.*

(d) [2 marks]

- **January to March is the height of the dry, dusty Harmattan season.** The hot, dry, dusty air **dries and injures the lining of the nose and the throat**, so the bacteria pass into the body far more easily (1 mark).
- **People also spend the nights in crowded, poorly aired rooms**, where droplets from coughing, sneezing and talking are easily breathed in by others (accept: more gatherings and travelling in the dry season) (1 mark).
- **From April the rains begin**, the dust settles and the air becomes moist, so fewer people are infected and the number of cases falls.
- *Marking note: 2 marks*

(e) [2 marks]

- **They show at once when the number of cases is rising**, so an outbreak is detected early and can be reported and controlled before it spreads.
- **They show the time of year and the places at greatest risk**, so the directorate can time its vaccination campaign and its health education and can send drugs, beds and staff where they are needed.
- Also accept: **they allow this year to be compared with past years**, and they are evidence when asking the assembly or the Ministry for support.
- *Marking note: 1 mark each for any TWO uses (2 marks). The use must be a decision the health service*

Question

(a) [2 marks]

- **Meningitis is the inflammation, or swelling, of the meninges - the thin coverings of the brain and the spinal cord** (1 mark).
- It is commonly caused by a **bacterium** (accept *Neisseria meningitidis*, the meningococcus; accept the note that some forms are viral) (1 mark).
- **Marking note: 2 marks, 1 each. The first mark needs the brain or spinal cord covering; 'a disease of the head' is not enough.**

(b) [3 marks]

- **A sudden high fever.**
- **A severe headache.**
- **Vomiting.**
- **A dislike of bright light** - the patient turns away from the window.
- Also accept: **confusion or drowsiness, convulsions (fits), a rash, and in a baby a bulging soft spot on the head with refusal to feed**.
- **Marking note: 1 mark each for any THREE symptoms (3 marks). Do not award the stiff neck, which the question excludes.**

(c) [2 marks]

- **Deafness, or damage to the hearing** - the commonest lasting effect.
- **Damage to the brain**, leading to difficulty in learning, loss of memory, epilepsy or paralysis.
- Also accept: **blindness**, **long absence from school**, and **death**, for meningitis can kill within a day or two.
- **Marking note: 1 mark each for any TWO effects (2 marks). Accept effects on the family, such as the cost of treatment, only as the second answer.**

(d) [3 marks]

- **Vaccination of the people at risk**, especially in the meningitis belt before the dry season.
- **Avoid overcrowded sleeping rooms and keep rooms and halls well aired**, with windows open.
- **Cover the mouth and the nose when coughing or sneezing**, and avoid sharing cups, bottles and spoons.
- Also accept: **report every suspected case to a health facility at once so that the patient is treated and close contacts are given preventive drugs**, and **health education through schools, radio and the information centre**.
- **Marking note: 1 mark each for any THREE measures (3 marks). Measur**

Question 5

(a) [3 marks]

- It is caused by a **bacterium** (accept *Neisseria gonorrhoeae*, the gonococcus) (1 mark).
- **Through unprotected sexual contact with a person who has the infection** (1 mark).
- **From an infected mother to her baby during birth**, when the baby passes through the birth canal (1 mark).
 - *Marking note: 3 marks: 1 for the bacterium and 1 for each of TWO routes. Refuse the mark for shaking hands, sharing a toilet seat, mosquito bites or eating together - these do NOT spread gonorrhoea, and they are the beliefs this lesson exists to correct.**

(b) [2 marks]

- **Pain or a burning feeling when passing urine**, often with the need to pass urine more often.
- **An unusual discharge** from the sexual organs.
- Also accept: **pain in the lower abdomen**, and the important point that **many infected women, and some men, show no symptoms at all** while still able to pass the infection on.
 - *Marking note: 1 mark each for any TWO symptoms (2 marks). Mark this part privately and do not read answers aloud.**

(c) [2 marks]

- **It can damage the reproductive organs and leave the person unable to have children** (infertility).
- **A baby born to an untreated mother may develop a severe infection of the eyes and become blind**.
- Also accept: **lasting pain in the lower abdomen**, **the spread of the infection to the joints and the blood**, and **a greater risk of catching other infections, including HIV**.
 - *Marking note: 1 mark each for any TWO effects (2 marks).**

(d) [2 marks]

- **He stopped the drug as soon as he felt better**, so the bacteria that survive are not killed; they multiply, become **resistant to the antibiotic**, and the illness returns in a form that is much harder to treat.
- **He was never examined or tested**, so the illness may not be gonorrhoea at all, and the drug the hawker sold may be the wrong one, the wrong dose, expired or fake.
- **His partner has not been treated**, so the two will keep passing the infection back to each other and on to others, and untreated gonorrhoea can leave him unable to have children.
 - *Marking note: 1 mark each*

Question

(a) [3 marks]

- **The individual** - washes the hands with soap, keeps food and the body clean, avoids crowded, airless rooms, goes early to a health facility when ill and **finishes the full course of the medicine** (1 mark).
- **The community** - organises communal labour to clear gutters and clean the market and the toilets, supports vaccination and education campaigns through the chief, the Assembly member and the information centre, and cares for patients without shaming them (1 mark).
- **The government** - supplies vaccines, drugs, clinics and trained health workers, provides safe water and sanitation

(b) [4 marks]

- **(i) A short title of a few words** that can be read from a distance, for example ***Cook it well, cover it, eat it hot*; *Wash your hands - save your stomach*; *Stop food poisoning today*** (1 mark).
- **(ii) Two short, practical messages** (1 mark each), for example: ***Wash your hands with soap before you eat*; *Cover your food against flies and dust*; *Do not eat food kept overnight without reheating*; *Buy from vendors with a health certificate*; *Drink only safe water***.
- **(iii) One non-word feature** (1 mark): **a large, clear picture or drawing** - a covered pan, a hand being washed, a f

(c) [3 marks]

- **Ask a health worker** - the nurse at the health centre, the community health officer or the Environmental Health Officer - and write down what is said.
- **Use reliable written sources** - the science textbook, a pamphlet from the health centre, or the website of the Ghana Health Service, the Food and Drugs Authority or the World Health Organization.
- **Compare at least two different sources** and keep only the facts on which they agree; reject what somebody merely heard in the street or saw forwarded on social media.
- Also accept: **show the draft poster to the science teacher or the nu**

Carbon Cycle

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	5.1 The Carbon Cycle as a Repeated Pattern...	Remember
2	D	5.1 The Carbon Cycle as a Repeated Pattern...	Remember
3	C	5.2 Photosynthesis as a Stage of the Carbo...	Remember
4	B	5.2 Photosynthesis as a Stage of the Carbo...	Remember
5	B	5.2 Photosynthesis as a Stage of the Carbo...	Remember
6	A	5.2 Photosynthesis as a Stage of the Carbo...	Understand
7	D	5.2 Photosynthesis as a Stage of the Carbo...	Understand
8	D	5.2 Photosynthesis as a Stage of the Carbo...	Remember
9	A	5.3 Respiration as a Stage of the Carbon C...	Remember
10	D	5.3 Respiration as a Stage of the Carbon C...	Remember
11	A	5.3 Respiration as a Stage of the Carbon C...	Remember
12	C	5.4 Burning and Fossil Fuels	Understand
13	C	5.4 Burning and Fossil Fuels	Understand
14	B	5.4 Burning and Fossil Fuels	Remember
15	B	5.4 Burning and Fossil Fuels	Understand
16	A	5.5 Decay and the Work of Decomposers	Remember
17	A	5.5 Decay and the Work of Decomposers	Remember
18	B	5.5 Decay and the Work of Decomposers	Remember
19	C	5.2 Photosynthesis as a Stage of the Carbo...	Understand
20	A	5.1 The Carbon Cycle as a Repeated Pattern...	Understand
21	D	5.1 The Carbon Cycle as a Repeated Pattern...	Analyse
22	B	5.1 The Carbon Cycle as a Repeated Pattern...	Understand
23	D	5.4 Burning and Fossil Fuels	Understand
24	C	5.6 The Carbon Cycle and the Environment	Analyse
25	B	5.6 The Carbon Cycle and the Environment	Understand
26	D	5.6 The Carbon Cycle and the Environment	Understand
27	A	5.6 The Carbon Cycle and the Environment	Apply
28	B	5.6 The Carbon Cycle and the Environment	Evaluate
29	C	5.6 The Carbon Cycle and the Environment	Apply
30	C	5.6 The Carbon Cycle and the Environment	Remember
31	B	5.6 The Carbon Cycle and the Environment	Remember
32	D	5.6 The Carbon Cycle and the Environment	Remember
33	A	5.6 The Carbon Cycle and the Environment	Apply
34	C	5.6 The Carbon Cycle and the Environment	Remember
35	C	5.2 Photosynthesis as a Stage of the Carbo...	Understand
36	D	5.1 The Carbon Cycle as a Repeated Pattern...	Analyse
37	A	5.1 The Carbon Cycle as a Repeated Pattern...	Analyse

Q	Ans	Curriculum Area	Cognitive Level
38	A	5.1 The Carbon Cycle as a Repeated Pattern...	Analyse
39	C	5.6 The Carbon Cycle and the Environment	Analyse
40	B	5.6 The Carbon Cycle and the Environment	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 6 (15%) | Apply: 3 (7%) | Evaluate: 2 (5%) | Remember: 17 (42%) | Understand: 12 (30%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 8 marks]****(a)** [2 marks]

- The carbon cycle is the **continuous movement (circulation)** of carbon between the air, living things, the soil and **back to the air again**.
- Carbon passes from carbon (IV) oxide in the air into green plants, then to animals, and back into the air, repeatedly.
- **Marking note:** 1 mark each for any TWO correct points (2 marks). Award 1 mark for 'carbon moving round and round' if the idea of movement between air and living things is present. Do NOT accept 'the cycle of carbon' alone.*

(b) [4 marks]

- **Photosynthesis**
- **Respiration**
- **Burning** (combustion)
- **Decay** (decomposition)
- **Marking note:** 1 mark each for all FOUR processes (4 marks). Accept 'combustion' for burning and 'decomposition' for decay. Do NOT accept 'eating' or 'food chain' as one of the four processes.*

(c) [2 marks]

- The **same carbon atoms are used again and again** - carbon is not destroyed or used up at any stage.
- The **processes repeat themselves over and over** in nature (carbon returns to the air and is taken up again).
- **Marking note:** 1 mark each for any TWO correct reasons (2 marks). Accept 'it goes round and round' only when the idea of repetition or of reuse is stated.*

Question 2 [Total: 10 marks]**(a)** [2 marks]

- **Carbon (IV) oxide** (carbon dioxide)
- **Water**
- **Marking note:** 1 mark each (2 marks). Do NOT accept oxygen or sunlight, which are needed but are not raw materials taken in by the plant.*

(b) [1 marks]

- **Oxygen**
- **Marking note:** 1 mark for oxygen.*

(c) [2 marks]

- The plant uses the carbon (IV) oxide with water to ****make food (carbohydrates)**** (1 mark).
- The ****carbon becomes part of the plant's body**** - it is stored in the food made, and oxygen is given out (1 mark).
 - **Marking note: 2 marks, 1 mark each. Accept 'the carbon is trapped in the plant' or 'the carbon changes into food'. Do NOT accept 'the carbon disappears'.*

(d) [2 marks]

- The carbon taken from the air is stored in the ****food made by the plant**** (1 mark).
- When the goat ****eats the plant****, the food passes into the goat, so the carbon becomes part of the ****goat's body**** (1 mark).
 - **Marking note: 2 marks, 1 mark each. Look for the idea that the carbon is in the food the plant made, and that eating transfers it to the animal.**

(e) [1 marks]

- It ****produces food**** for the plant and for animals (including humans), and it ****gives out oxygen**** for living things to breathe.
 - **Marking note: 1 mark for either point, or for both.**

(f) [2 marks]

- ****Little or no food**** will be made (1 mark).
- Reason: there is ****no light**** in the cupboard, and light is needed for photosynthesis (1 mark).
 - **Marking note: 2 marks, 1 mark for the correct prediction and 1 mark for the reason. Accept 'the plant will use up the food it has stored and may die' as the reason if light is named.**

Question 3 [Total: 9 marks]**(a)** [2 marks]

- Respiration is the process by which living things ****obtain energy from food****.
- It uses ****oxygen**** and gives out ****carbon (IV) oxide**** (and water) - it happens in all living cells, day and night.
 - **Marking note: 1 mark each for any TWO correct points (2 marks). Award 1 mark for 'getting energy from food' on its own; do NOT accept 'breathing' without the idea of energy from food.**

(b) [4 marks]

- Photosynthesis ****takes in carbon (IV) oxide**** and gives out oxygen; respiration ****takes in oxygen**** and gives out carbon (IV) oxide.
- Photosynthesis ****needs light**** and takes place only in ****green plants****; respiration takes place in ****all living things****, in light or in darkness, day and

night.

- Photosynthesis **stores energy** in food; respiration **releases energy** from food.

- **Marking note:** 2 marks each for any TWO correct differences (4 marks): 1 mark for each half of the comparison. Do NOT award a mark for a difference that describes only one process without contrasting it with the other.

(c) [2 marks]

- Any two of: **firewood (wood)**, charcoal, kerosene, petrol, diesel, natural gas (LPG), dry grass, crop residue or household refuse.

- **Marking note:** 1 mark each for any TWO correct fuels (2 marks). Accept any fuel really burnt in Ghana.

(d) [1 marks]

- It **adds a lot of carbon (IV) oxide** to the air (and destroys useful plants, soil organisms and the nutrients in the soil).

- **Marking note:** 1 mark for any one correct reason. Do NOT accept 'it is bad' or 'it can burn the farm' on its own.

Question 4 [Total: 6 marks]

(a) [2 marks]

- **Bacteria**

- **Fungi** (moulds, mushrooms, yeast)

- **Marking note:** 1 mark each (2 marks). Do NOT accept insects or earthworms as the main groups, although they help to break up the material.

(b) [2 marks]

- Any two of: **carbon (IV) oxide**, methane, water vapour (also accept 'heat' as energy given out).

- **Marking note:** 1 mark each for any TWO correct gases (2 marks). Do NOT accept oxygen or nitrogen - these are not given out during decay.

(c) [2 marks]

- Dead plants and animals would **pile up (not decay)** in the environment (1 mark).

- The **carbon trapped in their bodies would not return to the air**, so less carbon (IV) oxide would be added to the air and **the carbon cycle would not continue** (1 mark).

- **Marking note:** 2 marks, 1 mark each. Look for both the pile-up of dead matter and the break in the cycle.

Question**(a)** [4 marks]

- a - **photosynthesis**
 - b - **respiration**
 - c - **burning** (combustion)
 - d - **decay** (decomposition)
- *Marking note: 1 mark each (4 marks). The process must be matched to the correct letter. Do NOT accept 'breathing' for b.*

(b) [3 marks]

- e - **feeding** (the animals eat the green plants)
 - f - **decay** (decomposition) of dead plants and animals in the soil
 - g - **burning of fossil fuels** (combustion)
- *Marking note: 1 mark each (3 marks). Accept 'feeding' or 'eating' for e, and 'combustion of fossil fuels' for g.*

(c) [2 marks]

- The group is **fossil fuels** (coal, crude oil and natural gas) (1 mark).
 - Use: they **burn to give energy** - for cooking, for vehicles (petrol and diesel), for generating electricity, and in industry (1 mark).
- *Marking note: 2 marks: 1 mark for the name of the group, 1 mark for any correct use.*

(d) [1 marks]

- Because **photosynthesis takes carbon (IV) oxide out of the air** and uses the carbon to make food, so the amount of the gas in the air is reduced.
- *Marking note: 1 mark. The answer must say that the gas is taken from the air and used to make food.*

Question 6 [Total: 7 marks]**(a)** [1 marks]

- The **maize plant**
- *Marking note: 1 mark. Do NOT accept 'the rat' or 'the snake'.*

(b) [2 marks]

- Primary consumer - the **rat**
 - Secondary consumer - the **snake**
- *Marking note: 1 mark each (2 marks). The two must be matched to the correct terms; a reversed answer earns no mark.*

(c) [2 marks]

- The carbon is stored in the **body** (flesh and other tissues) of the rat (1 mark).
- When the snake **eats** (feeds on) the rat, the carbon passes into the snake's body (1 mark).
- *Marking note: 2 marks, 1 mark each. Accept 'the snake eats the rat, so it takes in the carbon that was in the rat' for full marks if the idea of eating is clear.*

(d) [2 marks]

- **Bacteria and fungi (decomposers)** break down the body of the dead snake (1 mark).
- Carbon (IV) oxide is **given out during the decay** and released into the air, so the carbon returns to the air (1 mark).
- *Marking note: 2 marks, 1 mark each. The two ideas required are the decomposers doing the breaking down and the carbon returning to the air as carbon (IV) oxide.*

Question 7 [Total: 10 marks]**(a)** [2 marks]

- The greenhouse effect is the **trapping (holding)** of heat from the sun by gases in the air.
- This heat is kept near the **earth's surface**, so the earth stays warm enough for life.
- *Marking note: 1 mark each for any TWO correct points (2 marks). Award 1 mark for 'gases trap heat' on its own; do NOT accept 'the earth is hot'.*

(b) [2 marks]

- Too much carbon (IV) oxide **traps too much heat** in the atmosphere (1 mark).
- The earth therefore becomes **warmer than normal (global warming)**, which changes the climate (1 mark).
- *Marking note: 2 marks, 1 mark each. The chain of reasoning (more gas - more heat trapped - earth warms) must be present for both marks.*

(c) [4 marks]

- Green plants **remove carbon (IV) oxide** from the air and **give out oxygen** during photosynthesis (1 mark).
- Respiration in living things, burning and decay **add carbon (IV) oxide** to the air and use up oxygen (1 mark).
- Because the two sets of processes **go on at the same time**, the gases that are removed are replaced (1 mark).
- This keeps the amounts of carbon (IV) oxide and oxygen in the air **fairly constant (balanced)**, which keeps life on earth going (1 mark).

- *Marking note: 4 marks, 1 mark each. Award marks for the pairs of processes named and for the idea of balance. A

(d) [2 marks]

- **Suggestion:** planting trees and keeping them on the farm (agroforestry), or using an improved (LPG) cooker instead of firewood, or **composting crop waste instead of burning it** (1 mark).

- **Benefit:** the trees remove carbon (IV) oxide from the air and give out oxygen - and they also give shade, fruit, fodder, firewood or extra income (1 mark).

- *Marking note: 2 marks: 1 mark for a sensible suggestion, 1 mark for a real benefit. This is a design task, so award method marks for any practical suggestion that would genuinely reduce carbon (IV) oxide in the air.*

CURRICULUM ASS

Climate Change

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	23.1 Climate Change and Its Signs	Remember
2	D	23.1 Climate Change and Its Signs	Remember
3	B	23.1 Climate Change and Its Signs	Understand
4	A	23.1 Climate Change and Its Signs	Understand
5	D	23.1 Climate Change and Its Signs	Understand
6	A	23.1 Climate Change and Its Signs	Understand
7	D	23.1 Climate Change and Its Signs	Evaluate
8	A	23.1 Climate Change and Its Signs	Analyse
9	C	23.2 Causes of Climate Change	Remember
10	B	23.2 Causes of Climate Change	Remember
11	B	23.2 Causes of Climate Change	Understand
12	C	23.2 Causes of Climate Change	Remember
13	C	23.2 Causes of Climate Change	Understand
14	B	23.2 Causes of Climate Change	Analyse
15	A	23.2 Causes of Climate Change	Understand
16	C	23.2 Causes of Climate Change	Understand
17	B	23.3 Effects of Climate Change on the Envi...	Understand
18	D	23.3 Effects of Climate Change on the Envi...	Understand
19	A	23.3 Effects of Climate Change on the Envi...	Understand
20	B	23.3 Effects of Climate Change on the Envi...	Understand
21	C	23.3 Effects of Climate Change on the Envi...	Understand
22	B	23.3 Effects of Climate Change on the Envi...	Analyse
23	D	23.3 Effects of Climate Change on the Envi...	Understand
24	C	23.3 Effects of Climate Change on the Envi...	Understand
25	B	23.3 Effects of Climate Change on the Envi...	Evaluate
26	C	23.3 Effects of Climate Change on the Envi...	Understand
27	A	23.4 Adapting to Climate Change	Remember
28	D	23.4 Adapting to Climate Change	Apply
29	A	23.4 Adapting to Climate Change	Analyse
30	B	23.4 Adapting to Climate Change	Understand
31	A	23.4 Adapting to Climate Change	Analyse
32	D	23.4 Adapting to Climate Change	Understand
33	D	23.4 Adapting to Climate Change	Apply
34	A	23.5 Mitigation and the Green Economy	Remember
35	A	23.5 Mitigation and the Green Economy	Apply
36	C	23.5 Mitigation and the Green Economy	Remember
37	B	23.5 Mitigation and the Green Economy	Understand

Q	Ans	Curriculum Area	Cognitive Level
38	C	23.5 Mitigation and the Green Economy	Understand
39	D	23.5 Mitigation and the Green Economy	Evaluate
40	D	23.5 Mitigation and the Green Economy	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 5 (12%) | Apply: 3 (7%) | Evaluate: 4 (10%) | Remember: 8 (20%) | Understand: 20 (50%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 10 marks]

(a) [3 marks]

- **P** - energy, that is heat and light, travelling from the sun to the surface of the earth (1 mark).
 - **Q** - heat given off by the warmed earth, escaping through the layer into space (1 mark).
 - **R** - heat that the layer of gases sends back down to the surface of the earth (accept 'heat trapped and returned by the gases') (1 mark).
- **Marking note:** 3 marks, 1 mark each. For P accept 'sunlight', 'solar energy' or 'radiation from the sun'. For R the answer must show the heat coming BACK to the earth; 'heat in the air' is not enough.*

(b) [2 marks]

- **Carbon dioxide** (1 mark).
 - **Methane** (1 mark).
 - Also accept: **water vapour**, **nitrous oxide**, **chlorofluorocarbons (CFCs)**.
- **Marking note:** 1 mark each for any TWO (2 marks). Do not accept oxygen or nitrogen; they are the commonest gases in the air but they do not trap heat in this way.*

(c) [3 marks]

- **The average temperature of the earth rises** - the earth becomes warmer. This is called global warming (1 mark).
 - **The thicker layer absorbs more of the heat that is leaving the earth**, so less heat escapes into space along arrow Q (1 mark).
 - **More of that heat is sent back down to the surface along arrow R**, so heat builds up at the surface and in the lower air (1 mark).
- **Marking note:** 3 marks. The first mark is for the rise in temperature; the other two for the mechanism - less heat escaping and more heat returned. An answer that says only 'the earth becomes hot' earns 1 mark.*

(d) [2 marks]

- **Burning fuels** - firewood, charcoal, petrol, diesel and gas - in kitchens, vehicles, generators and factories.
- **Cutting down and burning forests**, which both releases carbon dioxide and removes the trees that would have absorbed it.
- Also accept: **bush burning before the planting season**; **heaps of refuse left to rot**, and landfill sites, which give off methane; **keeping large herds of cattle**; **flooded rice fields**; **industry and mining**.

- *Marking note: 1 mark each for any TWO activities (2 marks). The activity must be one that people do; a volcano is natural and earns

Question 2 [Total: 10 marks]

(a) [2 marks]

- **1980 - 27.0 °C** (1 mark).
- **2020 - 28.4 °C** (1 mark).
- *Marking note: 2 marks, 1 each. The unit °C must appear at least once in the answer; do not take the mark off twice for the same omission.*

(b) [2 marks]

- **28.4 – 27.0** (1 mark for the correct working).
- **= 1.4 °C** (1 mark).
- *Marking note: 2 marks. Award the method mark for the correct subtraction even where the arithmetic slips. Insist on the unit in the final answer.*

(c) [2 marks]

- **The temperature rises steadily from one reading to the next**, by about 0.3 to 0.4 °C every ten years, and at no point does it fall (1 mark).
- **This suggests that the climate of the area is becoming warmer** - it is a sign of climate change, not merely a hot year (1 mark).
- *Marking note: 2 marks: 1 for describing the steady rise, 1 for the conclusion. 'The line goes up' alone earns the first mark only.*

(d) [2 marks]

- **The rains no longer begin and end when they used to**, and the rainy season has become shorter or less reliable.
- **Longer and hotter dry spells**, with streams, dams and wells drying up.
- Also accept: **heavier storms and more flooding when the rain does come**;
- **harvests failing, or crops that once grew well no longer doing so**;
- **grass and trees drying earlier**;
- **more dust and a longer Harmattan**.
- *Marking note: 1 mark each for any TWO signs (2 marks). The sign must be a long-term change, not a single storm or one hot afternoon.*

(e) [2 marks]

- **One station describes one place only**, and its reading can be affected by local changes - buildings, tarred roads or cleared trees round the station - rather than by the climate of the whole country (1 mark).
- **Collect and compare records from several stations and over as many years as possible**, set them beside the figures published by the Ghana Meteorological Agency or a world body, and add what the farmers and fishers of the area report about the seasons (1 mark).
- *Marking note: 2 marks: 1 for the weakness of a single record, 1 for a sensible

way of strengthening the evidence. T

Question 3 [Total: 12 marks]

(a) [3 marks]

- **Climate change** is a long-term change in the average pattern of temperature, rainfall and the seasons of a place, or of the earth as a whole, observed over many years (2 marks).
- **Weather** is the condition of the atmosphere at one place at one moment - today's sunshine, this afternoon's storm - **while climate** is the average of that weather taken over many years, usually thirty or more (1 mark).
- **Marking note:** 3 marks: 2 for the definition, which must contain the idea of a LONG-TERM change in the average, and 1 for the distinction. A candidate who says only 'the weather is changi

(b) [4 marks]

- **Burning petrol, diesel, charcoal and firewood** in vehicles, generators and kitchens - releases **carbon dioxide**.
- **Cutting down forests (deforestation)** - the carbon stored in the wood is released, and the **absorption of carbon dioxide by the trees is lost**.
- **Bush burning before planting** - releases **carbon dioxide** and destroys the grass and young trees that would absorb it.
- **Heaping refuse in dumps where it rots** - releases **methane**.
- Also accept: **keeping large herds of cattle** (methane); **industry, mining and cement works** (carbon dioxide and other gases); **f**

(c) [3 marks]

- **The rains no longer come when expected**, so seed planted at the usual time dies, or ripening crops are caught by an early dry spell, and harvests fail.
- **Longer and hotter dry seasons** dry up the streams, the dams and the pasture, so there is too little water for crops and animals.
- **Heavier storms and floods** wash away crops, seedlings and the topsoil itself.
- Also accept: **more pests and crop diseases** in the warmer conditions; **cocoa and other tree crops suffering where the dry season is longer and hotter**; **lower yields, so food prices rise and farmers earn less**.
- **Ma**

(d) [2 marks]

- **The rising sea floods the town and washes the land away**, destroying houses, roads and coconut farms, so that families have to move inland.
- **Salt water enters the wells, the lagoons and the farmland**, so the water is no longer fit to drink and crops will not grow.
- Also accept: **fish catches fall** as the warmer sea changes where the fish are found, so fishers earn less; **storms are stronger and the sea rougher**, so fishing is more dangerous.

- *Marking note: 1 mark each for any TWO effects (2 marks).*

Question 4 [Total: 10 marks]

(a) [4 marks]

- **Adaptation** is changing the way we live and work so that we can cope with the changes that are already happening (1 mark).
- **Example of adaptation**: planting early-maturing or drought-resistant maize; building a sea defence wall at Keta; harvesting rain water; changing the planting calendar; storing part of the harvest (1 mark).
- **Mitigation** is reducing the amount of greenhouse gas we release, or taking it back out of the air, so that the warming itself is slowed (1 mark).
- **Example of mitigation**: using solar panels instead of a diesel generator; planting trees; cooking on an i

(b) [3 marks]

- **Plant early-maturing and drought-resistant varieties** of maize, millet or cowpea, so that the crop ripens before the rains stop.
- **Change the planting calendar and follow the weather forecast** from the Meteorological Agency and the agricultural extension officer instead of the old fixed dates.
- **Build small dams and dugouts, harvest rain water and irrigate** so that there is water in the dry months.
- Also accept: **improved storage** so that part of the harvest is kept against a bad season; **mixed cropping and keeping livestock as well as crops**, so that one failure does not ruin th

(c) [3 marks]

- **Plant trees and protect those already standing**, so that carbon dioxide is taken out of the air and stored in the wood.
- **Stop bush burning** and use cleared weeds as compost or mulch instead of setting fire to them.
- **Change the way energy is used** - improved fuel-saving stoves, gas or solar instead of the open wood fire and the diesel generator.
- Also accept: **separate and compost organic waste** instead of dumping or burning it; **recycle plastic, metal and glass**; **walk, cycle or share transport**; **protect the forest reserve and the trees along the riverbank**.
- *Marking

Question 5 [Total: 8 marks]

(a) [2 marks]

- **A green economy is one in which people earn a living and the country develops without destroying the environment** (1 mark).
- **It releases little pollution and few greenhouse gases, uses natural resources carefully so that they are not used up, and still provides jobs and incomes** for the people (1 mark).
- Accept the short form: **growth that does not cost us the land, the water and the air.**
- **Marking note: 2 marks. An answer about the colour green, or about growing vegetables only, earns nothing.**

(b) [3 marks]

- **Installing and repairing solar panels and solar lanterns** - supplies electricity without burning fuel, so no greenhouse gas is released.
- **Raising seedlings in a nursery and planting trees** - the growing trees take carbon dioxide out of the air, hold the soil and bring back the forest.
- **Collecting and recycling plastic, metal and electronic waste** - keeps the waste out of the gutters, the land and the sea, and saves the energy that making new material would cost.
- Also accept: **organic farming and composting**; **making and selling improved fuel-saving stoves**; **beekeeping**; *

(c) [3 marks]

- **Plant trees on the compound and care for them** - shade, cleaner air and carbon dioxide absorbed.
- **Separate the waste and compost the food and garden waste** - the compost feeds the school garden and the refuse that would rot in a dump and release methane is kept out of it.
- **Stop burning refuse on the compound** - no poisonous smoke for the pupils to breathe and no carbon dioxide released.
- Also accept: **switch off lights and fans when rooms are empty, and fit solar panels or solar lamps**; **harvest rain water from the roof**; **a walking, cycling or transport-sharing day**; **a s**

Question 6 [Total: 10 marks]**(a)** [2 marks]

- **Growing trees take carbon dioxide out of the air and store the carbon in their wood**, so less of the gas is left to trap heat.
- **Trees cool and shade the land, hold the soil against erosion, slow run-off and flooding, and help to keep rainfall steady**; they also restore the habitats of plants and animals.
- **Marking note: 1 mark each for any TWO reasons (2 marks). The first reason is the central one; award the second mark for any genuine environmental benefit.**

(b) [3 marks]

- **Plan**: decide where on the compound the trees will stand and how many are needed, and obtain the permission of the head teacher (1 mark).
- **Obtain the right seedlings**: choose species suited to the place - shade, fruit or timber - and collect them from the Forestry Commission, the agricultural office or a local nursery (1 mark).
- **Plant and care for them**: dig the holes, plant early in the rainy season, water, mulch and fence or stake each tree, and **assign a named pupil or class to each tree** so that it is watered and protected after the ceremony (1 mark).
- **Marking note**: 1 mark

(c) [2 marks]

- **One real difficulty** (1 mark): the seedlings die for want of water in the dry season; goats, cattle or careless people break them; there is no money for seedlings or tools; nobody cares for them during the long holidays.
- **A matching solution** (1 mark): a written watering rota with mulching round each tree; a stake and a wire or palm-frond guard; free seedlings requested from the Forestry Commission or the assembly; the caretaker, or families living near the school, asked to water during the holidays.
- **Marking note**: 2 marks: 1 for the difficulty, 1 for a solution that actually answers

(d) [3 marks]

- **Why the law is needed** (1 mark): bush fires destroy trees, grass, crops and wildlife, burn out the humus and leave the soil bare to erosion, and they release large amounts of carbon dioxide while killing the very plants that would have absorbed it; one careless fire can burn a whole district's farms.
- **Form a fire volunteer squad and cut fire belts** round the farms and the woodlot before the dry season, and agree with the chief on when, if ever, burning is allowed (1 mark).
- **Educate and warn those most likely to start fires** - hunters, herders, charcoal burners, children and farmer

Conservation of Energy

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	16.1 Conservation of Energy and the Meanin...	Remember
2	A	16.1 Conservation of Energy and the Meanin...	Remember
3	D	16.2 Importance of Energy Conversion in Da...	Understand
4	B	16.5 Conserving Energy at Home, in School ...	Understand
5	A	16.1 Conservation of Energy and the Meanin...	Understand
6	D	16.2 Importance of Energy Conversion in Da...	Understand
7	A	16.3 Energy Waste, Efficiency and Househol...	Understand
8	D	16.3 Energy Waste, Efficiency and Househol...	Apply
9	B	16.3 Energy Waste, Efficiency and Househol...	Understand
10	C	16.5 Conserving Energy at Home, in School ...	Understand
11	C	16.5 Conserving Energy at Home, in School ...	Understand
12	D	16.3 Energy Waste, Efficiency and Househol...	Understand
13	A	16.5 Conserving Energy at Home, in School ...	Understand
14	B	16.3 Energy Waste, Efficiency and Househol...	Analyse
15	A	16.5 Conserving Energy at Home, in School ...	Evaluate
16	B	16.4 Impact of Energy Conversion on the En...	Understand
17	C	16.4 Impact of Energy Conversion on the En...	Remember
18	C	16.4 Impact of Energy Conversion on the En...	Understand
19	A	16.4 Impact of Energy Conversion on the En...	Understand
20	C	16.4 Impact of Energy Conversion on the En...	Understand
21	B	16.4 Impact of Energy Conversion on the En...	Understand
22	D	16.3 Energy Waste, Efficiency and Househol...	Understand
23	D	16.4 Impact of Energy Conversion on the En...	Analyse
24	A	16.5 Conserving Energy at Home, in School ...	Apply
25	B	16.1 Conservation of Energy and the Meanin...	Understand
26	B	16.5 Conserving Energy at Home, in School ...	Understand
27	D	16.5 Conserving Energy at Home, in School ...	Evaluate
28	A	16.3 Energy Waste, Efficiency and Househol...	Understand
29	A	16.3 Energy Waste, Efficiency and Househol...	Apply
30	C	16.3 Energy Waste, Efficiency and Househol...	Understand
31	B	16.1 Conservation of Energy and the Meanin...	Analyse
32	B	16.2 Importance of Energy Conversion in Da...	Analyse
33	C	16.5 Conserving Energy at Home, in School ...	Analyse
34	C	16.5 Conserving Energy at Home, in School ...	Evaluate
35	A	16.2 Importance of Energy Conversion in Da...	Understand
36	C	16.5 Conserving Energy at Home, in School ...	Understand
37	D	16.5 Conserving Energy at Home, in School ...	Evaluate

Q	Ans	Curriculum Area	Cognitive Level
38	D	16.4 Impact of Energy Conversion on the En...	Evaluate
39	B	16.1 Conservation of Energy and the Meanin...	Analyse
40	D	16.5 Conserving Energy at Home, in School ...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 6 (15%) | Apply: 3 (7%) | Evaluate: 6 (15%) | Remember: 3 (7%) | Understand: 22 (55%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 9 marks]****(a)** [2 marks]

- **Energy can neither be created nor destroyed** (1 mark);
- it can only be **changed from one form into another**, so the **total energy before a change equals the total energy after it** (1 mark).
- **Marking note:** 2 marks, 1 mark each. The first mark needs both 'not created' and 'not destroyed'; the second needs the change of form or the equality of the totals.*

(b) [2 marks]

- Energy conservation means **using energy carefully and avoiding waste** in the home, the school and the workplace (1 mark);
- so that **less fuel and electricity are used, costs are lower and the supply lasts longer** for the future (1 mark).
- **Marking note:** 2 marks, 1 mark each. Do NOT accept the law of conservation of energy as the answer here; the question asks for the everyday meaning.*

(c) [2 marks]

- **Energy conversion** is the **changing of energy from one form into another**, for example electrical energy into light in a lamp (1 mark).
- **Energy conservation** is **using energy carefully so that none is wasted**, for example switching the lamp off when it is not needed (1 mark).
- **Marking note:** 2 marks, 1 mark for each side of the difference. A candidate who explains only one of the two terms scores 1 mark.*

(d) [3 marks]

- It **puts energy into the form we actually need** - light for reading, heat for cooking, movement for transport and machines (1 mark).
- It makes the **free natural energy of the Sun, the wind and flowing water useful** to people (1 mark).
- It **powers industry, hospitals, schools and communication**, so it supports work, education, health and the economy (1 mark).
- **Marking note:** 1 mark each for any THREE sound points (3 marks). This answers the curriculum exemplar on classifying the importance of energy conversion in daily life.*

Question

(a) [1 marks]

- The **air conditioner**, which used **120 kWh**
- **Marking note:** 1 mark.*

(b) [2 marks]

- **120 + 90 + 45 + 30 + 15** (1 mark)
- **= 300 kWh** (1 mark)
- **Marking note:** 2 marks: 1 mark for the correct addition of all five values and 1 mark for the answer with the unit kWh.*

(c) [2 marks]

- **(120 ÷ 300) × 100** (1 mark)
- **= 40%** (1 mark)
- **Marking note:** 2 marks: 1 mark for the correct method and 1 mark for the answer. Follow through from a wrong total in (b) - award the method mark.*

(d) [1 marks]

- **90 – 15 = 75 kWh**
- **Marking note:** 1 mark. The unit must be given.*

(e) [3 marks]

- **Use the air conditioner for fewer hours, or use a fan instead**, and keep the doors and windows closed while it works.
- **Fit energy-saving (LED) bulbs and switch off lights that are not needed**, using daylight where possible.
- **Iron many clothes at one time**, **avoid opening the refrigerator too often**, and **switch off the television when nobody is watching**.
- **Marking note:** 1 mark each for any THREE practical ways (3 marks). The answer should relate to the appliances shown in Fig. 1.*

(f) [2 marks]

- Lighting used **45 kWh**, so the cost is **45 × GH¢1.50** (1 mark)
- **= GH¢67.50** (1 mark)
- **Marking note:** 2 marks: 1 mark for reading 45 kWh from the chart and multiplying, 1 mark for the answer in cedis. A candidate who uses the wrong bar but multiplies correctly earns the method mark only.*

(g) [1 marks]

- **Less electricity used means less fuel has to be burnt at the power station**, so **fewer harmful gases are released into the air** and fewer natural resources are used up.
- **Marking note:** 1 mark for the link between using less electricity and less fuel

burnt or less pollution.*

Question 3 [Total: 10 marks]

(a) [2 marks]

- **Electrical energy** is changed into **light energy** (1 mark)
- **and heat energy** (1 mark).
- **Marking note:** 2 marks, 1 mark each. A candidate who gives only 'electrical to light' scores 1 mark, since Fig. 2 shows that heat is also produced.*

(b) [2 marks]

- Useful form: **light energy (10 J)** (1 mark)
- Wasted form: **heat energy (90 J)**, which escapes to the surroundings (1 mark)
- **Marking note:** 2 marks, 1 mark each.*

(c) [2 marks]

- **$10\text{ J} + 90\text{ J} = 100\text{ J}$** , which is **exactly the energy supplied** to the lamp (1 mark).
- This shows that **no energy has been created or destroyed**: all of it has simply been **changed into other forms**, as the law of conservation of energy states (1 mark).
- **Marking note:** 2 marks, 1 mark each. The second mark needs the idea of conservation; simply writing '100 J' earns 1 mark.*

(d) [2 marks]

- **$(10 \div 100) \times 100$** (1 mark)
- **$= 10\%$** (1 mark)
- **Marking note:** 2 marks: 1 mark for the method and 1 mark for the answer with the per cent sign.*

(e) [2 marks]

- An LED lamp **turns a much greater part of the electrical energy into light and wastes far less as heat** (1 mark).
- The family therefore gets the **same brightness for much less electricity**, so the **bill is lower and less fuel is burnt at the power station** (1 mark).
- **Marking note:** 2 marks, 1 mark each.*

Question 4 [Total: 10 marks]

(a) [3 marks]

- **Burning fuels releases smoke and harmful gases** that pollute the air and cause coughs and chest diseases (1 mark).
- The **carbon dioxide released adds to global warming and changes the**

climate** (1 mark).

- **Cutting trees for firewood and charcoal destroys forests and causes soil erosion** (accept: dams flood farmland and displace people; oil spills pollute rivers and kill fish; generators make noise) (1 mark).

- *Marking note: 1 mark each for any THREE sound points (3 marks).*

(b) [3 marks]

- Burning coal, petrol, diesel and gas **releases large amounts of carbon dioxide into the atmosphere** (1 mark).

- The carbon dioxide **traps heat that the Earth would otherwise send back into space** (1 mark).

- As more of the gas collects, the **Earth becomes warmer and the climate changes** - unreliable rains, floods and droughts (1 mark).

- *Marking note: 3 marks, 1 mark each. Candidates need not use the term 'greenhouse effect', but the trapping of heat must be described for the second mark.*

(c) [2 marks]

- The **forest cover is lost, so the soil is exposed and washed away by the rain (erosion)***, and **animals lose their habitats**.

- There are **fewer trees to take in carbon dioxide and give out oxygen**, and the **rainfall of the area may fall, so streams dry up**.

- *Marking note: 1 mark each for any TWO effects (2 marks).*

(d) [2 marks]

- **Less fuel is burnt**, so there is **less air pollution and less carbon dioxide** released.

- **Fewer trees are cut for firewood and charcoal**, and **the fuels and other resources last longer for future generations** (accept: fewer new power stations and dams are needed).

- *Marking note: 1 mark each for any TWO benefits (2 marks).*

Question 5 [Total: 9 marks]

(a) [4 marks]

- **Switch off lights, fans and computers when the classroom or office is empty**, and at the close of school.

- **Use daylight in the classrooms** instead of switching on lamps during the day.

- **Fit energy-saving (LED) bulbs** and repair faulty switches and appliances.

- **Cook for the whole school at one time in covered pots**, and use an improved stove that needs less firewood.

- *Marking note: 1 mark each for any FOUR practical ways (4 marks).*

(b) [2 marks]

- **Less electricity, gas and charcoal are used, so the bills and the fuel bought each month cost less** (1 mark).
- **Appliances that are switched off and well maintained last longer**, so less money is spent on repairs and replacements (1 mark).
- **Marking note: 2 marks, 1 mark each.**

(c) [3 marks]

- **The habit:** leaving the **television, fan or security lights on when nobody needs them** (1 mark).
- **Why it wastes energy:** the appliance goes on **changing electrical energy into light, sound and heat that nobody uses**, and the bill rises (1 mark).
- **The correction:** **switch it off at the socket when it is not in use**, or fit a timer or sensor so that the lights come on only at night (1 mark).
- **Marking note: 3 marks as shown: 1 mark for naming a genuine habit, 1 mark for explaining the waste and 1 mark for a workable correction.**

Question 6 [Total: 10 marks]**(a)** [4 marks]

- A **clear title** and a short statement of **what energy conversion and energy conservation mean**.
- **Examples of energy changes in the school** (lamps, fans, the kitchen stove, the solar panel).
- **Practical tips for saving energy**, such as switching off lights and fans, and using daylight.
- **The effects of wasting energy** on the school's bills and on the environment, with a **simple diagram or chart and a short slogan**.
- **Marking note: 1 mark each for any FOUR suitable items (4 marks). This answers the curriculum exemplar that asks pupils to make a poster presentation on their f**

(b) [3 marks]

- A **rectangle representing the poster, divided into separate areas** (1 mark).
- **The title across the top and a diagram or chart in the middle** (1 mark).
- **A list of tips at the side and a slogan at the bottom**, with every area clearly named (1 mark).
- **Marking note: 3 marks as shown. Award the marks for a sensible, well-planned layout; artistic quality is not being marked. A plan with no labels scores a maximum of 1 mark.**

(c) [2 marks]

- Two sources: **science textbooks and the school library**, and the **internet** (accept newspapers, radio and television, the school's own electricity bills, or an officer from the electricity company) (1 mark for any two).

- Checking: **compare the information with a second source or a textbook, check how recent it is, and check that it comes from a known body** such as the Energy Commission or the electricity company (1 mark).

- **Marking note:** 2 marks: 1 mark for naming two sources and 1 mark for a sound way of checking reliability. This answers the exemplar on searching multimedia source

(d) [1 marks]

- **Display the poster on the notice board and explain it at the school assembly** (accept: talk to each class in turn, or present it at a science club meeting, an open day or a PTA meeting).

- **Marking note:** 1 mark for any one workable way of presenting the findings.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 4: Forces and Energy | **Sub-strand:** 3: Conversion and Conservation of Energy | **Class:** Basic 8 (JHS 2)

Content Standard: **B8/JHS2.4.3.1** - Evaluate the impact of conversion of energy and energy conservation on the environment.

> Source: NaCCA Science Common Co

Crop Production

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	7.1 Types of Seed Beds	Remember
2	B	7.2 Land Preparation and Planting Practices	Understand
3	C	7.1 Types of Seed Beds	Remember
4	A	7.1 Types of Seed Beds	Remember
5	A	7.1 Types of Seed Beds	Remember
6	C	7.1 Types of Seed Beds	Remember
7	B	7.1 Types of Seed Beds	Understand
8	A	7.1 Types of Seed Beds	Understand
9	D	7.1 Types of Seed Beds	Apply
10	B	7.3 Planting Materials and Transplanting	Remember
11	D	7.3 Planting Materials and Transplanting	Remember
12	A	7.3 Planting Materials and Transplanting	Remember
13	A	7.3 Planting Materials and Transplanting	Remember
14	C	7.3 Planting Materials and Transplanting	Remember
15	B	7.3 Planting Materials and Transplanting	Remember
16	A	7.3 Planting Materials and Transplanting	Remember
17	C	7.3 Planting Materials and Transplanting	Understand
18	C	7.3 Planting Materials and Transplanting	Understand
19	B	7.3 Planting Materials and Transplanting	Understand
20	A	7.3 Planting Materials and Transplanting	Remember
21	D	7.2 Land Preparation and Planting Practices	Remember
22	B	7.2 Land Preparation and Planting Practices	Understand
23	B	7.1 Types of Seed Beds	Understand
24	C	7.2 Land Preparation and Planting Practices	Remember
25	D	7.2 Land Preparation and Planting Practices	Apply
26	D	7.2 Land Preparation and Planting Practices	Understand
27	C	7.4 Comparing Crop Growth on Different See...	Analyse
28	D	7.5 Measuring, Recording and Presenting Re...	Understand
29	C	7.5 Measuring, Recording and Presenting Re...	Understand
30	D	7.5 Measuring, Recording and Presenting Re...	Apply
31	A	7.4 Comparing Crop Growth on Different See...	Understand
32	B	7.5 Measuring, Recording and Presenting Re...	Apply
33	A	7.4 Comparing Crop Growth on Different See...	Analyse
34	D	7.5 Measuring, Recording and Presenting Re...	Analyse
35	A	7.4 Comparing Crop Growth on Different See...	Analyse
36	C	7.4 Comparing Crop Growth on Different See...	Analyse
37	C	7.4 Comparing Crop Growth on Different See...	Apply

Q	Ans	Curriculum Area	Cognitive Level
38	B	7.1 Types of Seed Beds	Evaluate
39	B	7.1 Types of Seed Beds	Evaluate
40	D	7.5 Measuring, Recording and Presenting Re...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 5 (12%) | Apply: 5 (12%) | Evaluate: 3 (7%) | Remember: 15 (37%) | Understand: 12 (30%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 8 marks]

(a) [2 marks]

- A seed bed is a **piece of land** that has been prepared (cleared, dug, levelled) for planting.
- The soil in it is made **loose and fine** so that seeds can germinate and the young roots can grow easily.
- **Marking note:** 1 mark each for any TWO correct points (2 marks). Award 1 mark for 'land prepared for planting' alone. Do NOT accept 'a place where seeds are stored'.

(b) [3 marks]

- **Flat bed**
- **Raised bed**
- **Sunken bed**
- (Also accept: **nursery bed**, **ridges**, **mounds**)
- **Marking note:** 1 mark each for any THREE correct types (3 marks). Accept ridges and mounds, which are used in many farming communities in Ghana.

(c) [3 marks]

- To **loosen the soil** so that seeds germinate easily and roots can spread.
- To **remove weeds, stumps and stones** that would compete with the crops or obstruct the work.
- To **improve drainage and the movement of air** in the soil, and to **mix in manure** so the soil is more fertile.
- To make **planting, weeding and harvesting easier** for the farmer.
- **Marking note:** 1 mark each for any THREE correct reasons (3 marks). Do NOT accept 'to make the farm beautiful'.

Question 2 [Total: 10 marks]

(a) [3 marks]

- A - **flat bed** (the soil is level with the surrounding ground)
- B - **raised bed** (the soil is heaped above the surrounding ground)
- C - **sunken bed** (the soil is dug below the surrounding ground)
- **Marking note:** 1 mark each (3 marks). The name must be matched to the correct letter. Accept 'flat seed bed', 'raised seed bed' and 'sunken seed bed'.

(b) [2 marks]

- **B** - the raised bed (1 mark).
- Because the bed stands above the ground, **excess water drains away** and the roots **get air instead of rotting** in the water (1 mark).
- **Marking note:** 2 marks: 1 mark for the correct bed and 1 mark for the reason. If the bed named is wrong, no marks are awarded even if the reason is sound.*

(c) [2 marks]

- **C** - the sunken bed (1 mark).
- Because it lies below the ground it **holds water (and moisture)** around the roots for a longer time, which the crops need in dry weather (1 mark).
- **Marking note:** 2 marks: 1 mark for the correct bed and 1 mark for the reason. Accept 'it conserves moisture' for the reason mark.*

(d) [1 marks]

- It is **easy and quick to prepare** (little soil has to be moved), so a large area can be planted; it suits most crops on **well-drained land**.
- **Marking note:** 1 mark for any one correct advantage.*

(e) [2 marks]

- The soil of **B** is heaped above the ground while the soil of **C** is dug below the ground.
- **B** drains water away from the crops, whereas **C** holds water around the crops.
- **B** is used in wet or waterlogged places, while **C** is used in dry places.
- **Marking note:** 1 mark each for any TWO correct differences (2 marks). The answer must compare the two beds; a statement about only one of them earns no mark.*

Question 3 [Total: 12 marks]**(a)** [4 marks]

- (i) Cassava - **stem cuttings**
- (ii) Maize - **seeds** (grains)
- (iii) Plantain - **suckers**
- (iv) Yam - **setts** (pieces cut from the tuber)
- **Marking note:** 1 mark each (4 marks). Accept 'stem' for cassava, 'grains' for maize and 'tuber pieces' for yam. Do NOT accept 'seeds' for cassava, plantain or yam.*

(b) [2 marks]

- Transplanting is the **lifting of young seedlings from a nursery bed** (or seed box) (1 mark)
- and **planting them in the main field** or in their permanent position (1 mark).

- *Marking note: 2 marks, 1 mark each. Award 1 mark only for 'moving plants from one place to another' without reference to the nursery or the main field.*

(c) [3 marks]

- **Water the nursery bed before lifting** the seedlings, and lift them with **soil (a ball of earth) around the roots** so that the roots are not damaged.
- **Transplant in the evening or on a cool, cloudy day**, not in the hot sun.
- **Water the seedlings immediately after planting** and **provide shade** for a few days until they are established.
- **Firm the soil gently** around each seedling and keep the correct spacing.

- *Marking note: 1 mark each for any THREE correct precautions (3 marks). Award method marks for any sensible practice that protects the seedling from damage or from drought

(d) [3 marks]

- Tomato **seeds are very small** and need **close attention** (watering, shade and protection from pests), which is easier on a small nursery bed (1 mark).
- The **weak seedlings can be sorted out** and only the **strong, healthy ones transplanted** (1 mark).
- The seedlings can be planted in the field at the **correct spacing**, and **seed is not wasted**; the main field is also free for other use while the seedlings grow (1 mark).
- *Marking note: 3 marks, 1 mark each for any three sound reasons.*

Question 4 [Total: 11 marks]

(a) [1 marks]

- **18 cm**
- *Marking note: 1 mark. The unit must be given; award the mark for '18' with cm.*

(b) [1 marks]

- **Bed B**
- *Marking note: 1 mark. Accept 'B'.*

(c) [2 marks]

- Tallest = 26 cm (Bed B); shortest = 18 cm (Bed A) (1 mark)
- Difference = 26 cm - 18 cm = **8 cm** (1 mark)
- *Marking note: 2 marks: 1 mark for the correct working (26 - 18) and 1 mark for the answer with the unit. Award full marks for a correct answer of 8 cm even if no working is shown.*

(d) [3 marks]

- Bed B **drained better**, so the roots got **enough air** and did not rot (1 mark).
- The soil in Bed B was **deeper, looser and more fertile** (it may have had more manure), so the plants obtained more nutrients (1 mark).
- The **roots could spread more easily** in the loose soil and so took in more water and nutrients for growth (1 mark).
- *Marking note: 3 marks, 1 mark each for any three sound reasons. Accept 'fewer weeds competed with the crops on Bed B'.*

(e) [2 marks]

- The **number of flowers** on each plant and the **number of fruits** produced.
- The **number of leaves**, the **thickness (girth) of the stem** or the **weight of the fruits harvested**.
- *Marking note: 1 mark each for any TWO correct measurements (2 marks). The curriculum names height, size, number of flowers and number of fruits. Do NOT accept 'the colour of the plant', which is not a measurement.*

(f) [2 marks]

- So that **other pupils, teachers and farmers in the community learn** which seed bed gives the best growth and can **use the findings** on their own farms.
- So that the work can be **checked, questioned and repeated** by others, which is how scientists share and confirm results.
- *Marking note: 1 mark each for any TWO sound reasons (2 marks). Accept 'to practise explaining ideas clearly', which is the communication aim of the exemplar.*

Question 5 [Total: 9 marks]**(a)** [2 marks]

- The **same crop (same variety of pepper)** and the **same number of plants** at the **same spacing**.
- The **same amount of water and manure**, and the **same sunlight and weather** for both beds.
- *Marking note: 1 mark each for any TWO correct points (2 marks). Do NOT award a mark for 'the type of seed bed' - that is the thing being changed.*

(b) [1 marks]

- The **type of seed bed** - one bed is raised and the other is flat.
- *Marking note: 1 mark. This is the only thing that may be different in a fair test.*

(c) [2 marks]

- Stand the **metre rule** (or tape measure) upright beside the plant with the zero mark at the soil surface (1 mark).
- **Read the mark at the tip of the plant** with the eye level with the reading, and **record the height in centimetres** in the table (1 mark).
- **Marking note:** 2 marks, 1 mark each. Award method marks for any correct, careful procedure; look for measuring from the soil level and reading correctly.*

(d) [2 marks]

- **Plants do not all grow at the same rate**; a single plant may happen to be unusually tall or unusually short (1 mark).
- The **average of several plants is fairer and more reliable**, because it represents the whole bed and reduces the effect of one odd plant (1 mark).
- **Marking note:** 2 marks, 1 mark each. The key idea is the variation between plants and the reliability of the average.*

(e) [2 marks]

- A table with the **weeks (1, 2, 3, 4)** in the first column (1 mark)
- and a **column for each bed** - average height on the raised bed (cm) and average height on the flat bed (cm), with the **units given in the headings** (1 mark).
- **Marking note:** 2 marks. Award 1 mark for a ruled table with the time (weeks) as one heading and 1 mark for both beds shown with the unit (cm). A table drawn the other way round (beds as rows, weeks as columns) is equally correct. The table need not contain any readings.*

Question 6 [Total: 10 marks]**(a)** [4 marks]

- **Clear the land**: cut down the weeds and remove the rubbish and stones from the piece of land.
- **Mark out the bed** with pegs and a line to the size required (for example 1 m by 3 m).
- **Dig and loosen the soil** inside the marked area with a hoe, removing roots and stumps, and **mix in manure or compost**.
- **Heap the soil above the level of the ground** to form the raised bed, then **level and firm the top** of the bed and water it before planting.
- **Marking note:** 1 mark each for any FOUR sensible steps in a workable order (4 marks). This is a practical design task, so award method

(b) [2 marks]

- **Cutlass** - for **clearing the weeds and bushes** from the land.
- **Hoe** - for **digging, loosening and heaping up the soil** to form the bed.
- **Rake** - for **levelling the top of the bed** and removing stones and rubbish.
- **Watering can** - for **watering** the bed and the young plants.

- *Marking note: 1 mark for each tool correctly named WITH its use, maximum 2 marks. A tool named without a use earns no mark.*

(c) [2 marks]

- **Handle the cutlass and hoe carefully** and **stand well apart** while working so that nobody is struck.
- **Wear shoes or boots** (and gloves where possible), **do not play with the tools**, and **wash the hands with soap** after the work.
- **Report any cut or injury to the teacher at once**, and work in the cool of the morning or evening.

- *Marking note: 1 mark each for any TWO sensible precautions (2 marks). Accept any practical safety measure for garden work. Do NOT accept 'be careful' on its own.*

(d) [2 marks]

- **Water the crops regularly**, especially in the dry season, and **mulch the bed** with dry grass to keep the soil moist.
- **Weed the bed regularly**, **apply manure or compost**, **control pests and diseases**, and **fence the garden** to keep goats and fowls out.

- *Marking note: 1 mark each for any TWO correct practices (2 marks). Accept staking of tomato or pepper plants and thinning where it applies.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 2: Cycles | **Sub-strand:** 3: Crop Production | **Class:** Basic 8 (JHS 2)

Content Standard: **B8/JHS2.2.3.1** - Demonstrate

Ecosystem

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	11.1 The Ecosystem and its Parts	Remember
2	B	11.1 The Ecosystem and its Parts	Remember
3	A	11.4 Energy Transfer and the Role of the Sun	Remember
4	C	11.4 Energy Transfer and the Role of the Sun	Remember
5	A	11.2 Producers, Consumers and Feeding Terms	Remember
6	B	11.2 Producers, Consumers and Feeding Terms	Understand
7	D	11.2 Producers, Consumers and Feeding Terms	Remember
8	D	11.2 Producers, Consumers and Feeding Terms	Remember
9	A	11.2 Producers, Consumers and Feeding Terms	Understand
10	A	11.2 Producers, Consumers and Feeding Terms	Understand
11	C	11.3 Food Chains and Food Webs	Remember
12	A	11.3 Food Chains and Food Webs	Understand
13	A	11.3 Food Chains and Food Webs	Analyse
14	C	11.3 Food Chains and Food Webs	Remember
15	A	11.3 Food Chains and Food Webs	Understand
16	D	11.2 Producers, Consumers and Feeding Terms	Remember
17	B	11.2 Producers, Consumers and Feeding Terms	Remember
18	B	11.5 Interdependence and the Balance of an...	Remember
19	C	11.5 Interdependence and the Balance of an...	Understand
20	C	11.2 Producers, Consumers and Feeding Terms	Remember
21	A	11.2 Producers, Consumers and Feeding Terms	Understand
22	C	11.4 Energy Transfer and the Role of the Sun	Remember
23	D	11.4 Energy Transfer and the Role of the Sun	Analyse
24	C	11.4 Energy Transfer and the Role of the Sun	Understand
25	D	11.4 Energy Transfer and the Role of the Sun	Understand
26	D	11.5 Interdependence and the Balance of an...	Analyse
27	A	11.4 Energy Transfer and the Role of the Sun	Analyse
28	B	11.4 Energy Transfer and the Role of the Sun	Analyse
29	D	11.5 Interdependence and the Balance of an...	Analyse
30	D	11.5 Interdependence and the Balance of an...	Analyse
31	B	11.5 Interdependence and the Balance of an...	Understand
32	D	11.3 Food Chains and Food Webs	Understand
33	C	11.2 Producers, Consumers and Feeding Terms	Understand
34	C	11.2 Producers, Consumers and Feeding Terms	Remember
35	D	11.2 Producers, Consumers and Feeding Terms	Understand
36	C	11.3 Food Chains and Food Webs	Analyse
37	B	11.5 Interdependence and the Balance of an...	Evaluate

Q	Ans	Curriculum Area	Cognitive Level
38	B	11.3 Food Chains and Food Webs	Evaluate
39	A	11.1 The Ecosystem and its Parts	Apply
40	B	11.5 Interdependence and the Balance of an...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 8 (20%) | Apply: 1 (2%) | Evaluate: 3 (7%) | Remember: 15 (37%) | Understand: 13 (32%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 9 marks]

(a) [2 marks]

- An ecosystem is **all the living organisms** (plants, animals and micro-organisms) in a particular place (1 mark)
- **together with the non-living parts of their surroundings** - soil, water, air, light and temperature - all **interacting with one another** (1 mark).
- *Marking note: 2 marks, 1 mark each. Award 1 mark for the living things alone; the second mark needs the non-living surroundings or the interaction.**

(b) [6 marks]

- (i) **Producer** - an organism, usually a **green plant**, that makes its own food using energy from the Sun by photosynthesis (2 marks)
- (ii) **Primary consumer** - an animal that **feeds directly on producers** (green plants); it is a herbivore, for example a grasshopper or a goat (2 marks)
- (iii) **Secondary consumer** - an animal that **feeds on primary consumers**, for example a lizard that eats grasshoppers (2 marks)
- *Marking note: 2 marks for each term (6 marks): 1 mark for the basic idea and 1 mark for a correct example or for the link with the level below it. These three terms*

(c) [1 marks]

- **Water** (also accept air, soil, sunlight, rocks or temperature)
- *Marking note: 1 mark. The thing named must be non-living. Do NOT accept a dead animal or a fallen leaf, which were once living.**

Question 2 [Total: 12 marks]

(a) [2 marks]

- **Maize plant**
- **Grass**
- *Marking note: 1 mark each (2 marks). They are the producers because the arrows begin from them and they are green plants.**

(b) [2 marks]

- **maize plant → grasshopper → lizard → hawk**
- or **grass → rat → snake → hawk**
- or **maize plant → rat → snake → hawk**
- or **grass → grasshopper → lizard → snake**
- *Marking note: 2 marks: 1 mark for four organisms correctly taken from the web and 1 mark for the arrows pointing in the right direction (from the eaten to*

the eater). A chain written in reverse order scores a maximum of 1 mark.*

(c) [1 marks]

- **Grasshopper** (or **rat**)
- *Marking note:* 1 mark. Both feed directly on the producers. Do NOT accept the lizard, the snake or the hawk.*

(d) [2 marks]

- **The lizard** (also accept **the snake**) (1 mark).
- It **hunts and eats the grasshopper**, so it is a **predator**; but it is **itself** eaten by the snake and the hawk, so it is also a **prey** (1 mark).
- *Marking note:* 2 marks: 1 mark for a correct organism and 1 mark for the explanation. The explanation must show both roles. Do NOT accept the hawk, which nothing in the web eats, or the maize plant.*

(e) [2 marks]

- The arrows show the **direction in which food and energy flow** through the web (1 mark).
- Each arrow points **from the organism that is eaten towards the organism that eats it** (1 mark).
- *Marking note:* 2 marks, 1 mark each. The direction of the arrow must be explained correctly for the second mark; 'it shows who eats who' earns 1 mark only.*

(f) [2 marks]

- The number of grasshoppers would **increase** (1 mark),
- because the **lizards that feed on them** would no longer be there to control their numbers (1 mark). (Accept the further point that the grasshoppers would then destroy more maize and grass.)
- *Marking note:* 2 marks: 1 mark for the correct prediction and 1 mark for the reason. A reason without the prediction, or a prediction of 'decrease', earns no mark.*

(g) [1 marks]

- **Most organisms feed on more than one kind of food** (and are eaten by more than one kind of animal), and a web **shows all those links together**, so it is a truer picture of the ecosystem.
- *Marking note:* 1 mark for the idea of several feeding links rather than one.*

Question 3 [Total: 10 marks]

(a) [2 marks]

- P is the **producer** level (the green plants) (1 mark)
- Example: **maize plant** (accept grass, cassava, algae or any named green

plant) (1 mark)

- *Marking note: 2 marks, 1 mark each. The example must be a green plant. Do NOT accept a mushroom, which is not a producer.*

(b) [1 marks]

- **R**

- *Marking note: 1 mark. P is the producers, Q the primary consumers, R the secondary consumers and S the top consumers.*

(c) [1 marks]

- As **heat**

- *Marking note: 1 mark. Do NOT accept 'light' or 'sound'.*

(d) [2 marks]

- At each level the organisms **use most of the energy for respiration, movement, growth and other life activities**, and much of it **escapes to the surroundings as heat** (1 mark).

- Only the **small part that remains stored in their bodies is passed on** when they are eaten, so less and less energy is available at each higher level (1 mark).

- *Marking note: 2 marks, 1 mark each. Accept the point that parts of the organism (bones, fur, roots) are not eaten or digested.*

(e) [2 marks]

- **So much energy is lost at every stage** that after four or five levels **very little is left** (1 mark).

- There would **not be enough energy to feed and support another level** of consumers, so the chain ends (1 mark).

- *Marking note: 2 marks, 1 mark each. The answer must connect the loss of energy with the inability to support a further level.*

(f) [2 marks]

- They **trap the energy of the Sun and make food**, which every other organism in the ecosystem depends on, directly or indirectly.

- They **release oxygen** into the air for other organisms to breathe, and **provide shelter and habitats** for animals (and their roots hold the soil together).

- *Marking note: 1 mark each for any TWO correct points (2 marks). The food-making point should appear; award the second mark for oxygen, shelter or soil protection.*

Question 4 [Total: 10 marks]

(a) [3 marks]

- **Green plants absorb the sunlight** that falls on their leaves, using the green substance **chlorophyll** (1 mark).
- They use that energy, with **carbon dioxide and water**, to make food by **photosynthesis**, and the energy is **stored in the food** they make (1 mark).
- When **animals eat the plants**, the stored energy passes into their bodies, and so into the rest of the food chain (1 mark).
- *Marking note: 3 marks, 1 mark each. The word photosynthesis should appear; award the mark for a correct description even where the term is missing.**

(b) [3 marks]

- The Earth would become **extremely cold and dark**, and water would freeze, stopping the water cycle (1 mark).
- **Photosynthesis would stop**, so green plants would die and no new food would be made (1 mark).
- **Animals would starve** as the food chains broke down, and in time **almost all life on Earth would come to an end** (1 mark).
- *Marking note: 3 marks, 1 mark each for any three sound changes. This is the curriculum exemplar 'discuss how life on earth will be like without the sun'; accept any well-reasoned consequence.**

(c) [2 marks]

- They are the **only organisms that can trap the Sun's energy and make their own food** (1 mark).
- **Every other organism depends on them** for food, directly or indirectly, so the chain must start with them (1 mark).
- *Marking note: 2 marks, 1 mark each.**

(d) [2 marks]

- Animals **have no chlorophyll**, the green substance needed to absorb and trap sunlight (1 mark).
- They **cannot carry out photosynthesis** at all, so they must take in **ready-made food** by eating plants or other animals (1 mark).
- *Marking note: 2 marks, 1 mark each. Chlorophyll (or the inability to photosynthesise) must appear for full marks.**

Question 5 [Total: 9 marks]**(a)** [2 marks]

- Interdependence means that the organisms in an ecosystem **depend on one another for food, shelter, oxygen and other needs** (1 mark).
- None of them can live entirely on its own, so **a change that affects one organism affects the others too** (1 mark).
- *Marking note: 2 marks, 1 mark each.**

(b) [2 marks]

- Plants provide ****shelter, shade and nesting places**** for animals, while animals ****pollinate flowers and scatter seeds**** for the plants.
- Plants ****release oxygen**** that animals breathe, and animals ****release carbon dioxide**** that plants use; ****animal droppings and dead bodies enrich the soil**** for plants.
- **Marking note: 1 mark each for any TWO correct points that are NOT about feeding (2 marks). An answer about eating earns no mark here.**

(c) [2 marks]

- Decomposers ****break down dead plants and animals and their waste**** (1 mark),
- ****returning the nutrients to the soil**** so that green plants can use them again, and ****clearing away dead matter**** that would otherwise pile up (1 mark).
- **Marking note: 2 marks, 1 mark each. Named examples such as bacteria, fungi and mushrooms may support the answer but do not earn a mark on their own.**

(d) [3 marks]

- The ****number of rats would increase greatly****, because the snakes that fed on them are gone (1 mark).
- The rats would ****destroy much more maize****, so the farmer's ****harvest would fall**** (1 mark).
- ****Hawks and other animals that feed on snakes would lose a source of food**** and reduce in number, so the ****balance of the whole ecosystem is upset**** (1 mark).
- **Marking note: 3 marks, 1 mark each for any three sound effects. Credit any answer that traces a correct chain of consequences through the food web.**

Question 6 [Total: 10 marks]**(a)** [4 marks]

- ****Mark out a small area**** of the grass (for example one metre square) with pegs and string, so that the study area is clear.
- ****Search the area carefully**** - in the grass, under stones and leaves - and ****identify and list the organisms**** found there.
- ****Observe quietly over several days**** to see what each organism feeds on, looking for clues such as bitten leaves, droppings and remains of prey.
- ****Record the findings in a table**** (organism, what it feeds on, what feeds on it), then ****use the records to draw food chains and a food web****.
- **Marking note: 1 mark each for any FOUR sensible*

(b) [3 marks]

- A chain of ****four organisms beginning with a green plant****, for example ****grass → grasshopper → lizard → hawk**** (1 mark).
- ****Arrows drawn pointing from the organism eaten to the organism that eats it****

(1 mark).

- **Every organism clearly named (labelled)**, and the organisms chosen are ones that could really be found in grass behind a school (1 mark).

- **Marking note:** 3 marks as shown. A chain that begins with an animal loses the first mark; arrows drawn in the wrong direction lose the second. This is the 'illustrate with diagrams' exemplar.*

(c) [2 marks]

- **Do not touch or handle snakes, scorpions or any unfamiliar animal**, and **wear shoes and long sleeves** against bites and stings.

- **Do not taste any plant, fruit or seed**, **wash the hands with soap** after the study, and **stay with the teacher** throughout.

- **Marking note:** 1 mark each for any TWO sensible precautions (2 marks). Do NOT accept 'be careful' on its own.*

(d) [1 marks]

- **Every organism has its own part to play** in the ecosystem; if they are removed or killed the **feeding relationships and the balance of the ecosystem are upset**, and the organisms would die away from their habitat.

- **Marking note:** 1 mark for the idea of balance, or of respect for living things.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 3: Systems | **Sub-strand:** 3: Ecosystem | **Class:** Basic 8 (JHS 2)

Content Standard: **B8/JHS2.3.3.1** - Demonstrate an understanding of the interdependence of organisms in an ecosystem and their interaction.

> Source: NaCCA Scien

Electricity and Electr

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	15.1 Stages of Electricity Transmission	Remember
2	D	15.1 Stages of Electricity Transmission	Remember
3	D	15.1 Stages of Electricity Transmission	Understand
4	A	15.2 Transformers and High-Voltage Transmi...	Remember
5	C	15.2 Transformers and High-Voltage Transmi...	Understand
6	B	15.2 Transformers and High-Voltage Transmi...	Understand
7	B	15.1 Stages of Electricity Transmission	Remember
8	D	15.1 Stages of Electricity Transmission	Remember
9	B	15.1 Stages of Electricity Transmission	Remember
10	A	15.2 Transformers and High-Voltage Transmi...	Remember
11	B	15.3 Distribution, Use and Electrical Safety	Understand
12	C	15.3 Distribution, Use and Electrical Safety	Remember
13	A	15.3 Distribution, Use and Electrical Safety	Understand
14	A	15.3 Distribution, Use and Electrical Safety	Understand
15	D	15.3 Distribution, Use and Electrical Safety	Understand
16	C	15.2 Transformers and High-Voltage Transmi...	Analyse
17	D	15.1 Stages of Electricity Transmission	Understand
18	D	15.2 Transformers and High-Voltage Transmi...	Analyse
19	C	15.3 Distribution, Use and Electrical Safety	Evaluate
20	A	15.3 Distribution, Use and Electrical Safety	Evaluate
21	B	15.4 Resistors, Diodes and LEDs in Circuits	Remember
22	B	15.4 Resistors, Diodes and LEDs in Circuits	Remember
23	C	15.4 Resistors, Diodes and LEDs in Circuits	Remember
24	B	15.4 Resistors, Diodes and LEDs in Circuits	Remember
25	D	15.4 Resistors, Diodes and LEDs in Circuits	Understand
26	A	15.4 Resistors, Diodes and LEDs in Circuits	Understand
27	C	15.5 Charging and Discharging of a Capacitor	Remember
28	D	15.5 Charging and Discharging of a Capacitor	Remember
29	B	15.5 Charging and Discharging of a Capacitor	Remember
30	C	15.5 Charging and Discharging of a Capacitor	Understand
31	C	15.5 Charging and Discharging of a Capacitor	Understand
32	A	15.5 Charging and Discharging of a Capacitor	Understand
33	D	15.5 Charging and Discharging of a Capacitor	Understand
34	A	15.5 Charging and Discharging of a Capacitor	Analyse
35	A	15.5 Charging and Discharging of a Capacitor	Analyse
36	C	15.5 Charging and Discharging of a Capacitor	Apply
37	B	15.4 Resistors, Diodes and LEDs in Circuits	Analyse

Q	Ans	Curriculum Area	Cognitive Level
38	B	15.5 Charging and Discharging of a Capacitor	Analyse
39	D	15.5 Charging and Discharging of a Capacitor	Evaluate
40	A	15.4 Resistors, Diodes and LEDs in Circuits	Apply

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 6 (15%) | Apply: 2 (5%) | Evaluate: 3 (7%) | Remember: 15 (37%) | Understand: 14 (35%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 9 marks]

(a) [2 marks]

- Electricity transmission is the **carrying of electrical energy from the power station where it is generated** (1 mark)
- **through cables, pylons, transformers and substations to the places where it is used** - homes, schools and factories (1 mark).
- *Marking note: 2 marks, 1 mark each.**

(b) [4 marks]

- (i) **Step-up transformer** - it **raises the voltage** of the electricity leaving the power station (1 mark) **before it is sent along the transmission lines**, so that less energy is lost on the way (1 mark).
- (ii) **Pylon** - the **tall steel tower that carries the high-voltage cables high above the ground**, out of reach (1 mark).
- (iii) **Insulator** - it **holds the cable away from the metal pylon** so that the current cannot pass into the tower and down to earth (1 mark).
- *Marking note: 4 marks as shown. In (i), 'it changes the voltage' earns 1 mark only; the direction of the change is not required.**

(c) [3 marks]

- The cables have **resistance**, so some electrical energy is always **turned into heat in them** as the current flows (1 mark).
- For the same amount of power, a **higher voltage means a smaller current** flowing in the cables (1 mark).
- The heat lost depends on the size of the current, so the **smaller current wastes far less energy** over the long distance (1 mark).
- *Marking note: 3 marks, 1 mark each. No formula is required; award full marks for the reasoning 'higher voltage - smaller current - less heat lost'.**

Question 2 [Total: 12 marks]

(a) [2 marks]

- (i) J - the **step-up transformer** at the power station (1 mark)
- (ii) K - the **step-down transformer at the substation** (1 mark)
- *Marking note: 1 mark each (2 marks). The side notes in Fig. 1 say that the voltage is raised at J and lowered at K; accept 'substation' for K only where the lowering of the voltage is also mentioned.**

(b) [2 marks]

- Its **voltage** is raised at the power station and lowered again at the substation.
- It **travels** long distances along cables carried on pylons and is then shared out along the distribution lines (accept: some of the energy is lost as heat in the cables).
- **Marking note:** 1 mark each for any TWO correct points (2 marks).

(c) [2 marks]

- The transmission voltage is **far too high** for use in houses: it would be **extremely dangerous** and would destroy the appliances (1 mark).
- It must be **brought down** to the value that lamps, fans and other appliances are made to work on (about 230 V in Ghana) (1 mark).
- **Marking note:** 2 marks, 1 mark each. Accept the safety point or the appliance point for either mark.

(d) [4 marks]

- **Four boxes** drawn and joined by arrows pointing in one direction (1 mark).
- **Box 1:** the power station, where the electricity is generated (1 mark).
- **Box 2 and Box 3:** the step-up transformer with the transmission lines, and the substation with its step-down transformer (1 mark).
- **Box 4:** the school (the consumer), where the electricity is used (1 mark).
- **Marking note:** 4 marks as shown. This is the curriculum exemplar that asks for a flow chart from the point of generation to the point of consumption. A chart whose arrows run the wrong way, or whose stages are out of order,

(e) [2 marks]

- **Never touch or go near a fallen cable; keep others away and report it at once to the electricity company.**
- **Do not fly kites, climb trees or raise ladders and long metal objects near the lines, and do not build or tamper with transformers and substations under the wires.**
- **Marking note:** 1 mark each for any TWO sound rules (2 marks).

Question 3 [Total: 10 marks]**(a)** [6 marks]

- (i) **Resistor** - it **limits or controls the size of the current** flowing in the circuit (1 mark), and so **protects the other components from too large a current** (1 mark).
- (ii) **Diode** - it **allows current to pass through it in one direction only** (1 mark) and **blocks the current in the opposite direction** (1 mark).
- (iii) **LED (light emitting diode)** - it is a diode that **gives out light when current passes through it** (1 mark) **in the correct (forward) direction** (1

mark).

- *Marking note: 6 marks, 2 marks for each component as shown.*

(b) [2 marks]

- (i) Resistance - the **ohm** (1 mark)

- (ii) Capacitance - the **farad** (1 mark)

- *Marking note: 1 mark each (2 marks). Accept the symbols, and accept the microfarad as a practical unit of capacitance.*

(c) [2 marks]

- An LED has **very little resistance of its own**, so if it were connected alone **too large a current would flow through it and destroy it** (1 mark).

- The resistor **limits the current to a safe value**, so the LED lights properly and is not damaged (1 mark).

- *Marking note: 2 marks, 1 mark each.*

Question 4 [Total: 10 marks]

(a) [4 marks]

- (i) W - the **cell (battery)** (1 mark)

- (ii) X - the **switch** (shown open) (1 mark)

- (iii) Y - the **resistor** (1 mark)

- (iv) Z - the **capacitor** (1 mark)

- *Marking note: 1 mark each (4 marks). Accept 'dry cell' or 'battery' for W.*

(b) [2 marks]

- The **circuit is completed and a current flows** from the cell through the resistor, so **charge begins to collect on the plates of the capacitor** (1 mark).

- The **LED flashes (glows briefly)** while this charging current is flowing (1 mark).

- *Marking note: 2 marks, 1 mark each.*

(c) [1 marks]

- The current **falls steadily to zero**, so **no further current flows** in that branch of the circuit

- *Marking note: 1 mark. 'The current stops' is acceptable.*

(d) [3 marks]

- The **LED glows brightly for a moment and then fades out** (1 mark).

- The **charged capacitor now acts as the source** and **drives a current through the LED as it discharges** (1 mark).

- As the **stored charge is used up the current falls**, so the light grows dimmer and finally goes out (1 mark).

- *Marking note: 3 marks, 1 mark each. This answers the curriculum exemplar on the discharging action of a capacitor. A candidate who says the LED stays on

permanently earns no mark.*

Question 5 [Total: 9 marks]

(a) [2 marks]

- **Two metal plates placed close together but not touching** (1 mark),
- **separated by an insulating material (air, waxed paper, mica or plastic)**, with a connecting lead joined to each plate (1 mark).
- **Marking note:** 2 marks, 1 mark each. The insulator between the plates must be mentioned for the second mark.*

(b) [3 marks]

- When the switch is closed, **charge flows on to one plate and away from the other**, so one plate becomes **negatively charged and the other positively charged** (1 mark).
- The **current is large at first and falls steadily as the charge builds up** on the plates (1 mark).
- Charging stops when the **voltage across the capacitor is equal to the voltage of the supply**; the current then falls to zero (1 mark).
- **Marking note:** 3 marks, 1 mark each. No charge actually crosses the insulator between the plates; do not penalise a candidate who does not mention this, but credit it as an alternative.

(c) [2 marks]

- When the supply is removed and the plates are **joined through a conducting path**, the stored charge flows back round the circuit (in the opposite direction) (1 mark).
- This **gives a current that is large at first and falls to zero** as the capacitor empties, lighting an LED briefly if one is in the path (1 mark).
- **Marking note:** 2 marks, 1 mark each.*

(d) [2 marks]

- **Storing charge to give the flash in a camera**, and **smoothing the supply in a radio, television or phone charger**.
- **Keeping the clock or memory of a device running briefly when the power goes off**, **starting the motor of a fan or refrigerator**, or **timing the flashing of a lamp** (accept any two).
- **Marking note:** 1 mark each for any TWO correct uses (2 marks).*

Question 6 [Total: 10 marks]

(a) [4 marks]

- **Collect the components** - cell, switch, resistor, capacitor, LED and connecting wires - and **check that the cell is good**.
- **Connect the cell, the switch, the resistor and the capacitor in a loop**, joining the capacitor the correct way round (positive lead to the positive side).
- **Connect the LED across the capacitor**, with its longer leg towards the positive side, and **check every joint**.
- **Close the switch and observe the LED; then open the switch and observe it again**, recording what is seen each time.
- **Marking note:** 1 mark each for any FOUR workable steps in a sensible

(b) [2 marks]

- **On closing the switch:** the LED **flashes briefly and then goes out** as the capacitor charges up (1 mark).
- **On opening the switch:** the LED **glows again for a moment and fades away** as the capacitor discharges through it (1 mark).
- **Marking note:** 2 marks, 1 mark each.*

(c) [2 marks]

- **Use only the low-voltage cell provided - never the mains supply** - and **do not short-circuit the cell**.
- **Connect the capacitor and the LED the correct way round**, **discharge the capacitor before handling the leads**, and switch off when the work is finished.
- **Marking note:** 1 mark each for any TWO sound precautions (2 marks). 'Be careful' on its own earns no mark.*

(d) [2 marks]

- The **LED is connected the wrong way round**, so it cannot conduct (accept: the LED is damaged).
- The **cell is flat or wrongly connected, or a joint is loose so that the circuit is not complete** (accept: the resistor used is far too large).
- **Marking note:** 1 mark each for any TWO sensible faults (2 marks).*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 4: Forces and Energy | **Sub-strand:** 2: Electricity and Electronics |
Class: Basic 8 (JHS 2)

Content Standard: **B8/JHS2.4.2.1** - Demonstrate knowledge of electricity transmission. **B8/JHS2.4.2.2** - Demonstrate unde

Energy Conversion

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	A	13.1 Forms of Energy	Remember
2	A	13.1 Forms of Energy	Remember
3	D	13.1 Forms of Energy	Remember
4	D	13.1 Forms of Energy	Remember
5	A	13.1 Forms of Energy	Remember
6	C	13.1 Forms of Energy	Remember
7	D	13.1 Forms of Energy	Understand
8	B	13.2 Energy Conversion in Everyday Devices	Remember
9	C	13.2 Energy Conversion in Everyday Devices	Understand
10	C	13.2 Energy Conversion in Everyday Devices	Understand
11	B	13.2 Energy Conversion in Everyday Devices	Understand
12	A	13.2 Energy Conversion in Everyday Devices	Remember
13	B	13.2 Energy Conversion in Everyday Devices	Understand
14	A	13.2 Energy Conversion in Everyday Devices	Understand
15	D	13.2 Energy Conversion in Everyday Devices	Understand
16	C	13.2 Energy Conversion in Everyday Devices	Understand
17	C	13.2 Energy Conversion in Everyday Devices	Analyse
18	D	13.3 Producing Electricity from a Dammed R...	Remember
19	A	13.3 Producing Electricity from a Dammed R...	Remember
20	C	13.3 Producing Electricity from a Dammed R...	Remember
21	B	13.3 Producing Electricity from a Dammed R...	Understand
22	B	13.3 Producing Electricity from a Dammed R...	Understand
23	A	13.3 Producing Electricity from a Dammed R...	Analyse
24	D	13.3 Producing Electricity from a Dammed R...	Remember
25	C	13.3 Producing Electricity from a Dammed R...	Understand
26	D	13.4 Harnessing Natural Forms of Energy	Understand
27	A	13.4 Harnessing Natural Forms of Energy	Understand
28	D	13.4 Harnessing Natural Forms of Energy	Understand
29	B	13.4 Harnessing Natural Forms of Energy	Apply
30	C	13.5 Importance of Energy Conversion and E...	Understand
31	B	13.5 Importance of Energy Conversion and E...	Understand
32	D	13.5 Importance of Energy Conversion and E...	Analyse
33	B	13.5 Importance of Energy Conversion and E...	Analyse
34	B	13.2 Energy Conversion in Everyday Devices	Analyse
35	A	13.5 Importance of Energy Conversion and E...	Evaluate
36	C	13.3 Producing Electricity from a Dammed R...	Analyse

Q	Ans	Curriculum Area	Cognitive Level
37	A	13.4 Harnessing Natural Forms of Energy	Apply
38	B	13.4 Harnessing Natural Forms of Energy	Analyse
39	D	13.5 Importance of Energy Conversion and E...	Evaluate
40	C	13.5 Importance of Energy Conversion and E...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 7 (17%) | Apply: 2 (5%) | Evaluate: 3 (7%) | Remember: 12 (30%) | Understand: 16 (40%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 9 marks]****(a)** [2 marks]

- Energy conversion is the **changing of energy from one form into another form** (1 mark),
- for example **electrical energy changing into heat energy in a kettle**; the energy is **not destroyed - it only changes its form** (1 mark).
- **Marking note:** 2 marks, 1 mark each. The second mark may be earned either by a correct example or by the point that the energy is not destroyed.*

(b) [4 marks]

- (i) **Electrical energy to heat energy** (1 mark)
- (ii) **Chemical energy to electrical energy to light energy** (accept chemical to light) (1 mark)
- (iii) **Electrical energy to sound energy** (1 mark)
- (iv) **Light (solar) energy to chemical energy** (1 mark)
- **Marking note:** 1 mark each (4 marks). Both the starting form and the final form must be given; a single form earns no mark. In (ii) accept the two-step or the three-step chain.*

(c) [3 marks]

- **Heat energy** - from the cooking stove or the coal pot.
- **Light energy** - from the electric lamp or the lantern.
- **Sound energy** - from the radio or the television (accept electrical energy at the socket, or chemical energy in food and dry cells).
- **Marking note:** 1 mark for each correct pairing of a form of energy with a place where it is used or produced (3 marks). A form named without a place, or a place without a form, earns no mark.*

Question 2 [Total: 12 marks]**(a)** [1 marks]

- **Potential energy** (gravitational potential energy), because the water is stored at a high level
- **Marking note:** 1 mark. Do NOT accept kinetic energy; the stored water is not yet moving.*

(b) [4 marks]

- The **dam wall holds back the river**, so that a large volume of water is **stored at a high level** in the reservoir (1 mark).
- The water is released and **rushes down the penstock**, so its potential energy changes into **kinetic energy** (1 mark).
- The fast-moving water **strikes the blades of the turbine and makes it spin** (1 mark).
- The spinning turbine **drives the generator through the shaft**, and the generator **produces electricity**, which is carried away by the transmission lines (1 mark).
- **Marking note:** 1 mark for each correct step in the right order (4 marks). This is t

(c) [2 marks]

- (i) **Kinetic energy of the moving water to mechanical (kinetic) energy of the turbine** (1 mark)
- (ii) **Mechanical (kinetic) energy to electrical energy** (1 mark)
- **Marking note:** 1 mark each (2 marks). Both forms must be stated in each case.*

(d) [2 marks]

- The **higher the water is stored above the turbine, the greater its potential energy** (1 mark).
- As it falls, this becomes a **greater kinetic energy**, so the water **strikes the turbine with greater force and more electricity is produced** (1 mark).
- **Marking note:** 2 marks, 1 mark each. The link between the height, the energy and the output must be made; 'so that the water can flow' alone earns 1 mark at most.*

(e) [2 marks]

- The **water is renewable and free** once the dam has been built, so the **running cost is low** and no fuel has to be bought.
- **No fuel is burnt**, so there is **no smoke or air pollution** (accept: the reservoir also provides fish, water for irrigation and transport).
- **Marking note:** 1 mark each for any TWO (2 marks).*

(f) [1 marks]

- **People living in the valley lose their homes and farmland and have to be resettled** (accept: the still water breeds mosquitoes and bilharzia snails; the flow downstream is reduced, affecting farming and fishing).
- **Marking note:** 1 mark for any one sound problem.*

Question**(a) [4 marks]**

- (i) W - **solar energy** (light energy from the Sun) (1 mark)
 - (ii) X - **electrical energy** (1 mark)
 - (iii) Y - **chemical energy** stored in the battery (1 mark)
 - (iv) Z - **light energy** given out by the lamp (accept light and heat energy) (1 mark)
- **Marking note:** 1 mark each (4 marks). In (iii) the battery stores the energy as chemical energy; 'electrical energy' is not accepted there, because the arrow Y shows what the battery sends out after storing it - accept 'electrical energy' only where the candidate has clearly explained that the battery first stores it as chemical energy

(b) [1 marks]

- The **solar panel** on the roof
- **Marking note:** 1 mark.*

(c) [2 marks]

- The panel **produces electricity only while the Sun is shining** (1 mark).
 - The battery **stores the energy produced during the day and gives it out at night or on a cloudy day**, so the lamp can be used when there is no sunshine (1 mark).
- **Marking note:** 2 marks, 1 mark each.*

(d) [2 marks]

- **Heat energy** (1 mark).
 - The lamp is meant to **give light**, so the energy that escapes as heat **does no useful work and is wasted to the surroundings** (1 mark).
- **Marking note:** 2 marks, 1 mark each. 'Sound' is not accepted; a lamp gives out negligible sound.*

(e) [1 marks]

- The village obtains **light without waiting to be connected to the national grid**; the sunshine is **free and plentiful**, there is **no fuel to buy and no smoke**, and it is safer than a kerosene lamp.
- **Marking note:** 1 mark for any one sound advantage.*

Question 4 [Total: 10 marks]**(a) [3 marks]**

- The **moving air** (wind) strikes the blades of the wind turbine and makes them **spin** (1 mark).
- The spinning blades **turn a shaft that is connected to a generator** (1 mark).

- The generator **changes the mechanical energy into electrical energy**, which is stored in batteries or sent out along cables (1 mark).

- **Marking note:** 1 mark for each correct step in the right order (3 marks).*

(b) [3 marks]

- **Animal dung is mixed with water and fed into a closed tank (digester)** from which air is excluded (1 mark).

- **Bacteria break down the dung**, and the **gas produced collects at the top of the digester and is piped off** (1 mark).

- Use: the gas is **burnt for cooking or for lighting** (accept: to run a small generator; the slurry left behind is used as manure) (1 mark).

- **Marking note:** 3 marks: 2 marks for the process and 1 mark for a use. The exclusion of air or the action of bacteria must appear for the second process mark.*

(c) [2 marks]

- It **puts natural energy into a form people can actually use** - for lighting, cooking, pumping water, transport and industry.

- It **saves money and fuel and supports schools, clinics and businesses**, and reduces the cutting of firewood and the pollution that comes from burning fuels.

- **Marking note:** 1 mark each for any TWO sound reasons (2 marks).*

(d) [2 marks]

- Our **lamps, machines and appliances are made to work on particular forms of energy**, usually electrical energy or heat energy (1 mark).

- Sunlight, wind and flowing water are **not in those forms**, so a device such as a solar panel, a turbine or a generator **must first convert them** before the energy can be used (1 mark).

- **Marking note:** 2 marks, 1 mark each.*

Question 5 [Total: 9 marks]

(a) [3 marks]

- (i) **Kinetic (mechanical) energy to heat energy** (1 mark)

- (ii) **Potential energy to kinetic energy** (1 mark)

- (iii) **Chemical energy (in the boy) to kinetic energy to electrical energy to light energy** (accept kinetic to electrical to light) (1 mark)

- **Marking note:** 1 mark each (3 marks). Both forms must appear in each answer.*

(b) [2 marks]

- **No device changes all the energy supplied to it into the useful form**; some always escapes to the surroundings, mostly as **heat from friction or from the resistance of the wires** (1 mark).

- Example: a **filament bulb becomes very hot** while it is giving light, so part of

the electrical energy supplied leaves as heat (accept any suitable example) (1 mark).

- *Marking note: 2 marks: 1 mark for the explanation and 1 mark for a suitable example.*

(c) [2 marks]

- **Switch off lights, fans and the television when nobody is using them**, and **use energy-saving (LED) bulbs**.

- **Cover pots while cooking** and use the correct size of saucepan, dry clothes in the sun, and keep appliances well maintained.

- *Marking note: 1 mark each for any TWO practical ways (2 marks).*

(d) [2 marks]

- The LED **changes a much greater part of the electrical energy supplied into light** and **wastes far less of it as heat** (1 mark).

- For the **same brightness it therefore uses less electricity**, so the cells last longer and the electricity bill is lower (1 mark).

- *Marking note: 2 marks, 1 mark each. A candidate who writes only that the LED 'is better' or 'is modern' earns no mark.*

Question 6 [Total: 10 marks]

(a) [4 marks]

- **Cut the cardboard into vanes (blades) and fix them evenly round a cork or a cut plastic bottle** to form the wheel.

- **Push the knitting needle through the centre to act as the axle** and rest its ends on two supports (for example two tins) so that the wheel spins freely.

- **Tie one end of the string to the axle and the small load to the other end**, so the string winds on to the axle as the wheel turns.

- **Pour water steadily on to the vanes from a height and observe** the wheel turning and the load being lifted; **repeat with a faster flow and record** what happens.

- *Marking note:

(b) [3 marks]

- A clear sketch showing the **wheel with its vanes and the axle resting on the supports** (1 mark).

- The **string wound on the axle and carrying the load** shown correctly (1 mark).

- The **water being poured from above on to the vanes**, with **all the parts labelled** (1 mark).

- *Marking note: 3 marks as shown. Award the marks for a workable arrangement, however roughly drawn; an unlabelled diagram scores a maximum of 1 mark.*

(c) [2 marks]

- The **potential energy** of the water held above changes into **kinetic energy** as the water falls on to the vanes (1 mark).
- This **turns the wheel (mechanical energy)**, which **lifts the load**, so the energy becomes potential energy stored in the raised load (1 mark).
- *Marking note: 2 marks, 1 mark each. Accept a correctly written chain such as 'potential to kinetic to mechanical to potential'.*

(d) [1 marks]

- **Handle the knitting needle and the cutting tool carefully, keeping the sharp ends away from others** (accept: mop up spilt water at once so that nobody slips; keep water away from electrical sockets).
- *Marking note: 1 mark for any one sound precaution. 'Be careful' on its own earns no mark.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 4: Forces and Energy | **Sub-strand:** 1: Energy | **Class:** Basic 8 (JHS 2)

Content Standard: **B8/JHS2.4.1.1** - Demonstrate the skill to evaluate the conversion of energy from one form to another.

> Source: NaCCA Science Common Core Pr

Force and Motion

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	17.1 Magnets, Magnetic Materials and Simpl...	Remember
2	D	17.1 Magnets, Magnetic Materials and Simpl...	Remember
3	C	17.1 Magnets, Magnetic Materials and Simpl...	Understand
4	A	17.1 Magnets, Magnetic Materials and Simpl...	Remember
5	C	17.1 Magnets, Magnetic Materials and Simpl...	Understand
6	A	17.1 Magnets, Magnetic Materials and Simpl...	Understand
7	D	17.1 Magnets, Magnetic Materials and Simpl...	Understand
8	A	17.1 Magnets, Magnetic Materials and Simpl...	Remember
9	D	17.1 Magnets, Magnetic Materials and Simpl...	Understand
10	D	17.1 Magnets, Magnetic Materials and Simpl...	Analyse
11	C	17.1 Magnets, Magnetic Materials and Simpl...	Understand
12	D	17.1 Magnets, Magnetic Materials and Simpl...	Analyse
13	B	17.1 Magnets, Magnetic Materials and Simpl...	Understand
14	B	17.1 Magnets, Magnetic Materials and Simpl...	Remember
15	A	17.2 Magnetic Force in Domestic and Indust...	Remember
16	D	17.2 Magnetic Force in Domestic and Indust...	Understand
17	C	17.2 Magnetic Force in Domestic and Indust...	Understand
18	C	17.2 Magnetic Force in Domestic and Indust...	Understand
19	B	17.2 Magnetic Force in Domestic and Indust...	Understand
20	A	17.2 Magnetic Force in Domestic and Indust...	Understand
21	C	17.2 Magnetic Force in Domestic and Indust...	Understand
22	C	17.3 Magnetic Force and Newton's Second La...	Remember
23	B	17.3 Magnetic Force and Newton's Second La...	Remember
24	B	17.3 Magnetic Force and Newton's Second La...	Remember
25	D	17.3 Magnetic Force and Newton's Second La...	Apply
26	B	17.3 Magnetic Force and Newton's Second La...	Apply
27	C	17.3 Magnetic Force and Newton's Second La...	Analyse
28	D	17.3 Magnetic Force and Newton's Second La...	Analyse
29	A	17.3 Magnetic Force and Newton's Second La...	Analyse
30	B	17.3 Magnetic Force and Newton's Second La...	Analyse
31	B	17.3 Magnetic Force and Newton's Second La...	Analyse
32	C	17.4 Simple Machines Recalled	Understand
33	A	17.4 Simple Machines Recalled	Understand
34	D	17.4 Simple Machines Recalled	Understand
35	A	17.5 Complex Machines and the Quality of Life	Understand
36	B	17.5 Complex Machines and the Quality of Life	Understand

Q	Ans	Curriculum Area	Cognitive Level
37	B	17.5 Complex Machines and the Quality of Life	Analyse
38	D	17.5 Complex Machines and the Quality of Life	Analyse
39	A	17.5 Complex Machines and the Quality of Life	Evaluate
40	A	17.5 Complex Machines and the Quality of Life	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 9 (22%) | Apply: 2 (5%) | Evaluate: 2 (5%) | Remember: 9 (22%) | Understand: 18 (45%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 9 marks]****(a) [2 marks]**

- **Magnetic materials (any two):** **iron, steel, nickel, cobalt** (1 mark for two).
- **Non-magnetic materials (any two):** **copper, aluminium, wood, rubber, plastic, glass** (1 mark for two).
- **Marking note:** 2 marks: 1 mark for two correct magnetic materials and 1 mark for two correct non-magnetic materials. One correct material in a pair scores nothing for that pair.*

(b) [2 marks]

- **North near north: the magnets repel** - they push each other apart, because **like poles repel** (1 mark).
- **North near south: the magnets attract** - they pull together, because **unlike poles attract** (1 mark).
- **Marking note:** 2 marks, 1 mark each. Accept the general rule as the explanation.*

(c) [2 marks]

- A **permanent magnet keeps its magnetism for a long time**, while a **temporary magnet loses its magnetism as soon as the current or the magnetising force is removed**.
- A permanent magnet is usually made of **steel and is harder to magnetise**, while a temporary magnet is usually made of **soft iron and is magnetised easily** (accept: an electromagnet is a temporary magnet).
- **Marking note:** 1 mark each for any TWO valid differences (2 marks).*

(d) [3 marks]

- **Place the knitting needle flat on a table and hold it firmly**, then take a bar magnet and place **one pole at one end of the needle** (1 mark).
- **Stroke the needle from one end to the other in one direction only**, lifting the magnet well clear of the needle at the end of each stroke and returning to the same starting end; repeat **many times (about 30 to 50) with the same pole** (1 mark).
- **Test the needle** by seeing whether it can pick up pins or iron filings, or by suspending it freely to see whether it comes to rest pointing north and south (1 mark).
- **Marking note:** 3 marks as

Question 2

(a) [4 marks]

- **P** - coil of insulated copper wire wound round the nail (1 mark).
- **Q** - soft iron nail, the core (1 mark).
- **R** - dry cells / battery in a cell holder (1 mark).
- **S** - switch (1 mark).
- **Marking note:** 1 mark each (4 marks). Accept 'solenoid' for P, 'iron core' for Q and 'battery' for R. Do not accept 'wire' alone for P.*

(b) [1 marks]

- **An electromagnet** (a temporary magnet made by electricity).
- **Marking note:** 1 mark. Accept 'temporary magnet' only if 'electromagnet' is given.*

(c) [2 marks]

- **The paper clips fall off the nail** (1 mark).
- Opening the switch **stops the current, so the magnetic field dies away and the soft iron loses its magnetism immediately** (1 mark).
- **Marking note:** 2 marks: 1 mark for the observation and 1 mark for the reason. The reason mark needs the current stopping AND the loss of magnetism.*

(d) [2 marks]

- **Increase the number of turns of wire** wound round the nail.
- **Use more dry cells (a larger current)** in the circuit (accept: use a thicker soft iron core).
- **Marking note:** 1 mark each for any TWO correct ways (2 marks).*

(e) [2 marks]

- **Soft iron becomes a strong magnet as soon as the current flows and loses its magnetism as soon as the switch is opened** (1 mark).
- So the clips can be **picked up and released whenever the pupils wish**; **steel would stay magnetised** and would not let the clips go (1 mark).
- **Marking note:** 2 marks. A candidate who says only 'soft iron is a temporary magnet' scores 1 mark; the second mark needs the use made of that property.*

(f) [1 marks]

- **Lifting and dropping scrap iron and steel with a crane in a scrap yard or steel works** (accept: separating iron from other waste, electric bells, loudspeakers, electric motors).
- **Marking note:** 1 mark for any one correct industrial use.*

Question

(a) [2 marks]

- The **acceleration of a body is directly proportional to the force applied to it** and takes place **in the direction of the force** (1 mark).
- and is **inversely proportional to the mass of the body**; in symbols **$F = ma$** (1 mark).
- *Marking note: 2 marks, 1 mark each. Award the second mark for the symbol form $F = ma$ with the symbols explained.**

(b) [2 marks]

- **The larger the force applied, the larger the acceleration** - the two rise in the same ratio, so the **acceleration is directly proportional to the force** (1 mark).
- This is because the **mass of the trolley (2 kg) stays the same throughout**; in every row, $\text{force} \div \text{acceleration} = 2 \text{ kg}$ (1 mark).
- *Marking note: 2 marks. The first mark needs 'directly proportional' or a clear statement that doubling the force doubles the acceleration; the second needs the constant mass.**

(c) [3 marks]

- **$F = ma$, so $a = F \div m$** (1 mark)
- **$a = 20 \div 4$** (1 mark)
- **$a = 5 \text{ m/s}^2$** (1 mark)
- *Marking note: 3 marks: 1 mark for the correct formula, 1 mark for the correct substitution and 1 mark for the answer with the unit m/s^2 . A correct answer with no working scores 2 marks.**

(d) [3 marks]

- **The loaded truck has a much larger mass than the empty one** (1 mark).
- From **$F = ma$, $a = F \div m$** , so with the **same force the larger mass gives a smaller acceleration** (1 mark).
- So the loaded truck **picks up speed slowly**, and a **larger force** - a stronger push or a more powerful engine - is needed to start it as quickly as the empty truck (1 mark).
- *Marking note: 3 marks as shown. Accept any sound daily-life example, such as pushing a loaded truck at Kaneshie market, a loaded wheelbarrow of sand, or a trotro full of passengers.**

Question 4 [Total: 10 marks]**(a)** [4 marks]

- **Showing direction** - the **magnetic compass** used by travellers and surveyors.
- **Producing sound** - the **loudspeaker** of a radio, a television or a mobile

phone.

- **Giving a warning** - the **electric bell or the security alarm** on a door.
- **Lifting and separating scrap iron** - the **electromagnetic crane in a scrap yard**, or the separation of iron from other waste at a recycling plant (accept: magnetic door catch of a cupboard or refrigerator; electric motor in a fan, blender or corn mill; the bicycle dynamo).
- **Marking note:** 1 mark each for any FOUR applications (4 marks)

(b) [3 marks]

- The crane carries a **large electromagnet**; when the **current is switched on** it becomes strongly magnetised and attracts the pieces of iron and steel, which stick to it (1 mark).
- The crane **lifts and swings the load** across to the lorry or the heap where it is wanted (1 mark).
- The **current is then switched off**, the electromagnet **loses its magnetism** and the scrap drops at the right place, without anyone touching it (1 mark).
- **Marking note:** 3 marks as shown. Candidates who describe a permanent magnet lifting the scrap score a maximum of 1 mark, since the load could not be

(c) [3 marks]

- **Set-up:** place a **trolley carrying a steel bar on a smooth, level bench**, mark a start line and a finish line, and arrange to pull the trolley with a **strong magnet** (1 mark).
- **Method:** pull the trolley with **one magnet** and time how long it takes to reach the finish line; then repeat with **two or three magnets held together**, keeping the same trolley and the same distance, so that only the force is changed (1 mark).
- **Result:** the trolley reaches the finish line in a **shorter time** when the larger magnetic force is used, which shows that the **larger the force, the larger**

Question 5 [Total: 9 marks]

(a) [2 marks]

- A **simple machine** is a device with **few or no moving parts** that makes work easier by changing the size or the direction of the force applied (1 mark).
- A **complex (compound) machine** is a machine **made up of two or more simple machines working together** (1 mark).
- **Marking note:** 2 marks, 1 mark each.*

(b) [3 marks]

- **Lever** - a crowbar, a pair of scissors or the handles of a wheelbarrow.
- **Pulley** - the pulley used to draw water from a well or to lift blocks on a building site.
- **Inclined plane** - a ramp used to roll drums onto a lorry (accept: **wedge** - an axe or a knife; **screw** - a bolt or a bottle top; **wheel and axle** - a door

knob or a steering wheel).

- *Marking note: 1 mark each for any THREE simple machines, each with a correct example (3 marks). A name with no example scores half of nothing - award the mark only when both are given. This is the curriculum exemplar that recaps s

(c) [2 marks]

- A **simple machine** is one device with few or no moving parts, while a **complex machine** is built from two or more simple machines working together.

- A simple machine is usually **worked by the effort of a person and is cheap and easy to repair**, while a complex machine often **needs fuel or electricity, does much more work in a shorter time, and costs more to buy and maintain**.

- *Marking note: 1 mark each for any TWO valid differences (2 marks).*

(d) [2 marks]

- **Corn mill** - grinds maize into dough for banku and kenkey (accept **blender** - grinds pepper, tomatoes and onions).

- **Tractor** - ploughs and harrows the land on a farm (accept **sewing machine** - sews cloth into garments; **bicycle or motorcycle** - carries people and loads; **water pump** - lifts water for irrigation).

- *Marking note: 1 mark each for any TWO complex machines, each with its correct use (2 marks). Do not award a mark for a simple machine such as a cutlass, a ramp or a single pulley.*

Question 6 [Total: 10 marks]

(a) [4 marks]

- **J** - the brake lever on the handlebar (1 mark).

- **K** - the chain, running over the toothed wheels (sprockets) (1 mark).

- **L** - the pedal and crank (1 mark).

- **M** - the front wheel (accept: wheel and tyre) (1 mark).

- *Marking note: 1 mark each (4 marks).*

(b) [2 marks]

- **J** - a lever: the brake lever turns about a pivot so that a small effort of the fingers gives a large pull on the brake cable (1 mark).

- **M** - a wheel and axle: the large wheel turns on the small axle at the hub (1 mark).

- *Marking note: 2 marks, 1 mark each. The example need not be explained; the correct name of the simple machine earns the mark.*

(c) [2 marks]

- It is ****made up of several simple machines working together**** - levers (brake levers), wheels and axles (the wheels, the pedals and cranks), the chain and toothed wheels, and screws (the nuts and bolts) (1 mark).
- These parts ****all act together to change the push of the rider's legs into the turning of the wheels****, which no single simple machine could do (1 mark).
 - ***Marking note: 2 marks. A candidate who simply says 'it has many parts' scores 0; the simple machines must be identified for the first mark.***

(d) [2 marks]

- The pupil ****travels to school, to the farm and to the market far more quickly and with less effort****, so he or she ****arrives on time and less tired****, and ****saves the money that would be spent on transport fares****.
- ****Loads such as water, firewood and farm produce can be carried****, and the family can ****reach the clinic quickly in an emergency****; the bicycle also ****uses no fuel, so it does not pollute the air****.
 - ***Marking note: 1 mark each for any TWO sound benefits (2 marks). This answers the curriculum exemplar on how the functions of a complex machine can improve the quality of life.***

Human Body Sys

ANSWER KEY**SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	9.1 Regions of a Mammalian Tooth	Remember
2	B	9.1 Regions of a Mammalian Tooth	Remember
3	A	9.1 Regions of a Mammalian Tooth	Remember
4	A	9.1 Regions of a Mammalian Tooth	Remember
5	B	9.2 Internal Structure of a Tooth and the ...	Remember
6	B	9.2 Internal Structure of a Tooth and the ...	Understand
7	D	9.2 Internal Structure of a Tooth and the ...	Remember
8	B	9.2 Internal Structure of a Tooth and the ...	Remember
9	B	9.2 Internal Structure of a Tooth and the ...	Remember
10	C	9.2 Internal Structure of a Tooth and the ...	Understand
11	A	9.2 Internal Structure of a Tooth and the ...	Understand
12	A	9.2 Internal Structure of a Tooth and the ...	Understand
13	D	9.2 Internal Structure of a Tooth and the ...	Understand
14	C	9.3 Types of Teeth and their Functions in ...	Remember
15	A	9.3 Types of Teeth and their Functions in ...	Remember
16	D	9.3 Types of Teeth and their Functions in ...	Understand
17	D	9.3 Types of Teeth and their Functions in ...	Remember
18	C	9.3 Types of Teeth and their Functions in ...	Understand
19	A	9.3 Types of Teeth and their Functions in ...	Remember
20	D	9.3 Types of Teeth and their Functions in ...	Understand
21	C	9.3 Types of Teeth and their Functions in ...	Remember
22	A	9.3 Types of Teeth and their Functions in ...	Remember
23	C	9.3 Types of Teeth and their Functions in ...	Analyse
24	B	9.3 Types of Teeth and their Functions in ...	Analyse
25	B	9.4 Causes of Tooth Decay and Gum Disease	Remember
26	C	9.4 Causes of Tooth Decay and Gum Disease	Remember
27	C	9.4 Causes of Tooth Decay and Gum Disease	Understand
28	A	9.4 Causes of Tooth Decay and Gum Disease	Apply
29	D	9.4 Causes of Tooth Decay and Gum Disease	Understand
30	C	9.4 Causes of Tooth Decay and Gum Disease	Understand
31	D	9.5 Prevention of Tooth Decay and Care of ...	Evaluate
32	B	9.5 Prevention of Tooth Decay and Care of ...	Apply
33	A	9.5 Prevention of Tooth Decay and Care of ...	Understand
34	D	9.5 Prevention of Tooth Decay and Care of ...	Understand
35	B	9.5 Prevention of Tooth Decay and Care of ...	Apply
36	C	9.4 Causes of Tooth Decay and Gum Disease	Analyse

Q	Ans	Curriculum Area	Cognitive Level
37	D	9.5 Prevention of Tooth Decay and Care of ...	Analyse
38	A	9.5 Prevention of Tooth Decay and Care of ...	Evaluate
39	B	9.3 Types of Teeth and their Functions in ...	Evaluate
40	C	9.5 Prevention of Tooth Decay and Care of ...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 4 (10%) | Apply: 3 (7%) | Evaluate: 4 (10%) | Remember: 16 (40%) | Understand: 13 (32%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 9 marks]****(a)** [3 marks]

- **Crown**
- **Neck**
- **Root**

- *Marking note: 1 mark each (3 marks). These are the three regions named in the curriculum. Do NOT accept enamel, dentine and pulp - those are the layers, not the regions.*

(b) [2 marks]

- (i) Crown - the part **above the gum** that **bites, tears or grinds the food**
- (ii) Root - **fixes and holds the tooth firmly** in the socket of the jaw bone
- *Marking note: 1 mark each (2 marks). Accept 'the part used for chewing' for the crown and 'anchors the tooth' for the root.*

(c) [1 marks]

- **32**

- *Marking note: 1 mark. Do NOT accept 20, which is the number of milk teeth.*

(d) [2 marks]

- Milk teeth are the **first set and are 20 in number**, while permanent teeth are the **second set and are 32 in number**.
- Milk teeth are **smaller and fall out (are replaced)**, while permanent teeth are **larger and stronger and are not replaced** when they are lost.
- *Marking note: 1 mark each for any TWO correct differences (2 marks). The answer must compare the two sets; a statement about only one of them earns no mark.*

(e) [1 marks]

- **Enamel**

- *Marking note: 1 mark. Do NOT accept 'bone' or 'dentine'.*

Question 2 [Total: 11 marks]**(a)** [6 marks]

- I - **enamel**
- II - **dentine**
- III - **pulp cavity**
- IV - **gum**

- V - **cement**
- VI - **nerve and blood vessels**
 - **Marking note:** 1 mark each (6 marks). The name must be matched to the correct number. Accept 'pulp' for III, 'gums' for IV and 'cementum' for V. For VI accept either 'nerve' or 'blood vessels' alone.*

(b) [1 marks]

- The enamel is the **hard outer covering that protects the crown** of the tooth from wear, from damage during chewing and from decay.
 - **Marking note:** 1 mark for protection or hardness of the crown.*

(c) [2 marks]

- It contains the **nerves**, which make the tooth **sensitive to pain, heat and cold** (1 mark).
- It contains the **blood vessels**, which **supply food and oxygen** to the living cells of the tooth and keep it alive (1 mark).
 - **Marking note:** 2 marks, 1 mark each. Both the nerve function and the blood supply must appear for full marks.*

(d) [2 marks]

- The **pulp cavity contains the nerves** of the tooth, while the enamel and dentine outside it have no nerves (1 mark).
- When the decay reaches the pulp the **nerves are exposed and irritated** by food, acid, heat and cold, and the pulp may become infected, so **severe pain (toothache) is felt** (1 mark).
 - **Marking note:** 2 marks, 1 mark each. The first mark is for locating the nerves in the pulp; the second is for the exposure or irritation of those nerves.*

Question 3 [Total: 11 marks]

(a) [4 marks]

- W - **incisor**
- X - **canine**
- Y - **premolar**
- Z - **molar**
 - **Marking note:** 1 mark each (4 marks). The name must be matched to the correct letter. Accept 'eye tooth' for the canine.*

(b) [2 marks]

- (i) W (incisor) - **cuts and bites** off pieces of food
- (ii) Z (molar) - **grinds and chews** the food into small pieces
 - **Marking note:** 1 mark each (2 marks). Do NOT accept 'chewing' for the incisor or 'cutting' for the molar.*

(c) [2 marks]

- **X** is pointed (has one sharp cusp) while **Z** is broad with a large, ridged grinding surface.
- **X** is used for tearing food, while **Z** is used for grinding and chewing it.
- **X** has one root and stands near the front of the mouth, while **Z** has two or more roots and stands at the back.
- *Marking note: 1 mark each for any TWO correct differences (2 marks). The answer must compare the two teeth; a statement about only one of them earns no mark.*

(d) [2 marks]

- **A dog** (also accept a cat, lion, leopard or any named carnivore) (1 mark).
- It is a **flesh eater (carnivore)**, so it needs long, pointed canines to **hold its prey and tear flesh from the bones** (1 mark).
- *Marking note: 2 marks: 1 mark for a correct carnivore and 1 mark for the reason. A herbivore such as a cow or goat earns no mark, since its canines are poorly developed or absent.*

(e) [1 marks]

- The **jaw presses most strongly at the back**, so the broad molars can **grind the food thoroughly** after it has been cut and torn at the front of the mouth.
- *Marking note: 1 mark for any sound reason linked to the force of the jaw or to the order of cutting, tearing and then grinding.*

Question 4 [Total: 10 marks]**(a) [2 marks]**

- Tooth decay is the **gradual breaking down or destruction of the hard parts of a tooth** (enamel and dentine) (1 mark)
- by the **acids produced by bacteria** acting on food left on the teeth, so that **holes (cavities) are formed** (1 mark).
- *Marking note: 2 marks, 1 mark each. Award 1 mark for 'holes in the teeth' alone. The acid or bacteria must appear for the second mark.*

(b) [3 marks]

- **Sugar from the sweets remains on the teeth** and, together with bacteria and saliva, forms a sticky layer called **plaque** (1 mark).
- The **bacteria in the plaque feed on the sugar and produce acid** (1 mark).
- The **acid dissolves the enamel and then the dentine**, forming a hole (cavity) which may finally reach the pulp cavity and cause toothache (1 mark).
- *Marking note: 3 marks, 1 mark each, and the steps must be in a sensible order. The words plaque and acid should both appear; award the marks for the correct science even where the term plaque is missing.*

(c) [2 marks]

- The gums are **red, swollen and painful** and **bleed easily** when the teeth are brushed.
- There is **bad breath**, the **gums shrink away from the teeth**, and the **teeth become loose** in their sockets.
- **Marking note:** 1 mark each for any TWO correct signs (2 marks).

(d) [3 marks]

- The pain **prevents proper chewing**, so the pupil **eats poorly** and may become weak and undernourished (1 mark).
- The pupil **misses school or cannot concentrate** in class because of the toothache, so performance falls (1 mark).
- **Money is spent on treatment**, and **bad breath or missing front teeth** may make the pupil shy and lose confidence among friends (1 mark).
- **Marking note:** 3 marks, 1 mark each for any three sound effects. Accept loss of sleep. Do NOT award two marks for the same idea stated twice.

Question 5 [Total: 9 marks]**(a) [4 marks]**

- **Brush the teeth properly at least twice a day**, especially last thing before going to bed.
- **Reduce the eating of sweets, toffees and sugary drinks**, particularly between meals, and **rinse the mouth with clean water after eating**.
- **Eat foods rich in calcium** (milk, small fish eaten with the bones) and plenty of **fresh fruits and vegetables**.
- **Visit the dentist regularly** (at least twice a year) for check-ups, and **avoid using the teeth to open bottle tops or crack hard objects**.
- **Marking note:** 1 mark each for any FOUR correct ways (4 marks). Do NOT accept 'take medici

(b) [3 marks]

- **Wet the brush and put a little toothpaste on it** (or chew the end of the chewing stick until it is soft and brush-like) (1 mark).
- **Brush all the surfaces** - the outer, the inner and the biting surfaces - using **short up-and-down (and circular) strokes, brushing away from the gum**, not only across the front teeth (1 mark).
- **Brush for about two minutes**, also **clean the tongue**, then **rinse the mouth with clean water** and rinse the brush; replace the brush when the bristles spread (1 mark).
- **Marking note:** 3 marks, 1 mark each. This is the 'demonstrate proper ways of cleani

(c) [2 marks]

- **Milk** (and other dairy foods such as cheese and yoghurt)
- **Small fishes eaten with their bones**, eggs, beans, green leafy vegetables such as kontomire, and fresh fruits
- **Marking note:** 1 mark each for any TWO correct foods (2 marks). The food should be a source of calcium, vitamins or phosphorus. Do NOT accept sweets or soft drinks.*

Question 6 [Total: 10 marks]**(a) [4 marks]**

- **'Brush your teeth twice every day - and never miss bedtime.'**
- **'Cut down on toffees and fizzy drinks; rinse your mouth after eating.'**
- **'Eat milk, small fish and fruits for strong teeth.'**
- **'Visit the dentist twice a year; never use your teeth to open bottles.'** (with a clear drawing of a tooth or a brushing action)
- **Marking note:** 1 mark each for any FOUR sensible points (4 marks). This is a design task; award method marks for any message that would really help a Basic 4 pupil care for the teeth. A drawing or slogan may earn one of the marks.*

(b) [2 marks]

- A **toothbrush with toothpaste**
- A **chewing stick (sokodua)**; also accept **salt with clean water** or dental floss
- **Marking note:** 1 mark each for any TWO materials genuinely available locally (2 marks). The chewing stick is widely used in Ghana and must be accepted.*

(c) [3 marks]

- **Put a clean egg shell into a cup containing the fizzy drink**, and an identical shell into a cup of **clean water as a control** (1 mark).
- **Leave both cups for a day or two**, then remove the shells and compare them (1 mark).
- The shell from the fizzy drink is **soft, rough, stained and partly dissolved**, while the one from the water is unchanged - showing that the **acid in the drink attacks the hard material of the tooth (the enamel)** (1 mark).
- **Marking note:** 3 marks, 1 mark each. This is a design task, so award method marks for any workable procedure. The control (shell in wat

(d) [1 marks]

- **Do not allow the pupils to drink the liquid or taste the egg shells**, and **wash the hands with soap** after handling them (keep the cups covered and labelled).
- **Marking note:** 1 mark for any one sensible precaution. Do NOT accept 'be careful' on its own.*

CURRICULUM ASSESSMENT BLUEPRINT

****Strand:**** 3: Systems | ****Sub-strand:**** 1: The Human Body System | ****Class:**** Basic 8 (JHS 2)

****Content Standard:**** ****B8/JHS2.3.1.1**** - Demonstrate knowledge of parts of the mammalian tooth and the functions of the different types of teeth in relation to feeding in man.

> Source: NaCCA Sci

Human Health

ANSWER KEY**SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	A	20.1 Communicable Diseases and How They Sp...	Remember
2	A	20.1 Communicable Diseases and How They Sp...	Remember
3	C	20.1 Communicable Diseases and How They Sp...	Understand
4	D	20.1 Communicable Diseases and How They Sp...	Remember
5	A	20.1 Communicable Diseases and How They Sp...	Understand
6	D	20.1 Communicable Diseases and How They Sp...	Understand
7	C	20.1 Communicable Diseases and How They Sp...	Understand
8	B	20.2 Hepatitis B: Cause, Symptoms, Effects...	Remember
9	B	20.2 Hepatitis B: Cause, Symptoms, Effects...	Remember
10	C	20.2 Hepatitis B: Cause, Symptoms, Effects...	Remember
11	B	20.2 Hepatitis B: Cause, Symptoms, Effects...	Remember
12	A	20.2 Hepatitis B: Cause, Symptoms, Effects...	Understand
13	C	20.2 Hepatitis B: Cause, Symptoms, Effects...	Understand
14	B	20.2 Hepatitis B: Cause, Symptoms, Effects...	Understand
15	A	20.2 Hepatitis B: Cause, Symptoms, Effects...	Understand
16	A	20.2 Hepatitis B: Cause, Symptoms, Effects...	Understand
17	D	20.2 Hepatitis B: Cause, Symptoms, Effects...	Understand
18	B	20.2 Hepatitis B: Cause, Symptoms, Effects...	Apply
19	D	20.2 Hepatitis B: Cause, Symptoms, Effects...	Understand
20	B	20.3 HIV and AIDS	Remember
21	B	20.3 HIV and AIDS	Understand
22	C	20.3 HIV and AIDS	Understand
23	C	20.3 HIV and AIDS	Understand
24	D	20.3 HIV and AIDS	Analyse
25	D	20.3 HIV and AIDS	Evaluate
26	A	20.4 Measles and Other Common Communicable...	Remember
27	D	20.4 Measles and Other Common Communicable...	Understand
28	C	20.4 Measles and Other Common Communicable...	Understand
29	A	20.4 Measles and Other Common Communicable...	Understand
30	D	20.4 Measles and Other Common Communicable...	Understand
31	A	20.4 Measles and Other Common Communicable...	Apply
32	C	20.4 Measles and Other Common Communicable...	Remember
33	B	20.1 Communicable Diseases and How They Sp...	Apply
34	C	20.1 Communicable Diseases and How They Sp...	Understand
35	C	20.4 Measles and Other Common Communicable...	Analyse
36	A	20.5 Health Data and Planning to Minimise ...	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	D	20.5 Health Data and Planning to Minimise ...	Evaluate
38	B	20.5 Health Data and Planning to Minimise ...	Apply
39	D	20.5 Health Data and Planning to Minimise ...	Evaluate
40	B	20.5 Health Data and Planning to Minimise ...	Create

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 2 (5%) | Apply: 5 (12%) | Create: 1 (2%) | Evaluate: 3 (7%) | Remember: 10 (25%) | Understand: 19 (47%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 9 marks]

(a) [2 marks]

- A **communicable disease** is caused by a germ and can be passed from one person to another, directly or through food, water, air or a vector (1 mark).
- A **non-communicable disease** cannot be passed from person to person; it comes from the person's own body, diet or way of life, for example diabetes, hypertension or sickle-cell disease (1 mark).
- **Marking note:** 2 marks, 1 mark each. Accept 'infectious disease' for communicable.*

(b) [4 marks]

- **Through the air, in droplets from coughing and sneezing** - **measles** (accept tuberculosis, influenza) (1 mark).
- **Through contaminated food and water** - **cholera** (accept typhoid fever, dysentery) (1 mark).
- **Through infected blood and body fluids, including shared blades and needles** - **hepatitis B** (accept HIV) (1 mark).
- **Through the bite of a vector** - **malaria**, carried by the female Anopheles mosquito (accept mother-to-child transmission, or direct skin contact with ringworm or scabies) (1 mark).
- **Marking note:** 1 mark each for any FOUR routes, each with a correct answer.*

(c) [3 marks]

- **Wash the hands with soap under running water** before eating and after using the toilet, and **keep the toilets, the canteen and the compound clean with safe drinking water**.
- **Cover the mouth and the nose when coughing or sneezing**, keep the classrooms well aired, and **do not share cups, blades, needles or toothbrushes**.
- **Immunise the pupils, keep a sick pupil away from the others until he or she is well, and teach the pupils about the diseases** at assembly.
- **Marking note:** 1 mark each for any THREE sound measures (3 marks).*

Question 2 [Total: 12 marks]

(a) [1 marks]

- **Hepatitis B**, with 40 cases in all.
- **Marking note:** 1 mark.*

(b) [2 marks]

- **$24 + 16$** (1 mark)
- **$= 40$ cases** (1 mark)
- *Marking note: 2 marks: 1 mark for the addition of the male and female values and 1 mark for the answer.*

(c) [2 marks]

- **$40 \div 100 \times 100$** (1 mark)
- **$= 40\%$** (1 mark)
- *Marking note: 2 marks: 1 mark for the method and 1 mark for the answer with the per cent sign. Allow follow-through from a wrong total in (b).*

(d) [2 marks]

- **$24 - 16$** (1 mark)
- **$= 8$ more males than females** (1 mark)
- *Marking note: 2 marks: 1 mark for the subtraction and 1 mark for the answer.*

(e) [3 marks]

- **More men share razor blades, clippers and other sharp instruments** at barbering shops and during shaving, tattooing or tribal marking.
- **More men work where they meet blood or sharp instruments**, or may **have more sexual partners**, so they are exposed more often.
- **Women attend antenatal and child welfare clinics**, where they are screened, treated and advised earlier, so fewer of them come to be recorded as cases (accept: men report late to hospital, so their cases are recorded as full disease).
- *Marking note: 1 mark each for any THREE reasonable suggestions (3 marks). These a

(f) [2 marks]

- To **see which diseases are commonest and which group is most at risk**, and so **plan health education, screening and immunisation** for them.
- To **order the right drugs, vaccines and test kits**, to **compare one year with the next** to see whether the measures are working, and to report to the district health directorate for support.
- *Marking note: 1 mark each for any TWO sound uses (2 marks).*

Question 3 [Total: 10 marks]**(a)** [2 marks]

- The germ is the **hepatitis B virus** (1 mark).
- It attacks the **liver** (1 mark).
- *Marking note: 2 marks, 1 mark each. 'A virus' alone earns the first mark; 'bacterium' earns nothing.*

(b) [3 marks]

- Through **contact with the blood or body fluids of an infected person**, including **shared razor blades, needles, syringes, clippers and toothbrushes**.
- Through **transfusion with unscreened infected blood**, or the use of unsterilised instruments for injections, tattooing or tribal marks.
- Through **unprotected sexual intercourse with an infected person**, and **from an infected mother to her baby during pregnancy or at birth**.
- **Marking note: 1 mark each for any THREE correct routes (3 marks). Do not award a mark for routes that do not spread hepatitis B, such as sharing a desk, sh**

(c) [3 marks]

- **Yellowing of the eyes and the skin (jaundice)** and **dark, cola-coloured urine**.
- **Tiredness and general weakness**, with **loss of appetite, nausea and vomiting**.
- **Fever** and **pain in the abdomen**, especially on the right side below the ribs.
- **Marking note: 1 mark each for any THREE correct symptoms (3 marks). Note that many people carry the virus with no symptoms at all; a candidate who says so in addition is not penalised.**

(d) [2 marks]

- **On the patient:** the **liver is damaged**, which may lead to cirrhosis, liver cancer and even death; the patient becomes weak, misses school or work, and may remain a carrier for life.
- **On the family:** **money is spent on treatment and the family loses the patient's work and income**; other members of the household may be infected, and the family suffers worry and sometimes stigma.
- **Marking note: 1 mark each for any TWO effects (2 marks).**

Question 4 [Total: 10 marks]**(a) [1 marks]**

- **The liver**.
- **Marking note: 1 mark.**

(b) [2 marks]

- The condition is **jaundice** (1 mark).
- The **damaged liver can no longer remove the yellow bile pigment from the blood**, so it **collects in the blood and stains the eyes and the skin yellow** (1 mark).
- **Marking note: 2 marks: 1 mark for the name and 1 mark for the explanation. Accept 'the liver is not working properly, so the yellow colouring stays in the**

blood'.*

(c) [3 marks]

- **Immunisation with the hepatitis B vaccine** - the surest protection.
- **Never sharing razor blades, needles, syringes, clippers or toothbrushes**, and insisting on **sterile instruments** for injections, tattooing and tribal marks.
- **Using only screened blood for transfusion**, **avoiding unprotected sexual intercourse**, and **screening pregnant women** so that their babies are protected at birth.
- **Marking note:** 1 mark each for any THREE correct preventive measures (3 marks).*

(d) [2 marks]

- The vaccine makes the **body produce antibodies**, so the person is protected before ever meeting the virus, and it is **far cheaper and safer than treating the disease**.
- It **prevents the liver damage, cirrhosis and liver cancer** that can follow, and when **many people are vaccinated there are fewer carriers**, so the disease spreads less in the community.
- **Marking note:** 1 mark each for any TWO sound reasons (2 marks).*

(e) [2 marks]

- **Treat the classmate with kindness and respect, keep the matter confidential and continue to study, play and eat together**, since hepatitis B is not spread by ordinary contact (1 mark).
- **Encourage him or her to attend the clinic and take the treatment prescribed**, make sure **nobody shares blades, needles or toothbrushes**, and get themselves **vaccinated and tested** (1 mark).
- **Marking note:** 2 marks, 1 mark each. Any answer that suggests sending the classmate away, announcing the result publicly or refusing to sit with the pupil scores 0 for that point. Stigma is a learning outcome

Question 5 [Total: 9 marks]

(a) [2 marks]

- **HIV - human immunodeficiency virus** (1 mark).
- **AIDS - acquired immune deficiency syndrome** (1 mark).
- **Marking note:** 2 marks, 1 mark each. Accept 'immuno-deficiency' written as one or two words.*

(b) [3 marks]

- Through **unprotected sexual intercourse with an infected person**.
- Through **transfusion with unscreened infected blood**, or by **sharing needles, syringes, razor blades and other skin-piercing instruments**.
- From an **infected mother to her child during pregnancy, at birth or through**

breast milk**.

- *Marking note: 1 mark each for any THREE correct routes (3 marks).*

(c) [2 marks]

- By **shaking hands, hugging, playing together or sharing a desk, clothes, books or a toilet**.

- By **eating food prepared or shared with an infected person**, by **coughing, sneezing or breathing the same air**, or by **mosquito and other insect bites**.

- *Marking note: 1 mark each for any TWO correct answers (2 marks). This part is included to correct the fear and stigma that surround HIV.*

(d) [2 marks]

- The child suffers **high fever, a rash over the body, red watery eyes, a runny nose and a cough**, and becomes **weak and off food**.

- Measles may lead to **blindness, ear infection, severe diarrhoea, pneumonia, malnutrition and even death**, and the child misses school for a long time.

- *Marking note: 1 mark each for any TWO effects (2 marks).*

Question 6 [Total: 10 marks]

(a) [4 marks]

- **Find out how common the disease is** from the records of the health centre, and set a clear target for the plan (1 mark).

- **Organise health education** at assembly, in the churches and mosques, at the market and on the information centre or local radio, using posters, drama and the class presentation (1 mark).

- **Arrange a screening and vaccination day with the health centre** for pupils and community members, and encourage pregnant women to attend antenatal clinic (1 mark).

- **Campaign against the sharing of blades, needles, clippers and toothbrushes**, and teach the barbers and haird

(b) [2 marks]

- **The nurse or community health officer at the health centre** - to **screen, vaccinate, treat and give correct information** (1 mark).

- **The chief, the assembly member, religious leaders or the P.T.A.** - to **call the community together and support the message** (accept the barbers and hairdressers, who must change their practice) (1 mark).

- *Marking note: 1 mark each for any TWO people or groups, each with how they would help (2 marks). A name with no role scores nothing.*

(c) [2 marks]

- Two sources: **the health centre and its nurses**, and **the Ghana Health Service, the textbook, the school library or the internet** (1 mark for any two).

- Checking: **compare the information with a second source, show it to the nurse or the teacher, and use only material from known health bodies, checking how recent it is** (1 mark).

- **Marking note:** 2 marks: 1 mark for naming two sources and 1 mark for a sound way of checking. This part carries the digital-literacy competency.*

(d) [2 marks]

- **Compare the number of hepatitis B cases recorded at the health centre before the plan and after six months or a year**, and **count how many people were screened and vaccinated** (1 mark).

- **Check whether the barbers and hairdressers now use a new blade for each customer**, and **test what people know with a short questionnaire before and after the education** (1 mark).

- **Marking note:** 1 mark each for any TWO ways that can actually be measured or compared (2 marks). A general hope that 'people will be healthy' scores 0.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 5: Humans

Life Cycle Anoph

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	6.1 The Anopheles Mosquito and Malaria	Remember
2	B	6.1 The Anopheles Mosquito and Malaria	Remember
3	D	6.1 The Anopheles Mosquito and Malaria	Remember
4	A	6.3 The Adult Mosquito: Parts and Feeding	Understand
5	B	6.2 The Life Cycle of the Anopheles Mosquito	Remember
6	C	6.2 The Life Cycle of the Anopheles Mosquito	Remember
7	B	6.2 The Life Cycle of the Anopheles Mosquito	Remember
8	B	6.2 The Life Cycle of the Anopheles Mosquito	Remember
9	C	6.2 The Life Cycle of the Anopheles Mosquito	Remember
10	B	6.2 The Life Cycle of the Anopheles Mosquito	Remember
11	A	6.2 The Life Cycle of the Anopheles Mosquito	Understand
12	C	6.2 The Life Cycle of the Anopheles Mosquito	Remember
13	D	6.2 The Life Cycle of the Anopheles Mosquito	Understand
14	B	6.6 Management and Control of the Mosquito	Apply
15	A	6.6 Management and Control of the Mosquito	Apply
16	C	6.6 Management and Control of the Mosquito	Evaluate
17	D	6.6 Management and Control of the Mosquito	Apply
18	D	6.6 Management and Control of the Mosquito	Remember
19	C	6.3 The Adult Mosquito: Parts and Feeding	Understand
20	B	6.4 Economic Importance of the Mosquito	Understand
21	C	6.5 Impact of the Mosquito on Humans	Understand
22	C	6.5 Impact of the Mosquito on Humans	Remember
23	D	6.3 The Adult Mosquito: Parts and Feeding	Remember
24	A	6.3 The Adult Mosquito: Parts and Feeding	Understand
25	A	6.3 The Adult Mosquito: Parts and Feeding	Understand
26	C	6.3 The Adult Mosquito: Parts and Feeding	Understand
27	A	6.3 The Adult Mosquito: Parts and Feeding	Analyse
28	B	6.2 The Life Cycle of the Anopheles Mosquito	Understand
29	D	6.2 The Life Cycle of the Anopheles Mosquito	Remember
30	D	6.2 The Life Cycle of the Anopheles Mosquito	Apply
31	C	6.6 Management and Control of the Mosquito	Apply
32	A	6.6 Management and Control of the Mosquito	Evaluate
33	D	6.6 Management and Control of the Mosquito	Understand
34	D	6.2 The Life Cycle of the Anopheles Mosquito	Analyse
35	B	6.6 Management and Control of the Mosquito	Analyse
36	B	6.1 The Anopheles Mosquito and Malaria	Understand

Q	Ans	Curriculum Area	Cognitive Level
37	A	6.1 The Anopheles Mosquito and Malaria	Understand
38	A	6.5 Impact of the Mosquito on Humans	Analyse
39	C	6.6 Management and Control of the Mosquito	Evaluate
40	A	6.5 Impact of the Mosquito on Humans	Understand

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 4 (10%) | Apply: 5 (12%) | Evaluate: 3 (7%) | Remember: 14 (35%) | Understand: 14 (35%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 8 marks]****(a)** [2 marks]

- A vector is an **organism** (insect or other animal) that carries a disease-causing organism from one host to another.
- The vector itself may not become sick; it only carries the germ to the person or animal it bites.
- *Marking note: 1 mark each for any TWO correct points (2 marks). Award 1 mark for 'an insect that carries disease' if it is not developed further. Do NOT accept 'an animal that causes disease' on its own - the vector carries the germ, it does not make the germ.**

(b) [4 marks]

- **Egg**
- **Larva** (wiggler)
- **Pupa**
- **Adult** (imago)
- *Marking note: 1 mark each for all FOUR stages in the correct order (4 marks). The order must be correct for full marks; an order that puts the pupa before the larva earns a maximum of 2 marks and a comment.**

(c) [2 marks]

- **Water:** the egg, the larva and the pupa
- **Land/air:** the adult mosquito
- *Marking note: 1 mark for naming the water stages and 1 mark for the adult stage (2 marks). Accept 'the young stages (egg, larva, pupa) live in water'.**

Question 2 [Total: 9 marks]**(a)** [4 marks]

- (i) **eggs** (laid on the surface of water)
- (ii) **larva** (wiggler)
- (iii) **pupa**
- (iv) **adult mosquito**
- *Marking note: 1 mark each (4 marks). The stage must be matched to the correct number. Accept 'wiggler' for (ii) and 'imago' for (iv).**

(b) [1 marks]

- **Complete metamorphosis**
 - **Marking note:** 1 mark. Do NOT accept 'incomplete metamorphosis' or 'metamorphosis' alone.*

(c) [2 marks]

- The **larva lives in water** and has **no wings**; the **adult lives on land** and **has wings**.
- The larva **breathes through a tube** at the water surface and feeds on tiny water plants or animals; the adult feeds on **nectar (male) or blood (female)**.
- The larva **crawls (wriggles)** while the adult **flies**.
 - **Marking note:** 1 mark each for any TWO correct differences (2 marks). The difference must be a comparison between (ii) and (iv); a statement about only one of them earns no mark.*

(d) [2 marks]

- The **young stages (egg, larva and pupa)** can only develop in **water** - they need water to live (1 mark).
- The larva and pupa **breathe air at the surface** of the water, so the eggs must be laid where there is water (1 mark).
 - **Marking note:** 2 marks, 1 mark each. Accept 'the young ones cannot live outside water' for the first mark and 'they need water to breathe and grow' for the second.*

Question 3 [Total: 10 marks]**(a)** [4 marks]

- I - **proboscis**
- II - **antenna** (antennae)
- III - **wings**
- IV - **abdomen**
 - **Marking note:** 1 mark each (4 marks). The label must be matched to the correct part. Do NOT accept 'nose' for I or 'tail' for IV.*

(b) [2 marks]

- The proboscis **pierces the skin** and is used for **sucking blood** (1 mark).
- In the female Anopheles it also **passes the plasmodium (the malaria germ)** into the **blood** of the person she bites (1 mark).
 - **Marking note:** 2 marks, 1 mark each. Award the first mark for piercing/sucking; the second mark is for the link with malaria. Accept 'it is the mouth part of the mosquito' only with piercing or sucking.*

(c) [2 marks]

- Only the **female** bites human beings and takes blood (1 mark).
- She needs the blood for the **development of her eggs**, and the **plasmodium** in her body is passed into the person's blood when she bites (1 mark).
- **Marking note:** 2 marks, 1 mark each. Do NOT award a mark for 'the male has no proboscis' - that statement is wrong.*

(d) [1 marks]

- It rests with its **body at an angle (sloping) to the surface**, not flat on it.
- **Marking note:** 1 mark. The key idea is the angled (tilted) resting posture; accept a correct drawing that shows the tilted body.*

(e) [1 marks]

- The **adult has wings and flies (lives on land)**; the larva has **no wings, lives in water and crawls**.
- **Marking note:** 1 mark for any one correct difference. Do NOT accept 'the adult is bigger' on its own.*

Question 4 [Total: 12 marks]**(a) [2 marks]**

- The **economic importance** of an organism is the way it **affects people's money, work and wealth** (whether it helps or harms them).
- It is the **value** of the organism to humans: what it costs or gives in money, work and health, and to farming and industry.
- **Marking note:** 1 mark each for any TWO correct points (2 marks). Award 1 mark for 'how it is useful or harmful to people'. Do NOT accept 'how many there are'.*

(b) [2 marks]

- Mosquito **larvae and adults are food** for fishes, birds, bats and other animals in the **food chain**.
- **Marking note:** 1 mark for each correct point, maximum 2 marks. Accept 'they are part of the food chain' with an example. Do NOT accept 'they make honey' or 'they add nutrients to the soil'.*

(c) [3 marks]

- They **spread malaria** (the female Anopheles carries the plasmodium).
- Their **bites are painful and disturb sleep**, and they are a **nuisance** at home.
- Time and money are **lost through sickness and treatment** - people miss work, school and farming.
- **Marking note:** 1 mark each for any THREE correct points (3 marks). Do NOT accept 'they are bad' or 'they bite' without a stated harm.*

(d) [3 marks]

- The pupil ****stays away from school**** while sick, so lessons are ****missed**** (1 mark).
- Weakness and fever make ****concentration hard****, so performance in class and in examinations ****falls**** (1 mark).
- Money is spent on ****treatment and medicine**** instead of on school needs such as books or fees; repeated sickness may even make the pupil ****drop out**** (1 mark).
- **Marking note: 3 marks, 1 mark each for any three sound points. Look for the loss of lessons, the effect on performance, and the cost or drop-out risk.**

(e) [2 marks]

- ****Workers and farmers stay away from work and farms****, so less is produced and national income falls.
- ****Large sums are spent on treating and preventing the disease**** (drugs, nets, hospital care), money that could build schools, roads and hospitals.
- **Marking note: 1 mark each for any TWO correct points (2 marks). The answer must name an effect on work, production or money.**

Question 5 [Total: 10 marks]**(a) [2 marks]**

- The stages found in water are the ****egg, the larva and the pupa**** (1 mark).
- They are ****concentrated in the water**** and ****cannot fly away****, so they can be killed or poured away easily - unlike the adult mosquitoes that fly about (1 mark).
- **Marking note: 2 marks, 1 mark each. Accept 'the young stages live in water' for the first mark.**

(b) [4 marks]

- ****Drain or pour away stagnant water**** and fill up the hollows where water collects.
- ****Cover all water storage containers**** (barrels, pots, tanks) so that mosquitoes cannot lay eggs in them.
- ****Clear the bushes and weeds**** around the house, and ****cut the grass**** short.
- ****Spray the rooms with insecticide**** (or use mosquito coils) and fix ****wire mesh**** on the windows.
- ****Sleep under a treated mosquito net****, and visit the hospital early when you feel feverish.
- Put ****fishes (such as tilapia)**** in ponds and wells to eat the larvae.
- **Marking note: 1 mark each for any FOUR correct methods.**

(c) [2 marks]

- Certain ****fishes (for example tilapia) are put into the pond**** and they ****feed on the mosquito larvae and pupae**** in the water (1 mark).
- This reduces the number of larvae that develop into adult mosquitoes, so fewer

mosquitoes bite people (1 mark).

- *Marking note: 2 marks, 1 mark each. The method is called **biological control**; accept the term with the correct explanation.*

(d) [2 marks]

- Prevention is **cheaper and safer** than cure - treating malaria costs money and the illness can become serious, causing **death or long absence from work and school** (1 mark).

- When mosquitoes are controlled, **the whole family stays healthy**, so no money or time is lost; the measures (net, clearing water, spraying) protect everyone, not only the sick person (1 mark).

- *Marking note: 2 marks, 1 mark each for any two sound points. Accept 'prevention is better than cure because you do not fall sick at all' if supported with a reason such as cost, time lost or danger of death.*

Question 6 [Total: 11 marks]

(a) [3 marks]

- **Walk round** the school compound and the nearby houses **looking for water that is standing still** (drains, pits, open barrels, wells, old tyres, bottles, broken pots).

- **Check each place carefully** (using a spoon or a small net) to see whether there are **wrigglers (larvae) or pupae** in the water.

- **Record the findings** in a table or a map of the compound: where the water is, what container holds it, and whether young mosquitoes were found.

- *Marking note: 1 mark each for any THREE sensible steps (3 marks). This is a design task, so award method marks for any practical step that

(b) [3 marks]

- **Collect eggs or wrigglers** from a pond or gutter with a little water, put them in a **glass jar** and cover the jar with a piece of cloth (to keep the adults in).

- **Observe the jar every day**, and **draw and label** each stage as it appears (egg, larva, pupa, adult), recording the changes in a table with dates.

- Alternatively, **draw and label the four stages on a chart** and arrange them in a circle with arrows to show the cycle, and present it to the class.

- *Marking note: 3 marks: 1 mark for the breeding/observation method, 1 mark for drawing and labelling each stage, 1 mark for

(c) [3 marks]

- A **drawing of the four stages** of the life cycle with their names.

- The **breeding places** (stagnant water, open barrels, old tyres, drains) shown crossed out or marked with a big X.

- The **control methods** in pictures: pouring away water, covering containers, sleeping under a net, spraying, clearing bushes.

- A **short clear message and a title** (for example: "No stagnant water, no mosquitoes, no malaria!").

- **Marking note:** 1 mark each for any THREE sensible features (3 marks). This is a design task, so award method marks for any feature that would really teach someone how to cont

(d) [2 marks]

- **Cover the breeding jar always** so that adult mosquitoes cannot escape and bite the class.

- **Wash hands with soap after handling pond water** or the young stages, and **do not drink or taste** the water; wear a long-sleeved shirt when going round the compound.

- **Marking note:** 1 mark each for any TWO sensible precautions (2 marks). The key idea is protection from bites and from dirty water. Do NOT accept 'do not fall in the water' on its own.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 2: Cycles | **Sub-strand:** 2: Life Cycle of Organisms | **Class:** Basic 8 (JHS 2)

Con

Living Cells

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	3.1 The Cell, Its Parts and Their Functions	Remember
2	B	3.1 The Cell, Its Parts and Their Functions	Understand
3	A	3.1 The Cell, Its Parts and Their Functions	Remember
4	A	3.1 The Cell, Its Parts and Their Functions	Remember
5	D	3.1 The Cell, Its Parts and Their Functions	Understand
6	B	3.2 Plant Cells and Animal Cells	Remember
7	A	3.2 Plant Cells and Animal Cells	Understand
8	B	3.2 Plant Cells and Animal Cells	Remember
9	B	3.2 Plant Cells and Animal Cells	Remember
10	C	3.1 The Cell, Its Parts and Their Functions	Remember
11	B	3.3 Prokaryotic and Eukaryotic Cells	Apply
12	D	3.3 Prokaryotic and Eukaryotic Cells	Understand
13	C	3.3 Prokaryotic and Eukaryotic Cells	Apply
14	C	3.3 Prokaryotic and Eukaryotic Cells	Understand
15	D	3.5 Micro-Organisms, Health and Safety Mea...	Remember
16	C	3.5 Micro-Organisms, Health and Safety Mea...	Understand
17	D	3.5 Micro-Organisms, Health and Safety Mea...	Understand
18	C	3.5 Micro-Organisms, Health and Safety Mea...	Apply
19	D	3.1 The Cell, Its Parts and Their Functions	Remember
20	B	3.1 The Cell, Its Parts and Their Functions	Remember
21	C	3.2 Plant Cells and Animal Cells	Understand
22	A	3.2 Plant Cells and Animal Cells	Understand
23	C	3.1 The Cell, Its Parts and Their Functions	Understand
24	A	3.2 Plant Cells and Animal Cells	Understand
25	D	3.3 Prokaryotic and Eukaryotic Cells	Understand
26	C	3.3 Prokaryotic and Eukaryotic Cells	Understand
27	A	3.3 Prokaryotic and Eukaryotic Cells	Analyse
28	A	3.2 Plant Cells and Animal Cells	Remember
29	B	3.2 Plant Cells and Animal Cells	Understand
30	D	3.4 Classifying Organisms by Cell Type	Apply
31	A	3.3 Prokaryotic and Eukaryotic Cells	Analyse
32	B	3.3 Prokaryotic and Eukaryotic Cells	Understand
33	A	3.3 Prokaryotic and Eukaryotic Cells	Understand
34	C	3.3 Prokaryotic and Eukaryotic Cells	Apply
35	D	3.5 Micro-Organisms, Health and Safety Mea...	Understand
36	A	3.5 Micro-Organisms, Health and Safety Mea...	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	D	3.3 Prokaryotic and Eukaryotic Cells	Analyse
38	D	3.4 Classifying Organisms by Cell Type	Apply
39	B	3.1 The Cell, Its Parts and Their Functions	Apply
40	C	3.1 The Cell, Its Parts and Their Functions	Understand

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 3 (7%) | Apply: 8 (20%) | Remember: 11 (27%) | Understand: 18 (45%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 8 marks]

(a) [2 marks]

- A cell is the **basic** (smallest) unit of all living things.
- **All living things are made up of one or more cells.**
 - *Marking note: 1 mark each for any TWO correct points (2 marks). Award 1 mark for 'the smallest part of a living thing' if the idea of a unit of life is present. Do NOT accept 'a small thing' on its own.**

(b) [3 marks]

- (i) Nucleus - **controls all the activities of the cell** (and contains the genetic material).
- (ii) Cell membrane - **controls the movement of substances into and out of the cell**.
- (iii) Cytoplasm - the **living material** in which the other organelles are found; most cell activities take place there.
 - *Marking note: 1 mark each (3 marks). The function must be matched to the correct part; a correct statement attached to the wrong part earns no mark.**

(c) [3 marks]

- (i) **Chloroplast** (accept 'chloroplasts')
- (ii) **Cell wall**
- (iii) **Vacuole** (accept 'large central vacuole')
 - *Marking note: 1 mark each (3 marks). Do NOT accept 'cell membrane' for (ii) or 'cytoplasm' for (iii).**

Question 2 [Total: 8 marks]

(a) [3 marks]

- **Chloroplasts**
- **Cell wall**
- **Large central vacuole** (accept 'large vacuole')
 - *Marking note: 1 mark each for any THREE (3 marks). Accept 'plastids' for chloroplasts. Do NOT accept nucleus, cytoplasm or cell membrane - these are found in both cells.**

(b) [1 marks]

- **Centriole** (accept 'small vacuoles' or 'centrosome')
 - *Marking note: 1 mark. Do NOT accept 'mitochondria', which are found in both cells.**

(c) [2 marks]

- Both have a **cell membrane** and **cytoplasm**.
- Both have a **nucleus** and other organelles such as mitochondria and ribosomes.
- Both are **living** and carry out life processes such as respiration.
- **Marking note:** 1 mark each for any TWO correct similarities (2 marks). The similarity must be true of BOTH cells; a difference given as a similarity earns no mark.*

(d) [2 marks]

- It **stores cell sap** (water with dissolved sugars and salts).
- It helps to **keep the cell firm (turgid)** and therefore gives the plant support.
- **Marking note:** 1 mark each for any TWO correct functions (2 marks). Do NOT accept 'it makes food' - that is the work of chloroplasts.*

Question 3 [Total: 10 marks]**(a)** [4 marks]

- I - **cell wall**
- II - **cell membrane**
- III - **cytoplasm** (accept 'cytoplasm with ribosomes')
- IV - **nuclear material** (accept 'DNA', 'loop of DNA', 'genetic material')
- **Marking note:** 1 mark each (4 marks). The label must be matched to the correct part. Do NOT accept 'nucleus' for IV - a prokaryotic cell has no true nucleus.*

(b) [2 marks]

- I (cell wall): **gives the cell its fixed shape** and **protects** the cell (1 mark).
- II (cell membrane): **controls the movement of substances** into and out of the cell (1 mark).
- **Marking note:** 2 marks, 1 mark each. The function must be matched to the correct label.*

(c) [1 marks]

- **Prokaryotic organisms** (accept 'prokaryotes')
- **Marking note:** 1 mark for the correct group.*

(d) [3 marks]

- A bacterial cell has **no true nucleus** - its genetic material lies loose in the cytoplasm.
- It has **no membrane-bound organelles** such as mitochondria and chloroplasts.
- It has **no chloroplasts and no large central vacuole**, and some bacteria have a capsule and flagellum.
- **Marking note:** 1 mark each for any THREE correct differences (3 marks).

Accept 'it is much smaller' or 'it is unicellular' paired with a structural difference, but do not award a mark for size or 'unicellular' alone.*

Question 4 [Total: 10 marks]

(a) [3 marks]

- V - **nucleus**
- VI - **chloroplast**
- VII - **mitochondrion**
- *Marking note: 1 mark each (3 marks). Do NOT accept 'cell' for V or 'vacuole' for VII.*

(b) [1 marks]

- The chloroplast **traps light energy** (through chlorophyll) and uses it to **make food** during photosynthesis.
- *Marking note: 1 mark for either point, or for 'it makes food for the plant by photosynthesis'.*

(c) [2 marks]

- It has a **true nucleus** enclosed by a **nuclear membrane** (1 mark).
- It has **membrane-bound organelles** such as mitochondria, chloroplasts and a vacuole (1 mark).
- *Marking note: 2 marks, 1 mark each. Accept 'it has a nucleus' for the first mark only if the enclosed/true nucleus idea is clear.*

(d) [2 marks]

- Plants **make their own food** by photosynthesis, and the chloroplast (with chlorophyll) is needed to **trap light energy** (1 mark).
- Animals **feed on ready-made food**, so they have no need for chloroplasts (1 mark).
- *Marking note: 2 marks, 1 mark each. The answer must contrast how plants and animals obtain food.*

(e) [2 marks]

- **Water enters the cell** (through the partially permeable membrane) and the cell becomes **firm and swollen (turgid)** (1 mark).
- Explanation: water moves from where it is in plenty into the cell, and the **strong cell wall prevents the cell from bursting** (1 mark).
- *Marking note: 2 marks, 1 mark each. Accept 'the vacuole fills up with water and pushes on the cell wall' as the explanation. Do NOT accept 'the cell bursts' for a plant cell.*

Question 5**(a) [4 marks]**

- A **prokaryotic cell** is a cell with **no true nucleus**: its genetic material lies loose in the cytoplasm (1 mark), and it has **no membrane-bound organelles** (1 mark).
- A **eukaryotic cell** has a **true nucleus enclosed by a nuclear membrane** (1 mark) and **membrane-bound organelles** such as mitochondria and chloroplasts (1 mark).
- **Marking note:** 4 marks, 1 mark for each of the four ideas. Accept 'a bacterium is prokaryotic' for one mark only if the structural idea is given as well.*

(b) [6 marks]

- **Prokaryotic cell:** a bacterium; a blue-green alga.
- **Eukaryotic cell:** Amoeba; a mushroom; a human cheek cell; a maize leaf cell.
- **Marking note:** 1 mark for each organism placed in the correct column (6 marks). The columns must be correct: a bacterium and a blue-green alga are the ONLY prokaryotic organisms listed. Do NOT award a mark for an organism placed in the wrong column.*

(c) [2 marks]

- A prokaryotic cell has **no true nucleus**; a eukaryotic cell has a **true nucleus**.
- A prokaryotic cell has **no membrane-bound organelles**; a eukaryotic cell has **membrane-bound organelles**.
- Prokaryotes are usually **unicellular** small cells; eukaryotes may be unicellular or multicellular and are larger.
- **Marking note:** 1 mark each for any TWO correct differences (2 marks). The difference must be stated as a comparison.*

Question 6 [Total: 6 marks]**(a) [2 marks]**

- Any two of: **cholera, tuberculosis, typhoid fever, tetanus, whooping cough, leprosy** (also accept 'boils' or 'food poisoning').
- **Marking note:** 1 mark each for any TWO correct diseases (2 marks). Do NOT accept malaria, measles or sleeping sickness - these are not caused by bacteria.*

(b) [3 marks]

- **Wash your hands with soap** under running water, especially before eating and after visiting the toilet.
- **Boil drinking water** (or treat it) and **cover food** to keep flies away.
- **Keep the surroundings clean**, use a toilet properly and bury or dispose of waste well.
- **Cook food well** and wash fruits and vegetables before eating them.

- **Visit the hospital** for treatment when you are sick so that you do not spread the disease.

- **Marking note:** 1 mark each for any THREE sensible measures (3 marks). The measure must really prevent the spread of bacterial disease. Do NOT accept

(c) [1 marks]

- Bacteria are useful in **fermenting milk into yoghurt** (also accept: fermenting maize dough for kenkey; making cheese; producing some medicines; helping to decompose waste to form manure).

- **Marking note:** 1 mark for any one correct use. Do NOT accept 'they are used for food' on its own.*

Question 7 [Total: 6 marks]

(a) [2 marks]

- **Prokaryotic organisms** - made up of cells with **no true nucleus**.

- **Eukaryotic organisms** - made up of cells with a **true nucleus**.

- **Marking note:** 2 marks: 1 mark for naming the two groups and 1 mark for a correct description of at least one of them. Accept 'prokaryotes and eukaryotes'.*

(b) [2 marks]

- **Observe and list** the organisms (or pictures of them) to be studied.

- Look at the **type of cell** each organism is made of (with a microscope or from charts), **record the results in a table**, and place each organism in the correct group.

- **Marking note:** 1 mark each for any TWO sensible steps (2 marks). Look for observing, recording in a table, and using the cell type as the criterion.*

(c) [2 marks]

- A plant cell has a **cell wall**, **cell membrane** and **cytoplasm**.

- It also has a **nucleus**, **chloroplasts** and a **large central vacuole**.

- **Marking note:** 2 marks: 1 mark for any three correctly named parts, 1 mark for the remaining parts. Do NOT award a mark for parts that belong to animal cells only.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 1: Diversity of Matter | **Sub-strand:** 2: Living Cells | **Class:** Basic 8 (JHS 2)

Content Standard: **B8/JHS2.1.2.1** - Demonstrate an understanding of the types of cells and their structure in relation to different

Mixtures and Separation

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	1.1 Mixtures: Definition and Classification	Remember
2	A	1.1 Mixtures: Definition and Classification	Understand
3	C	1.1 Mixtures: Definition and Classification	Understand
4	D	1.1 Mixtures: Definition and Classification	Understand
5	C	1.2 Solutions, Colloids and Suspensions	Remember
6	A	1.2 Solutions, Colloids and Suspensions	Understand
7	B	1.2 Solutions, Colloids and Suspensions	Remember
8	D	1.2 Solutions, Colloids and Suspensions	Understand
9	C	1.3 Light Scattering in Mixtures (Tyndall ...	Understand
10	A	1.2 Solutions, Colloids and Suspensions	Analyse
11	B	1.1 Mixtures: Definition and Classification	Apply
12	D	1.2 Solutions, Colloids and Suspensions	Apply
13	C	1.4 Separation Methods: Filtration, Decant...	Remember
14	B	1.4 Separation Methods: Filtration, Decant...	Remember
15	A	1.4 Separation Methods: Filtration, Decant...	Remember
16	D	1.4 Separation Methods: Filtration, Decant...	Apply
17	A	1.4 Separation Methods: Filtration, Decant...	Apply
18	B	1.4 Separation Methods: Filtration, Decant...	Understand
19	D	1.4 Separation Methods: Filtration, Decant...	Understand
20	C	1.5 Separation Methods: Evaporation, Disti...	Remember
21	C	1.5 Separation Methods: Evaporation, Disti...	Analyse
22	A	1.5 Separation Methods: Evaporation, Disti...	Remember
23	D	1.5 Separation Methods: Evaporation, Disti...	Apply
24	B	1.5 Separation Methods: Evaporation, Disti...	Remember
25	A	1.4 Separation Methods: Filtration, Decant...	Understand
26	D	1.5 Separation Methods: Evaporation, Disti...	Remember
27	B	1.1 Mixtures: Definition and Classification	Understand
28	C	1.6 Choosing and Performing Separation Met...	Apply
29	B	1.3 Light Scattering in Mixtures (Tyndall ...	Analyse
30	A	1.6 Choosing and Performing Separation Met...	Apply
31	C	1.2 Solutions, Colloids and Suspensions	Apply
32	D	1.4 Separation Methods: Filtration, Decant...	Apply
33	B	1.1 Mixtures: Definition and Classification	Understand
34	D	1.6 Choosing and Performing Separation Met...	Evaluate
35	A	1.4 Separation Methods: Filtration, Decant...	Remember
36	C	1.4 Separation Methods: Filtration, Decant...	Apply
37	A	1.2 Solutions, Colloids and Suspensions	Understand

Q	Ans	Curriculum Area	Cognitive Level
38	B	1.2 Solutions, Colloids and Suspensions	Understand
39	C	1.2 Solutions, Colloids and Suspensions	Analyse
40	D	1.5 Separation Methods: Evaporation, Disti...	Remember

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 4 (10%) | Apply: 10 (25%) | Evaluate: 1 (2%) | Remember: 12 (30%) | Understand: 13 (32%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 12 marks]****(a) [2 marks]**

- A mixture is a combination of **two or more substances** that are **not chemically joined together**.
- The substances in a mixture **keep their own properties** and can be **separated by physical means** (filtration, evaporation, sieving, etc.).
- The substances of a mixture can be mixed in **any ratio**.
- **Marking note:** 1 mark each for any TWO correct points (2 marks). Award 1 mark for 'two or more substances mixed together' only if the idea that they are not chemically combined is present. Do NOT accept 'two things put together' on its own.*

(b) [2 marks]

- In a **mixture** the substances are **not chemically joined**; in a **compound** the elements are **chemically joined**.
- A mixture can be **separated by physical means**; a compound can only be separated by **chemical means**.
- The parts of a mixture **keep their own properties**; a compound has **new properties** different from its elements.
- The parts of a mixture can be mixed in **any ratio**; the elements of a compound combine in a **fixed ratio**.
- **Marking note:** 1 mark each for any TWO correct differences (2 marks). The difference must be stated as a comparison, not as a description.*

(c) [4 marks]

- (i) air - **homogeneous**
- (ii) sand and water - **heterogeneous**
- (iii) salt solution - **homogeneous**
- (iv) groundnut and rice mixture - **heterogeneous**
- **Marking note:** 1 mark each (4 marks). The wording and the letter of the mixture must be matched correctly before the mark is given.*

(d) [4 marks]

- (i) air - **fractional distillation** (accept 'cooling the air', accept 'distillation')
- (ii) sand and water - **filtration** (accept 'decantation' or 'sedimentation')
- (iii) salt solution - **evaporation** (accept 'distillation' or 'crystallisation')
- (iv) groundnut and rice mixture - **sieving** (accept 'hand-picking')
- **Marking note:** 1 mark each (4 marks). Accept any method that is scientifically suitable for the mixture named, even if it is not the commonest one.*

Question**(a)** [6 marks]

- (i) **Medium-sized** particles (between about 1 nm and 1000 nm)
 - (ii) **Does not settle** (remains permanently mixed)
 - (iii) **Does not settle**
 - (iv) **Settles on standing** (particles settle at the bottom)
 - (v) **Does not scatter** light
 - (vi) **Scatters** light
- **Marking note:** 1 mark each for all SIX blanks correctly filled (6 marks). The blank number must be matched to the correct answer. Accept clear equivalents, e.g. 'intermediate size' for (i) and 'never settles' for (ii).*

(b) [1 marks]

- Solution particles are **smaller than 1 nm** (about 1 nm or less).
- **Marking note:** 1 mark. Accept 'less than 1 nm' or 'about 1 nanometre'.*

(c) [3 marks]

- (i) Solution: salt solution, sugar solution, sea water, soft drink, vinegar, air, brass.
 - (ii) Colloid: milk, blood, fog, smoke, starch solution, jelly, paint, mayonnaise, Milo drink.
 - (iii) Suspension: muddy water, chalk and water, sand and water, tomato juice, kerosene and water.
- **Marking note:** 1 mark each for any TWO correct examples of each group (3 marks). Do NOT give a mark for an example repeated from another part of the question. Accept any correct example from the local environment.*

(d) [4 marks]

- (i) **29 g** of sugar
 - (ii) The mass of sugar that dissolves **increases** as the temperature of the water rises (the higher the temperature, the more sugar dissolves).
 - (iii) **No**, not all the sugar will dissolve.
 - (iii) Reason: only **36 g** of sugar can dissolve in 100 cm³ of water at 80 °C, so the remaining **4 g** stays at the bottom as undissolved sugar.
- **Marking note:** (i) 1 mark; (ii) 1 mark; (iii) 1 mark for the correct decision and 1 mark for the reason (4 marks in all). In (iii) do not award the reason mark for repeating the question; the number 36 g or 4 g must appear.*

Question 3 [Total: 10 marks]

(a) [3 marks]

- A: the path of the light is **not visible** - the light passes straight through.
 - B: the **path of the light is clearly visible** inside the liquid as a bright track.
 - C: the path of the light is **not clearly visible** - the light is blocked and the mixture is opaque.
- *Marking note: 1 mark each (3 marks). The answer must name the mixture or its letter; an answer that gives only 'visible' or 'not visible' with no link to A, B or C earns no mark.*

(b) [2 marks]

- The particles of milk are **large enough to scatter the light** in all directions (1 mark).
 - The scattered light reaches the eye from the side, so the **path of the beam becomes visible** (1 mark).
- *Marking note: 2 marks, 1 mark for each point. Accept 'the particles reflect the light' if scattering in all directions is implied. Do NOT accept 'milk is white' on its own.*

(c) [3 marks]

- A - **solution**
 - B - **colloid**
 - C - **suspension**
- *Marking note: 1 mark each (3 marks). All three must be correctly matched to A, B and C; a correct list in the wrong order earns no mark.*

(d) [1 marks]

- The **Tyndall effect** (accept 'the scattering of light').
- *Marking note: 1 mark for the correct name.*

(e) [1 marks]

- Any one of: blood, fog or mist, smoke, starch solution, jelly, Milo drink, hair spray, paint.
- *Marking note: 1 mark for any correct example. Do NOT accept salt solution, sea water or muddy water - these are not colloids.*

Question 4 [Total: 12 marks]**(a)** [4 marks]

- I - **thermometer**
 - II - **round-bottom flask**
 - III - **Liebig condenser**
 - IV - **beaker** (accept 'receiver' or 'conical flask')
- *Marking note: 1 mark each (4 marks). Accept 'flask' for II and 'condenser' for III. Do NOT accept 'flask' for IV or 'test tube' for IV.*

(b) [2 marks]

- I (thermometer): **measures the temperature** of the vapour or liquid so that the boiling point can be checked (1 mark).
- III (condenser): **cools the vapour** and changes it back into a liquid - it condenses the vapour (1 mark).
- *Marking note: 2 marks, 1 mark each. The function must be linked to the correct part; a correct statement attached to the wrong letter earns no mark.**

(c) [1 marks]

- **Evaporation** (accept 'vaporisation').
- *Marking note: 1 mark. If a candidate writes 'boiling', award 0 here but allow the mark in (d) if the reason given refers to the boiling point.**

(d) [2 marks]

- **Ethanol** is collected first in IV (1 mark).
- Reason: ethanol has a **lower boiling point** (about 78 °C) than water (100 °C), so it vapourises first (1 mark).
- *Marking note: 2 marks. Award the second mark only if a comparison of boiling points is given. Do NOT accept 'because ethanol is lighter'.**

(e) [1 marks]

- To **cool (condense) the vapour** into a liquid; also to prevent the vapour from escaping into the air.
- *Marking note: 1 mark for any one correct reason.**

(f) [1 marks]

- The mixture in the flask contains **two liquids**; the liquid in IV contains **only one** (pure) liquid.
- The mixture boils over a **range of temperatures**; the liquid in IV boils at a **fixed temperature**.
- *Marking note: 1 mark for any one correct difference. The answer must be a comparison, not a description of only one of them.**

(g) [1 marks]

- The water would **evaporate and escape** into the air instead of being collected, and the **salt would be left behind** in the flask.
- Distillation is used to recover the **liquid** from a mixture, not the solid; to obtain salt, **evaporation to dryness** should be used instead.
- *Marking note: 1 mark for any answer that identifies that the salt would be left in the flask and that evaporation is the correct method.**

Question 5

(a) [4 marks]

- (i) Iron filings and sand - **magnetic separation**
 - (ii) Ground pepper and groundnut soup - **filtration**
 - (iii) Salt and water - **evaporation**
 - (iv) Salt, sand and water - **filtration, then evaporation** (both must be named)
- **Marking note:** 1 mark each (4 marks). In (iv) both processes must be named for the mark. Do NOT accept 'distillation' for (iii) unless the candidate shows that the water is to be collected.*

(b) [4 marks]

- (i) **Iron is magnetic** (attracted by a magnet) while sand is not, so the iron filings are picked out and the sand remains.
- (ii) **Pepper is insoluble** in the soup and its particles are large enough to be caught on the filter paper as the residue, while the soup passes through as the filtrate.
- (iii) **Salt is dissolved** in the water and cannot be caught by a filter; on heating, the water escapes as vapour and the salt is left behind.
- (iv) The sand is **insoluble**, so filtration removes it first; the salt is **dissolved**, so the filtrate must then be **evaporated** to recover the s

(c) [2 marks]

- **Filtration:** pour the mixture through **filter paper** in a **funnel**. The **sand** is trapped as the residue and the salt solution passes through as the filtrate.
 - **Evaporation:** pour the filtrate into an **evaporating dish** and heat gently (or leave it in the sun). The water evaporates and **salt crystals** are left behind. Dry the sand and the salt.
- **Marking note:** 2 marks: 1 mark for each stage correctly described, and the order must be filtration first. Accept 'leave the filtrate in the sun' instead of heating. Maximum 2 marks.*

(d) [2 marks]

- Pour a little of the drink into a **clean, transparent glass** and shine a **torch or beam of light** through it (1 mark).
 - In a **true solution** the path of the light is **not visible** and the drink stays clear; if it were a **suspension**, the light path would be **visible (scattered)** and **particles would settle** on standing (1 mark).
- **Marking note:** 2 marks. Accept a 'leave it to stand and observe whether particles settle' test if it is fully described. Look for the comparison between what is seen in a solution and in a suspension.*

CURRICULUM ASSESSMENT BLUEPRINT****St**

Renewable Energy and

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	14.1 Renewable and Non-Renewable Sources o...	Remember
2	D	14.1 Renewable and Non-Renewable Sources o...	Understand
3	C	14.1 Renewable and Non-Renewable Sources o...	Remember
4	B	14.1 Renewable and Non-Renewable Sources o...	Remember
5	D	14.1 Renewable and Non-Renewable Sources o...	Remember
6	A	14.1 Renewable and Non-Renewable Sources o...	Remember
7	A	14.1 Renewable and Non-Renewable Sources o...	Remember
8	B	14.1 Renewable and Non-Renewable Sources o...	Understand
9	C	14.1 Renewable and Non-Renewable Sources o...	Understand
10	B	14.2 Producing Energy from a Renewable Source	Understand
11	C	14.2 Producing Energy from a Renewable Source	Remember
12	D	14.2 Producing Energy from a Renewable Source	Understand
13	B	14.1 Renewable and Non-Renewable Sources o...	Analyse
14	B	14.2 Producing Energy from a Renewable Source	Remember
15	A	14.2 Producing Energy from a Renewable Source	Apply
16	A	14.1 Renewable and Non-Renewable Sources o...	Understand
17	D	14.3 Managing Renewable Energy Sources Sus...	Understand
18	C	14.3 Managing Renewable Energy Sources Sus...	Understand
19	B	14.3 Managing Renewable Energy Sources Sus...	Analyse
20	D	14.3 Managing Renewable Energy Sources Sus...	Apply
21	D	14.3 Managing Renewable Energy Sources Sus...	Analyse
22	B	14.3 Managing Renewable Energy Sources Sus...	Apply
23	B	14.3 Managing Renewable Energy Sources Sus...	Evaluate
24	C	14.2 Producing Energy from a Renewable Source	Analyse
25	D	14.4 Differences between Heat and Temperature	Remember
26	B	14.4 Differences between Heat and Temperature	Remember
27	A	14.4 Differences between Heat and Temperature	Remember
28	D	14.4 Differences between Heat and Temperature	Remember
29	A	14.4 Differences between Heat and Temperature	Remember
30	D	14.5 The Relationship between Heat and Tem...	Understand
31	C	14.5 The Relationship between Heat and Tem...	Understand
32	C	14.4 Differences between Heat and Temperature	Understand
33	B	14.5 The Relationship between Heat and Tem...	Analyse
34	D	14.5 The Relationship between Heat and Tem...	Analyse
35	A	14.5 The Relationship between Heat and Tem...	Analyse
36	C	14.4 Differences between Heat and Temperature	Remember
37	A	14.4 Differences between Heat and Temperature	Apply

Q	Ans	Curriculum Area	Cognitive Level
38	C	14.5 The Relationship between Heat and Tem...	Analyse
39	A	14.5 The Relationship between Heat and Tem...	Evaluate
40	A	14.5 The Relationship between Heat and Tem...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 8 (20%) | Apply: 4 (10%) | Evaluate: 3 (7%) | Remember: 14 (35%) | Understand: 11 (27%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 9 marks]

(a) [4 marks]

- (i) A **renewable source of energy** is one that is replaced by nature as fast as it is used, so it does not run out (1 mark); examples are **solar, wind, hydro and biogas** (1 mark).
- (ii) A **non-renewable source** took millions of years to form and is being used up far faster than it can be replaced (1 mark); examples are **coal, crude oil and natural gas** (1 mark).
- **Marking note:** 4 marks: 1 mark for each definition and 1 mark for at least one correct example in each case.*

(b) [3 marks]

- **Renewable sources:** wind, hydro, biogas
- **Non-renewable sources:** coal, crude oil, natural gas
- **Marking note:** 3 marks: award 1 mark for each pair of correct entries (6 correct entries = 3 marks; 4 correct = 2 marks; 2 correct = 1 mark). A candidate who writes the answers as two lists instead of a table still earns the marks. The list of sources is taken from the curriculum exemplar.*

(c) [2 marks]

- They are **replaced by nature and will not run out**, while fossil fuels will one day be finished.
- They **do not pollute the air with smoke and harmful gases**, and the source itself is **free**, so the running cost is low (accept: they can serve villages far from the national grid).
- **Marking note:** 1 mark each for any TWO (2 marks).*

Question 2 [Total: 12 marks]

(a) [4 marks]

- (i) P - the **inlet (mixing) tank**, through which the dung and water are fed into the plant (1 mark)
- (ii) Q - the **digester (digestion tank)**, in which the dung and water are broken down (1 mark)
- (iii) R - the **gas outlet pipe**, which carries the gas to the kitchen (1 mark)
- (iv) S - the **outlet (slurry outlet pipe and pit)**, through which the used slurry leaves the plant (1 mark)
- **Marking note:** 1 mark each (4 marks). Accept any wording that shows the correct function - for example 'feeding tank' for P and 'gas pipe' for R.*

(b) [2 marks]

- **Biogas** (mainly **methane**) (1 mark)
- Use: it is **burnt in the school kitchen for cooking** (accept: for lighting the lamps) (1 mark)
- **Marking note:** 2 marks, 1 mark each.*

(c) [3 marks]

- **Animal dung is mixed with water and fed in through the inlet tank** (1 mark).
- The mixture is held in the **closed digester, from which air is kept out** (1 mark).
- **Bacteria break down the dung**, and the **gas formed collects above the slurry and is piped off** through R (1 mark).
- **Marking note:** 3 marks, 1 mark each. The exclusion of air or the action of bacteria must appear for the second or third mark; an answer that says only 'the dung rots' earns 1 mark.*

(d) [1 marks]

- The **slurry is used as manure on the school farm** (accept: the compound is kept clean and flies are reduced; the school saves the money spent on firewood, charcoal or LPG; fewer trees are cut for firewood).
- **Marking note:** 1 mark for any one correct benefit other than the gas itself.*

(e) [2 marks]

- The **dung and organic waste are produced every day** for as long as the animals are kept and the farm is worked (1 mark).
- The source is therefore **replaced as fast as it is used and cannot be used up**, which is what makes a source renewable (1 mark).
- **Marking note:** 2 marks, 1 mark each. The idea of continuous replacement must appear.*

Question 3 [Total: 10 marks]**(a)** [4 marks]

- (i) **Solar:** there is **no sunshine at night and less on cloudy days**, so storage is needed; the **panels and batteries are costly to buy and to replace**, and **dust on the panels reduces the output** (2 marks).
- (ii) **Hydro:** the **water level in the reservoir falls during a long drought**, so less electricity is produced; the **reservoir silts up with eroded soil**, the dam is **very costly to build**, and **people have to be resettled** (2 marks).
- **Marking note:** 1 mark each for any TWO challenges under each source (4 marks). This answers the curriculum exemplar that asks pupils

(b) [3 marks]

- **Plant trees to replace those cut for firewood, and protect the forests along the rivers that feed the dam.**
- **Maintain the equipment** - clean the solar panels, service the turbines and replace worn-out batteries - and **train local people to carry out repairs**.
- **Use energy sparingly**, fit energy-saving appliances, and **check erosion so that the reservoir does not silt up**.
- *Marking note: 1 mark each for any THREE workable steps (3 marks).*

(c) [3 marks]

- Fossil fuels are **non-renewable** and will one day be used up, so a country that depends on them alone will be left without power (1 mark).
- **Burning them pollutes the air** with smoke and harmful gases and adds to climate change (1 mark).
- Their **prices rise and fall on the world market**, so electricity becomes costly and uncertain, and money must be spent on importing them (1 mark).
- *Marking note: 3 marks, 1 mark each for any three sound reasons.*

Question 4 [Total: 10 marks]**(a)** [6 marks]

- **What it is** - Heat: a **form of energy that flows from a hotter body to a cooler one**; Temperature: the **degree of hotness or coldness of a body** (2 marks).
- **SI unit** - Heat: the **joule (J)**; Temperature: the **kelvin (K)** (commonly measured in degrees Celsius) (2 marks).
- **How it is measured** - Heat: with a **calorimeter** (or found from the mass and the temperature change); Temperature: with a **thermometer** (2 marks).
- *Marking note: 6 marks, 1 mark for each correct entry in the table. Accept 'degree Celsius' for the unit of temperature, but the kelvin is the SI unit.*

(b) [2 marks]

- **When heat is supplied to a body its temperature rises; when heat is taken away the temperature falls** (1 mark).
- For the same substance, the **greater the heat supplied the greater the rise in temperature**, though the rise also depends on the **mass of the body** (1 mark).
- *Marking note: 2 marks, 1 mark each.*

(c) [1 marks]

- From the **body at the higher temperature to the body at the lower temperature** (from hotter to cooler)
- *Marking note: 1 mark.*

(d) [1 marks]

- $27 + 273 = 300 \text{ K}$

- *Marking note: 1 mark for 300 K. Accept the working $27 + 273$ shown with the correct answer; the unit K should be written.*

Question 5 [Total: 9 marks]

(a) [1 marks]

- *Cup A*, whose thermometer reads 90°C

- *Marking note: 1 mark.*

(b) [3 marks]

- *Bucket B* (1 mark).

- Although its *temperature is lower*, it holds a *far greater mass of water - 2000 cm^3 against 100 cm^3 * (1 mark).

- The *heat energy in a body depends on its mass as well as on its temperature*, so the total heat energy stored in B is greater (1 mark).

- *Marking note: 3 marks: 1 mark for naming B and 2 marks for the reason. Naming B with no reason earns 1 mark only.*

(c) [2 marks]

- *Temperature measures only the degree of hotness* of the water and *does not depend on how much water there is* (1 mark).

- *Heat is the total energy in the water and depends on the mass as well as on the temperature*, so the hotter container is not necessarily the one with more heat energy (1 mark).

- *Marking note: 2 marks, 1 mark each. This is the key distinction in the standard.*

(d) [3 marks]

- Its *temperature falls* (1 mark),

- because the water is *hotter than its surroundings, so heat flows out of it into the cup and the surrounding air* (1 mark),

- and it goes on losing heat *until the water and the surroundings are at the same temperature*, that is room temperature (1 mark).

- *Marking note: 3 marks, 1 mark each. A candidate who writes that 'coldness enters the water' loses the second mark; heat flows out, cold does not flow in.*

Question 6 [Total: 10 marks]

(a) [3 marks]

- A *beaker (or small saucepan) containing water*

- A *thermometer*

- A **source of heat** (Bunsen burner, spirit lamp or stove), with a **tripod stand** and **wire gauze**, and a **clock or stop-watch**
- **Marking note:** 1 mark each for any THREE suitable items (3 marks). The thermometer and the heat source should appear.*

(b) [4 marks]

- **Pour a measured volume of water into the beaker** and stand it on the gauze on the tripod.
- **Place the thermometer in the water** with its bulb well under the surface but **not touching the bottom**, and **read the starting temperature**.
- **Light the burner and start the stop-watch at the same moment.**
- **Read and record the temperature every one minute**, stirring gently, until the water boils; then **switch off the heat and compare the readings**.
- **Marking note:** 1 mark each for any FOUR correct steps in a workable order (4 marks). The correct placing of the thermometer bulb and

(c) [2 marks]

- A **ruled table with the headings 'Time (minutes)' and 'Temperature of the water (°C)'** (1 mark).
- **Units shown in the headings and blank rows provided** for the readings (1 mark).
- **Marking note:** 2 marks as shown. A table without units in the headings scores 1 mark.*

(d) [1 marks]

- **Handle the hot beaker with a cloth or tongs and keep away from the flame and the boiling water** (accept: do not stir with the thermometer; tie back loose clothing; stand clear of the steam).
- **Marking note:** 1 mark for any one sound precaution. 'Be careful' on its own earns no mark.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 4: Forces and Energy | **Sub-strand:** 1: Energy | **Class:** Basic 8 (JHS 2)

Content Standard: **B8/JHS2.4.1.2** - Show an understanding of the sources of renewable energy and how to manage these sources in a sustainable manner.
B8/JHS2.4.1.3 -

Science and Indu

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	22.1 Science, Technology and Innovation an...	Remember
2	C	22.1 Science, Technology and Innovation an...	Remember
3	D	22.1 Science, Technology and Innovation an...	Remember
4	D	22.1 Science, Technology and Innovation an...	Understand
5	B	22.1 Science, Technology and Innovation an...	Understand
6	C	22.1 Science, Technology and Innovation an...	Understand
7	A	22.1 Science, Technology and Innovation an...	Understand
8	B	22.1 Science, Technology and Innovation an...	Apply
9	A	22.1 Science, Technology and Innovation an...	Understand
10	B	22.2 Technology at Work in Ghanaian Life	Understand
11	D	22.2 Technology at Work in Ghanaian Life	Understand
12	A	22.2 Technology at Work in Ghanaian Life	Remember
13	B	22.2 Technology at Work in Ghanaian Life	Understand
14	D	22.2 Technology at Work in Ghanaian Life	Remember
15	C	22.2 Technology at Work in Ghanaian Life	Understand
16	A	22.2 Technology at Work in Ghanaian Life	Understand
17	D	22.2 Technology at Work in Ghanaian Life	Understand
18	C	22.2 Technology at Work in Ghanaian Life	Analyse
19	C	22.3 Technological Advancement and the Gha...	Remember
20	D	22.3 Technological Advancement and the Gha...	Understand
21	D	22.3 Technological Advancement and the Gha...	Remember
22	A	22.3 Technological Advancement and the Gha...	Understand
23	A	22.3 Technological Advancement and the Gha...	Evaluate
24	D	22.3 Technological Advancement and the Gha...	Understand
25	B	22.3 Technological Advancement and the Gha...	Understand
26	B	22.3 Technological Advancement and the Gha...	Understand
27	D	22.3 Technological Advancement and the Gha...	Remember
28	B	22.3 Technological Advancement and the Gha...	Understand
29	C	22.3 Technological Advancement and the Gha...	Evaluate
30	C	22.3 Technological Advancement and the Gha...	Evaluate
31	B	22.4 Innovation, Responsible Use and Judgi...	Understand
32	A	22.4 Innovation, Responsible Use and Judgi...	Understand
33	A	22.4 Innovation, Responsible Use and Judgi...	Evaluate
34	D	22.4 Innovation, Responsible Use and Judgi...	Evaluate
35	C	22.4 Innovation, Responsible Use and Judgi...	Analyse
36	B	22.4 Innovation, Responsible Use and Judgi...	Understand

Q	Ans	Curriculum Area	Cognitive Level
37	A	22.4 Innovation, Responsible Use and Judgi...	Understand
38	C	22.4 Innovation, Responsible Use and Judgi...	Apply
39	B	22.4 Innovation, Responsible Use and Judgi...	Apply
40	A	22.4 Innovation, Responsible Use and Judgi...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 2 (5%) | Apply: 3 (7%) | Evaluate: 6 (15%) | Remember: 8 (20%) | Understand: 21 (52%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 10 marks]****(a) [3 marks]**

- **P - science** (the careful, orderly study of nature to find out how and why things happen) (1 mark).
- **Q - technology** (the use of that knowledge to make tools, machines and methods) (1 mark).
- **R - innovation** (a new or improved product, method or idea that is actually put to use) (1 mark).
- **Marking note:** 3 marks, 1 mark each. The single word is enough; the explanation in brackets is not required. Do not accept 'invention' for P or Q.*

(b) [3 marks]

- **Science** supplies the knowledge, and **technology** puts that knowledge to work to meet a human need (1 mark).
- **Technology** in turn supplies science with instruments - the microscope, the thermometer, the computer - so that scientists can observe and measure what they could not reach before, and so discover more (1 mark).
- **A correct example** (1 mark), for example: the scientific finding that an electric current produces a magnetic field was used to build the electric motor and the fan; the study of germs led to vaccines and to the refrigerator used to store them; the study of light le

(c) [2 marks]

- **The needs and the problems of the people** decide what is studied and what is made. Where malaria is common, scientists work on malaria; where there is no mains electricity, inventors work on solar lamps (1 mark).
- **So the relationship works in both directions:** science and technology change the way people live, and the way people live, their needs, their money and their government's choices, decide what science and technology do next (1 mark).
- **Marking note:** 2 marks. The first mark is for society setting the agenda; the second for recognising that the relationship is two-way. An an

(d) [2 marks]

- **Mobile money** - people send and receive money, pay school fees and buy goods **without travelling to a bank**, which many villages do not have.
- Also accept: **the solar lantern or solar home system** (light and charging where there is no mains electricity); **the solar-powered vaccine refrigerator** at a rural clinic; **the cassava or palm-fruit processing machine** (less of the harvest is wasted); **the solar fish dryer** (clean, fast drying, free of flies); **the**

drone used to deliver blood and medicines**; **plastic waste recycled into pavement blocks**; **the borehole with a hand pu

Question 2 [Total: 10 marks]

(a) [4 marks]

- **Agriculture** - the tractor, the irrigation pump, the knapsack sprayer, improved seed.
- **Health care** - the X-ray machine, the solar-powered vaccine refrigerator, the laboratory microscope.
- **Communication** - the mobile telephone, the radio, the internet.
- **Transport** - the lorry, the motor bicycle, the aeroplane.
- Also accept: **industry** (processing and packaging machines); **the home** (the refrigerator, the gas cooker, the electric iron); **education** (the computer, the projector); **banking and trade** (mobile money, the ATM); **water supply** (the borehole and pump).
-

(b) [3 marks]

- **Work is done faster and with less effort** , so one person can do in a day what once took a week.
- **More food and more goods are produced** , and less of the harvest is wasted, so there is food to sell as well as to eat.
- **Illness is found and treated earlier** , and vaccines, clean water and light at night have lengthened life.
- Also accept: **messages and money move in seconds** ; **new jobs, trades and businesses have been created** ; **pupils can learn from sources far beyond the school** .
- *Marking note: 1 mark each for any THREE benefits (3 marks). The benefit must be to people, n

(c) [3 marks]

- **Pollution** - smoke and fumes in the air, waste water and chemicals in the rivers, oil and refuse on the land.
- **Destruction of the natural environment** - forests cleared, farmland dug up and rivers silted by mining and building.
- **Mountains of waste** - plastic sachets and bottles that block the gutters and cause floods, and old phones, televisions and refrigerators that nobody can safely destroy.
- Also accept: **some workers lose their jobs to machines** ; **accidents and injuries from machines and vehicles** ; **noise from mills, generators and loudspeakers** ; **less exercise and mo

Question

(a) [3 marks]

- **X** - smoke, or fumes, rising from the burning cables (1 mark).
- **Y** - the stream, or river, running past the yard (1 mark).
- **Z** - the heap of electronic waste (e-waste) - old televisions, computers, refrigerators and telephones (accept 'scrap' or 'old electrical appliances') (1 mark).
- **Marking note:** 3 marks, 1 mark each. For Z accept any answer that shows the candidate recognises discarded electrical or electronic goods; 'rubbish' alone is too vague.

(b) [2 marks]

- **It is breathed in.** The smoke carries poisonous gases and very fine particles that irritate the nose, throat and lungs, causing coughing, chest pain, asthma and, over years, lasting damage to the lungs.
- **It carries metal poisons such as lead.** These enter the blood and can damage the brain and the nerves, and they are especially harmful to young children and to unborn babies.
- Also accept: **the smoke settles on food, water and washing** in the nearby houses; **it causes burning eyes and headaches**; **it hides the air pollution that worsens heart disease**.
- **Marking note:** 1 mark

(c) [2 marks]

- **Rain washes acids, oils and metal poisons out of the broken appliances** and the run-off carries them down into the stream (1 mark).
- **The stream becomes polluted:** the water is no longer safe to drink, cook or bathe in; fish and other water life are killed; and the poisons pass into people who drink the water, eat the fish or eat crops watered from it (1 mark).
- **Marking note:** 2 marks: 1 for the poisons reaching the water, 1 for the consequence to the water life or to the people. 'It makes the water dirty' alone earns 1 mark at most.

(d) [2 marks]

- **The gain is small and private, but the harm is large and shared.** A few workers earn a little, while the smoke, the polluted stream and the poisoned soil damage the health of the whole neighbourhood, including children who earn nothing from the yard.
- **The damage lasts far longer than the income.** Metal poisons stay in the soil and the water for many years and the cost of treating illness falls on the families and on the state.
- **The choice is not between this yard and no work at all.** The same metals can be recovered safely at a proper recycling plant, with tools, gloves, masks and

(e) [3 marks]

- **Provide, or licence, a proper recycling plant** where the metals are recovered by machine instead of by open burning.
- **Ban open burning on the site and enforce the ban**, through the Environmental Protection Authority and the assembly's by-laws.
- **Train the workers and supply protective equipment** - gloves, boots, masks - and register them so that their health is checked.
- Also accept: **move the yard away from houses and from the stream**;
require importers and sellers of electrical goods to take back old appliances;
provide collection points and public education so that e-wa

Question 4 [Total: 10 marks]**(a) [2 marks]**

- **A single excavator moves in one day what a hundred people with shovels could not move in months**, so a far larger area of forest and farmland is cleared and dug up in a very short time.
- **The machines work in and beside the river itself and use chemicals such as mercury on a large scale**, so the water is muddied and poisoned far faster than the river can clean itself.
- Also accept: **machines reach deep and remote places hand tools never could**, and **the pits and heaps they leave behind are too large for the people to fill in again**.
- **Marking note: 1 mark each for any TWO reasons**

(b) [3 marks]

- **The rivers become muddy and are poisoned with mercury**, so the water is unsafe and costs far more to treat for town supply.
- **Farmland and forest are destroyed** and left as pits and heaps of sand, so cocoa and food crops are lost and farmers lose their livelihood.
- **People fall ill** - mercury poisoning, skin and stomach disease from dirty water - and the open pits fill with water in which children drown and mosquitoes breed.
- Also accept: **fish and other water life are killed**; **the soil is no longer fertile**; **school attendance falls as children are drawn into the pits**.
-

(c) [2 marks]

- **Minerals are produced in quantity** and sold abroad, earning foreign exchange and taxes that pay for roads, schools, clinics and electricity.
- **Jobs and incomes are created** for miners, drivers, mechanics, traders and food sellers, and skills are learnt.
- Also accept: **raw materials for industry**, and **development of roads, power and water in mining districts**.
- **Marking note: 1 mark each for any TWO benefits (2 marks). This part exists**

to keep the answer balanced; a candidate who refuses to admit any benefit earns nothing here.*

(d) [3 marks]

- **Licence, inspect and police the mines**, so that only registered miners work and the law against mining in river beds and forest reserves is enforced.
- **Require every miner to restore the land** - to refill the pits, replace the topsoil and replant trees or crops when the gold is finished - and to lodge money in advance to pay for it.
- **Treat the waste water in settling ponds and ban the careless use of mercury**, replacing it with safer recovery methods.
- Also accept: **train and equip small-scale miners with cleaner methods**; **monitor the rivers regularly and publish the results**

Question 5 [Total: 8 marks]

(a) [2 marks]

- **The problem is stated plainly and is a real one** in a school or a community (1 mark).
- **The goal is stated**: what the device or method must actually do - for example, *it must sound a warning when the tank is full*, or *it must dry fifty kilogrammes of maize in two days without the rain touching it* (1 mark).
 - **Marking note**: 2 marks. This part carries the competency CP 6.1, the ability to define a goal towards solving a problem. A candidate who describes a gadget but never says what problem it solves earns nothing here.*

(b) [2 marks]

- **A list of suitable materials that can really be obtained locally** - for example empty bottles, a float, wire, a bell or buzzer, an old dry cell, bamboo, wire mesh, a polythene sheet, scrap wood (1 mark).
- **A simple labelled sketch** showing the main parts and where each material goes (1 mark).
 - **Marking note**: 2 marks. Award the sketch mark for any drawing with at least two sensible labels. No marks are taken off for untidy drawing; this is a science paper, not an art paper.*

(c) [2 marks]

- **The explanation follows a clear order** - what happens first, next and last (1 mark).
- **It rests on a correct scientific idea** (1 mark): the float rises because it is lighter than water and closes the circuit, so the buzzer sounds; the glass traps heat so the air inside the dryer becomes hotter than the air outside and carries the moisture away; the dark surface absorbs more radiation than a shiny one.
 - **Marking note**: 2 marks. The science need not be advanced, but it must be

correct. A design that could not possibly work as described earns the first mark only, for the orderly explanation.

(d) [2 marks]

- **One benefit** (1 mark): water, money, time, food or fuel saved; work made safer or easier; health improved; waste reduced.
- **One difficulty** (1 mark): the cost of the materials; parts that wear out or rust and must be replaced; the device may fail in rain or harmattan dust; somebody must be trained to use and repair it; it may be stolen or damaged.
- **Marking note:** 2 marks, 1 each. Reward honesty: a candidate who can name a real weakness in his or her own design is thinking like an engineer. A candidate who writes 'there is no difficulty at all' earns nothing for the second mark.*

Question 6 [Total: 10 marks]

(a) [3 marks]

- **The science teacher, a laboratory technician or a qualified engineer or electrician** in the community.
- **The science textbook, an encyclopaedia or the school or public library.**
- **The website or the office of a recognised body** - the Ghana Standards Authority, the Energy Commission, the Food and Drugs Authority, the CSIR, a university or the Ministry of Energy.
- Also accept: **a published scientific article or report**, and **the manufacturer's own documents**, read critically alongside an independent source.
- **Marking note:** 1 mark each for any THREE sources (3 marks). 'Social media'

(b) [3 marks]

- **Who wrote it, and what do they know?** Is the author a scientist, an institution or an unnamed person forwarding a message?
- **Does any other independent source agree?** A claim that appears in two or three trusted places separately is far stronger than one that appears everywhere in exactly the same forwarded words.
- **Is evidence given?** Are there measurements, a test anyone can repeat, or only promises and pictures?
- Also accept: **how recent the information is**; **whether the writer gains money if you believe it**; **whether the claim contradicts well-established science** - energy

(c) [4 marks]

- **(i) Two helps** (1 mark each): quick contact with parents and friends, especially in an emergency; access to information, dictionaries, past questions and lessons; sending and receiving money for fees and provisions; a torch, a calculator, a clock and a camera in one pocket.
- **(ii) Two harms** (1 mark each): hours lost from study, chores and sleep; false

and harmful information believed and passed on; bullying, insults and contact with strangers; the cost of data and the risk of theft; eye strain, poor posture and less exercise and play.

- *Marking note: 4 marks in all: 2 + 2. Mark onl

Soils and Rocks

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	A	24.1 Soil and Its Physical Properties	Remember
2	D	24.1 Soil and Its Physical Properties	Understand
3	B	24.1 Soil and Its Physical Properties	Remember
4	B	24.1 Soil and Its Physical Properties	Remember
5	C	24.1 Soil and Its Physical Properties	Understand
6	A	24.1 Soil and Its Physical Properties	Understand
7	B	24.1 Soil and Its Physical Properties	Understand
8	A	24.1 Soil and Its Physical Properties	Apply
9	B	24.2 Soil Types, Water Retention and the S...	Remember
10	C	24.2 Soil Types, Water Retention and the S...	Understand
11	C	24.2 Soil Types, Water Retention and the S...	Remember
12	A	24.2 Soil Types, Water Retention and the S...	Remember
13	B	24.2 Soil Types, Water Retention and the S...	Understand
14	B	24.2 Soil Types, Water Retention and the S...	Understand
15	C	24.2 Soil Types, Water Retention and the S...	Understand
16	D	24.2 Soil Types, Water Retention and the S...	Apply
17	D	24.2 Soil Types, Water Retention and the S...	Understand
18	B	24.3 Soil Water, Osmosis and Crop Production	Remember
19	C	24.3 Soil Water, Osmosis and Crop Production	Remember
20	A	24.3 Soil Water, Osmosis and Crop Production	Understand
21	D	24.3 Soil Water, Osmosis and Crop Production	Understand
22	C	24.3 Soil Water, Osmosis and Crop Production	Remember
23	A	24.3 Soil Water, Osmosis and Crop Production	Apply
24	C	24.3 Soil Water, Osmosis and Crop Production	Analyse
25	D	24.4 Rocks and Their Visible Characteristics	Remember
26	A	24.4 Rocks and Their Visible Characteristics	Remember
27	B	24.4 Rocks and Their Visible Characteristics	Understand
28	B	24.4 Rocks and Their Visible Characteristics	Remember
29	D	24.4 Rocks and Their Visible Characteristics	Understand
30	D	24.4 Rocks and Their Visible Characteristics	Remember
31	C	24.4 Rocks and Their Visible Characteristics	Analyse
32	D	24.4 Rocks and Their Visible Characteristics	Understand
33	D	24.4 Rocks and Their Visible Characteristics	Apply
34	A	24.5 Weathering: How Rocks Become Soil	Remember
35	C	24.5 Weathering: How Rocks Become Soil	Understand
36	A	24.5 Weathering: How Rocks Become Soil	Understand
37	B	24.5 Weathering: How Rocks Become Soil	Understand

Q	Ans	Curriculum Area	Cognitive Level
38	D	24.5 Weathering: How Rocks Become Soil	Understand
39	A	24.5 Weathering: How Rocks Become Soil	Analyse
40	C	24.5 Weathering: How Rocks Become Soil	Understand

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 3 (7%) | Apply: 4 (10%) | Remember: 14 (35%) | Understand: 19 (47%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 10 marks]

(a) [2 marks]

- **Aim:** to find out how much water each of the three soils holds back, or lets through (1 mark).
- **The property compared is the water-retaining ability (water retention) of the soils**, which is the other side of how freely they drain (1 mark).
- Accept also: **to compare the drainage of the three soils**.
- **Marking note:** 2 marks, 1 each. 'To test the soils' is too vague. Accept either retention or drainage for the second mark, since each implies the other.*

(b) [3 marks]

- **Soil A:** $100 - 42 = 58 \text{ cm}^3$ (1 mark).
- **Soil B:** $100 - 25 = 75 \text{ cm}^3$ (1 mark).
- **Soil C:** $100 - 8 = 92 \text{ cm}^3$ (1 mark).
- **Marking note:** 3 marks, 1 each. The working 100 minus the volume collected must be seen or implied. Insist on cm^3 at least once; do not penalise the missing unit three times over.*

(c) [2 marks]

- **A is the sandy soil**: it let the most water through (42 cm^3), because its particles are large and the spaces between them are wide, so water drains freely (1 mark).
- **C is the clay soil**: it let only 8 cm^3 through and held back 92 cm^3 , because its particles are very fine and tightly packed, so the water is held in the tiny spaces (1 mark).
- It follows that **B is the loamy soil**, which lies between the two.
- **Marking note:** 2 marks: 1 for each soil correctly identified WITH a reason drawn from the particle size or the spaces. A bare letter with no reason earns nothing.*

(d) [3 marks]

- **The same mass (or volume) of soil in each funnel**.
- **The same volume of water poured into each funnel** - here 100 cm^3 .
- **The same funnels, and the same filter paper or cotton wool** at the neck of each.
- Also accept: **all three soils dried in the same way before the test**; **the water poured at the same rate**; **the same time allowed for the water to drain before reading**; **the soils packed to the same tightness**.
- **Marking note:** 1 mark each for any THREE controlled variables (3 marks). The answer must be something that is KEPT THE SAME; naming the soil type,

which is the t

Question 2 [Total: 12 marks]

(a) [3 marks]

- **Texture** - how coarse or fine the soil feels, which depends on the size of its particles.
- **Colour** - dark brown or black soils are usually rich in humus, while pale or red soils are usually poorer in it.
- **Water-retaining ability and drainage** - how much water the soil holds and how fast the rest passes through.
- Also accept: **air content (porosity)**; **structure**, that is the way the particles cling together into crumbs; **the amount of humus**.
- **Marking note:** 1 mark each for any THREE physical properties (3 marks). The fertility or the pH of a soil is not a physical property

(b) [4 marks]

- **Size of the particles** - sandy: **large (coarse)**; clay: **very small (fine)** (1 mark for the row).
- **Drainage of water** - sandy: **very fast, free drainage**; clay: **very slow, often waterlogged** (1 mark).
- **Amount of water retained** - sandy: **little**; clay: **a great deal** (1 mark).
- **Ease of working with a hoe** - sandy: **light and easy to work**; clay: **heavy and hard to work - sticky when wet, hard and cracked when dry** (1 mark).
- **Marking note:** 4 marks, 1 for each complete row. Both cells of a row must be correct for the mark; a row with one cell right and one wrong

(c) [3 marks]

- **It holds enough water for the crop but allows the surplus to drain**, so the plant is neither thirsty nor waterlogged.
- **It keeps air in its spaces**, so the roots and the soil organisms can respire.
- **It is usually rich in humus and plant nutrients**, and it holds those nutrients instead of letting them wash away.
- Also accept: **it is easy to work and the roots spread through it freely**; **it does not cake or crack like clay, nor dry out like sand**.
- **Marking note:** 1 mark each for any THREE reasons (3 marks). 'It is the best soil' repeats the question and earns nothing.

(d) [2 marks]

- **The light, gritty (sandy) field**: dig in **organic manure, compost, poultry droppings or green manure**, and mulch the surface, so that the soil holds more water and more nutrients (1 mark).
- **The heavy, sticky (clay) field**: work in **sand and well-rotted manure or compost** to open the soil, and **make ridges, mounds or drains** so that the

surplus water can run off (1 mark).

- *Marking note: 2 marks, 1 each. The treatment must match the field; a candidate who adds sand to the sandy field earns nothing for that mark.*

Question 3 [Total: 10 marks]

(a) [4 marks]

- **W - the topsoil** (1 mark).

- **X - the subsoil** (1 mark).

- **Y - the weathered or broken rock**, the layer of loose, angular fragments (accept 'partly weathered rock') (1 mark).

- **Z - the parent rock, or bedrock**, the solid unbroken rock (1 mark).

- *Marking note: 4 marks, 1 each. Accept 'horizon A', 'horizon B', 'horizon C' and 'horizon D' only where the matching ordinary name is also given. The whole set of layers is called the soil profile.*

(b) [2 marks]

- **W is darker, because it is rich in humus; X is paler and contains little humus.**

- **W is loose and crumbly and holds most of the plant roots, earthworms and other soil organisms**, while **X is more compact and stony, and few roots and organisms reach it**.

- Also accept: **X receives the minerals washed down from W by rain water**, and **W is the thinner of the two layers in this cut**.

- *Marking note: 1 mark each for any TWO differences (2 marks). A difference must compare the two layers; a statement about W alone earns nothing.*

(c) [2 marks]

- **It is the layer in which the crop is planted and in which almost all the feeding roots grow** (1 mark).

- **It holds the humus, the plant nutrients, most of the soil water and air, and the earthworms, bacteria and fungi that keep the soil alive**, so the yield of the farm depends on it (1 mark).

- *Marking note: 2 marks. An answer that says only 'it is on top' earns nothing.*

(d) [2 marks]

- **Clearing and burning the vegetation and leaving the land bare**, so that the rain washes the topsoil away and the sun burns out the humus - prevent it by **keeping a cover crop, mulching, or planting trees as a windbreak** (1 mark).

- **Ploughing or making ridges straight up and down a slope**, which lets the run-off carry the soil downhill - prevent it by **ploughing and ridging along the contour, or by terracing the slope** (1 mark).

- Also accept: **sand winning, mining and road building that strip the topsoil** - require that the topsoil be set aside and replaced afterwards; **overgraze

Question

(a) [3 marks]

- **Igneous rock** - formed when **molten rock material (magma or lava) cools and becomes solid** (1 mark).
- **Sedimentary rock** - formed when **layers of sand, mud, shell and other sediment are deposited, pressed and cemented together over a very long time** (1 mark).
- **Metamorphic rock** - formed when **an existing rock is changed by great heat and pressure inside the earth**, without melting completely (1 mark).
- **Marking note:** 3 marks, 1 for each type correctly named AND matched with its formation. A name alone, with no formation, earns nothing.*

(b) [3 marks]

- **Igneous: granite** (accept basalt, pumice, obsidian) (1 mark).
- **Sedimentary: limestone** (accept sandstone, shale, mudstone, conglomerate) (1 mark).
- **Metamorphic: marble** (accept slate, quartzite, schist, gneiss) (1 mark).
- Note for the teacher: granite is widely quarried in Ghana for building stone and chippings, and limestone is quarried at Nauli and Bupe for cement.
- **Marking note:** 3 marks, 1 each. The example must be matched to the right type; a correct rock named under the wrong heading earns nothing.*

(c) [2 marks]

- **Colour**, and whether the colour is even or mixed.
- **Hardness** - whether it can be scratched with a nail or scratches glass; and whether it crumbles in the hand.
- **Texture** - the size of the grains and whether the surface feels rough or smooth, glassy or dull.
- **Whether it shows layers, bands or separate crystals**, which helps to place it as sedimentary, metamorphic or igneous.
- Also accept: **how heavy it feels for its size**; **where and when it was collected**; **a sketch or photograph**.
- **Marking note:** 2 marks: award 1 mark for any TWO correct features and 2 marks for FOU

(d) [2 marks]

- **The mineral part of every soil comes from rock that has been broken down**: weathering reduces the rock beneath to smaller and smaller pieces, which mix with the humus from decayed plants and animals to make soil (1 mark).
- **The kind of rock therefore decides the kind of soil formed** - its particles, its colour and the minerals it contains. This is why the rock under a soil is called its parent material (1 mark).
- **Marking note:** 2 marks. The first mark is for the breaking down of rock into

the soil particles; the second for the link between the type of rock and the type of soil.*

Question 5 [Total: 10 marks]

(a) [2 marks]

- **Weathering is the breaking down of rock into smaller and smaller pieces where the rock lies**, by heat and cold, water, chemicals or living things (1 mark).
- **In weathering the pieces are not carried away; in erosion the broken material or the soil IS carried away** by running water, wind, ice or gravity (1 mark).
- *Marking note: 2 marks. The distinction is the one examiners look for: weathering breaks down in place, erosion transports.*

(b) [4 marks]

- **Heating and cooling**: the sun heats the rock by day and it cools at night, so the rock expands and contracts again and again until it cracks and pieces flake off the surface.
- **The action of water**: rain water seeps into the cracks and, being slightly acidic, slowly dissolves some of the minerals, weakening the rock; running water also wears the surface away.
- **The roots of the tree and other plants**: as a root grows in a crack it widens it and forces the pieces of rock apart; lichens and mosses also attack the surface chemically.
- **Wind-blown sand and grit** wear and scratch the

(c) [4 marks]

- **Stage 1** - the **solid rock is attacked by heat, cold, water and plant roots and begins to crack** (1 mark).
- **Stage 2** - the **cracks widen and the rock breaks into boulders, stones and gravel** (1 mark).
- **Stage 3** - **continued weathering grinds these pieces into fine particles - sand, silt and clay** (1 mark).
- **Stage 4** - **lichens, mosses and small plants grow in the particles, die and decay; their remains and those of animals add humus**, and the mixture of mineral particles, humus, water and air forms a soil deep enough for crops (1 mark).
- Teachers should note that this

Question 6 [Total: 8 marks]

(a) [2 marks]

- **The soil water round the root hair is a very weak solution, while the cell sap inside the root hair is a stronger, more concentrated solution** (1 mark).

- **Water therefore passes through the thin wall and membrane of the root hair into the cell by osmosis**, from the weaker to the stronger solution; the many long, thin root hairs give a very large surface for this to happen (1 mark).

- **Marking note:** 2 marks. The word osmosis alone earns 1 mark; the second mark needs the difference in concentration. Dissolved mineral salts are carried in with the water, but they also enter by other means.

(b) [2 marks]

- **(i) Too dry**: there is too little water in the soil for the roots to replace what the leaves are losing, so the cells lose their firmness and the plant droops (1 mark).

- **(ii) Waterlogged**: the water fills the air spaces in the soil, so **the roots cannot get the oxygen they need to respire**; they begin to rot and can no longer take in water, so the plant wilts even though it is standing in water (1 mark).

- **Marking note:** 2 marks, 1 each. Part (ii) is the discriminating one; the mark requires the lack of oxygen at the roots.*

(c) [2 marks]

- **Set-up** (1 mark): take three identical tins or pots with drainage holes, label them sandy, loamy and clay, fill each with the same volume of its soil, and plant the same number of seeds of the same crop - maize or beans - at the same depth in each.

- **Fair test and measurement** (1 mark): give each pot the same volume of water at the same times, keep all three in the same place with the same sunlight, and then **measure the height of the seedlings and count the leaves every three days for two or three weeks**, recording the results in a table before comparing them.

- **Marking note:** 2 marks

(d) [2 marks]

- **Mulch the surface** with grass, straw, leaves or crop residues, so that less water evaporates from the soil.

- **Dig in compost or organic manure**, which helps the soil to hold water like a sponge.

- Also accept: **plant a cover crop**; **plant trees as shade or as a windbreak**; **weed the field, since weeds take up the water**; **make ridges or tied ridges to trap the rain where it falls**; **irrigate from a dugout or small dam**; **stop burning the crop residues**.

- **Marking note:** 1 mark each for any TWO measures (2 marks).*

CURRICULUM ASSESSMENT BLUEPRINT

****Strand:** 5: H**



Solar System

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	10.1 Composition of the Solar System	Remember
2	A	10.1 Composition of the Solar System	Remember
3	A	10.1 Composition of the Solar System	Remember
4	C	10.1 Composition of the Solar System	Remember
5	D	10.1 Composition of the Solar System	Remember
6	A	10.1 Composition of the Solar System	Understand
7	B	10.2 Identifying the Outer Planets	Remember
8	C	10.2 Identifying the Outer Planets	Understand
9	C	10.2 Identifying the Outer Planets	Remember
10	A	10.3 Properties Peculiar to Each Outer Planet	Remember
11	D	10.2 Identifying the Outer Planets	Remember
12	C	10.2 Identifying the Outer Planets	Understand
13	C	10.3 Properties Peculiar to Each Outer Planet	Remember
14	D	10.2 Identifying the Outer Planets	Understand
15	A	10.2 Identifying the Outer Planets	Analyse
16	D	10.2 Identifying the Outer Planets	Understand
17	C	10.3 Properties Peculiar to Each Outer Planet	Understand
18	C	10.3 Properties Peculiar to Each Outer Planet	Remember
19	B	10.3 Properties Peculiar to Each Outer Planet	Understand
20	D	10.3 Properties Peculiar to Each Outer Planet	Understand
21	B	10.3 Properties Peculiar to Each Outer Planet	Analyse
22	A	10.3 Properties Peculiar to Each Outer Planet	Remember
23	C	10.3 Properties Peculiar to Each Outer Planet	Understand
24	C	10.3 Properties Peculiar to Each Outer Planet	Understand
25	D	10.4 Why There is No Life on the Outer Pla...	Understand
26	B	10.4 Why There is No Life on the Outer Pla...	Analyse
27	D	10.4 Why There is No Life on the Outer Pla...	Analyse
28	B	10.4 Why There is No Life on the Outer Pla...	Analyse
29	B	10.1 Composition of the Solar System	Understand
30	A	10.1 Composition of the Solar System	Remember
31	A	10.1 Composition of the Solar System	Understand
32	B	10.5 Modelling and Studying the Outer Sola...	Apply
33	A	10.5 Modelling and Studying the Outer Sola...	Apply
34	C	10.5 Modelling and Studying the Outer Sola...	Apply
35	D	10.2 Identifying the Outer Planets	Apply
36	B	10.2 Identifying the Outer Planets	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	D	10.3 Properties Peculiar to Each Outer Planet	Evaluate
38	A	10.4 Why There is No Life on the Outer Pla...	Evaluate
39	D	10.5 Modelling and Studying the Outer Sola...	Evaluate
40	B	10.2 Identifying the Outer Planets	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 5 (12%) | Apply: 5 (12%) | Evaluate: 4 (10%) | Remember: 13 (32%) | Understand: 13 (32%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 9 marks]****(a)** [2 marks]

- The solar system is the **Sun** together with all the bodies that move round it (1 mark).
- These bodies are the **eight planets and their moons, the asteroids, comets and meteors** (1 mark).
 - *Marking note: 2 marks, 1 mark each. Award 1 mark for 'the Sun and the planets'; the second mark needs other members such as moons, asteroids or comets.*

(b) [4 marks]

- **Jupiter**
- **Saturn**
- **Uranus**
- **Neptune**
- *Marking note: 1 mark each, and the order must be correct for full marks (4 marks). All four named but in the wrong order scores a maximum of 2 marks.*

(c) [1 marks]

- The **asteroid belt**
- *Marking note: 1 mark. Accept 'belt of asteroids'.*

(d) [2 marks]

- The planets **do not produce light of their own**, unlike the Sun and the other stars (1 mark).
- They are seen because they **reflect the light of the Sun** that falls on them towards the Earth (1 mark).
 - *Marking note: 2 marks, 1 mark each. The word **reflect** (or 'the Sun's light bounces off them') must appear for the second mark.*

Question 2 [Total: 12 marks]**(a)** [4 marks]

- I - **Jupiter**
- II - **Saturn**
- III - **Uranus**
- IV - **Neptune**
- *Marking note: 1 mark each (4 marks). The name must be matched to the correct number. Candidates should use the clues in the figure: I is the largest*

and II carries the rings.*

(b) [2 marks]

- The **great red spot** (1 mark)
- It is a **huge storm** whirling in the atmosphere of the planet, larger than the Earth and lasting for hundreds of years (1 mark).
 - *Marking note: 2 marks, 1 mark each. Accept 'red spot' for the name. The second mark requires the idea of a storm; 'a big red mark' earns no mark.*

(c) [2 marks]

- The **rings** (of Saturn) (1 mark)
- They are made up of **many pieces of ice, dust and rock** moving round the planet (1 mark).
 - *Marking note: 2 marks, 1 mark each. Do NOT award the second mark for 'solid metal' or 'a solid band' - the rings are not solid.*

(d) [1 marks]

- **IV** - Neptune
- *Marking note: 1 mark. Accept either the number or the name.*

(e) [2 marks]

- The inner planets are **small and rocky with solid surfaces**, while the outer planets are **very large and made mainly of gases**.
- The inner planets are **nearer the Sun and warmer**, while the outer planets are **far from the Sun and very cold**.
- The inner planets have **few or no moons**, while the outer planets have **many moons** (and rings).
 - *Marking note: 1 mark each for any TWO correct differences (2 marks). The answer must compare the two groups; a statement about only one group earns no mark.*

(f) [1 marks]

- They are **extremely large (giant)** and are made up **mainly of gases**, not of solid rock.
 - *Marking note: 1 mark. Both the size and the gas must appear; 'they are big' alone earns no mark.*

Question 3 [Total: 10 marks]

(a) [1 marks]

- **143 thousand km** (about 143 000 km)
- *Marking note: 1 mark. The unit must be given; accept 143 000 km.*

(b) [1 marks]

- **IV** (Neptune)
- *Marking note: 1 mark. Accept either the number or the name.*

(c) [2 marks]

- Difference = $143 - 49$ (1 mark)
- = **94 thousand km** (about 94 000 km) (1 mark)
- *Marking note: 2 marks: 1 method mark for the subtraction and 1 mark for the answer with the unit. A bare correct answer scores both marks.*

(d) [2 marks]

- $143 \div 51$ (1 mark)
- = about 2.8, that is **roughly 3 times** as wide (1 mark)
- *Marking note: 2 marks: 1 method mark for the division and 1 mark for the answer. Accept any answer between 2.5 and 3 described as 'about 3 times'. The answer has no unit, since it is a comparison.*

(e) [2 marks]

- They are all **much larger than the Earth** and are **made up mainly of gases** (no solid surface to stand on).
- They are all **very far from the Sun and very cold**, all have **many moons**, and each takes **many Earth years** to go round the Sun once.
- *Marking note: 1 mark each for any TWO correct shared properties (2 marks). Do NOT accept 'they are all planets', which is given in the question.*

(f) [2 marks]

- It means that the **sizes of the planets and the distances between them are not in their true proportions** in the drawing (1 mark).
- It is necessary because the **real distances and sizes are far too vast** to be drawn in proportion on one page - the small planets would be invisible (1 mark).
- *Marking note: 2 marks, 1 mark each. The first mark is for explaining the phrase; the second is for the reason.*

Question 4 [Total: 10 marks]**(a)** [4 marks]

- (i) Jupiter - the **largest planet** in the solar system, with the **great red spot** (a giant storm)
- (ii) Saturn - famous for its **bright, wide rings** of ice and rock
- (iii) Uranus - **rotates on its side**, so that it appears to roll along its orbit; it looks blue-green
- (iv) Neptune - the **farthest planet from the Sun**, deep blue, with the strongest winds and the longest orbit
- *Marking note: 1 mark each (4 marks). The property must be peculiar to that

planet. Do NOT award a mark for 'it is big and cold', which is true of all four.*

(b) [2 marks]

- They are **very far away from the Sun** compared with the Earth (1 mark).
- So **very little of the Sun's heat and light reaches them** - the heat spreads out and is very weak at that distance (1 mark).
- *Marking note: 2 marks, 1 mark each. Do NOT accept 'they are covered with ice' as the reason - the ice is a result of the cold, not its cause.*

(c) [2 marks]

- Neptune is **much farther from the Sun** than Jupiter (1 mark).
- So its **orbit (path round the Sun) is very much longer**, and it has a far greater distance to cover to complete one journey (1 mark).
- *Marking note: 2 marks, 1 mark each. Accept the additional point that it also travels more slowly along its orbit. Do NOT accept 'it is lazy' or 'it is heavier'.*

(d) [2 marks]

- A moon (natural satellite) is a **natural body that moves round a planet** (1 mark).
- The outer planets are **very large and have a strong pull (force of gravity)**, so they can hold many moons in orbit round them (1 mark).
- *Marking note: 2 marks, 1 mark each. Accept 'satellite' for moon. Do NOT accept 'a moon is a small star'.*

Question 5 [Total: 9 marks]

(a) [3 marks]

- **Air (oxygen) to breathe**
- **Water**
- **Food** and a **suitable, moderate temperature** (not extremely hot or cold)
- A **suitable surface or habitat** to live on
- *Marking note: 1 mark each for any THREE correct conditions (3 marks).*

(b) [3 marks]

- It is **extremely cold**, because it is very far from the Sun.
- It has **no solid surface** and **no liquid water** - it is made up mainly of gases.
- Its **atmosphere has no oxygen** to breathe (it is mainly hydrogen and helium, with poisonous gases), and **violent storms** blow there constantly.
- *Marking note: 1 mark each for any THREE correct reasons (3 marks). The reasons must be about conditions on the planet. Do NOT accept 'it is too far to travel to' - that is about reaching it, not about life existing there.*

(c) [2 marks]

- Saturn is made up ****mainly of gases and has no solid surface****, so there is nothing firm to stand on and the astronaut would ****sink into the planet**** (1 mark).
- Besides, the ****extreme cold, the crushing pressure and the absence of oxygen**** would make survival impossible (1 mark).
 - ***Marking note: 2 marks, 1 mark each. The absence of a solid surface must appear for the first mark.***

(d) [1 marks]

- The ****journey takes many years and the conditions would kill a human being****; a probe needs ****no air, food or water****, is cheaper and safer, and can send back pictures and readings without returning.
- ***Marking note: 1 mark for any one sound reason.***

Question 6 [Total: 10 marks]**(a) [2 marks]**

- ****Balls of different sizes**** (or moulded clay, paper mâché, calabash pieces) - to ****represent the four planets and the Sun****.
- ****Wire, string or thin sticks and a cardboard or wooden base**** - to ****hold the planets in position**** at their different distances; ****paint and paper labels**** to colour and name them.
- ***Marking note: 1 mark for each material correctly named WITH its use, maximum 2 marks. A material named without a use earns no mark. Accept any local material that would really work.***

(b) [4 marks]

- ****Find out the correct order, relative sizes and distances**** of Jupiter, Saturn, Uranus and Neptune from books or the internet.
- ****Make or choose balls of the right relative sizes**** - the biggest for Jupiter, a slightly smaller one for Saturn, and two smaller ones for Uranus and Neptune.
- ****Paint each planet in its colour**** and add a card ring round Saturn.
- ****Fix the planets with wire or string to the base in the correct order****, spacing them out from the Sun, then ****label each one, write a title and display the model**** for the class to discuss.
- ***Marking note: 1 mark each for any FOU**

(c) [2 marks]

- The ****correct order**** of the planets from the Sun, with each in ****its own separate orbit****.
- The ****relative sizes**** - Jupiter the largest, then Saturn, with Uranus and Neptune much smaller - and the ****rings round Saturn****.
- ***Marking note: 1 mark each for any TWO correct points (2 marks).***

(d) [2 marks]

- The **sizes and the distances are not to true scale** - the real distances are far too vast to show on a table.
- The planets in the model are **held by wire and do not move**, whereas the real planets **move freely in orbit and rotate**; the model also cannot show the gases, the storms, the many moons or the extreme cold.

- **Marking note:** 1 mark each for any TWO correct limitations (2 marks). This item rewards candidates who understand that a model is only a representation; accept any sound limitation.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 3: Systems | **Sub-strand:** 2:

Waste Management

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	19.1 Waste and the Types of Waste	Remember
2	C	19.1 Waste and the Types of Waste	Remember
3	D	19.1 Waste and the Types of Waste	Remember
4	B	19.1 Waste and the Types of Waste	Remember
5	D	19.1 Waste and the Types of Waste	Remember
6	B	19.1 Waste and the Types of Waste	Understand
7	A	19.1 Waste and the Types of Waste	Understand
8	C	19.1 Waste and the Types of Waste	Remember
9	D	19.1 Waste and the Types of Waste	Remember
10	D	19.1 Waste and the Types of Waste	Understand
11	A	19.1 Waste and the Types of Waste	Remember
12	B	19.2 Sustainable Approaches to Waste Manag...	Understand
13	B	19.2 Sustainable Approaches to Waste Manag...	Understand
14	D	19.2 Sustainable Approaches to Waste Manag...	Understand
15	C	19.2 Sustainable Approaches to Waste Manag...	Remember
16	B	19.2 Sustainable Approaches to Waste Manag...	Understand
17	D	19.2 Sustainable Approaches to Waste Manag...	Remember
18	C	19.2 Sustainable Approaches to Waste Manag...	Understand
19	D	19.2 Sustainable Approaches to Waste Manag...	Understand
20	B	19.2 Sustainable Approaches to Waste Manag...	Understand
21	C	19.3 Effects of Poor Waste Management on H...	Understand
22	A	19.3 Effects of Poor Waste Management on H...	Understand
23	C	19.3 Effects of Poor Waste Management on H...	Understand
24	B	19.3 Effects of Poor Waste Management on H...	Understand
25	A	19.3 Effects of Poor Waste Management on H...	Understand
26	B	19.3 Effects of Poor Waste Management on H...	Understand
27	C	19.3 Effects of Poor Waste Management on H...	Analyse
28	A	19.2 Sustainable Approaches to Waste Manag...	Understand
29	A	19.2 Sustainable Approaches to Waste Manag...	Evaluate
30	B	19.2 Sustainable Approaches to Waste Manag...	Apply
31	D	19.4 Surveying and Reporting on a Communit...	Apply
32	C	19.4 Surveying and Reporting on a Communit...	Apply
33	A	19.4 Surveying and Reporting on a Communit...	Apply
34	D	19.4 Surveying and Reporting on a Communit...	Understand
35	A	19.4 Surveying and Reporting on a Communit...	Evaluate
36	B	19.5 Managing and Evaluating Waste Managem...	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	A	19.5 Managing and Evaluating Waste Managem...	Evaluate
38	A	19.5 Managing and Evaluating Waste Managem...	Remember
39	D	19.5 Managing and Evaluating Waste Managem...	Evaluate
40	C	19.5 Managing and Evaluating Waste Managem...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 1 (2%) | Apply: 5 (12%) | Evaluate: 5 (12%) | Remember: 11 (27%) | Understand: 18 (45%)

Waste**SECTION B: THEORY — MARKING SCHEME****Question 1 [Total: 9 marks]****(a) [2 marks]**

- **Waste** is **any material that is no longer wanted or useful to the person or place that produced it** (1 mark).
- **Waste management** is the **collection, sorting, treatment and safe disposal of waste**, including reducing, reusing and recycling it (1 mark).
- **Marking note: 2 marks, 1 mark each.**

(b) [4 marks]

- **Solid waste** - plastic bottles, sachet-water bags, broken bottles, sweepings.
- **Liquid waste** - bath water, water from the kitchen sink, waste water from a factory.
- **Gaseous waste** - smoke from a chimney, exhaust fumes from a vehicle.
- **Medical waste** - used syringes, bandages and expired drugs (accept domestic, agricultural, industrial or electronic waste with a correct example).
- **Marking note: 1 mark each for any FOUR types, each with a correct example (4 marks). A type named with no example, or with a wrong example, scores nothing for that item. Candidates may use either**

(c) [3 marks]

- To **keep the community clean and free from offensive smells**, and to **prevent diseases** such as cholera, typhoid and malaria.
- To **stop gutters from choking and causing floods**, and to **protect the soil, the streams and the air from pollution**.
- To **recover useful materials and manure** from the waste, and to keep the community attractive to visitors and to business.
- **Marking note: 1 mark each for any THREE sound reasons (3 marks).**

Question 2 [Total: 12 marks]**(a) [5 marks]**

- **A - reduce** (prevention: avoid producing the waste in the first place) (1 mark).
- **B - reuse** (1 mark).
- **C - recycle** (1 mark).
- **D - recover** - composting, or recovering energy from the waste (1 mark).
- **E - disposal** - landfill or incineration (1 mark).
- **Marking note: 1 mark each (5 marks). The order must be as shown. Accept 'refuse' or 'prevent' for A, 'composting' for D and 'dumping in a sanitary landfill'**

for E.*

(b) [2 marks]

- **Waste that is never produced does not have to be collected, treated or disposed of at all**, so no land, fuel or money is spent on it and it cannot pollute the environment (1 mark).
- It also **saves the raw materials and the energy** that would have been used to make the goods in the first place (1 mark).
- **Marking note:** 2 marks, 1 mark each. A candidate who merely writes 'because it is at the top' scores 0.*

(c) [3 marks]

- **A (reduce):** **carry a basket to the market instead of collecting plastic bags**, or write and photocopy on both sides of the paper (1 mark).
- **B (reuse):** **wash a glass bottle and use it again to store drinking water**, or use empty tins as pots for seedlings in the school garden (1 mark).
- **C (recycle):** **collect plastic bottles and sachet bags and send them to a recycling factory** to be made into new plastic items (1 mark).
- **Marking note:** 1 mark each (3 marks). The example must match the level: a candidate who gives recycling as an example of reuse scores 0 for that mark.*

(d) [2 marks]

- **Nothing useful is recovered** - the materials, the plant nutrients and the energy in the waste are all lost (1 mark).
- It **takes up land**, and dumps and open burning **pollute the soil, the water and the air and breed flies, rats and mosquitoes** (1 mark).
- **Marking note:** 1 mark each for any TWO sound reasons (2 marks).*

Question 3 [Total: 10 marks]

(a) [4 marks]

- **Diseases such as cholera, typhoid and dysentery spread**, because flies settle on the refuse and then on food, and dirty water enters the wells and streams.
- **Mosquitoes breed in the stagnant water** held in discarded tins and containers and spread **malaria**; rats and cockroaches also multiply.
- **Gutters choke with plastics and the town floods** when it rains.
- **The air, the soil and the streams are polluted** by smoke, bad smells and the liquid that drains from the dump; **animals swallow plastics and die**, and the community looks dirty.
- **Marking note:** 1 mark each for any FOU

(b) [3 marks]

- **Plastics do not rot**, so they **collect in the gutters and block them** (1 mark).
- When it rains the **water cannot flow away**, so it overflows and floods the

streets and houses** (1 mark).

- The **flood water carries rubbish and human waste into homes and wells, spreading cholera and typhoid**, and the **pools left behind breed mosquitoes that spread malaria** (1 mark).

- *Marking note: 3 marks as shown - blocking, flooding and the spread of disease. A candidate who stops at the flooding scores 2 marks.*

(c) [3 marks]

- Open burning gives off **thick smoke, soot, carbon dioxide and poisonous gases**, especially when **plastics and rubber** are burnt (1 mark).

- The smoke **pollutes the air**, so the air is dirty to breathe and visibility is poor (1 mark).

- People living nearby suffer **coughing, catarrh, sore eyes and chest diseases such as asthma and bronchitis**, and the fire may spread and destroy property (1 mark).

- *Marking note: 3 marks as shown; 1 mark each for the gases, the pollution and the effect on health.*

Question 4 [Total: 10 marks]

(a) [1 marks]

- **Taking the waste to a public container**, used by **80 households**.

- *Marking note: 1 mark.*

(b) [2 marks]

- **36 + 24** (1 mark)

- **= 60 households** (1 mark)

- *Marking note: 2 marks: 1 mark for adding the two correct values and 1 mark for the answer.*

(c) [2 marks]

- **20 ÷ 200 × 100** (1 mark)

- **= 10%** (1 mark)

- *Marking note: 2 marks: 1 mark for the correct method and 1 mark for the answer with the per cent sign.*

(d) [2 marks]

- **Burning** - it gives off **smoke and poisonous gases that pollute the air** and cause chest diseases (1 mark).

- **Dumping in a gutter or the bush** - it **chokes the gutters and causes floods**, breeds flies and mosquitoes and litters the land (1 mark). (Accept **burying** in an unprepared pit, since plastics do not rot and the drainage can pollute the groundwater.)

- *Marking note: 2 marks: 1 mark for each harmful method correctly named

WITH its reason. A method with no reason scores nothing.*

(e) [3 marks]

- ****Provide more public containers and have them emptied regularly****, and extend the collection service to more houses.
- ****Teach households to separate their waste**** and to ****compost the biodegradable part**** for the farms and gardens, while plastics and metals are collected for recycling.
- ****Stop open burning and dumping in gutters**** through education and the enforcement of the sanitation bye-laws, supported by a regular community clean-up day.
- ***Marking note: 1 mark each for any THREE workable recommendations (3 marks). The recommendation must follow from the survey results; a general st**

Question 5 [Total: 9 marks]

(a) [4 marks]

- ****Find out what waste the school produces and where it comes from****, then ****agree on the aim and plan the activity with the head teacher**** (1 mark).
- ****Provide and clearly label separate bins**** for food waste, paper, plastics and metals, and ****place them at the classrooms, the kitchen and the canteen**** (1 mark).
- ****Educate the pupils at assembly**** on which waste goes into which bin, and ****appoint a sanitation prefect or duty group**** for each week (1 mark).
- ****Empty the bins regularly - compost the food waste in the school garden and hand the plastics and metals to the recyclers**** - and ke

(b) [3 marks]

- ****Compare the compound before and after**** - count the heaps of litter or photograph the same spots each week.
- ****Measure the waste****: see whether the ****number of bags sent to the dump each week has fallen****, and how much ****compost and how many sacks of plastics have been recovered****.
- ****Check whether the bins are being used correctly**** and ****ask the pupils, teachers and cleaners for their views****; look also at whether the cost of waste collection has fallen.
- ***Marking note: 1 mark each for any THREE genuine ways of evaluating (3 marks). The way must allow a comparison or a measurement;**

(c) [2 marks]

- ****Talk about the project at assembly and at the P.T.A. meeting, and put up posters and slogans**** on the notice board; form a ****sanitation club**** and reward the cleanest class each month.
- ****Invite the chief, the assembly member and the environmental health officer****, hold a ****community clean-up day****, and spread the message through the

information centre, the local radio station or social media.

- *Marking note: 1 mark each for any TWO practical ways (2 marks).*

Question 6 [Total: 10 marks]

(a) [4 marks]

- **Choose a shaded corner of the garden and clear a space or dig a shallow pit** (1 mark).
- **Collect biodegradable waste** - kitchen peelings, vegetable remains, grass, leaves and animal droppings - and **build the heap in layers**: plant waste, then droppings or old manure, then a thin layer of topsoil, until it is about a metre high (1 mark).
- **Water the heap lightly and cover it** with leaves, grass or an old sack to keep in the moisture and the heat (1 mark).
- **Turn the heap every two weeks** to let in air; after about **two to three months** the compost is **dark, crumbly and sweet

(b) [2 marks]

- Composting **turns the waste into free manure that adds nutrients and humus to the soil** and saves the money spent on fertiliser, while burning destroys it all.
- Composting **produces no smoke**, so it **does not pollute the air or harm health**, and it keeps the waste out of the gutters and off the dump.
- *Marking note: 1 mark each for any TWO advantages (2 marks).*

(c) [2 marks]

- **Plastics, glass, tins and other metals** - they are **non-biodegradable, so they do not rot** and would remain in the compost and in the soil (1 mark).
- **Medical waste, batteries and other chemical waste** (accept diseased plants or meat scraps) - they are **dangerous, poison the soil, and attract rats and flies** (1 mark).
- *Marking note: 2 marks: 1 mark for each kind of waste correctly named WITH a reason.*

(d) [2 marks]

- **Wash and reuse the bottles and containers** to store water, kerosene or seeds, or cut them open to serve as **pots for seedlings in the school garden**.
- **Collect and sell the sachet bags and bottles to a recycling company**, where they are made into new bottles, buckets, chairs, roofing sheets or paving blocks (accept: make bags, mats and craft items from clean sachet bags).
- *Marking note: 1 mark each for any TWO practical uses (2 marks).*

CURRICULUM ASSESSMENT BLUEPRINT

****Strand:**** 5: Humans and the Environment | ****Sub-strand:**** 1: Waste Management | ****Class:**** Basic 8 (J



Integrated Science Glossary — Basic 8

This glossary contains key terms and definitions that appear throughout this book. Use it as a quick reference while studying and preparing for examinations.

Acceleration

The rate of change of velocity. Measured in meters per second squared (m/s^2).

Alloy

A mixture of two or more metals, or a metal and another element.

Ampere

The unit of electric current. One ampere equals one coulomb per second.

Antibody

A protein produced by the immune system to fight infection.

Atmosphere

The layer of gases surrounding the earth.

Bacteria

Single-celled microorganisms that can be beneficial or harmful.

Balanced Diet

A diet containing all essential nutrients in the right proportions.

Blood

The fluid that circulates in the body, carrying oxygen, nutrients, and waste products.

Boiling Point

The temperature at which a liquid changes to a gas. Water boils at 100°C at standard pressure.

Carbohydrate

An organic compound that provides energy to living organisms.

Catalyst

A substance that speeds up a chemical reaction without being consumed.

Circuit

A closed path through which electric current flows.

Conductor

A material that allows electricity or heat to flow through it easily.

Contamination

The presence of unwanted substances in a material or environment.

Corrosion

The gradual destruction of metals by chemical reaction with the environment.

Current

The flow of electric charge through a conductor. Measured in amperes (A).

Density

Mass per unit volume. Measured in kilograms per cubic meter (kg/m³).

Disease

A disorder of structure or function in a living organism.

Dissolving

The process by which a solute forms a solution with a solvent.

Echo

A reflected sound wave.

Electrolysis

The decomposition of a substance using electric current.

Electron

A negatively charged particle that orbits the nucleus of an atom.

Enzyme

A biological catalyst that speeds up chemical reactions in living organisms.

Evolution

The gradual change in species over many generations.

Fat

An organic compound that stores energy and insulates the body.

Food Chain

A sequence showing how energy flows from one organism to another.

Frequency

The number of vibrations per second. Measured in hertz (Hz).

Fungus

A group of organisms including mushrooms, molds, and yeasts.

Global Warming

The long-term increase in Earth's average temperature.

Greenhouse Effect

The trapping of heat in the atmosphere by greenhouse gases.

Hemoglobin

The protein in red blood cells that carries oxygen.

Hereditary

Characteristics passed from parents to offspring through genes.

Immunity

The ability of the body to resist and fight disease.

Insulator

A material that does not allow electricity or heat to flow through it easily.

Ion

An atom or molecule with a positive or negative electric charge.

Light

Electromagnetic radiation that can be seen by the human eye.

Magnetic Field

The region around a magnet where magnetic forces can be detected.

Melting Point

The temperature at which a solid changes to a liquid. Ice melts at 0°C.

Metamorphism

The change in the structure of a rock due to heat and pressure.

Mineral

A naturally occurring inorganic solid with a specific chemical composition.

Molecule

The smallest unit of a compound that retains its chemical properties.

Newton

The unit of force. One newton accelerates one kilogram at one meter per second squared.

Nutrition

The process by which organisms obtain and use food for growth and maintenance.

Oxidation

A chemical reaction involving the loss of electrons or addition of oxygen.

Ozone

A molecule made of three oxygen atoms (O₃) that protects Earth from UV radiation.

Parasite

An organism that lives on or in another organism and benefits at the host's expense.

pH Scale

A scale from 0 to 14 that measures how acidic or basic a substance is.

Potential Energy

Energy stored in an object due to its position or state.

Power

The rate at which work is done. Measured in watts (W).

Pressure

Force per unit area. Measured in pascals (Pa) or newtons per square meter (N/m²).

Protein

An organic compound made of amino acids, essential for growth and repair.

Radiation

Energy transmitted in the form of waves or particles.

Reflection

The bouncing back of light, sound, or other waves from a surface.

Refraction

The bending of light as it passes from one medium to another.

Reproduction

The biological process by which organisms produce offspring.

Resistance

The opposition to the flow of electric current. Measured in ohms (Ω).

Rock Cycle

The continuous process by which rocks change from one type to another.

Salt

An ionic compound formed when an acid reacts with a base.

Satellite

An object that orbits a planet or other celestial body.

Sound

A form of energy produced by vibrations and transmitted through a medium.

Speed

The distance traveled per unit time. Measured in meters per second (m/s).

Static Electricity

Electric charge that builds up on the surface of an object.

Sublimation

The change from a solid directly to a gas without passing through the liquid state.

Sustainable

Meeting present needs without compromising the ability of future generations to meet their needs.

Temperature

A measure of the average kinetic energy of particles. Measured in °C or K.

Thermal Energy

The total kinetic energy of particles in a substance.

Thermometer

An instrument used to measure temperature.

Tissue

A group of similar cells that work together to perform a specific function.

Transformer

A device that changes the voltage of alternating current.

Transpiration

The loss of water vapor from plant leaves.

Ultrasound

Sound waves with frequencies above the human hearing range (above 20,000 Hz).

Vaccine

A substance that stimulates the immune system to produce immunity to a disease.

Velocity

Speed in a specific direction. Measured in meters per second (m/s).

Virus

A microscopic infectious agent that can only replicate inside living cells.

Vitamin

An organic compound required in small amounts for normal body function.

Voltage

The electrical potential difference between two points. Measured in volts (V).

Wavelength

The distance between successive crests of a wave. Measured in meters (m).

Weight

The force of gravity acting on an object. Measured in newtons (N).

Curriculum Reference

This book aligns with the following curriculum standards:

- Ghana Education Service (GES)
Standards-Based Curriculum for Junior High Schools (2019)
- National Council for Curriculum and Assessment (NaCCA)
Curriculum Framework for Basic Education
- West African Examinations Council (WAEC)
BECE Syllabus for Integrated Science / Mathematics

The content has been reviewed to ensure alignment with:

- Strand and sub-strand organization
- Learning indicators and objectives
- Cognitive levels (Knowledge, Understanding, Application, Analysis)
- Core competencies (Critical Thinking, Problem Solving, Communication)

Teachers and students should use this book in conjunction with the official curriculum documents available from the Ghana Education Service.

For more information:

Ghana Education Service: www.ges.gov.gh

NaCCA: www.nacca.gov.gh

WAEC: www.waec.org

About Beacon Educational Consult

Beacon Educational Consult is a Ghanaian educational publishing company dedicated to producing high-quality learning materials for Junior High School students across Ghana.

Our mission is to support students in achieving academic excellence through comprehensive, curriculum-aligned resources that prepare them for the Basic Education Certificate Examination (BECE) and beyond.

We believe that every student deserves access to excellent educational materials, and we are committed to making quality textbooks affordable and accessible to schools and families throughout Ghana.

Our books are:

- Written by experienced Ghanaian educators
- Aligned with the national curriculum
- Reviewed for accuracy and quality
- Designed to support both classroom learning and independent study

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